Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 — Functional materials and green chemistry · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26CY151A · Unit_04: Functional materials and green chemistry
2. 2 · Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 2)
3. 3 · Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1)
4. 4 · Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials
5. 5 · Polymers: Introduction, types of polymerizations, definition of glass transition temperature
6. 6 · Structure-property relationship
7. 7 · Synthesis and properties of epoxy resin
8. 8 · Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1)
9. 9 · Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2))
10. 10 · Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same
11. 11 · Conducting polymers - examples, types of doping
12. 12 · Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism
13. 13 · Green chemistry: Introduction, twelve principles of green chemistry with examples
14. 14 · Source slide deck: Commercial polymers synthesis and properties of Plexiglass
15. 15 · Source slide deck: Experiential learning Based on nanomaterials
16. 16 · Historical question-source pages · unit mapping candidates
17. Appendix A · Symbol glossary
18. Appendix B · Sources, coverage and gaps
19. Appendix C · Revision and retrieval practice

M1

Unit_04 · Functional materials and green chemistry

1 UE26CY151A · Unit_04: Functional materials and green chemistry

This unit master organizes source material for Engineering Chemistry, Unit_04, officially titled Functional materials and green chemistry. It includes 12 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26CY151A, Unit_04 (Functional materials and green chemistry). Bucket folder present: yes. Videos: 12; slide decks: 12.

EV0000001

2 Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 2)

This lesson is indexed from the source video titled Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 2). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000002 · EV0000003

Graphs and diagrams

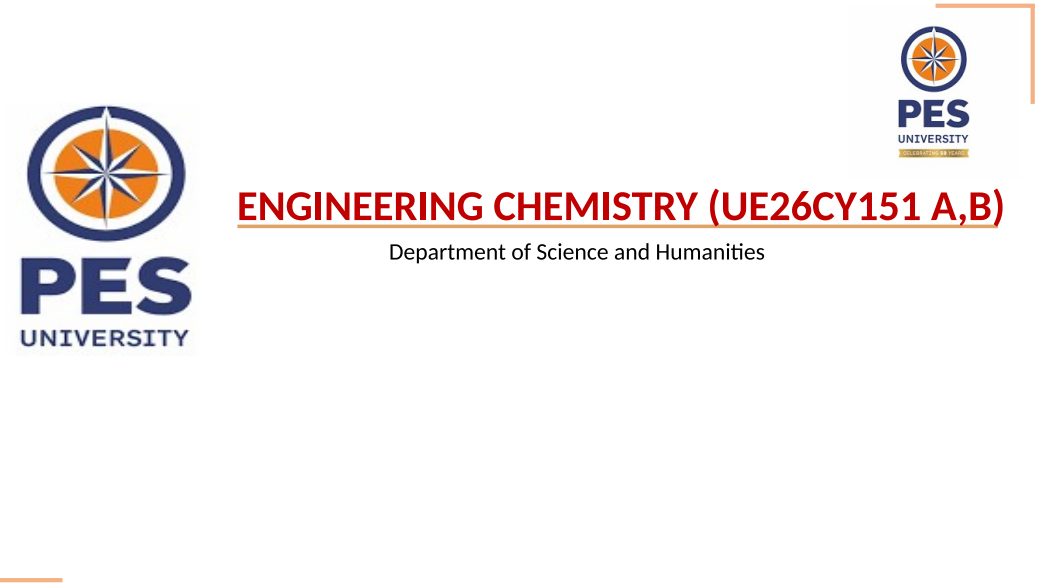


Figure G000001. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 1.

EV0000195

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

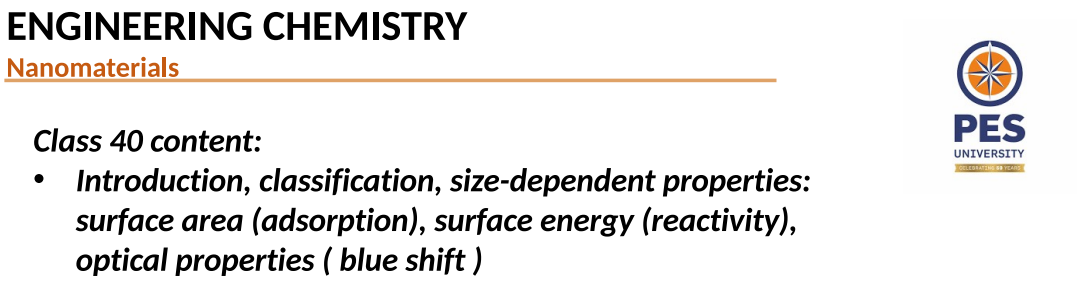


Figure G000002. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 2.

EV0000198

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

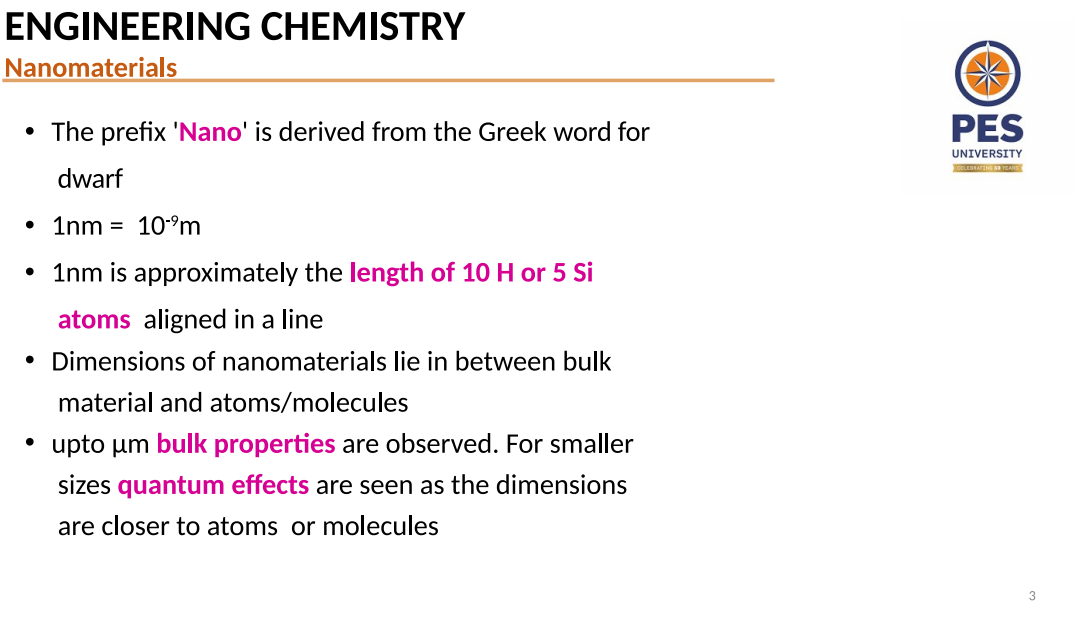


Figure G000003. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 3.

EV0000201

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

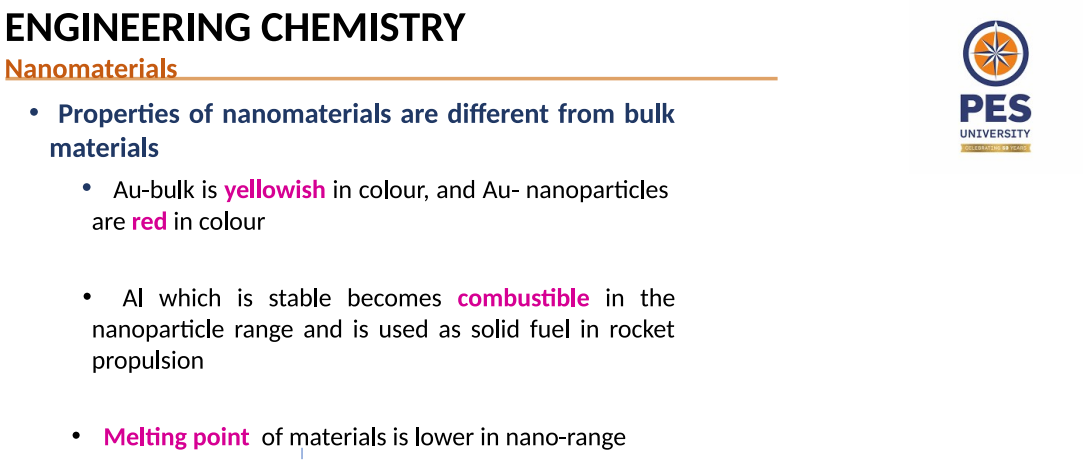


Figure G000004. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 4.

EV0000204

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

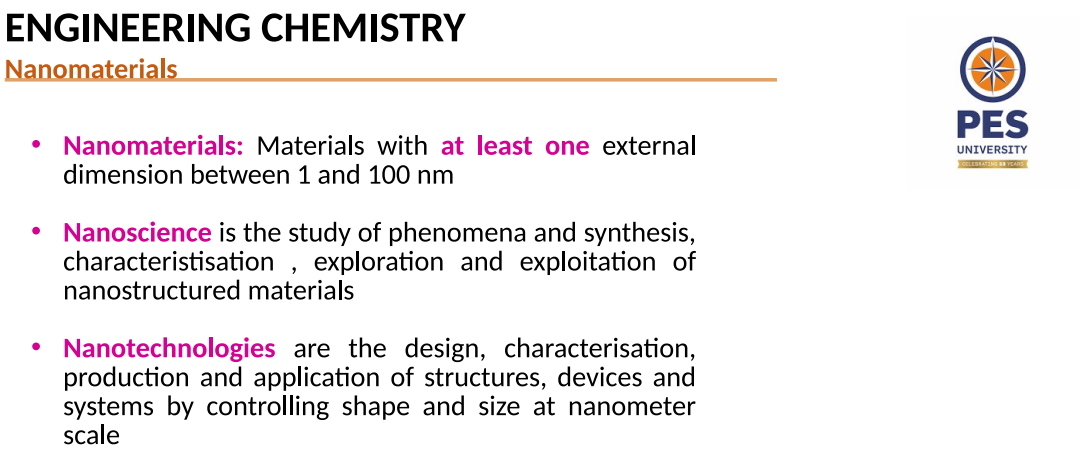


Figure G000005. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 5.

EV0000207

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

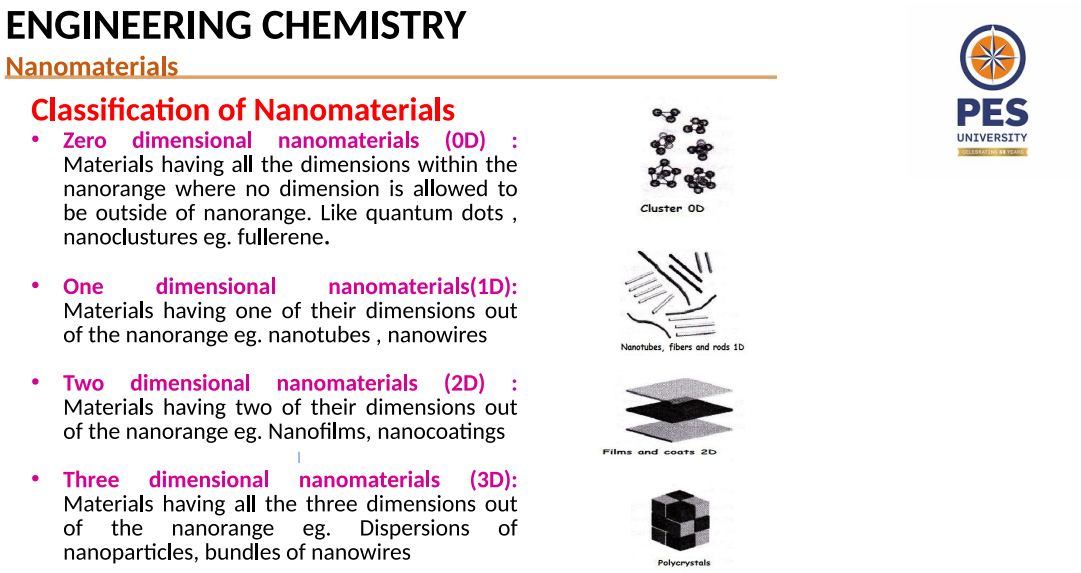


Figure G000006. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 6.

EV0000210

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Nanomaterials

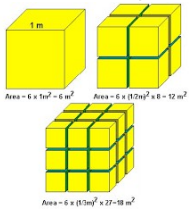
Nanomaterials possess specific properties which maybe significantly different from bulk properties

Reasons

Large fraction of surface atoms / unit volume:

Properties are dictated by surface atoms rather than bulk atoms

Eg Pt sheet versus Pt powder



Source:https://nanosense.sri.com/activities/sizematters/introduction/SM_Lesson1Student.pdf

Large surface energy : Surface atoms are held with less force as compared to bulk. Surface activity high .Eg Al

Figure G000007. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 7.

EV0000213

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

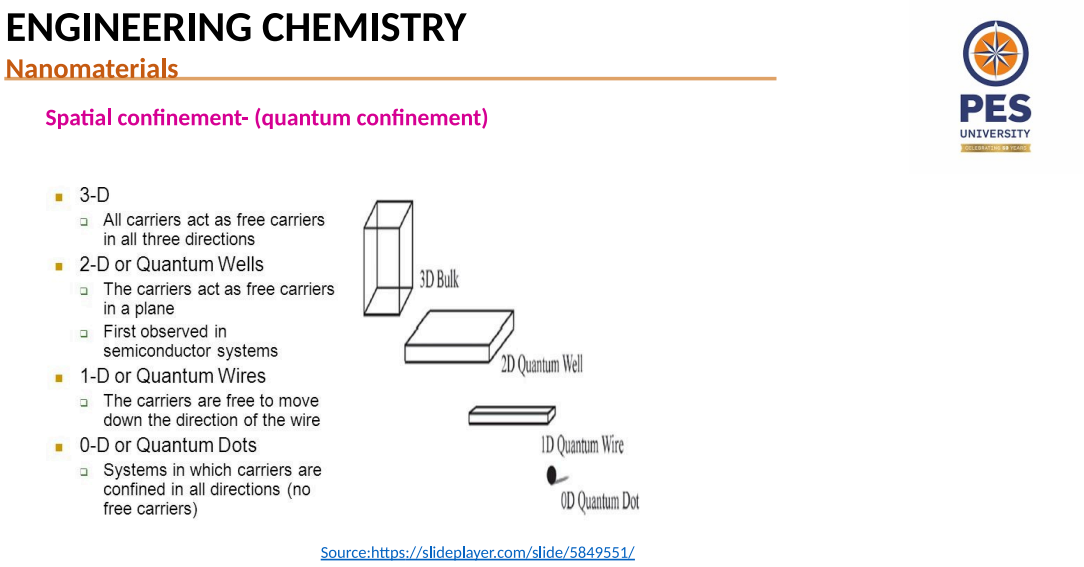


Figure G000008. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 8.

EV0000216

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

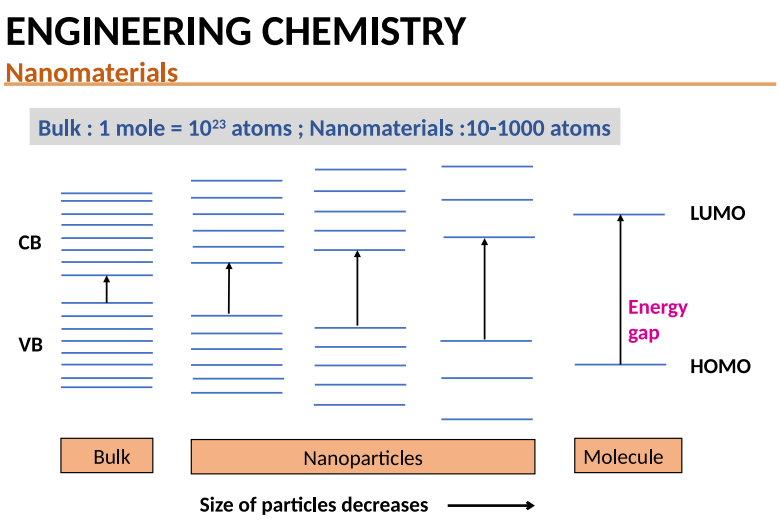


Figure G000009. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 9.

EV0000219

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

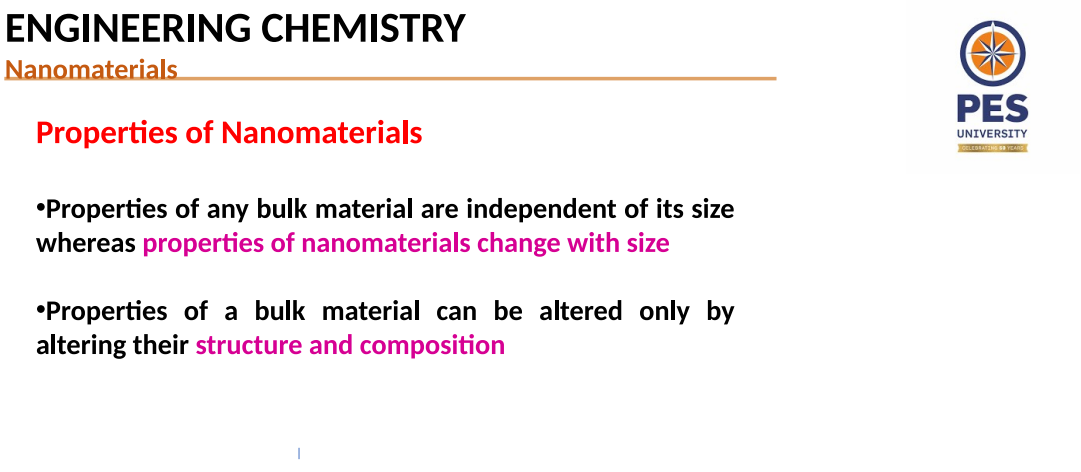


Figure G000010. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 10.

EV0000222

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

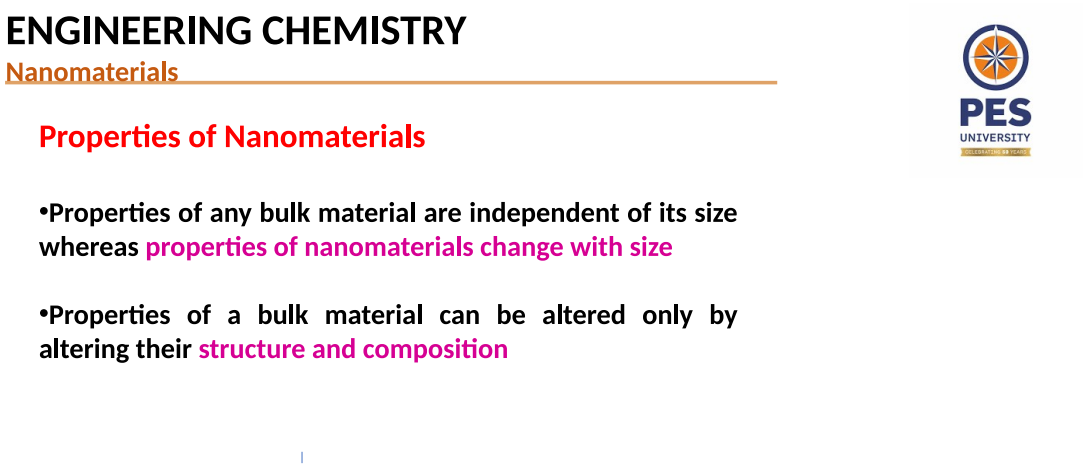


Figure G000011. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 11.

EV0000225

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

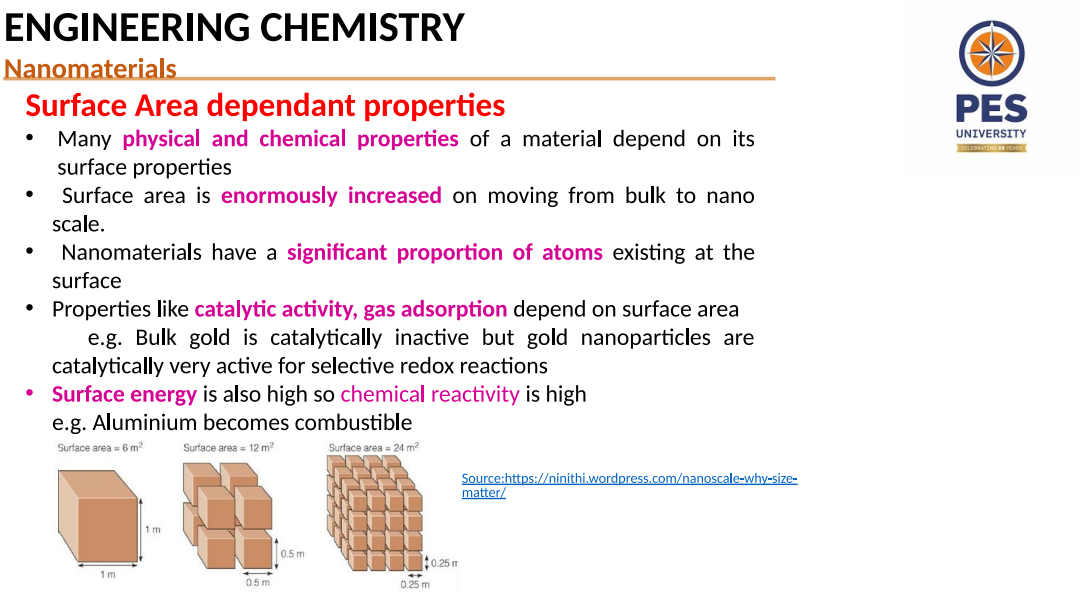


Figure G000012. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 12.

EV0000228

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

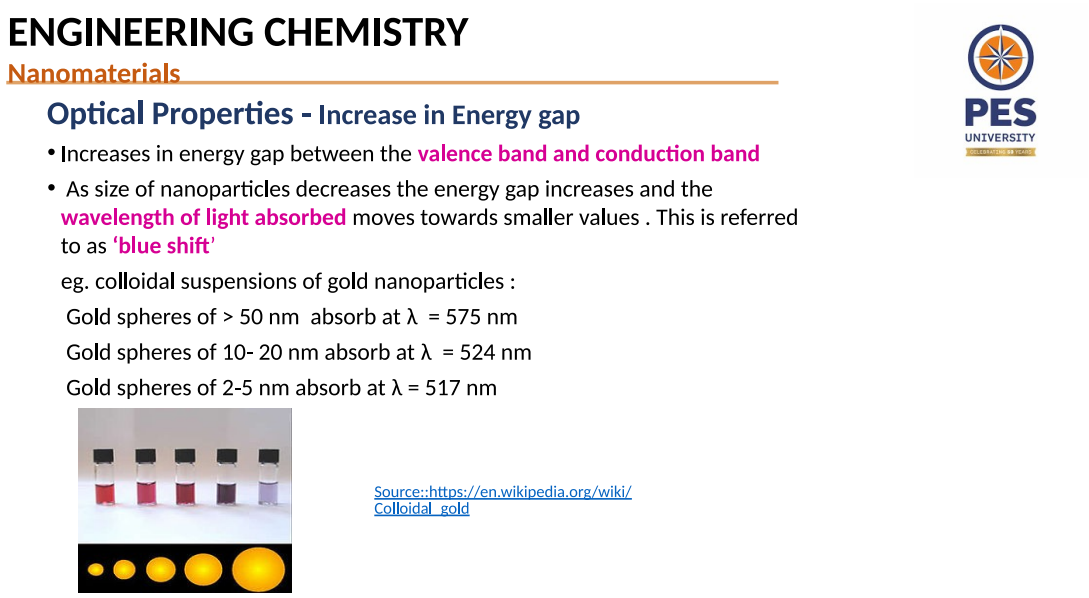


Figure G000013. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 13.

EV0000231

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000014. Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea · source slide page 14.

EV0000234

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000194 · EV0000195 · EV0000196
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Class 40 content: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties (blue shift) EV0000197 · EV0000198 · EV0000199
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Nanomaterials The prefix 'Nano' is derived from the Greek word for dwarf 1nm = 10-9m 1nm is approximately the length of 10 H or 5 Si atoms aligned in a line Dimensions of nanomaterials lie in between bulk material and atoms/molecules upto μm bulk properties are observed. For smaller sizes quantum effects are seen as the dimensions are closer to atoms or molecules 3 EV0000200 · EV0000201 · EV0000202
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Properties of nanomaterials are different from bulk materials Au-bulk is yellowish in colour, and Au- nanoparticles are red in colour Al which is stable becomes combustible in the nanoparticle range and is used as solid fuel in rocket propulsion Melting point of materials is lower in nano-range EV0000203 · EV0000204 · EV0000205

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Nanomaterials: Materials with at least one external dimension between 1 and 100 nm Nanoscience is the study of phenomena and synthesis, characteristisation , exploration and exploitation of nanostructured materials Nanotechnologies are the design, characterisation, production and application of structures, devices and systems by controlling shape and size at nanometer scale EV0000206 · EV0000207 · EV0000208
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Classification of Nanomaterials Zero dimensional nanomaterials (0D) : Materials having all the dimensions within the nanorange where no dimension is allowed to be outside of nanorange. Like quantum dots , nanoclustures eg. fullerene. One dimensional nanomaterials(1D): Materials having one of their dimensions out of the nanorange eg. nanotubes , nanowires Two dimensional nanomaterials (2D) : Materials having two of their dimensions out of the nanorange eg. Nanofilms, nanocoatings Three dimensional nanomaterials (3D): Materials having all the three dimensions out of the nanorange eg. Dispersions of nanoparticles, bundles of nanowires EV0000209 · EV0000210 · EV0000211
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Nanomaterials possess specific properties which maybe significantly different from bulk properties Reasons Large fraction of surface atoms / unit volume: Properties are dictated by surface atoms rather than bulk atoms Eg Pt sheet versus Pt powder
Source:https://nanosense.sri.com/activities/sizematters/introduction/SM_Lesson1Student.pdf Large surface energy : Surface atoms are held with less force as compared to bulk. Surface activity high .Eg Al EV0000212 · EV0000213 · EV0000214
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Spatial confinement- (quantum confinement) [Image text, OCR] Quantum Confinement 3-D All carriers actasfree carriers inallthreedirections 2-DorQuantumWells 3DBulk Thecarriersactasfreecarriers inaplane First observed in semiconductorsystems 2DQuantum Well 1-DorQuantumWires Thecarriersarefreetomove downthedirectionofthewire O-DorQuantumDots IDQuantumWire Systemsinwhichcarriersare confined in all directions(no OD Quantum Dot free carriers)
Source:https://slideplayer.com/slide/5849551/ EV0000215 · EV0000216 · EV0000217
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Bulk : 1 mole = 1023 atoms ; Nanomaterials :10-1000 atoms LUMO HOMO CB VB Energy gap Bulk Molecule Nanoparticles Size of particles decreases EV0000218 · EV0000219 · EV0000220
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Properties of Nanomaterials Properties of any bulk material are independent of its size whereas properties of nanomaterials change with size Properties of a bulk material can be altered only by altering their structure and composition EV0000221 · EV0000222 · EV0000223
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Properties of Nanomaterials Properties of any bulk material are independent of its size whereas properties of nanomaterials change with size Properties of a bulk material can be altered only by altering their structure and composition EV0000224 · EV0000225 · EV0000226

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Surface Area dependant properties Many physical and chemical properties of a material depend on its surface properties Surface area is enormously increased on moving from bulk to nano scale. Nanomaterials have a significant proportion of atoms existing at the surface Properties like catalytic activity, gas adsorption depend on surface area e.g. Bulk gold is catalytically inactive but gold nanoparticles are catalytically very active for selective redox reactions Surface energy is also high so chemical reactivity is high e.g. Aluminium becomes combustible [Image text, OCR] Surfacearea=6m² Surfacearea=12m² Surfacearea=24m² 1m 0.5m 0.25m 1m 0.5m 0.25m Source:https://ninithi.wordpress.com/nanoscale-why-size-matter/ EV0000227 · EV0000228 · EV0000229
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Optical Properties - Increase in Energy gap Increases in energy gap between the valence band and conduction band As size of nanoparticles decreases the energy gap increases and the wavelength of light absorbed moves towards smaller values . This is referred to as 'blue shift' eg. colloidal suspensions of gold nanoparticles : Gold spheres of > 50 nm absorb at λ = 575 nm Gold spheres of 10- 20 nm absorb at λ = 524 nm Gold spheres of 2-5 nm absorb at λ = 517 nm Source::https://en.wikipedia.org/wiki/Colloidal_gold EV0000230 · EV0000231 · EV0000232
14. Source slide page 14 (extracted text): THANK YOU Department of Science and Humanities EV0000233 · EV0000234 · EV0000235

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.388] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module two on nanomaterials and in this class we are going to talk about properties of nanomaterials, surface area dependent properties, electrical properties and optical properties. [00:00:19.044] Properties of nanomaterials are very different from that of the bulk materials. We find that for bulk material the properties are independent of size whereas when we come to nanomaterials they change with size. For example, if we have a block of gold [00:00:36.816] Then it is it has a particular colour that is uh called golden colour. Now even if you make it into smaller and smaller size, like for example, the block of gold which was say hundred grams, now we make it into a five gram piece, by breaking it into smaller pieces, we find that. [00:00:54.384] There is no change in the colour, it still continues to remain in that golden colour. But that is not true when we come to the nanoparticle range. Gold, when we get it into the nanoparticle range, we find that as we decrease the size, the colour of gold also. [00:01:11.952] So changes. So what does it tell us? That the optical properties are getting altered with the dimension of the particles. So this [00:01:23.408] Is what happens for all most of the properties of nanomaterials. They change with size. Now, if at all we want to change the bulk material properties, it can be done only if we change the structure and the composition. Whereas in nanomaterials, the material remains the same. [00:01:41.104] The structure composition remains the same, it's only the size which is changing, and that can alter the properties. [00:01:49.284] Let us look at the most important property of nanomaterials, the surface area. We know that many physical and chemical properties of a material are going to depend on surface properties, and when we come to nanomaterials, the surface area has enormously increased. [00:02:07.972] Where in a bulk, let us look at this picture here. We have a bulk material over here, it is being broken down into smaller particles. We know that the surface area has increased so much over here. Now if you make it further smaller, again we find a large increase in nano in the [00:02:25.808] The surface

area. Now if you go to nanometer range, then definitely the surface area is going to be very high. And we know that since in nanomaterials the surface atoms are much more than what the atoms at inside the bulk of the material and therefore [00:02:45.808] All the properties are mostly dictated by the surface atoms. And since we have such a large surface area, such a large number of surface atoms, therefore the properties which depend on surface area definitely get affected when you take. [00:03:04.832] the particle when take the material from bulk to nanocyte. So properties like catalytic activity we know that whenever we use a catalyst we prefer to take it in the powder form because we want to increase the surface area and catalytic catalysis is a surface phenomena. So definitely [00:03:22.480] If we take it in nano range, nano particle range, the catalytic activity is going to increase enormously. Similarly, gaps gas absorption also depends on surface area. For example, bulk gold is found to be catalytically inactive, but gold nanoparticles. [00:03:40.096] Are found to be catalytically very active for selective redox reactions. Now, another important fact which we discussed. [00:03:49.508] In the previous class, that the surface energy of these nanoparticles is very high. Why? Because there are a lot of surface atoms which are held much less tightly as compared to the bulk atoms. So therefore, the energy of the surface is going to be very [00:04:07.280] very high and every system tries to lower its energy so we find that it can do so by uh by uh the reactivity we find that this gets projected in the high chemical reactivity of the nanoparticles so because of this [00:04:24.848] high surface energy, the chemical reactivity is also very high, and this uh I have quoted before that aluminium, though it is in bulk, it is found to be passive, when it comes in the nanoparticle range, it is found to be combustible. [00:04:41.604] Let us talk of the next property that is electrical property. For nanoparticles, it is found that electrical conductivity decreases as compared to the bulk material. The [00:04:53.744] One important reason for that is spatial confinement. We talked about spatial confinement also, and we understood that as we go into the nanoparticle range.

EV0000002 · EV0000003 · EV0000004 · EV0000005

Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.744] The energy levels become more and more discrete, and the band gap increases or the energy gap increases. What are we talking about here? Now when we have metals, we know that the [00:05:20.064] uh valence band and the conductance band are uh overlapping and therefore the [00:05:25.968] Electrons can easily move into the conduction band and we find that metals are good conductors. Whereas in insulators the energy gap is very high and therefore they are not found to conduct electricity. Whereas in semiconductors the band gap is less. [00:05:43.664] Energy gap is less now, and therefore, it is possible that if we give the right kind of energy, the electrical conduction is possible. [00:05:52.272] Now the same semiconductors when we get into the nanoparticle range, as I said, due to spatial confinement, the energy levels become more discrete and the band gap increases. You can see the same energy. [00:06:07.268] Semiconductor now the band gap has become more and obviously when the band gap is more than the conductivance properties are going to get altered. So some metals which are found to be good conductors in bulk now they become semiconductors when you get [00:06:25.040] So, we will get it into the nanoparticle size and whereas some semiconductors, [00:06:30.160] Uh you can see here this the band gap is so high, so obviously the conductance has become less and they can almost be acting like insulators. So this is how the electrical property uh gets affected when we get the particles into the nano range. [00:06:46.948] Another important reason is due to surface scattering. Now, electrical conductivity is also found to decrease due to surface scattering. So let us understand what's happening here. Now electrons have a mean free

part. What is a mean free part for an electron? [00:07:05.984] the distance which the electrons move before they undergo some kind of collusion. Now when we have a bulk material [00:07:16.132] We know that the electron has a mean prepared and there's no obstruction. But when we have the same bulk material into the nanoparticle range, now we know that the nanoparticle has become has a very small dimension. Now it is so small that if the na if the dimension [00:07:33.904] of this nanoparticle is smaller than the mean free part of the electron, then when the electron is moving, obviously it will undergo scattering. [00:07:44.464] Now this scattering could be elastic scattering or inelastic scattering on the surface. So this is called elastic or inelastic surface scattering. Now if it is elastic scattering, then it is just [00:07:58.848] Changing its direction and therefore there is conservation of energy as well as momentum. So there's no uh the uh conductivity does not get affected. Whereas if it is inelastic scattering, then when the electron is scattered in randomly in all directions. [00:08:16.528] We find that the scattered electron now loses loses its uh velocity, and when the electron loses its velocity, it's understood that the [00:08:27.440] Mobility is now less and therefore the electrical conductivity decreases. So for nanoparticles it is found that compared to the bulk, electrical conductivity is less. Coming to the next property that is optical properties. [00:08:44.772] Nanomaterials show unique optical properties. [00:08:49.060] As I mentioned, the colour depends on the size of the particles. In the bulk material, there is no change in colour even if you decrease the size. But when you come into the nano range, it is found that as we change the size of the particles, there is a change in colour. Now this can be [00:09:06.832] explained by two factors. One is due to the surface plasmon resonance and the other is due to increase in band gap. Let's understand what is the surface plasmon resonance. [00:09:23.428] If you look at metals. Now metals have these positive lattice points, and there's a sea of electrons around it. Now these lattice points are fixed and the sea of the electrons are moving. This is the picture of a metal. [00:09:43.108] Now when radiation falls, these surface electrons are getting polarized. What happens is the [00:09:53.824] Electric field which is fluctuating, the electromagnetic radiation which is formed falling on the surface, it has an electric field. Now this electric field is fluctuating, and what happens is the positive ions. Now you can see in the picture here that these positive lattice points.

EV0000006 · EV0000007 · EV0000008 · EV0000009

Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.536] Points. Now the electrons have moved slightly away. Now the electric field is moving this way, and the electrons have moved slightly away from the lattice points, and this is known as polarization, and this happens on the surface. So there is a surface polarization of these electrons. Now [00:10:28.496] Because the lattice point is positive, these is an electron cloud which is getting formed, there is an attraction which is going to pull these electrons back. Now when they come back again, now they will overshoot and then again try to come back. So this kind of an oscillation which sets [00:10:46.128] Up in the in the system. This collective oscillation of these electrons is what is known as the plasmon. So these electrons now, the electron cloud rather is now oscillating with a particular frequency and these. [00:11:03.696] Collective oscillations of these electrons is what is known as plasma. Now these are oscillating with a particular frequency. Now this frequency when it matches with the frequency of the radiation which is falling on the metal, the [00:11:20.452] Then there is going to be resonance. When resonance happens, we know that the radiation is going to be absorbed, and when radiation is absorbed in the visible light, then the material appears to be colored. So plasmon frequency now. [00:11:38.224] This is going to depend on the size, shape and

nature of the metal. [00:11:45.028] Now it is not just the uh just this happens uh because there's an electron cloud and a positive uh metal side. It is not the same for all the metals. It is going to depend on the nature of the metal, it also depends on the size of the particle. [00:12:02.864] Why is it dependent on the size of the particles? Because we know that the electric field there is a polarization. This electric field is definitely going to depend on the size, not only on the size, it also depends on the shape. Whether the particle is spherical or whether it's rod shaped or it's a platelet shape, then definitely the [00:12:20.496] Electric field is going to be different each time, and therefore the plasma frequency is different. And if the plasmon frequency is different, definitely the light absorbed is going to also be different. Therefore, as the size of the particle changes, the color of the [00:12:38.064] The nanoparticle also changes. Another reason is the increase in the energy gap. This is again what we have understood before as spatial confinement or quantum confinement. [00:12:53.184] We know that as we go into the nanoparticle range, when we go to smaller and smaller sizes, the energy levels become more discrete and the energy gap [00:13:05.808] increases. Now if energy gap increases, obviously the wavelength of the light absorbed is going to be different and therefore [00:13:15.280] As we increase as the energy gap goes on increasing, then the the energy required is going to be more, so the wavelength is going to be less. And that is exactly the reason why we find that as the size of the nanoparticles becomes [00:13:32.944] Smaller and smaller, we find a blue shift which happens in the in the particles. Okay. So increase in energy gap is we are talking between the valence band and the conductors band. Now as the size of the nanoparticles. [00:13:50.000] Is decreasing the energy gap increase. This we understood uh in when we were talking about spatial confinement and the wavelength of the light absorbed, as I mentioned just now, goes towards smaller values, and this is referred to as blue shift. Now, if we have color. [00:14:07.664] Suspensions of gold nanoparticles, as you can see here, as the size is increasing, or if we say as the size is decreasing, we find there is a change in color. This is very very characteristic of nanoparticles. So let us take the example of gold. [00:14:25.232] Nanoparticles gold spheres which have greater than 50 nanometers they absorb at a wavelength of 575 nanometers now gold spheres 10 to 20 nanometers absorb at 524 nanometers whereas 2 to 5 nanometers absorb at 517 nanometers what does it mean? [00:14:42.800] The wavelength has become less, that means the frequency is more, that means the energy gap is more. As we decrease the size, we find that. [00:14:54.564] the energy gap is becoming more and more and the light now is now moving the wavelength uh is now moving more towards the uh violet so we call it as the blue shift

EV0000010 · EV0000011 · EV0000012 · EV0000013

Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.988] So in the next class we are going to talk about some more properties of nanomaterials like the mechanical properties, the thermal properties, and magnetic properties. Thank you.

EV0000014 · EV0000015 · EV0000016 · EV0000017

Full transcript: 2244 normalized words, 61 timestamped segments, 923.264 seconds duration. Low-confidence segments flagged: 0.

EV0000002 · EV0000003 · EV0000004 · EV0000005

3 Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface

energy (reactivity), optical properties -blue shift (Part 1)

This lesson is indexed from the source video titled Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000018 · EV0000019

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.900] Hello everyone, welcome back to this course on engineering chemistry. [00:00:06.116] We will start with a new module today on nanomaterials. The content will be nanomaterials, classification and deviation of properties from bulk material, and we will look at different properties of nanoparticles. [00:00:24.388] In today's class we will study of nanomaterials, about nanomaterials, the classification of nanoparticles, and the reasons for deviation in properties of nanomaterials from the bulk material. Let us first understand what we mean by a nanoparticle. [00:00:43.460] The prefix nano is derived from the Greek word dwarf meaning very tiny. Now we know that one nanometer is equal to ten raised to the power of minus nine meters. [00:00:56.496] So how small is one nanometer when we talk in terms of atoms and molecules? One nanometer would be approximately equal when we have ten hydrogen atoms placed one next to the other or five silicon atoms. [00:01:14.624] aligned in a line. So that would be equal to one nanometer. So we can understand that the dimensions of these nanometer particles or nanomaterials are lying between that of the [00:01:29.712] bulk material and atoms and molecules. So when we have one atom, one molecule, compared to that in a nanoparticle we have just about [00:01:39.456] ten hydrogen atoms or five silicon atoms for one nanometer. So we understand that the properties will be more closer to those of atoms and molecules. Now in the [00:01:51.792] Previous module we studied about molecular spectroscopy and we understood that when we study about molecules or atoms we need to use the quantum mechanical treatment. [00:02:02.532] So that is what we notice that up to a micrometer level, when the particle size is up to micrometer, bulk properties are observed. But when the dimension becomes close to the nanometer, then we are going to observe [00:02:21.700] Because the dimension are now, dimensions are closer to those of atoms or molecules. So why so much interest in nanomaterials? [00:02:35.652] because their properties are very very different from that of the bulk materials. For example, gold. All of us know that bulk gold as we see it is yellowish in colour. In fact the a name golden colour has been given to gold. [00:02:52.772] Now the same gold when we take in the form of nanoparticles, then the color changes to red. That means there is some change in the optical property. [00:03:05.380] Aluminium as a second example, which is known to be passive in nature, it becomes combustible in the nanoparticle range. That means the activity has become so high something involved which is [00:03:21.136] passive has become so active that it is combustible to the extent that it is used as a solid fuel in rocket propulsion. [00:03:31.588] Another property of

importance is melting point of the materials when they are taken in the nanoparticle range becomes lower. [00:03:41.504] Now we all understand that melting point is very characteristic of a material. If we want to check purity of the material, we always quote the melting point. [00:03:53.284] Now when the same material is taken in the nano range, we find there is a decrease in the melting point. So we notice that all properties will be different, or rather many of the properties become different when you take particles in the nano range. [00:04:14.212] So first let us define what would be called as a nanomaterial how what would be cla what materials would you classify as a nanomaterial? So materials with at least one of the dimension is between one and hundred nanometers. [00:04:33.988] So at least one of the dimensions, if it is lying within one and hundred nanometers, then that particles, the particle if the particle size is that small then [00:04:46.848] This material will be classified as a nanomaterial. And what would be nano science? It is the study of the phenomena and synthesis, characterization, exploration, and exploitation of these nanostructure materials.

EV0000018 · EV0000019 · EV0000020 · EV0000021

Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.280] Now when we apply this nanosigns [00:05:07.268] to design characterization production and use in structures, devices and systems by controlling the shape and size in the nanomaterial scale, then that is referred to as a nanotechnology. [00:05:25.604] Before studying the reasons why nanomaterials behave so differently from the bulk material, let us talk about classification of nanomaterials based on their dimensions. Now the first classification would be [00:05:45.764] it will be called as a zero dimensional nanomaterial. What does it mean? That all the dimensions, all the three dimensions X, Y, and Z [00:05:57.776] are within the nano range and just now we define what is the nano range from one nanometer to hundred nanometer if all the three dimensions are within the [00:06:09.328] nano range and none of the dimensions are out of the nano range we call it as a zero dimensional nanomaterial and the example would be these clusters, nano clusters where the dimension of none of the dimensions is exceeding uh the nano [00:06:26.960] Range exceeding 100 nanometers. Example, as I said, nano clusters or quantum dots fall into this category of nanomaterials. Now coming to one dimensional nanomaterials where [00:06:40.800] in in these materials the particles one of their dimensions is out of the nano range the other two dimensions will be within the nano range but one of the dimensions is going to be outside of the nano range [00:06:56.496] An example would be nanotubes and nanovires we can understand look at the picture here. So here in twos of the dimensions it is still in the nano range within hundred nanometers, but one dimension is now out of the nano range. So such materials are [00:07:14.128] So called nanomaterials, but they will be classified as one dimensional nanomaterials. Similarly, we come to two dimensional nanomaterials. It's understood now that two of the dimensions are allowed to be outside of the nanorange. So you can see in the picture here. So when you have nano coatings, nano. [00:07:31.696] films these will be classified as two dimensional nanomaterials only one dimension is now within the nano range but still we call it as a nanomaterial. [00:07:43.024] And then coming to three dimensional nanomaterials where all the three dimensions are outside of the nano range. Then how is it a nanomaterial only specific types of materials where you have like say bundles of nanovires. So each component is still uh behaving like a [00:08:00.688] Nanomaterial, therefore, these overall will look like as if all the three dimensions are outside of the nanorange. So such [00:08:09.760] uh materials are also classified as three dimensional nanomaterials where we have all the three dimensions outside of the nanorange like for example dispersions of nanoparticles or we

have bundles of nanovires. [00:08:24.932] Now we come to the interesting part where we are going to look at reasons why nanomaterials behave so different from the bulk material. [00:08:39.236] First and foremost reason which we should discuss here is that the large fraction of surface atoms per unit volume. Let us understand what it is. Now when we have a bulk material, we have few [00:08:56.640] Atoms on the surface, but major of the atoms are in the bulk of the material. So whatever is the property of the material is governed by the bulk atoms. But when we come to nanomaterials, what happens? The particle size has [00:09:14.352] Become so small now that on the surface you have a large number of surface atoms, whereas in the bulk we have very few atoms now. So the [00:09:26.928] Properties now are strictly governed by the surface atom because they are larger in number as compared to the bulk atoms. So you will find that many of these properties are now dictated by the surface atoms rather than the bulk atoms. For example, [00:09:44.656] When we have a platinum sheet as compared, which is uh we can say is bulk, whereas compared to platinum powder, but to a nanopowder. So what has happened now? Now the surface area, each particle now, the surface area has is more, and therefore overall surface area.

EV0000022 · EV0000023 · EV0000024 · EV0000025

Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.224] has become so high that the number of surface atoms are very very large and therefore we will find that the properties now will be dictated by these atoms rather than the bulk atoms. So let us see look at this picture here. We have a box like this. [00:10:19.872] Now you started dividing this box into smaller uh sizes then we know now this surface area was only this much but now each of these boxes now exposed to the surface the atoms on the surface are exposed so we have larger number of surface atoms and similarly if you go on [00:10:37.520] Dividing it further and further, the surface area is definitely going to increase and the number of surface atoms will go on increasing. So you can imagine if you make it into a nano particle size, then the surface area definitely is going to be very very high. [00:10:53.696] So the properties now are dictated by the surface atoms. Now another reason for the naroparticle behavior being different is the large surface energy. [00:11:06.212] Now we know that we have a large number of surface atoms, and these surface atoms are held with much less force as compared to the bulk atoms. We all know that the bulk atoms are surrounded by [00:11:20.064] atoms on all sides and therefore there's a uh force of attraction bit and repulsion between these uh bulk atoms. But when we look at the surface atoms they are being held [00:11:32.992] or uh or pulled inside or held together with only one on on only one side that other side other side is exposed to the uh environment therefore there is they are held with less force and therefore there is more energy on the surface and when uh any uh [00:11:52.752] System has more energy tries to it tries to reduce its energy and therefore we'll find that the surface activity will be very high. And just now I gave an example of aluminium where we find that aluminium in bulk is passive in nature, whereas when it is taken in the form of [00:12:10.448] Of nanoparticles, we find that the the same aluminium becomes so combustible that it is being used as rocket uh fuel. So therefore this [00:12:23.428] The reason being that there are large number of surface atoms and they are being held with less force, so surface energy is very high. Now coming to [00:12:34.948] Another reason which is very very important and if we understand this, all properties, especially electrical and uh optical properties, can be explained very easily. [00:12:49.296] by the use of the concept of spatial confinement or quantum confinement. Now let us understand what we mean by spatial confinement. Now suppose you have a bulk material [00:13:03.812] We know that uh let us take the example of a metal. We have these metal at the lattice points and we have a sea of electrons. Now

when we take a bulk material, we know that the electrons are free to move in all the directions. [00:13:19.204] That is for a bulk. But now suppose you make this bulk material so thin, so thin, so thin. You make it so small and so thin that it has now become into a two dimensional material. [00:13:35.392] What would happen? Now you have restricted the motion of the electrons into two directions. So that is you are spatially confining the uh the carriers, the electrons, into two dimension. [00:13:50.976] when you make it into a two dimensional quantum well say. So it is now become just into two dimensional and when you reduce it further, like you make it into a one dimensional uh quantum wire say [00:14:05.264] So what will happen now? Your electrons can move only in one direction. In the other two directions, they are completely confined. You have changed, you have made it so thin that now it is allowed to move only in one uh di dimension. So you are confining the electron. [00:14:22.896] In the holes to move only in one direction, and when you come to zero dimension, which are called quantum dots, you have restricted the motion completely, and it's almost like within a very small area, you are confining. [00:14:40.208] The electron and the poles. So this is going to bring about a lot of changes in the properties of these nanomaterials. Let us understand this further. When we have a bulk material, we know that. [00:14:59.204] Say it is one hole of the uh material or the metal we are considering. One hole would mean to the 10 to the power of 23 atoms.

EV0000026 · EV0000027 · EV0000028 · EV0000029

Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.772] compared to nanomaterials where we'll have a maximum of ten to thousand atoms. Now that is a big difference between the bulk and the nanomaterials. [00:15:20.868] Now here, when we say one mole, that means we have 10 to the power 23 atoms, that means so many electrons, and also means that there are so many energy levels. Now these energy levels [00:15:34.768] will be now almost overlapping with each other. They'll be very close and almost continuous. They will be continuous to the extent that in metals we know that the balance band and the conductance band are overlapping. [00:15:50.148] When we come to semiconductors, of course there is a small gap, nevertheless, these levels and the levels in the valence band and the condensed band, there are so many levels because there are so many atoms. And these levels will be very close to one another that they will be continuous. [00:16:07.920] But when we come to nano range, what happens is now we have reduced the number of atoms. There are fewer atoms, fewer electrons, so the density of states falls down. And what we observe, say we are in a two dimensional. [00:16:27.344] uh quantum well then we'll find that the energy levels now because there are less number of atoms, less number of electrons, less number of energy stays. So these energy levels start becoming discrete. What was like a continuous uh levels in the bulk now has [00:16:45.008] become uh discrete levels you can see these energy levels separate they become discrete and also now the band gap increases and when we come to uh two dimensional [00:16:58.368] uh nano particles and if you are making it into one dimensional further spatial confinement now the energy levels become more and more discrete they become farther apart and also we'll find that the band gap also increases the energy gap increases [00:17:16.048] And then when we come to uh zero dimensional [00:17:20.356] nanomaterial the energy levels now are completely discrete and the energy gap has increased. Now understand that these are all nanoparticles. If you compare them with a molecule in a molecule we know we have absolutely discrete energy levels. [00:17:37.264] Right, we studied that the transition happens in electronic spectroscopy, electronic transitions will happen from the homo. [00:17:45.360] to the lumino, the highest occupied molecular orbital to the lowest unoccupied molecular orbital, and this

is the energy gap. So when you come to the molecule we find the energy gap is very high. But as these become more and more in number, if you go this way now [00:18:02.992] So, as the particles become more and more in number, or the size of the particles goes on increasing, then we'll find that more number of energy states will be possible, and this band gap is decreasing. When we come to bulk, we find that the valence band, there are so many. [00:18:20.560] Many energy levels now because of so many atoms there. So we'll find that the balance band and the conduct conductance band in metals it's almost overlapping and becomes these are continuous now. So that is what is understood by quantum confinement. [00:18:38.128] or the spatiment. So what is the result of all this? What do we understand? That due to spatient or quantum confinement, the energy levels become more and more discrete and the energy gap increases. [00:18:57.668] This is the result of quantum confinement, and this is going to have a strong bearing on the properties, especially the electronic properties, electrical properties, and the optical properties. And one more reason. [00:19:15.728] for the bulk material behaving differently from the nanomaterial, but the nanomaterial behaves different from the bulk, is that in the bulk material we find that there are lattice imperfections like wherever there are dislocations, kinks, etcetera, are found in the [00:19:34.608] uh lattice. Why? Because the metal lattice is formed in such a way that layer after layer after layer is getting deposited of the metal. And many times we do not get a perfect layer. Rather there are some imperfections uh which are like dislocations or kinks are possible. [00:19:52.240] in the metal structure, it's a bulk metal structure. But when you come to the nanomaterial range, what has happened now you just have to they're very small in size the particles. Therefore very small lattice has to be formed and therefore there is less possibility of imperfections.

EV0000030 · EV0000031 · EV0000032 · EV0000033

Transcript block 5 (source interval 001200-001499 seconds): [00:20:09.808] Now reduced imperfections is also going to have an effect on the properties of these nano. [00:20:18.400] materials. So these are the various reasons why the nanomaterial behaves differently from the bulk material. And in the next class we are going to study how exactly these reasons are affecting the thermal properties, optical properties and [00:20:36.048] so on so forth of the nanomaterials. Thank you.

EV0000034 · EV0000035 · EV0000036 · EV0000037

Full transcript: 2867 normalized words, 83 timestamped segments, 1243.477333 seconds duration. Low-confidence segments flagged: 0.

EV0000018 · EV0000019 · EV0000020 · EV0000021

4 Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials

This lesson is indexed from the source video titled Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials. Its complete timestamped

transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000038 · EV0000039

Graphs and diagrams

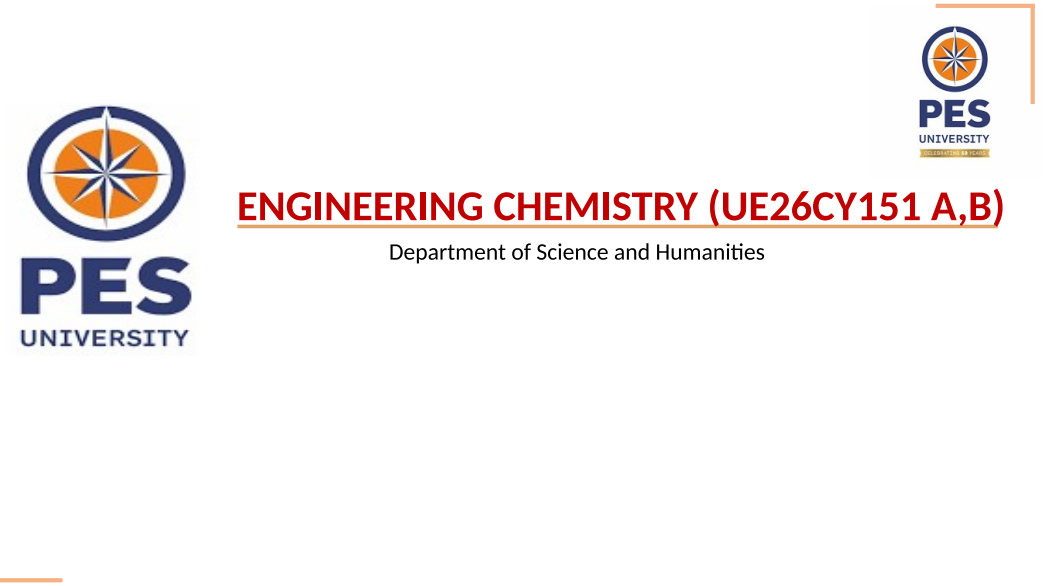


Figure G000015. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 1.

EV0000237

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

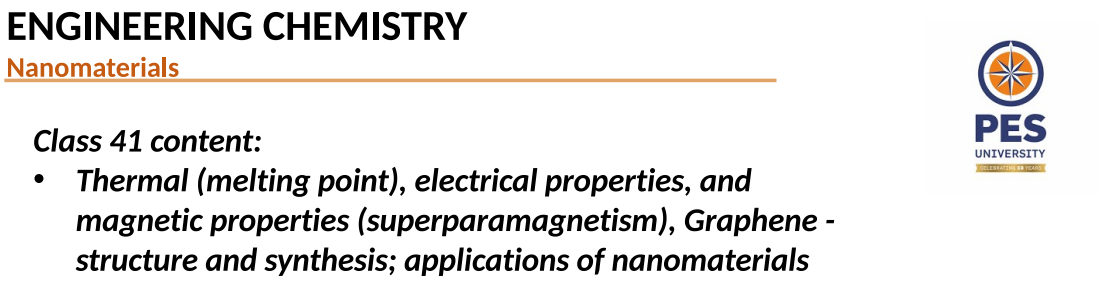


Figure G000016. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 2.

EV0000240

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

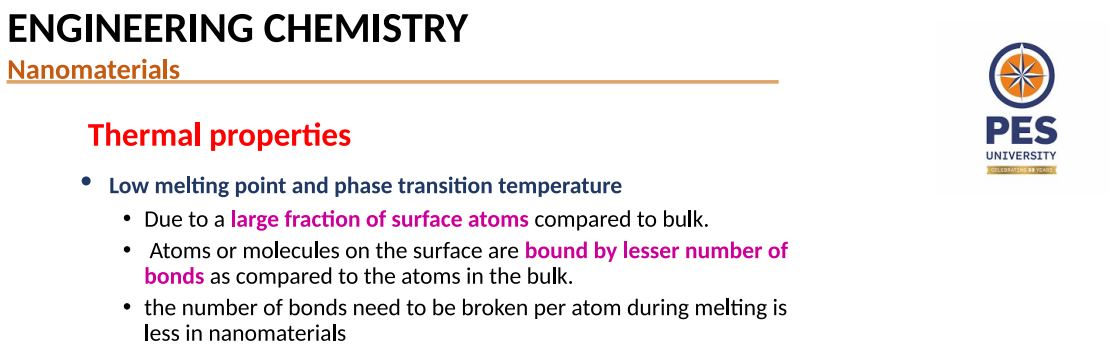


Figure G000017. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 3.

EV0000243

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

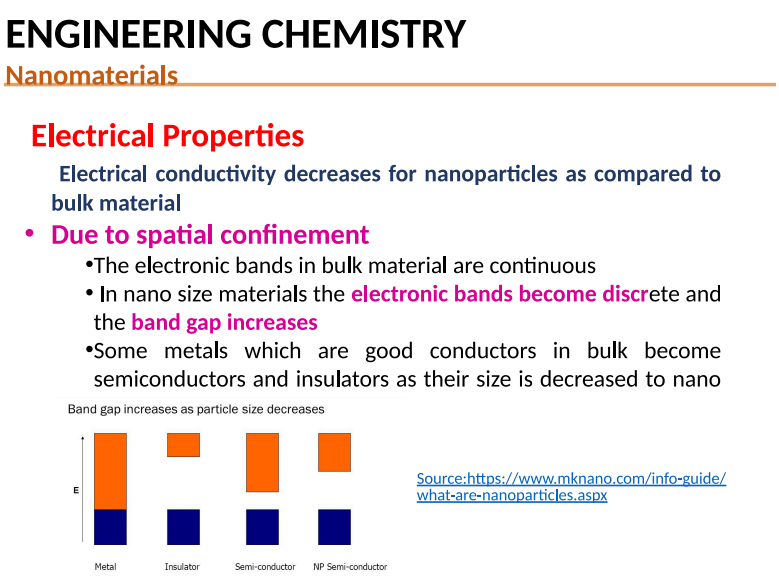


Figure G000018. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 4.

EV0000246

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

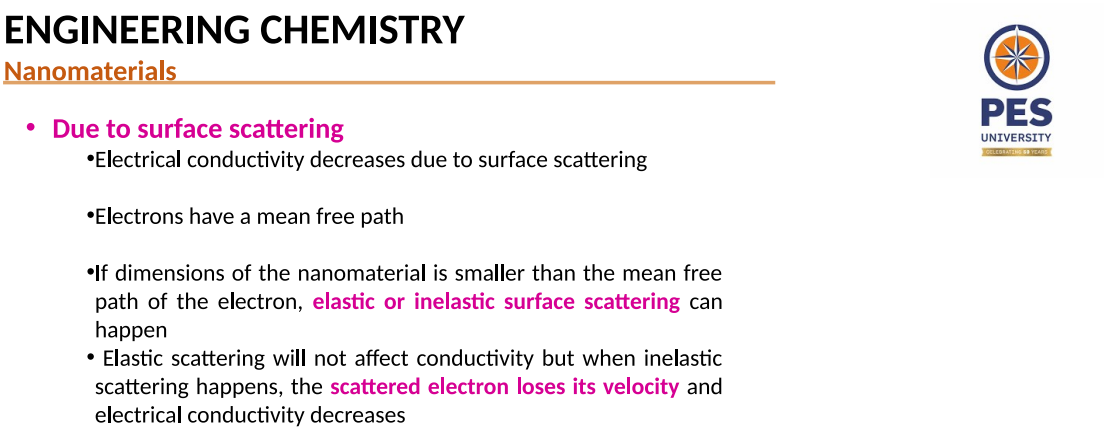


Figure G000019. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 5.

EV0000249

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

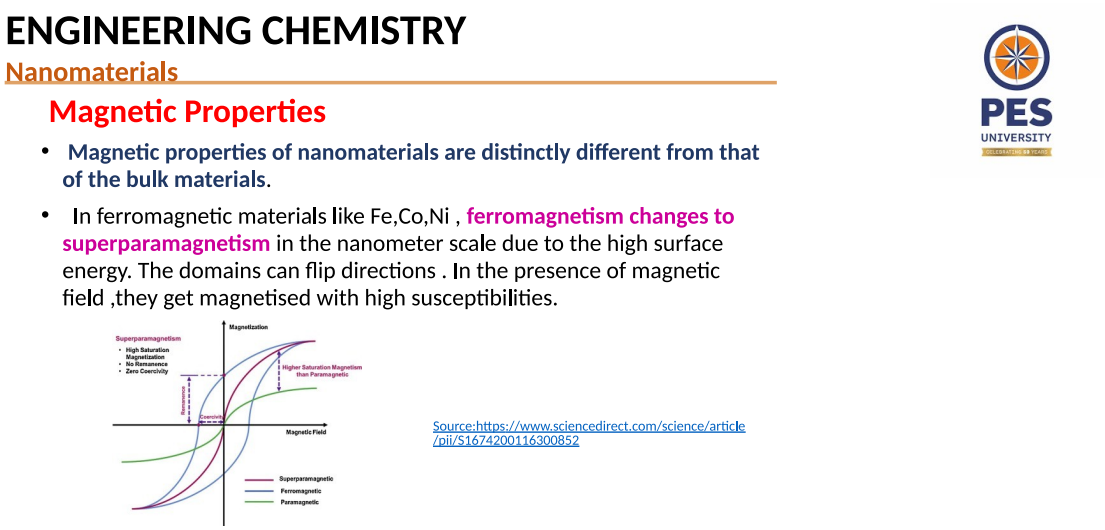


Figure G000020. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 6.

EV0000252

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

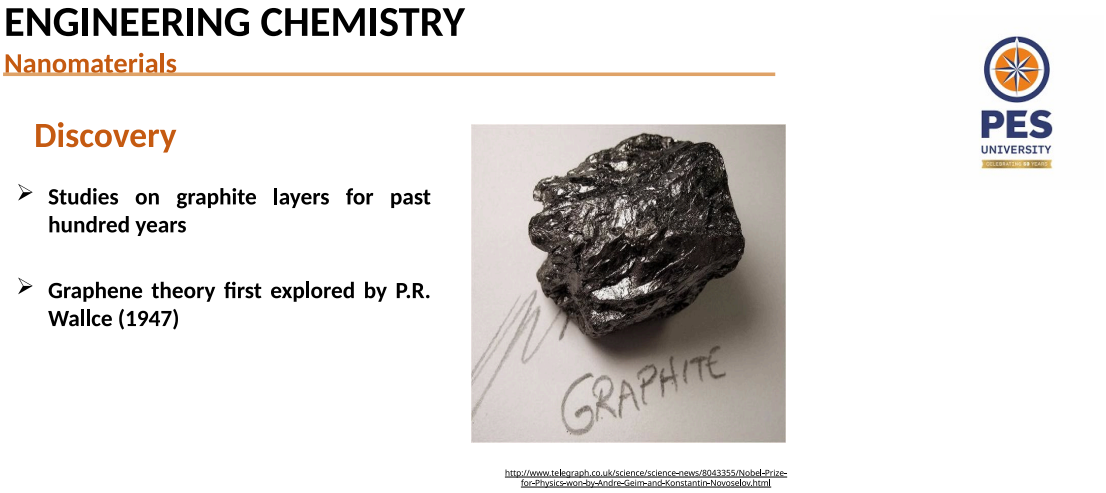


Figure G000021. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 7.

EV0000255

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

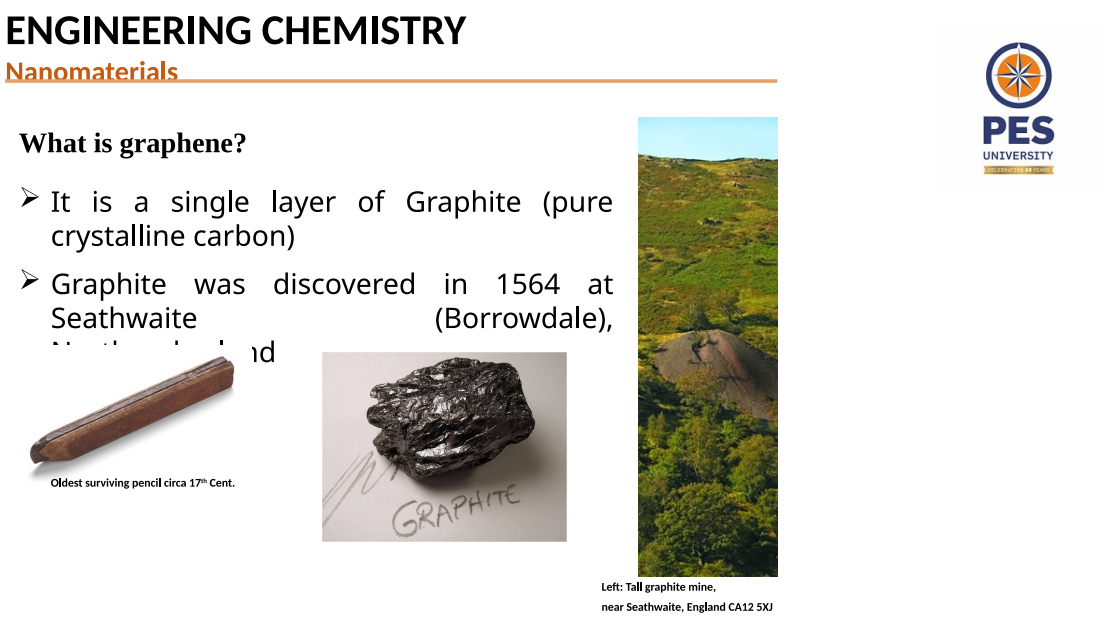


Figure G000022. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 8.

EV0000258

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

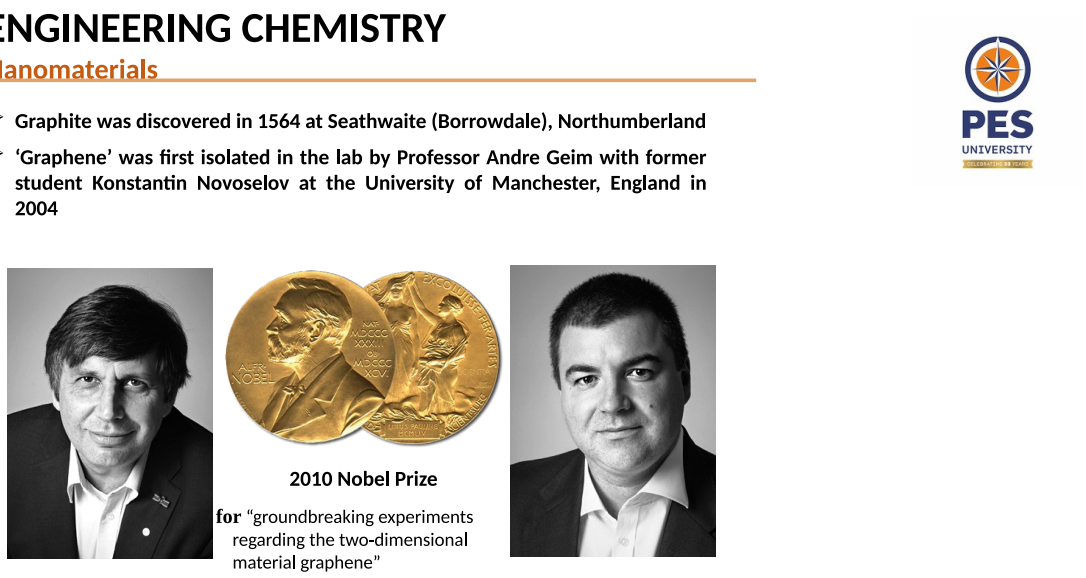


Figure G000023. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 9.

EV0000261

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

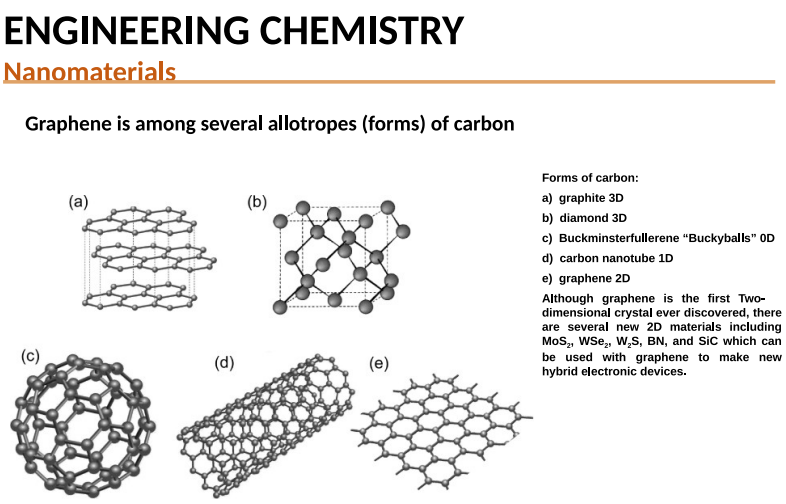


Figure G000024. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 10.

EV0000264

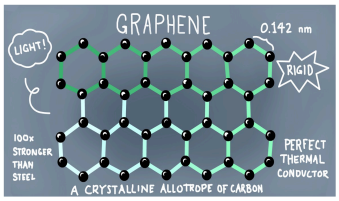
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Nanomaterials

Structure

- Graphene is a crystalline allotrope of carbon with 2-dimensional properties. Its carbon atoms are densely packed in a regular atomic-scale chicken wire (hexagonal) pattern.
- Each atom has four bonds, one σ bond with each of its three neighbors and one π-bond that is oriented out of plane. The atoms are about 1.42 Å apart
- Graphene can self-repair holes in its sheets, when exposed to molecules containing carbon, such as hydrocarbons. Bombarded with pure carbon atoms, the atoms perfectly align into hexagons, completely filling the holes



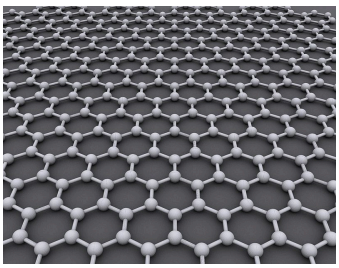


Figure G000025. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 11.

EV0000267

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

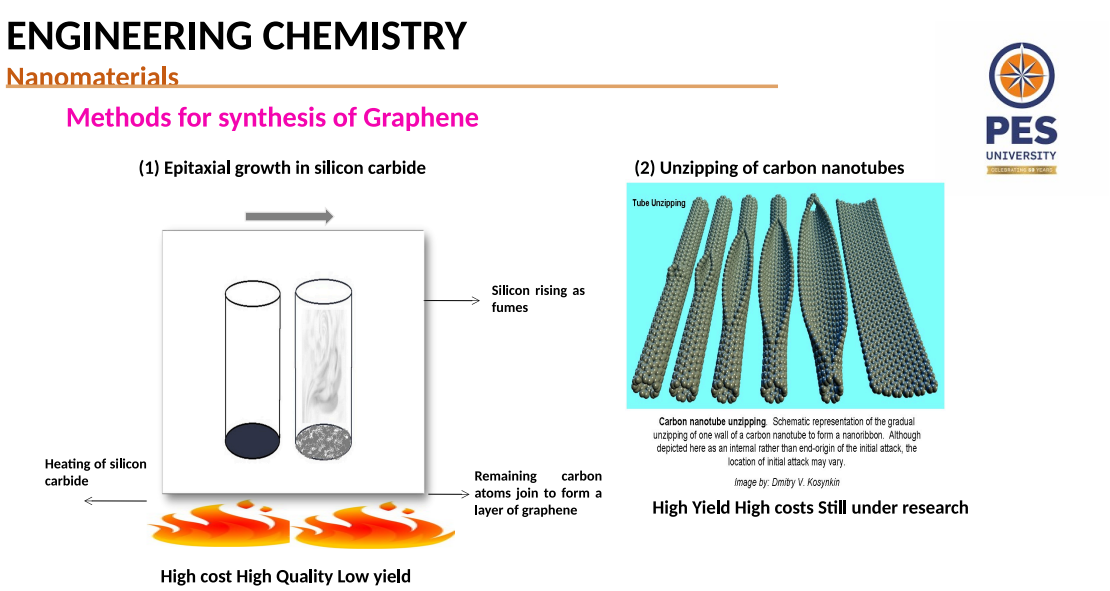


Figure G000026. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 12.

EV0000270

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

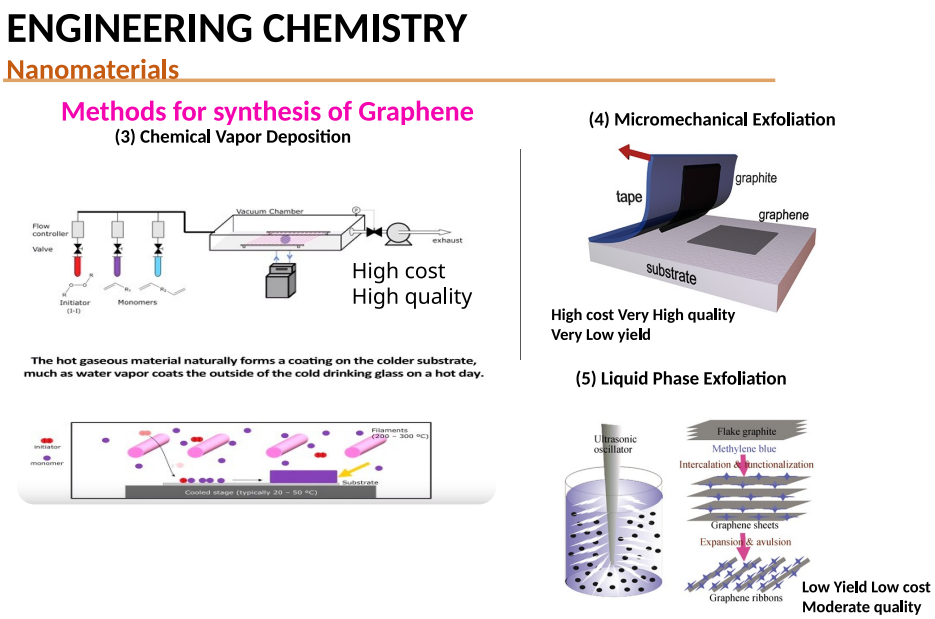


Figure G000027. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 13.

EV0000273

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

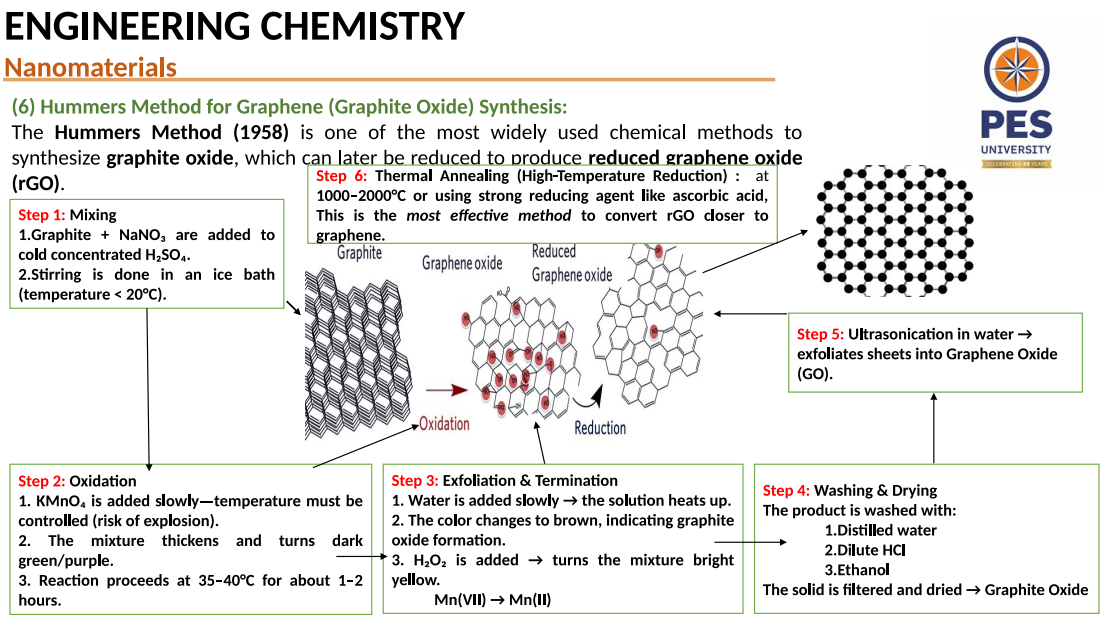


Figure G000028. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 14.

EV0000276

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

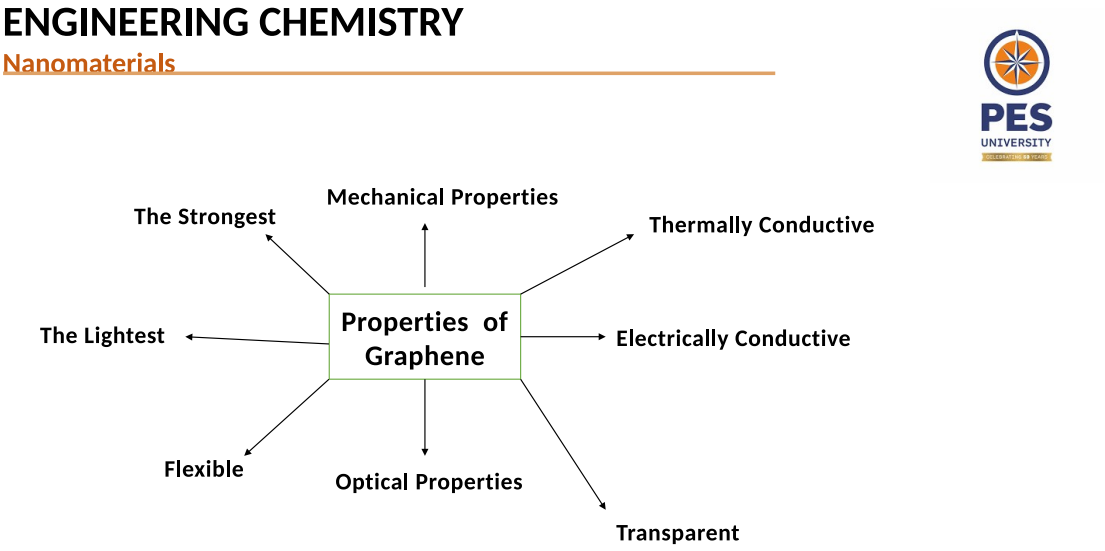


Figure G000029. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 15.

EV0000279

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

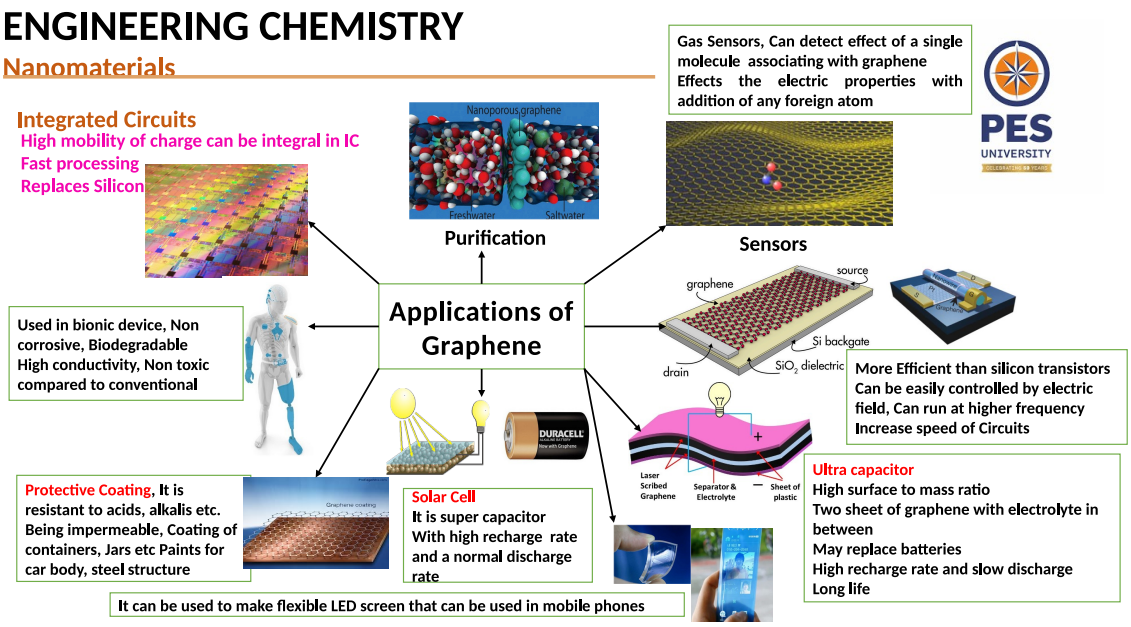


Figure G000030. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 16.

EV0000282

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

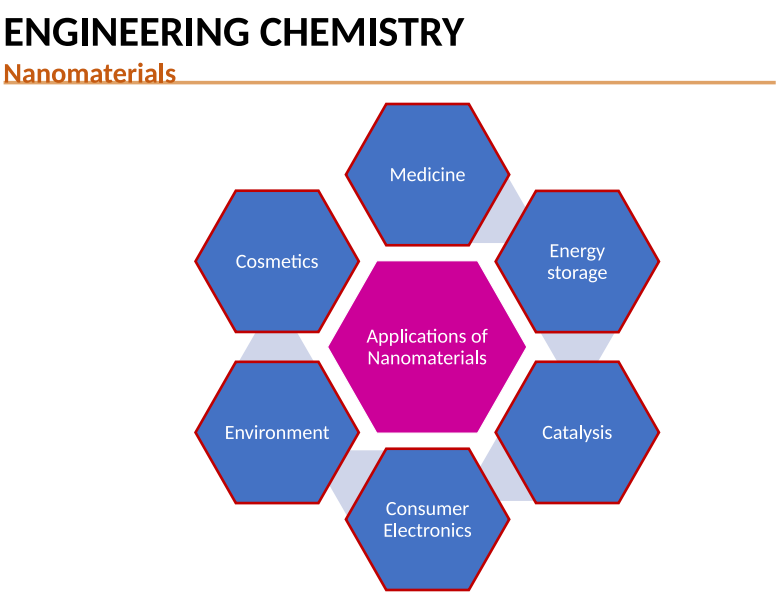


Figure G000031. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 17.

EV0000285

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

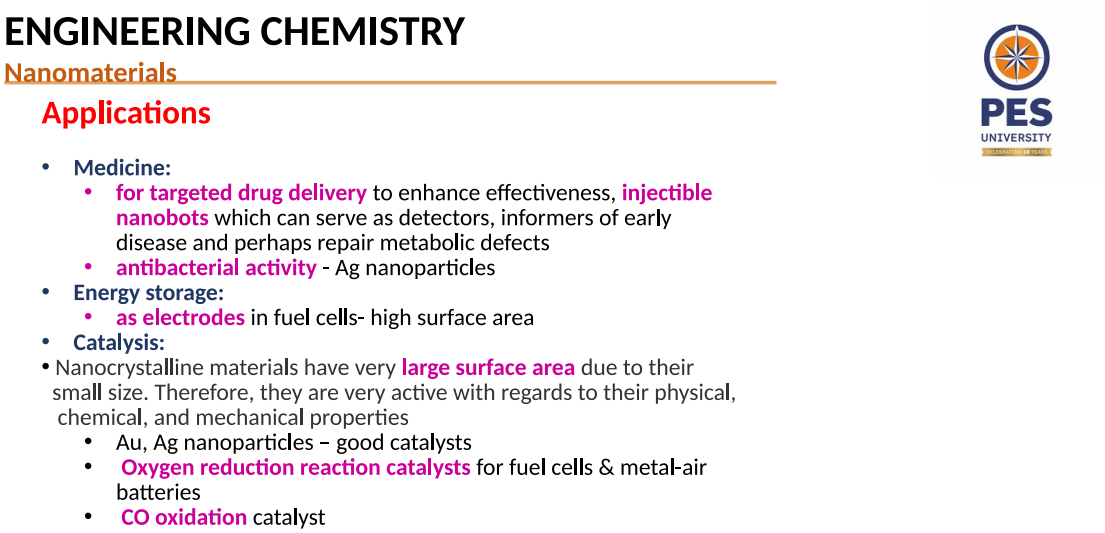


Figure G000032. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 18.

EV0000288

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

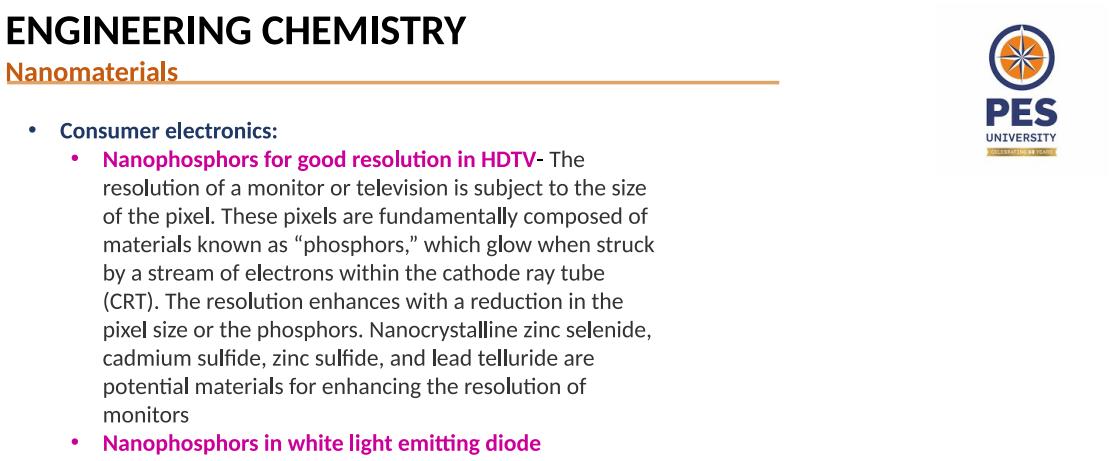


Figure G000033. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 19.

EV0000291

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

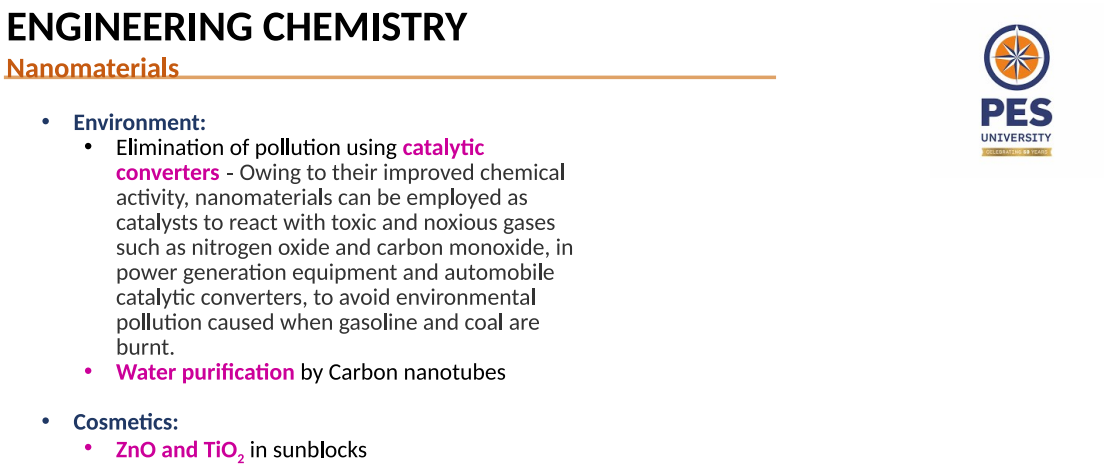


Figure G000034. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 20.

EV0000294

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000035. Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s · source slide page 21.

EV0000297

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000236 · EV0000237 · EV0000238
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Class 41 content: Thermal (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials EV0000239 · EV0000240 · EV0000241
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Thermal properties Low melting point and phase transition temperature Due to a large fraction of surface atoms compared to bulk. Atoms or molecules on the surface are bound by lesser number of bonds as compared to the atoms in the bulk. the number of bonds need to be broken per atom during melting is less in nanomaterials EV0000242 · EV0000243 · EV0000244
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Electrical Properties Electrical conductivity decreases for nanoparticles as compared to bulk material Due to spatial confinement The electronic bands in bulk material are continuous In nano size materials the electronic bands become discrete and the band gap increases Some metals which are good conductors in bulk become semiconductors and insulators as their size is decreased to nano range [Image text, OCR] Electrical Properties Bandgap increases asparticle sizedecreases E Metal Insulator Semi-conductor NP Semi-conductor SHINE:Seattle'sHubfor Industry-drivenNanotechnologyEducation 18 Source:https://www.mknano.com/info-guide/what-are-nanoparticles.aspx EV0000245 · EV0000246 · EV0000247

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Due to surface scattering Electrical conductivity decreases due to surface scattering Electrons have a mean free path If dimensions of the nanomaterial is smaller than the mean free path of the electron, elastic or inelastic surface scattering can happen Elastic scattering will not affect conductivity but when inelastic scattering happens, the scattered electron loses its velocity and electrical conductivity decreases EV0000248 · EV0000249 · EV0000250
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Magnetic Properties Magnetic properties of nanomaterials are distinctly different from that of the bulk materials. In ferromagnetic materials like Fe,Co,Ni , ferromagnetism changes to superparamagnetism in the nanometer scale due to the high surface energy. The domains can flip directions . In the presence of magnetic field ,they get magnetised with high susceptibilities. [Image text, OCR] Magnetization Superparamagnetism High Saturatlon Magnetization No Remanence .Zero Coerclvity 1Higher Saturation Magnetism than Paramagnetic Coercivity Magnetic Field Superparamagnetic Ferromagnetic Paramagnetic
   Source:https://www.sciencedirect.com/science/article/pii/S1674200116300852 EV0000251 · EV0000252 · EV0000253
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Discovery [Image text, OCR] GRAPHITE Studies on graphite layers for past hundred years Graphene theory first explored by P.R. Wallce (1947) http://www.telegraph.co.uk/science/science-news/8043355/Nobel-Prize-for-Physics-won-by-Andre-Geim-and-Konstantin-Novoselov.html EV0000254 · EV0000255 · EV0000256
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Nanomaterials What is graphene? It is a single layer of Graphite (pure crystalline carbon) Graphite was discovered in 1564 at Seathwaite (Borrowdale), Northumberland Oldest surviving pencil circa 17th Cent. Left: Tall graphite mine, near Seathwaite, England CA12 5XJ EV0000257 · EV0000258 · EV0000259
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Graphite was discovered in 1564 at Seathwaite (Borrowdale), Northumberland ‘Graphene’ was first isolated in the lab by Professor Andre Geim with former student Konstantin Novoselov at the University of Manchester, England in 2004 [Image text, OCR] AT MDCCC XXX OB- ALFR· MDCCC NOBEL XCV. 2010 Nobel Prize for “groundbreaking experiments regarding the two-dimensional material graphene” EV0000260 · EV0000261 · EV0000262
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Graphene is among several allotropes (forms) of carbon Forms of carbon: a) graphite 3D b) diamond 3D c) Buckminsterfullerene “Buckyballs” 0D d) carbon nanotube 1D e) graphene 2D Although graphene is the first Two- dimensional crystal ever discovered, there are several new 2D materials including MoS2, WSe2, W2S, BN, and SiC which can be used with graphene to make new hybrid electronic devices. [Image text, OCR] (c) (d) (e) EV0000263 · EV0000264 · EV0000265

11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Structure [Image text, OCR] GRAPHENE 0.142 nm LIGHT! RIGID 100x PERFECT STRONGER THERMAL THAN CONDVLTOR STEEL ALLOTRDPE OF CARBON A CRYSTALLINE Graphene is a crystalline allotrope of carbon with 2-dimensional properties. Its carbon atoms are densely packed in a regular atomic-scale chicken wire (hexagonal) pattern. Each atom has four bonds, one σ bond with each of its three neighbors and one π-bond that is oriented out of plane. The atoms are about 1.42 Å apart Graphene can self-repair holes in its sheets, when exposed to molecules containing carbon, such as hydrocarbons. Bombarded with pure carbon atoms, the atoms perfectly align into hexagons, completely filling the holes EV0000266 · EV0000267 · EV0000268
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Methods for synthesis of Graphene (1) Epitaxial growth in silicon carbide (2) Unzipping of carbon nanotubes [Image text, OCR] TubeUnzipping Carbon nanotube unzipping. Schematic representation of the gradual unzipping of one wall of a carbon nanotube to form a nanoribbon. Although depicted here as an internal rather than end-origin of the initial attack, the location of initial attack may vary. Imageby:DmitryV.Kosynkin Heating of silicon carbide Silicon rising as fumes Remaining carbon atoms join to form a layer of graphene High Yield High costs Still under research High cost High Quality Low yield EV0000269 · EV0000270 · EV0000271
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Methods for synthesis of Graphene (4) Micromechanical Exfoliation (3) Chemical Vapor Deposition [Image text, OCR] graphite tape graphene substrate [Image text, OCR] Vacuum Chamber Flow G controller exhaust Valve Initiator Monomers (I-I) High cost High quality High cost Very High quality Very Low yield [Image text, OCR] Thehotgaseousmaterialnaturallyformsacoatingonthecoldersubstrate, muchaswatervaporcoatstheoutsideofthecolddrinkingglassonahotday. Filaments (200-3000C) initiator monomer Substrate Cooled stage (typically 20 -50 C) (5) Liquid Phase Exfoliation [Image text, OCR] Flake graphite Ultrasonic oscillator Methyleneblue Intercalation&functionalization Graphenesheets Expansion&avulsion Grapheneribbons Low Yield Low cost Moderate quality EV0000272 · EV0000273 · EV0000274
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Nanomaterials (6) Hummers Method for Graphene (Graphite Oxide) Synthesis: The Hummers Method (1958) is one of the most widely used chemical methods to synthesize graphite oxide, which can later be reduced to produce reduced graphene oxide (rGO). Step 6: Thermal Annealing (High-Temperature Reduction) : at 1000–2000°C or using strong reducing agent like ascorbic acid, This is the most effective method to convert rGO closer to graphene. Step 1: Mixing Graphite + $NaNO_3$ are added to cold concentrated H_2SO_4. Stirring is done in an ice bath (temperature $< 20°C$). [Image text, OCR] Graphite Reduced Grapheneoxide Grapheneoxide HO HO HO HO HO OH HO Oxidation Reduction Step 5: Ultrasonication in water → exfoliates sheets into Graphene Oxide (GO). Step 2: Oxidation 1. $KMnO_4$ is added slowly—temperature must be controlled (risk of explosion). 2. The mixture thickens and turns dark green/purple. 3. Reaction proceeds at 35–40°C for about 1–2 hours. Step 3: Exfoliation & Termination 1. Water is added slowly → the solution heats up. 2. The color changes to brown, indicating graphite oxide formation. 3. H_2O_2 is added → turns the mixture bright yellow. Mn(VII) → Mn(II) Step 4: Washing & Drying The product is washed with: Distilled water Dilute HCl Ethanol The solid is filtered and dried → Graphite Oxide EV0000275 · EV0000276 · EV0000277

15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Mechanical Properties The Strongest Thermally Conductive Properties of Graphene The Lightest Electrically Conductive Flexible Optical Properties Transparent EV0000278 · EV0000279 · EV0000280
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Gas Sensors, Can detect effect of a single molecule associating with graphene Effects the electric properties with addition of any foreign atom Integrated Circuits High mobility of charge can be integral in IC Fast processing Replaces Silicon Purification Sensors Applications of Graphene Used in bionic device, Non corrosive, Biodegradable High conductivity, Non toxic compared to conventional More Efficient than silicon transistors Can be easily controlled by electric field, Can run at higher frequency Increase speed of Circuits Ultra capacitor High surface to mass ratio Two sheet of graphene with electrolyte in between May replace batteries High recharge rate and slow discharge Long life Protective Coating, It is resistant to acids, alkalis etc. Being impermeable, Coating of containers, Jars etc Paints for car body, steel structure Solar Cell It is super capacitor With high recharge rate and a normal discharge rate It can be used to make flexible LED screen that can be used in mobile phones EV0000281 · EV0000282 · EV0000283
17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Nanomaterials EV0000284 · EV0000285 · EV0000286
18. Source slide page 18 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Applications Medicine: for targeted drug delivery to enhance effectiveness, injectible nanobots which can serve as detectors, informers of early disease and perhaps repair metabolic defects antibacterial activity - Ag nanoparticles Energy storage: as electrodes in fuel cells- high surface area Catalysis: Nanocrystalline materials have very large surface area due to their small size. Therefore, they are very active with regards to their physical, chemical, and mechanical properties Au, Ag nanoparticles – good catalysts Oxygen reduction reaction catalysts for fuel cells & metal-air batteries CO oxidation catalyst EV0000287 · EV0000288 · EV0000289
19. Source slide page 19 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Consumer electronics: Nanophosphors for good resolution in HDTV- The resolution of a monitor or television is subject to the size of the pixel. These pixels are fundamentally composed of materials known as “phosphors,” which glow when struck by a stream of electrons within the cathode ray tube (CRT). The resolution enhances with a reduction in the pixel size or the phosphors. Nanocrystalline zinc selenide, cadmium sulfide, zinc sulfide, and lead telluride are potential materials for enhancing the resolution of monitors Nanophosphors in white light emitting diode EV0000290 · EV0000291 · EV0000292
20. Source slide page 20 (extracted text): ENGINEERING CHEMISTRY Nanomaterials Environment: Elimination of pollution using catalytic converters - Owing to their improved chemical activity, nanomaterials can be employed as catalysts to react with toxic and noxious gases such as nitrogen oxide and carbon monoxide, in power generation equipment and automobile catalytic converters, to avoid environmental pollution caused when gasoline and coal are burnt. Water purification by Carbon nanotubes Cosmetics: ZnO and TiO2 in sunblocks EV0000293 · EV0000294 · EV0000295
21. Source slide page 21 (extracted text): THANK YOU Department of Science and Humanities EV0000296 · EV0000297 · EV0000298

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.964] Hello everyone, welcome back to this course on engineering chemistry. We continue with module two on nanomaterials and this will be the last class of this module. [00:00:13.412] So we are going to today study some more properties of nanomaterials, the mechanical properties, thermal properties, and magnetic properties. And finally, we will look at some applications of nanomaterials. [00:00:29.092] When we talk of mechanical properties, the most important thing we need to understand here that the strength of the nanomaterials is found to be greater than the bulk material. What does it mean? When we take a substance in the form of a bulk and [00:00:46.692] it has a certain strength. But when you come to the nanomaterial range, it is found that the strength increases to a great extent. Now it is found that it almost reaches the theoretical strength. [00:01:02.064] Now that would be one or two orders more than the uh the strength which is found in the bulk materials. Now let us take an example of copper. Bulk copper we know can be bent with certain uh amount of ease. But when [00:01:21.776] We take these copper particles in the nano range, like when you get it to below 50 nanometers, it is found that they become super hard in nature. So they are considered as super hard materials. The same copper which in the form of bulk could be bent cannot. [00:01:39.440] We bend now and it becomes super hard and it loses all its malleability and deductivity, which we find in the case of bulk copper. So what would be the reason for that? The reason is that when you take a particle in the nano range, the copper. [00:01:57.008] in the nano range the number of defects becomes very less there is less probability of defects occurring now what do we mean by this [00:02:06.944] When we have large lattice lattices as in the case of bulk material, we know one layer above the other is getting formed when the uh metal is being uh when we have a metal. There are one after the other layers are getting formed. Now [00:02:21.024] In these layers we will find that there could be imperfections like dislocations, kinks, and vacancies. And because of these imperfections or these kind of these kind of vacancies and dislocations, etcetera, we find that the strength of the material. [00:02:38.736] [ASR low confidence; check audio] material becomes less, but when we [00:02:41.168] Have particles in the nano range, we know that we are forming very small lattice which has a smaller cross section, and therefore the probability of finding these imperfections is very less, and therefore we find that when we go to this nano range of such a small particle. [00:02:58.800] Of less than 15 nanometer, definitely the number of imperfections is less. We have a very good lattice, and that results in making copper, for example, a super hard material. [00:03:13.732] Let us talk of thermal properties. The first property we need to discuss is the melting point. [00:03:20.528] Now we find that when we have in the nanorange, when we have particles in the nanorange, the melting point gets reduced and also the phase transition temperature also becomes less. What would be phase transition temperature when you change from solid to uh liquid. Now [00:03:38.192] That temperature is also reduced, or other phase transitions could also happen with two different types of solids. Also, the the same solid existing in polymorphic forms that also would be a phase transition. Now it is found that the phase transition temperature also decreases the melting. [00:03:55.760] Point also decreases. Now we know that under a certain set of conditions, the melting point as well as the phase transition temperatures are fixed values. Now these cannot change. [00:04:07.248] They are very characteristic. In fact, melting point, if you want to check the purity of a metal the fur uh or of any material, the first thing we do is check the melting point. And that would be characteristic of the material. And we know if the material if it is showing uh this uh melting point. [00:04:24.880] Point which is listed, then we know that the material we have taken is pure. But the same material in the nano range now the melting

point becomes less. What is the reason for that? It's very simple to understand. We know that. [00:04:43.428] There are a lot of surface atoms in the nanomaterials because the particles are so small, most of the atoms are on the surface. Now these surface atoms are held with less force. So when you are trying to melt a compound of

EV0000038 · EV0000039 · EV0000040 · EV0000041

Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.200] Obviously, you need to break the bonds. Now, this now we know in nanomaterials the properties are dictated by the surface atom. Now these surface atoms are held with less energy, so less energy will be required to break these bonds, and therefore we have lesser bonds, so the less energy. [00:05:18.768] energy will be required to break these parts and hence we find that the melting point is going to uh decrease. So because of these large number of surface atoms, we find that the melting point of [00:05:34.672] materials in the nano range becomes less. Another uh property which changes [00:05:41.472] is the thermal conductivity. It is found that there is a decrease in thermal conductivity. Now we know when we go to the nanorange the particle size is very small and what is thermal conductivity due to? It is due to the vibration of these lattices. [00:05:59.120] Points. Now these lattice points are when they vibrate, they generate a wave which is referred to as the phonons. Now if the part length of these phonons is larger than the size of the particle, what is going to happen? [00:06:16.688] obviously it will result in scattering which is known as phonon scattering and once this scattering happens that means it is going to result in a decrease in the thermal conductivity. [00:06:29.764] Let us talk of magnetic properties. We find that the magnetic properties of nanomaterials are distinctly different from that of the bulk materials. You take a material in the bulk form, and when you go to the nano form [00:06:46.528] Range of the same material, the magnetic properties are found to be very different. Let us take up the example of [00:06:55.460] Iron cobalt and nickel. Now iron cobalt and nickel are known to be ferromagnetic in nature. But the uh iron cobalt and nickel when they are taken in the nano range, we find that this ferromagnetism changes to superparamagnetism. [00:07:13.232] in the nanometer uh range. Now why would this happen? [00:07:18.980] Let us first understand what exactly is a ferromagnetic material from this graph and what is a paramagnetic material so that we can uh understand what is that change. Now we know that ferromagnetic materials when you are [00:07:33.504] uh when you put them in the magnetic field they get magnetized and the magnet uh the magnetic susceptibility is very high and when you reduce the magnetic field they do not come back to zero or when you are when you remove the magnetic field we don't find that they come back to zero [00:07:51.248] Uh the magnetic uh susceptibility does go to zero, but in fact we find that there is a certain memory and that is given by the hysteresis. So the magnetic property does not come to uh zero. The magnetism doesn't come to zero when we [00:08:08.816] uh remove the magnetic field in the case of a ferromagnetic materials. But the same material now I don't when it is taken in the nano range [00:08:20.560] We find that they change into paramagnetism. They shall start showing paramagnetic behavior. Now what is paramagnetic behavior? The [00:08:29.444] material gets aligned in a magnetic field but to a very small extent. So the susceptibility is magnetic susceptibility is low for a paramagnetic material, which you can see from the graph here. But in the case of nanomaterials [00:08:45.936] The material which was ferromagnetic does become paramagnetic, that means it loses its magnetism to a great extent. [00:08:54.992] but in the presence of the magnetic field we find that the magnetic susceptibility is very high. So this is called superparamagnetism. [00:09:06.272] Now why does this happen? Because we know that the surface

energy is very high. Because the surface energy is high, now these uh magnetic uh dipoles they can flip. [00:09:22.288] These will flip in the in the presence of the magnetic field, they will they can flip at room temperature itself, they can flip directions and therefore they lose their ferromagnetic nature. But nevertheless, when there is a magnetic field, they will get [00:09:39.952] Magnetized like a paramagnetic um material. [00:09:44.784] But the magnetization is going to be very high, so that the magnetic susceptibility is going to be very high in this case. And when you remove the magnetic field, they come back to zero. So this behavior is referred to as superparamagnetic.

EV0000042 · EV0000043 · EV0000044 · EV0000045

Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.000] behavior so from ferromagnetic it changed to superparamagnetic behavior why the reason is the high surface energy so the domains at room temperature itself flip directions and therefore they lose their uh alignment [00:10:18.448] with the ma uh alignment and therefore when the magnetic field is applied they do get some small uh amount of alignment uh like paramagnetic behavior uh behavior but the susceptibility is now very high. They get magnetized to a great extent and that is called superparamagnetic [00:10:36.080] uh behavior but the g uh the interesting part is that when the magic field is removed uh they don't behave like ferromagnetic beh uh material which can retain they have memory but here it becomes zero. So this is [00:10:52.576] what is observed when we take this iron cubernical in the nanomaterial range. Now another property is the another example is where we have bulk, gold and platinum. They are [00:11:05.936] non-magnetic, but then in the nano size range they start behaving like magnetic particles. So what we understand is that magnetic properties will be sufficiently different, distinctly different from that of the bulk materials. [00:11:22.372] Now let us talk of some applications of nanomaterials. We know that nanomaterials have a range of applications in in many fields. So some of them we have here is medicine. [00:11:37.200] Then energy storage, catalysis, then consumer electronics, environmental [00:11:43.376] and cosmetics. Now let us look at each one of them and understand how these nanomaterials are being uh used. Now in medicine it is used for targeted drug delivery to enhance the effectiveness. For example, when we have [00:12:00.640] these cancer cells and some drugs are given, we know that the they don't not only affect the cancer cells, but uh the cell around are also getting affected. So if we use these nanomaterials then [00:12:15.344] and have a targeted drug delivery, it is going to target only the cancer cells and that is going to be very uh helpful. And then we can also have injectable nanopods which can serve as detectors, informers of early disease and perhaps repair metabolic [00:12:34.416] defects. So these nanobots can be injected into the bloodstream wherein they can be controlled [00:12:42.096] uh from uh externally and there could you can the directions can be given and they can inform if there is a disease or they could probably even repair any uh defects which are found inside the body. And apart from that we know that the there's uh [00:12:59.392] Silver nanoparticles are known for their antibacterial activities. Now these are just some of the nanoparticles examples which uh have been taken here. There are several more applications in medicine. Now another uh application would be in the in [00:13:17.072] The region of energy storage. [00:13:21.168] The electrodes which are used in fuel cells. Now, of course, we understand nanomaterials, nanoparticles, they have very large surface area, and that is very important in the case of fuel cells, especially the electrodes. The electrodes need to have very high. [00:13:38.896] uh surface area because that is where the actual reaction takes place. The electrode [00:13:45.668] is in contact with the fuel or the oxidant, it is also in contact with the catalyst. The catalyst being is being impregnated onto the electrode and it's also in contact with the electrolyte.

Therefore a high surface in the or area of the electrode which would help in [00:14:02.064] uh making the fuel cells very efficient. Then coming to catalysis. Now nano crystalline materials again they have a very large surface area because they are in they have a very small size and most of the atoms we find on the surface and these [00:14:19.696] atoms will also be uh very uh active so with regard to their physical, chemical and mechanical properties, so we know that they can serve as very good catalysts. If we use the material in as a nano uh particle [00:14:37.264] In the nanoparticle range, we find that they will serve as a very good catalyst. For example, we know gold and silver nanoparticles are being used as catalysts in several experiments. And in fuel cells, again, oxygen reduction reaction is a very slow reaction. So we have to use a [00:14:56.708] high surface area uh material there and a good catalyst over there. So this catalyst if it has high surface area definitely it will make the reaction happen at a faster rate. So in fuel cells and also in metal and batteries where catalysts are used. These process

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 4 (source interval 000900-001199 seconds): [00:15:14.480] while make uh make for very good catalyst and also uh as a catalyst for carbon monoxide oxidation. Now coming to consumer electronics. Now we know that nanophosphors [00:15:28.176] for good resolution in HD TV. Now what happens for uh in HD TVs we have uh a very high resolution and um for the monitor or the television and it depends on the size of the pixel. So what are these pixels basically [00:15:45.856] They are composed of materials which are known as phosphors which glow when they are struck with a stream of electrons from the within the catchment rate two. Now the resolution enhances with the with the reduction in the pixels size. [00:16:03.472] Size or the phosphorus. So obviously, if we take nanophosphors, then the size is sufficiently reduced, and therefore we find the number of pixels is more, and therefore we will find that the resolution is going to be much much higher. So nanocrystal. [00:16:21.040] materials like zinc selenite, then cadmium sulphide, zinc sulfide and lead sul tellurite. So we have these are the various materials which are being uh used [00:16:34.848] uh and they are very potential for enhancing the resolution of monitors. And nanophosphors are also being employed in white light emitting diodes. Now coming to environment. Now in the for removing pollution [00:16:53.584] Nanomaterials can be very very useful and are being employed in catalytic converters. Now because they have high chemical activity and due to the high surface area, therefore, nanomaterials are being used as catalysts to remove the toxicity. [00:17:11.344] And the noxious gases such as the nitrogen oxide and carbon monoxide in automobiles as well as for power generation equipment. So they are used as catalytic converters and these can be [00:17:28.368] uh used to remove uh environmental produ uh pollution rather reduce uh environmental pollution uh especially where gasoline and coal are being burnt and we also know that carbon carbon nanotubes are being used for water purification [00:17:46.032] Now let us look at this last application in cosmetics. Now here zinc oxide and titanium oxide we know are used as sun blocks, but now they are being used in the the particles are nanoparticles which are being used. Now if you [00:18:03.492] Uh no, earlier uh when sun blocks used to be used they would be white in colour. So whoever is using like especially in cricketers old time if you have seen old matches, you will see cricketers put this white kind of thing on their face and it uh that was due to the the sun block. But now because [00:18:21.264] Of the use of these nanomaterials, now these same zinc oxide and titanium oxide, they are taken in the nanoparticle range, and it is found that they now because of the nanoparticle range they become transparent. So everybody uses sun blom these days, and they you cannot maintain. [00:18:38.832]

out you don't get that whitish uh color on the face and they are now transparent. So this way nanopaterial nanomaterials are being used in cosmetics. These are just some of the applications which we have listed over here but yes they have a several [00:18:56.400] Applications in the electronics industry and or many applications are found for nanomaterials. [00:19:06.052] So that brings us to the end of module two on nanomaterials and also the end of unit one. So in the next class we are going to start with unit two where we will study phase equilibrium. Thank you.

EV0000050 · EV0000051 · EV0000052 · EV0000053

Full transcript: 2790 normalized words, 78 timestamped segments, 1162.986667 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:02:38.736

EV0000038 · EV0000039 · EV0000040 · EV0000041

5 Polymers: Introduction, types of polymerizations, definition of glass transition temperature

This lesson is indexed from the source video titled Polymers: Introduction, types of polymerizations, definition of glass transition temperature. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000054 · EV0000055

Graphs and diagrams

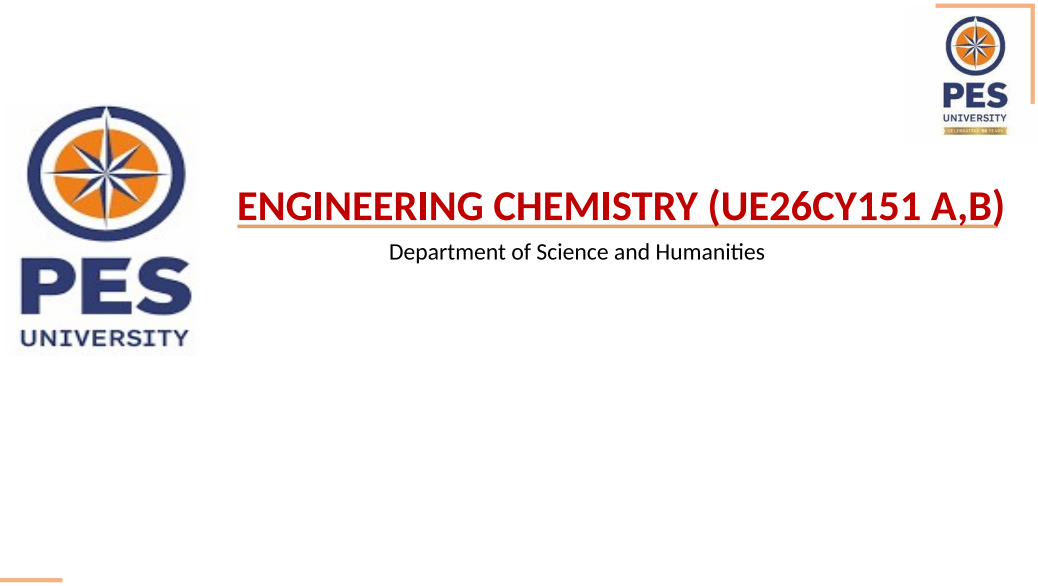


Figure G000036. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 1.

EV0000300

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

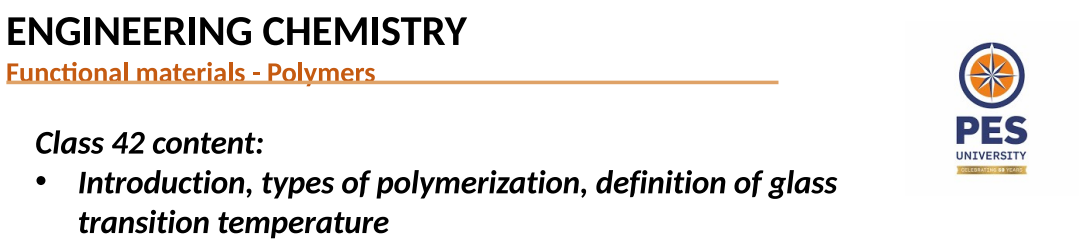


Figure G000037. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 2.

EV0000303

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

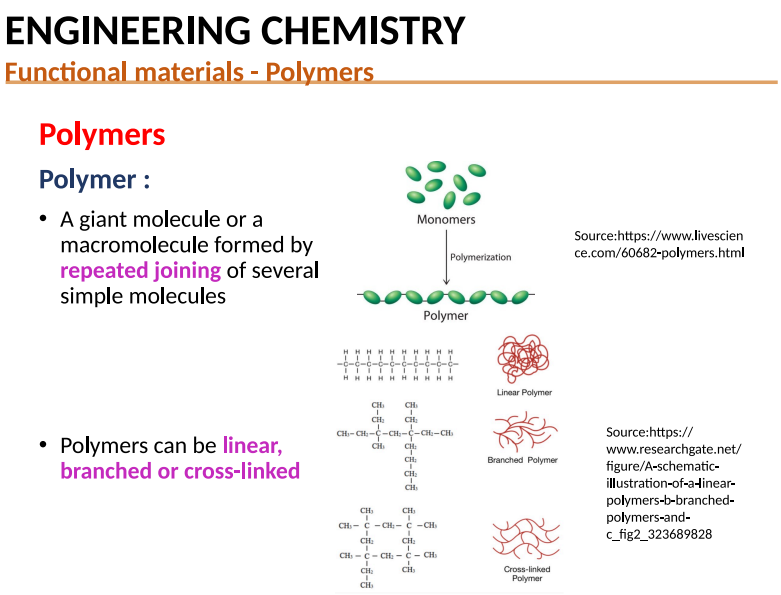
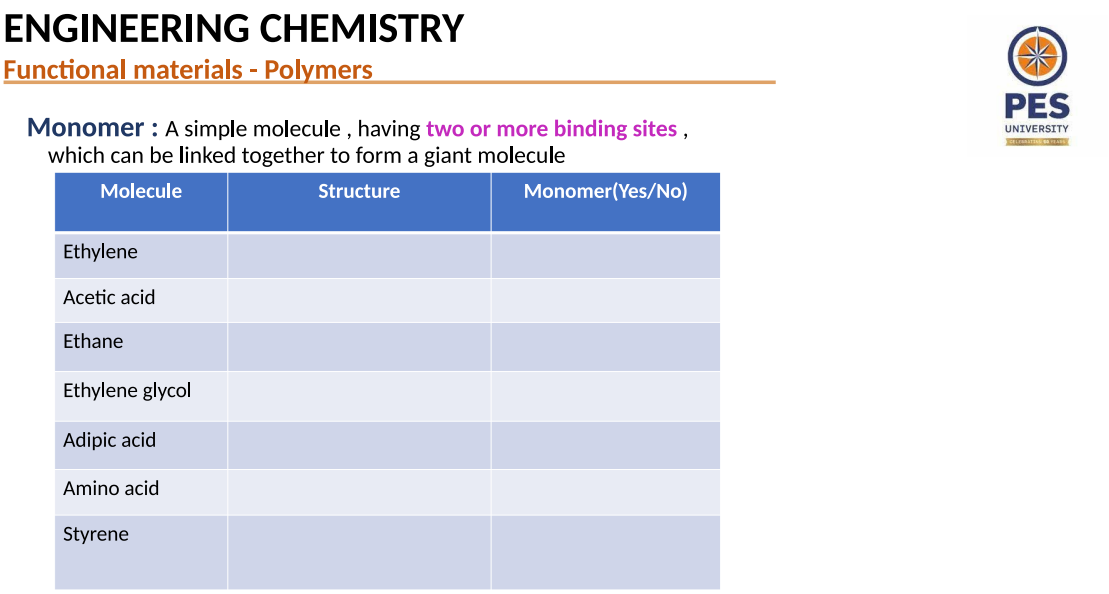


Figure G000038. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 3.

EV0000306

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



ENGINEERING CHEMISTRY

Functional materials - Polymers

Monomer : A simple molecule , having **two or more binding sites** , which can be linked together to form a giant molecule

Molecule	Structure	Monomer(Yes/No)
Ethylene		
Acetic acid		
Ethane		
Ethylene glycol		
Adipic acid		
Amino acid		
Styrene		

Figure G000039. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 4.

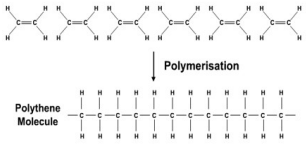
EV0000309

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Polymerisation: The process by which monomers of low molecular weight are converted to polymers of high molecular weight



Source:https://www.assignmentpoint.com/science/chemistry/addition-polymerisation.html

Functionality : The total number of functional groups or bonding sites present in a monomer

Monomer	Structure	Functionality
Ethene		bifunctional
Ehtylene glycol		bifunctional
Glycerol		trifunctional
Amino acid		bifunctional

Figure G000040. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 5.

EV0000312

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

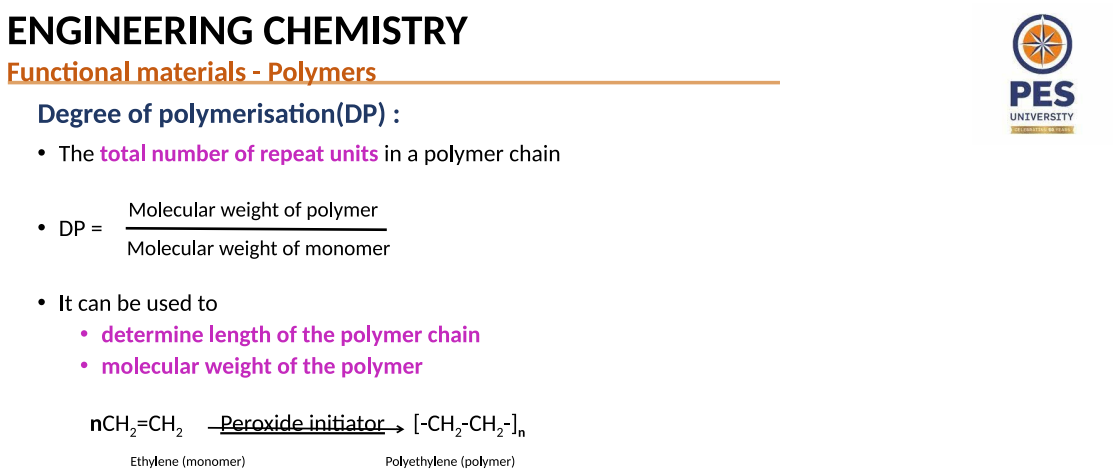


Figure G000041. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 6.

EV0000315

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

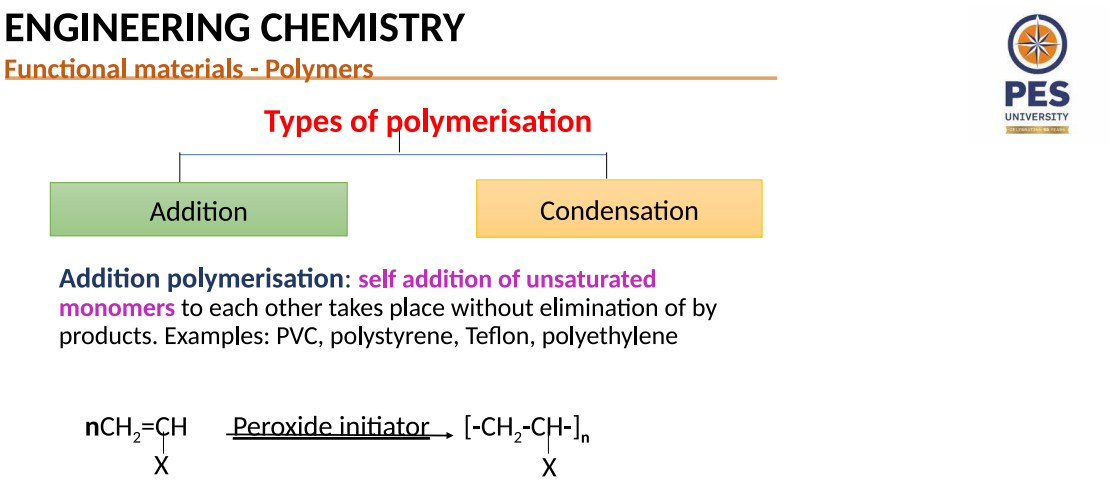


Figure G000042. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 7.

EV0000318

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

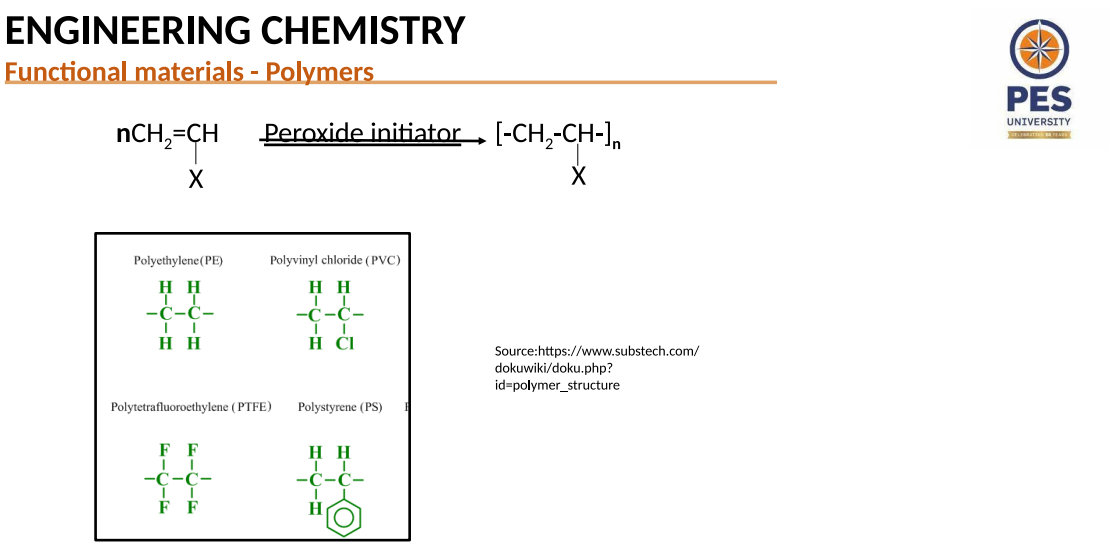


Figure G000043. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 8.

EV0000321

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

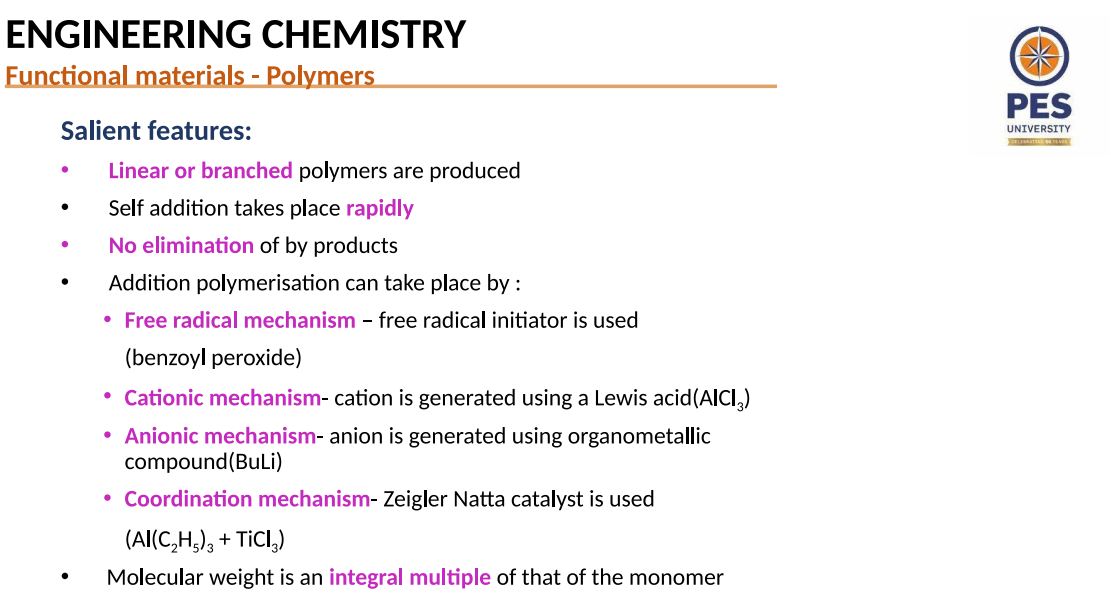


Figure G000044. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 9.

EV0000324

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

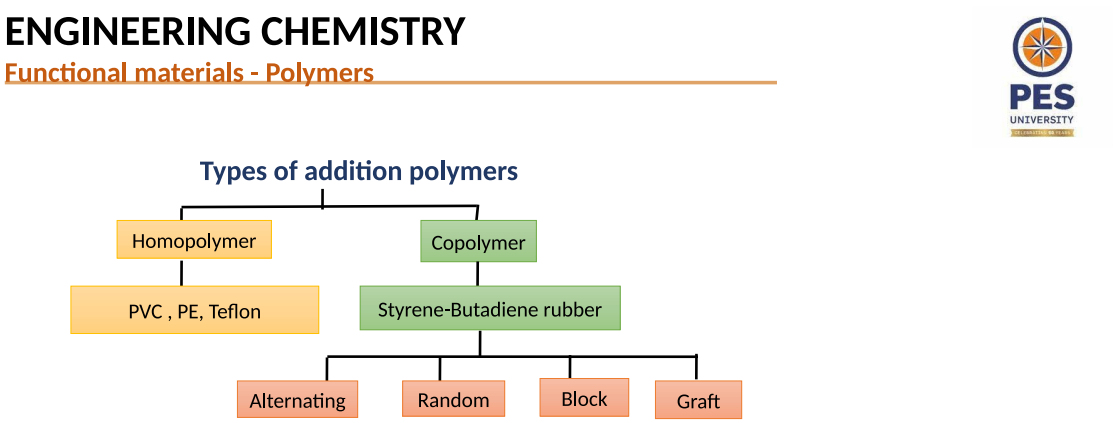


Figure G000045. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 10.

EV0000327

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

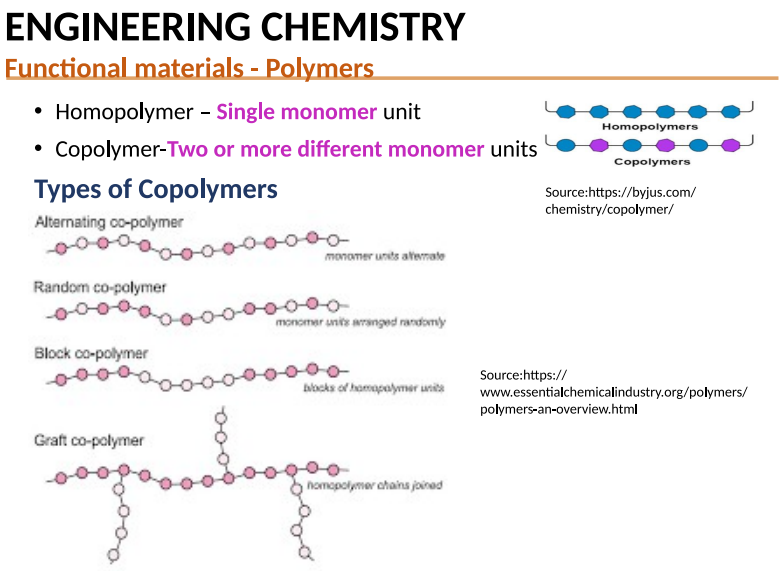


Figure G000046. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 11.

EV0000330

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

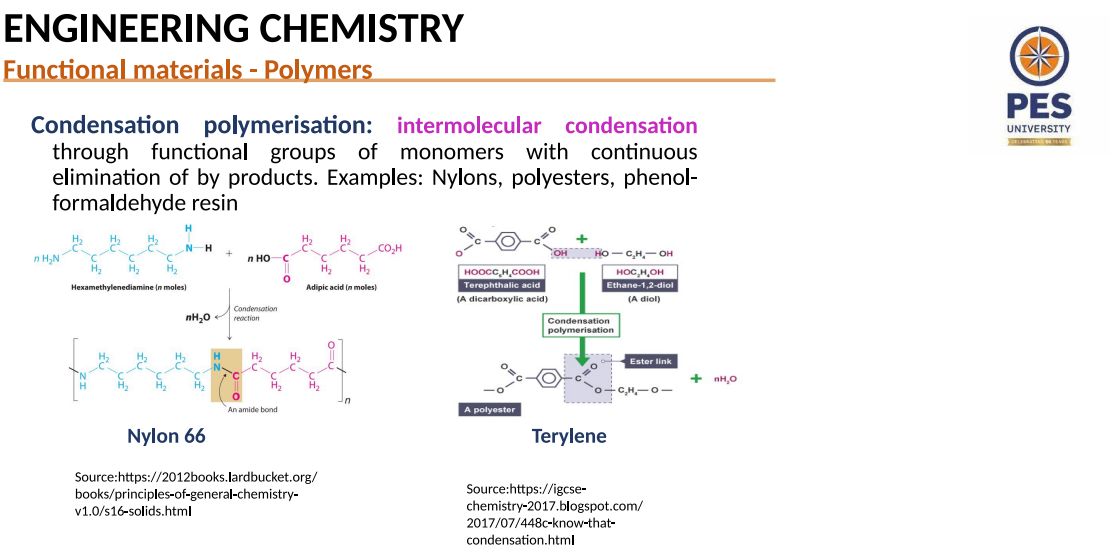


Figure G000047. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 12.

EV0000333

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

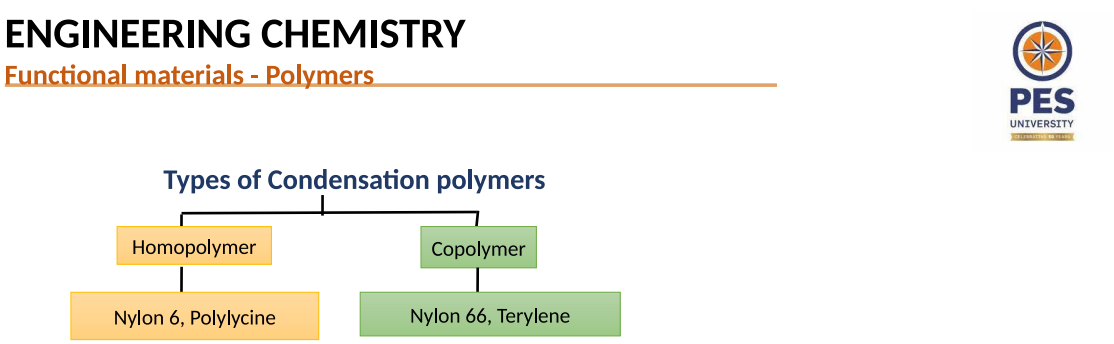


Figure G000048. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 13.

EV0000336

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

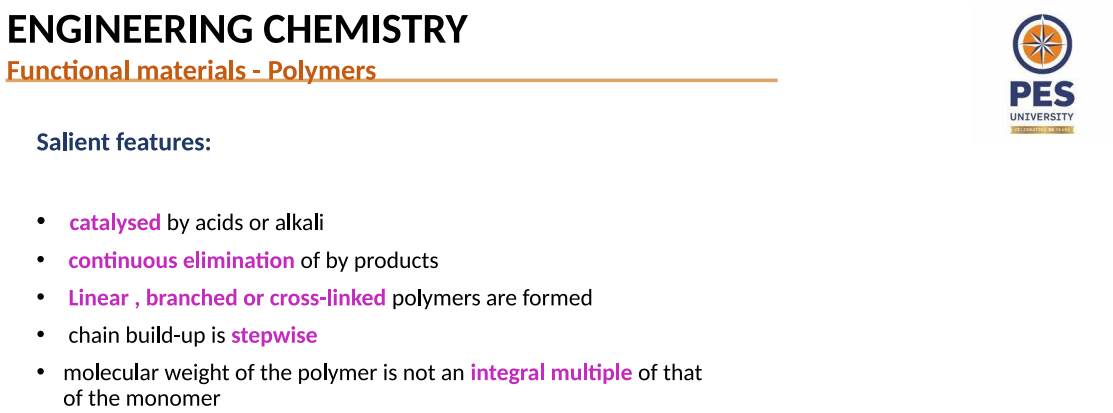
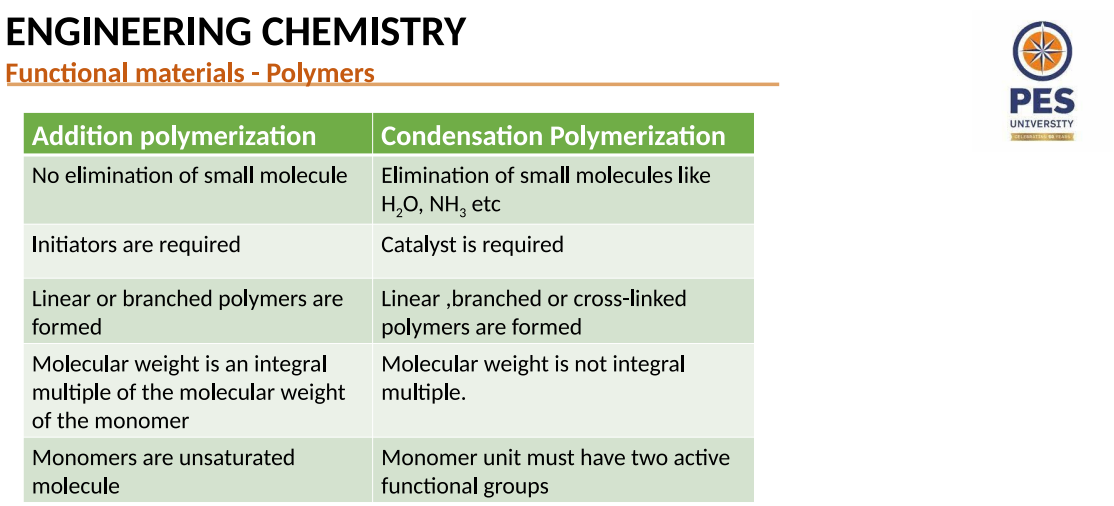


Figure G000049. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 14.

EV0000339

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



ENGINEERING CHEMISTRY

Functional materials - Polymers

Addition polymerization	Condensation Polymerization
No elimination of small molecule	Elimination of small molecules like H_2O, NH_3 etc
Initiators are required	Catalyst is required
Linear or branched polymers are formed	Linear ,branched or cross-linked polymers are formed
Molecular weight is an integral multiple of the molecular weight of the monomer	Molecular weight is not integral multiple.
Monomers are unsaturated molecule	Monomer unit must have two active functional groups

Figure G000050. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 15.

EV0000342

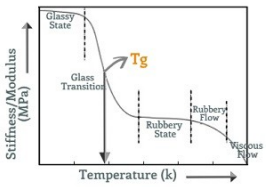
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Glass Transition temperature

- All amorphous polymers, when cooled below a certain temperature, become **stiff, hard, brittle and glassy** but above this temperature they are **soft, flexible and rubbery**
- This transition temperature of a polymer from glassy state to rubbery state is called its **glass transition temperature** and is denoted by T_g
- The rubbery state on heating melts and starts flowing. This temperature is the **melting point, T_m**



Source:https://omnexus.specialchem.com/polymer-properties/properties/glass-transition-temperature

Figure G000051. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 16.

EV0000345

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

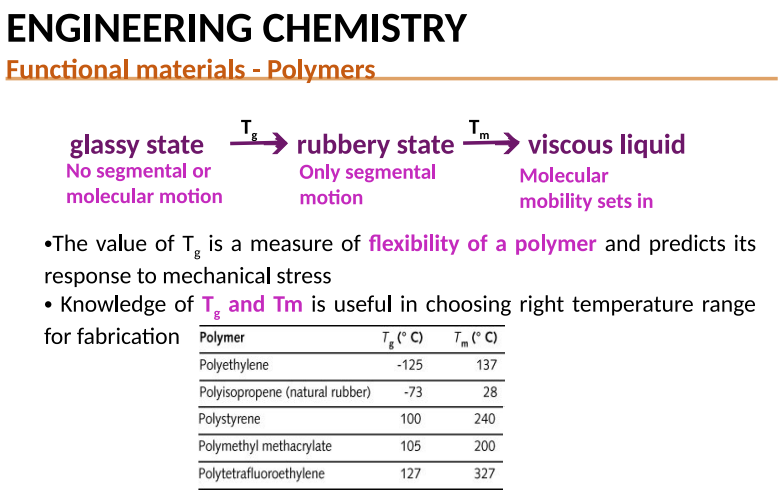


Figure G000052. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 17.

EV0000348

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

THANK YOU

Department of Science and Humanities

Figure G000053. Polymers Introduction types of polymerizations definition of glass transition temperature · source slide page 18.

EV0000351

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000299 · EV0000300 · EV0000301
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 42 content: Introduction, types of polymerization, definition of glass transition temperature EV0000302 · EV0000303 · EV0000304
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Polymers Polymer : A giant molecule or a macromolecule formed by repeated joining of several simple molecules Polymers can be linear, branched or cross-linked Source:https://www.livescience.com/60682-polymers.html [Image text, OCR] H H H H H H H H (a) H H H H H H H H Linear Polymer CH3 CH; CH2 CH2 (b) CH-CH2-C- CH-C-CH-CH; CH: CH2 CH2 Branched Polymer CH2 CH; CH; CH; CH;- C -CH2- C -CH; (c) 1 CH2 CH2 CH-C- CH-C- CH; CH2 CH3 Cross-linked Polymer CH3 Source:https://www.researchgate.net/figure/A-schematic-illustration-of-a-linear-polymers-b-branched-polymers-and-c_fig2_323689828 EV0000305 · EV0000306 · EV0000307
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Monomer : A simple molecule , having two or more binding sites , which can be linked together to form a giant molecule Molecule | Structure | Monomer(Yes/No) Ethylene | | Acetic acid | | Ethane | | Ethylene glycol | | Adipic acid | | Amino acid | | Styrene | | EV0000308 · EV0000309 · EV0000310

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Polymerisation: The process by which monomers of low molecular weight are converted to polymers of high molecular weight Functionality : The total number of functional groups or bonding sites present in a monomer [Image text, OCR] Polymerisation H H H H H H H H H H Polythene Molecule H H H H H Source:https://www.assignmentpoint.com/science/chemistry/addition-polymerisation.html Monomer | Structure | Functionality Ethene | | bifunctional Ehtylene glycol | | bifunctional Glycerol | | trifunctional Amino acid | | bifunctional EV0000311 · EV0000312 · EV0000313
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Degree of polymerisation(DP) : The total number of repeat units in a polymer chain DP = It can be used to determine length of the polymer chain molecular weight of the polymer Molecular weight of polymer Molecular weight of monomer nCH2=CH2 Peroxide initiator [-CH2-CH2-]n Ethylene (monomer) Polyethylene (polymer) EV0000314 · EV0000315 · EV0000316
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Types of polymerisation Addition polymerisation: self addition of unsaturated monomers to each other takes place without elimination of by products. Examples: PVC, polystyrene, Teflon, polyethylene Condensation Addition nCH2=CH Peroxide initiator [-CH2-CH-]n X X EV0000317 · EV0000318 · EV0000319
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers nCH2=CH Peroxide initiator [-CH2-CH-]n X X [Image text, OCR] Monomerstructures ofdifferentpolymers Polyethylene(PE) Polyvinyl chloride (PVC) Polypropylene(PP) HI HL H H H H C C C C C -C H H H Cl H CH3 Polytetrafluoroethylene(PTFE) Polystyrene (PS) Polymethyl metacrylate (PMMA) Ol H H H H C C -C-O-CH3 H CH3 www.substech.com Source:https://www.substech.com/dokuwiki/doku.php?id=polymer_structure EV0000320 · EV0000321 · EV0000322
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Salient features: Linear or branched polymers are produced Self addition takes place rapidly No elimination of by products Addition polymerisation can take place by : Free radical mechanism – free radical initiator is used (benzoyl peroxide) Cationic mechanism- cation is generated using a Lewis acid(AlCl3) Anionic mechanism- anion is generated using organometallic compound(BuLi) Coordination mechanism- Zeigler Natta catalyst is used (Al(C2H5)3 + TiCl3) Molecular weight is an integral multiple of that of the monomer EV0000323 · EV0000324 · EV0000325
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Types of addition polymers Homopolymer Copolymer PVC , PE, Teflon Styrene-Butadiene rubber Alternating Random Block Graft EV0000326 · EV0000327 · EV0000328
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Homopolymer – Single monomer unit Copolymer-Two or more different monomer units Types of Copolymers Source:https://byjus.com/chemistry/copolymer/ [Image text, OCR] Alternating co=polymer -0-00-0- monomer unsits affemate Random co-polymer monomer units arsnged rardomy Block co=polymer blocks of horoopolymer unilts Graft co-polymer nomcpolymer etans jbihed Source:https://www.essentialchemicalindustry.org/polymers/polymers-an-overview.html EV0000329 · EV0000330 · EV0000331

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Condensation polymerisation: intermolecular condensation through functional groups of monomers with continuous elimination of by products. Examples: Nylons, polyesters, phenol-formaldehyde resin [Image text, OCR] H H H H2N nHO H H H H Hexamethylenediamine(nmoles) Adipicacid (nmoles) Condensation nH2O reaction H N N C H H H H Anamidebond Nylon [Image text, OCR] OH HO-CH-OH HOOCC,H,COOH HOC,HOH Terephthalicacid Ethane-1,2-diol (A dicarboxylicacid) (A diol) Condensation polymerisation Esterlink nH,O CH-O- Apolyester Nylon 66 Terylene Source:https://2012books.lardbucket.org/books/principles-of-general-chemistry-v1.0/s16-solids.html Source:https://igcse-chemistry-2017.blogspot.com/2017/07/448c-know-that-condensation.html EV0000332 · EV0000333 · EV0000334
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Types of Condensation polymers Homopolymer Copolymer Nylon 6, Polylycine Nylon 66, Terylene EV0000335 · EV0000336 · EV0000337
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Salient features: catalysed by acids or alkali continuous elimination of by products Linear , branched or cross-linked polymers are formed chain build-up is stepwise molecular weight of the polymer is not an integral multiple of that of the monomer EV0000338 · EV0000339 · EV0000340
15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Addition polymerization | Condensation Polymerization No elimination of small molecule | Elimination of small molecules like H2O, NH3 etc Initiators are required | Catalyst is required Linear or branched polymers are formed | Linear ,branched or cross-linked polymers are formed Molecular weight is an integral multiple of the molecular weight of the monomer | Molecular weight is not integral multiple. Monomers are unsaturated molecule | Monomer unit must have two active functional groups EV0000341 · EV0000342 · EV0000343
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Glass Transition temperature All amorphous polymers, when cooled below a certain temperature, become stiff, hard, brittle and glassy but above this temperature they are soft, flexible and rubbery This transition temperature of a polymer from glassy state to rubbery state is called its glass transition temperature and is denoted by Tg The rubbery state on heating melts and starts flowing. This temperature is the melting point, Tm [Image text, OCR] Glassy State Tg Transition Glass (ew) iRubbery Rubbery Flow State Wisdous OW Temperature (k) Source:https://omnexus.specialchem.com/polymer-properties/properties/glass-transition-temperature EV0000344 · EV0000345 · EV0000346
17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers glassy state ☒ rubbery state ☒ viscous liquid The value of Tg is a measure of flexibility of a polymer and predicts its response to mechanical stress Knowledge of Tg and Tm is useful in choosing right temperature range for fabrication Tg Tm No segmental or molecular motion Only segmental motion Molecular mobility sets in EV0000347 · EV0000348 · EV0000349
18. Source slide page 18 (extracted text): THANK YOU Department of Science and Humanities EV0000350 · EV0000351 · EV0000352

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.932] Hello everyone, welcome back to this course on engineering chemistry. Today we'll start with a new module on functional materials. [00:00:10.756] In that module we are going to study about polymers, the types of polymerization, structure property relationship, commercial polymers will take up butile rubber, biodegradable polymers, epoxy resin, polymer composites, cablar and carbon fiber [00:00:27.216] Conducting polymers and we will look at molecular weight of polymers. Then we'll also study organic LEDs. So today's class let us start with an introduction to polymers, the types of polymerization, and about the glass transition temperature. [00:00:48.996] First, we will take up some definitions. Let us understand what is a polymer. A giant molecule or a macromolecule, which is formed by repeated joining of several simple molecules. If you look at the picture here, we can see some small [00:01:06.960] molecules over here and they have been joined together by a process which is called polymerization to give this long chain and this chain of these simple molecules is known as polymer. What we have to know is between each of these [00:01:22.672] small simple molecules the bond is a covalent bond now these polymer chains they could be linear they could be branched like this or they there could be a cross link network type of polymer [00:01:43.780] What is a monoma then? A simple molecule having two or more binding sites which can be linked together to form a giant molecule. [00:01:54.212] When we say simple molecule, do all simple molecules classify as small modes? Let us take up some examples and understand this. Ethylene. What is the structure of ethylene? CH2, double bond, CH2. Yes. [00:02:09.728] What is the criteria? There should be two or more binding sites. Yes, we find that there are two binding sites across the double bond. The double bond can open and we get two carbons wherein the bonding can happen. So this we will say yes. Now acetic acid. [00:02:26.400] CH3 COOH. Now here we have just one functional group, so therefore it cannot bind with several molecules. It can maximum m bind with one more ovastic acid and you just end up with a dimer. You definitely cannot get a polymer. [00:02:44.900] Now ethane CH3, CH3. No, there are no binding sites at all. So there is no possibility of formation of a polymer. Ethylene glycol. What is ethylene glycol? CH2OH. CH2OH. So now here we have [00:03:04.160] Two OH groups, yes, if the binding can happen on both sides, therefore we could get a polymer from here. Adipic acid. What is adipic acid? We have COOH, and here we have CH2 four times. [00:03:21.024] And again we have COOH. So definitely there are two uh carboxylas carboxylic acid groups here, therefore it will act as a monument. Amino acid we have NH2 and this [00:03:37.124] CH2 could be N times, and then we have the C O O H. Yes, here also we have two different binding functional groups. Therefore, this also will qualify as a monomer. Styrene. What is styrene? CH [00:03:56.612] to CH and there is a benzene ring here. So phenyl group, therefore, yes. Where does the binding take place? Now here the binding is across the double bond. The moment you have unsaturation then we know that it is possible that you [00:04:15.200] This double bond can open up and you have two binding sites and therefore this also qualifies as a monomer. So not all simple molecules can act as monomers unless they have this two or more binding sites, the molecule cannot be used as a monomer. [00:04:32.676] The next definition is of polymerization. The process by which monomers of low molecular weight are converted to polymers of high molecular weight. Let us take this example of ethene. You can see there are double bonds, so many ethene uh molecules are there, and when they are polymerized, you get a [00:04:50.448] uh chain like this and this is polyethylene. Now when we say molecule for a polymer please understand that we are referring to a full chain. One chain will be referred to as a polymer molecule.

EV0000054 · EV0000055 · EV0000056 · EV0000057

Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.932] Now another term we need to understand is functionality. The total number of functional groups or bonding sites which are present in a monoma. Now let us take up some examples again. Ethene. Now ethene would be CH2. [00:05:19.568] Double bond CH2. Now since there are two binding sites here, we say that it is bifunctional. Ethylene glycol. Just now we saw it is CH2OH. [00:05:31.716] CH2OH. Therefore, again it is bifunctional. Now what would be glycerol? Glycerol is CH2OH. [00:05:43.460] CHOH and then again CH2OH. Therefore, we know that there are three functional groups through which binding can happen. So glycerol is going to be trifunctional in nature. Now coming to amino acid. [00:06:00.848] Just now we saw it has an NH2 group and then CH two it could be N number and then we have COOH. Again there are two functional groups, so it is again bifunctional in nature. Degree of polymerization. [00:06:18.416] So the total number of repeat units in a polymer chain. How would you find out the degree of polyzation? If you know the molecular weight of the polymer and you also know the molecular weight of the monomer which you have used, we can calculate the degree of polymerization, molecular weight of polymer divided by [00:06:36.048] molecular weight of the monoma. What would be the use of knowing the degree of polymerization? We can determine the length of the polymer chain and we can also find out what is the molecular weight of the polymer. So suppose we have a reaction like this where ethylene is [00:06:53.040] uh undergoing polymerization and giving you polyethylene then we have n of these ethylene molecules and of course initiator is a must whenever we do uh polymerization initiator or a catalyst is required without that polymerization cannot happen. So whenever we write a [00:07:10.704] voluminization reaction it is essential that we write the initiator or the catalyst. So here we see that this double bond opened up and we have a [00:07:23.632] long chain of how many units there will be n such units. So what would be the degree of polymerization in the above for the above reaction? That would be n because how did you define degree of polymerization? The total number of repeat units. So in this particular reaction it is going to be n. [00:07:44.036] Now there are two types of polymerization which are possible. One is the addition polymerization and the other is the condensation polymerization. Let us take them up one by one. [00:07:55.024] Addition polymerization would be self addition of unsaturated monomers to each other takes place without the elimination of any byproducts. So here such an addition polymerization is also known as chain polymerization. [00:08:16.260] chain polymerization and examples would be PVC polystyrene teflon polyethene. So how exactly it is happening? Let us see the reaction here. So we have N of CH2 double bond CHX. C x is some other [00:08:33.028] Entity except uh so it could be H also, it is just some substitute substitution substituted group over here. So peroxide indicator, and we get that what happens, the double bond opens and the X remains intact over here. So CH2 single bond CHX. [00:08:50.800] Of course the bonding is happening on both the directions and therefore we have n to n of these units. Now let us see how using this reaction we can get different types of polymers. [00:09:06.020] So here we saw that the substituent is X. Now just by changing this substituent, we can get different types of polymers. For example, if we have hydrogen as the substituent, we end up with polyethylene. [00:09:24.868] Now, if we change the substituent to Cl, then we find that the polymer now becomes polyvinyl chloride. [00:09:34.224] Just by changing the substituent to Cl, we got polyvinyl chloride, which is totally different from polyethylene in properties. Now, if all these hydrogens were to be substituted, as in the case of Teflon. [00:09:50.992] What do we get? We get is all four are fluorine atoms now. Instead of hydrogens, we are using fluorine atoms and what we end up will be Teflon, which is very very uh strong polymer.

EV0000058 · EV0000059 · EV0000060 · EV0000061

Transcript block 3 (source interval 000600-000899 seconds): [00:10:06.852] and has exceptional uh properties and it is known as wonder plastic. Now another example would be if we replace the X by a C6H5. [00:10:18.724] Then what we get is polystyrene. So what do we understand here? Just by changing the hydrogen atom. [00:10:27.808] You could change one atom, you could change all the four, you could change any number of hydrogen atoms, each time it is going to give you a polymer with totally different properties. [00:10:42.628] Now what are the salient features for addition polymerization? We can have linear or branched polymers. [00:10:51.268] Self addition takes place very rapidly, as I mentioned, it's also known as chain polymerization. There's no elimination of byproducts. Addition polymerization can take place by [00:11:03.492] Different mechanisms. It could happen by free radical mechanism. So in that case, the initiator used is a free radical benzoyl peroxide. So free radical mechanism, what does it do? It is going to generate a free radical. [00:11:20.068] in the monomer and that is going to help in the propagation of the reaction. So that is what is free radical mechanism. So if we want to carry out the polymerization through free radical mechanism the initiator used has to be some uh [00:11:37.680] free radical generator and that example would be benzoyl peroxide. That is not the only mechanism which can be followed for addition polymerization, it is also possible to do by cationic mechanism. Now in this case [00:11:53.136] A cation is generated just like in the free radical mechanism, free radical is generated. Here we are going to generate a cation. So what would be the initiator in this case? It is a Lewis acid. [00:12:06.404] That would be example would be ALCl3. Now if the mechanism is anionic mechanism, then obviously now we understand anion has to be generated and therefore we are going to use some other agent now and that would be an organometallic compound. So you know common [00:12:24.176] Only known organic organometallic compounds have Wrignax reagents. So here we are going to use butile lithium. That would be one example. Or it could also happen through coordination mechanism. What is a coordinate bond where the covalent bond where the electrons are shared by one of the [00:12:42.368] uh atoms. So if we have coordinate bond formation then this will be known as coordination mechanism. If the polymerization is happening through this mechanism, what would be used here? We are going to use Ziegler Nata catalyst and example of that would be [00:12:59.424] uh aluminium tri ethyl aluminium plus titanium chloride ti Cl3 or TICL four is also used sometimes. So that would be coordination mechanism. Now the good thing about Ziegler Not a catalyst is that [00:13:12.784] This is happening catalyst when we say it always happens on the surface. The reaction is going to happen on the surface. So when we use Ziegler N catalyst and the mechanism is a coordination mechanism, all the monomers are going to come onto the catalytic surface. So the [00:13:30.448] Polymer which is produced is going to be a very stereorcular polymer. So by changing the mechanism, we can get different types of polymers. Now [00:13:44.464] For addition polymerization, we should know that molecular weight is an integral multiple of that of the monoma. The reason being that there is no elimination of any byproducts. Now amongst the addition polymerization. [00:13:59.972] We can have different types. It could be a homopolymer where you have just one type of monoma being used, or it could be a copolymer where you will have [00:14:11.456] Two or more different types of monomers being used. Now when you talk of homopolymers, best examples would be PVC, polyethylene, teplon, where only one type of monomer. Like for example, polyethylene, we are using only ethylene as the monomer, and PVC we are using vinyl fluoride. [00:14:29.104] So that would be a homopolymer, but what would be a copolymer where we use two different monomer units. So here, for example, SBR, you know, styrene, butadine, rubber. Now here there are two

monomers, one is styrene and one is butadine. Now when we use two different monomers. [00:14:47.492] we could again get different types of co-polymers. [00:14:52.132] Depending on how these monomers are arranged. It's not necessary that they have to be alternate. So they could be alternating, they could be random, they could be block, or they could be graft. So these are different types of copolymers now.

EV0000062 · EV0000063 · EV0000064 · EV0000065

Transcript block 4 (source interval 000900-001199 seconds): [00:15:07.780] Let us look at them. When we say a single monomer unit, then it's called homopolymer. When we have two or more different monomer units, it's called a copolymer. So you can see here same unit being used, that is a homopolymer. Now in different units, two different units are being used, monomers are being used. [00:15:25.552] it is called a copolymer. Now as I mentioned you these copolymers when they are formed, the monomers arrangement could be different in different types of copolymers. Now if [00:15:38.000] It is alternating. You can see one is say M1, the other is M two, and similarly they are alternating throughout the chain. That would be an alternating co-polymer. Or it could be random means. [00:15:50.628] The copoly the two monomer units are just appearing randomly. There's no particular arrangement that would be a random copoly mark. Then we can also have a block copolymer where we have a block of one type of monomer, and then we have a block of another type of monoma. Then again, this block. [00:16:08.400] Is coming up and then we have another block. So this is known as a block copolymer, or we could also have a graft copolymer where we have the backbone of one monomer and the other monomer is being grafted onto this backbone. So such a [00:16:26.304] uh copolymer could be called as a graft copolymer. So these are the different types of copolymer which are possible in addition polymerization. [00:16:36.388] Talking about condensation, polymerization, it would be intermolecular condensation through functional group of the monomers with continuous elimination of byproducts. And common examples would be nylons, polyester, and [00:16:49.792] uh phenol formaldehyde resins all these are condensation polymers. Now let us take up nylon sixty six. Here we have the adipic acid and we have hexamethylene diamine. [00:17:05.124] Hexamethlene diamine, when we use that, there is an elimination of [00:17:11.364] the uh so here is sorry this is hexamethylene diamine and this is the uh adipic acid we see that COOH group here so OH from here and the hydrogen from the uh hexamethylene diamine is being used so [00:17:27.488] hydrogen of the amine group. So here we have elimination of water molecule. So when this molecule gets eliminated, the kind of linkage which we get is NHCO. So this is known as an amide linkage. The linkage here is an amide linkage. [00:17:47.428] CONH group is known as the MI group. So here nylon is 66 is formed by condensation of hexamy, diamine, and uh [00:17:59.236] Adipic acid and elimination of water molecule is going to give you this kind of bonding between the uh two uh molecules it will take place and this rip unit is going to be repeated throughout the [00:18:14.948] polymer chain. Now another example would be pterylene, where we have terithalic acid, again we have COOH here and ethylene glycol. So here we have [00:18:26.656] uh OH over A here and also OH over here. So the it is to be written in this way so that the bonding will take place on both directions. So when this condensation happens again there is a elimination of water molecule. But what is the linkage we are getting here? Now it is a [00:18:43.600] ester linkage COO group so it is an ester linkage which is getting formed over here so we have this condensation between the two molecules water is getting eliminated and get C O [00:18:59.428] as the linkage. So this will be a polyester. We call it a polyester and the common name for using when you use terithalic acid and ethylene glycol is

terylin. So this is condensation polymerization. [00:19:12.112] examples. Now what would be the salient features here? The as acids or alkali are being used as catalysts. In chain in chain polymerization, that is in addition polymerization we used initiators. [00:19:28.352] We use the initiators like pre radicals, etcetera. Here it is cataly here we use a catalyst and this is also known as step polymerization. Chain polymerization is addition polymerization. Condensation polymerization is called step polymerization. Here [00:19:46.144] Because the chain buildup happens stepwise, and there is continuous elimination of the byproducts. And one thing to note here is that linear branched or even cross-linked polymers are possible for condensation polymers. And of course, the molecular weight of the polymer is not an integral multiple.

EV0000066 · EV0000067 · EV0000068 · EV0000069

Transcript block 5 (source interval 001200-001499 seconds): [00:20:05.860] of the monoma because continuous elimination of these small molecules is happening. So these would be [00:20:15.300] Differences between addition and condensation polymerization. So there's no elimination of any molecule. Here there are small molecules are getting eliminated. Here initiators are required. Here we have catalyst and then linear or branch polymer are produced. Here we have cross linked also apart from linear and branched molecules. [00:20:33.072] Weight is an integral multiple because there's no elimination of molecules, but small molecules here it is not an integral multiple of the monoma. And then monomers are unsaturated molecules are used as monomers and here they must have two active functional groups. [00:20:53.764] Now we'll come to glass transition temperature. All amorphous polybles when they are cooled below a certain temperature, they are found to become stiff, they become hard, brittle, and that state is known as the glassy state. [00:21:13.572] But above the temperature, when it is heated, they become soft and flexible, and that is referred to as the rubbery state. So this transition temperature [00:21:25.892] at which the polymer changes from the glassy state to the rubbery state is known as the glass transition temperature. And that is given by Tg. [00:21:40.932] Now you come to the rubbery state. This rubbery state when you heat it further then the [00:21:46.976] Polymer will start flowing, it gets into a viscous flow, and this temperature is known as the melting point of the polymer. So in a polymer there will be two temperatures which are very important. One when it changes, you can see the graph here, one when it changes from the glass state. [00:22:04.624] To the rubbery state. So as you increase the temperature, it is changing here, and that point where it changes from glassy to rubbery state is known as the Tg, and this rubbery state further. [00:22:17.412] it when you heat it, when you're heating it inside, then it becomes into a viscous liquid and starts flowing, and when that change happens that is known as the melting point. [00:22:27.748] So we have the glassy state going to the rubbery state at Tg and then at TM it changes to viscous liquid. So what exactly is happening? Why is it changing like this? Now in the glass state, we know that polymer is made up of these long chains. [00:22:43.776] When we go to the very low temperature, I mean below the Tg, it is found that there is absolutely no segmental motion, neither the molecular motion. So it's like very stiff and therefore it behaves almost like a glass. [00:22:59.824] Is that the glassy stint. Now when you heat it beyond Tg, what happens? You are giving energy to these molecules. So some of the force of attraction between the polymer chains is overcome, and we find that some kind of [00:23:15.376] segmental motion will set in though the polymer is still not completely force of attraction has not been completely removed uh overcome. So some small segmental motion will start [00:23:31.424] But nevertheless, no molecular motion is here. So therefore only segmental motion. And this would be referred to as a rubbery state. But when you

heat more, that means you will overcome all the forces of attraction uh between the polymer chains and the molecules now become absolutely [00:23:50.112] free like this and therefore it change it becomes into a viscous liquid. So this is what happens at the glass transition temperature. So the glass transition temperature gives us an idea about the flexibility of a polymer and predicts also its response to mechanical stress. Now [00:24:07.792] If you want to fabricate a polymer or use it for a particular application, both TG and TM values are very important. We should know what are these values. So you can see some of the polymers listed here, the TG and the TM values. [00:24:25.360] Now polyethylene if you see minus one twenty five degrees centigrade. So that means at room temperature we find that it is above the T G value and it is [00:24:37.632] Soft, whereas coming to polystyrene, we know polystyrene what it looks like, and there we see it is 100. So that means. [00:24:46.448] At room temperature it is still in the glassy state. So it is still very hard and brittle, whereas polyethylene will be soft and flexible. So we can make out from the glass transition temperature that [00:24:59.460] Where is the what would be the state at room temperature, and of course, when you know TG and TM you can fabricate and use this polymer for different applications. So the TG values give us an idea of thermal expansion, heat capacity, refractive index, electrical, and mechanical.

EV0000070 · EV0000071 · EV0000072 · EV0000073

Transcript block 6 (source interval 001500-001799 seconds): [00:25:17.232] Properties of the polymer. So this is about the glass transition temperature. In the next class, we are going to talk about the structure property relation between [00:25:31.328] uh the in the in the polymers the the relation between the structure and property of a polymer. Thank you.

EV0000074 · EV0000075 · EV0000076 · EV0000077

Full transcript: 3636 normalized words, 101 timestamped segments, 1539.1 seconds duration. Low-confidence segments flagged: 0.

EV0000054 · EV0000055 · EV0000056 · EV0000057

6 Structure-property relationship

This lesson is indexed from the source video titled Structure-property relationship. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000078 · EV0000079

Graphs and diagrams

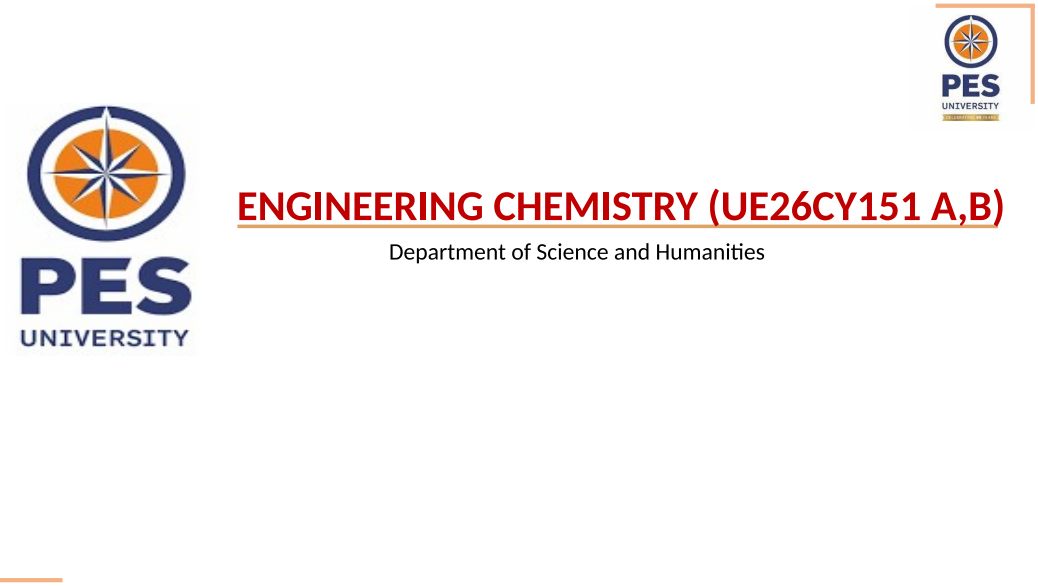


Figure G000054. Structure property relationship · source slide page 1.

EV0000354

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000055. Structure property relationship · source slide page 2.

EV0000357

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

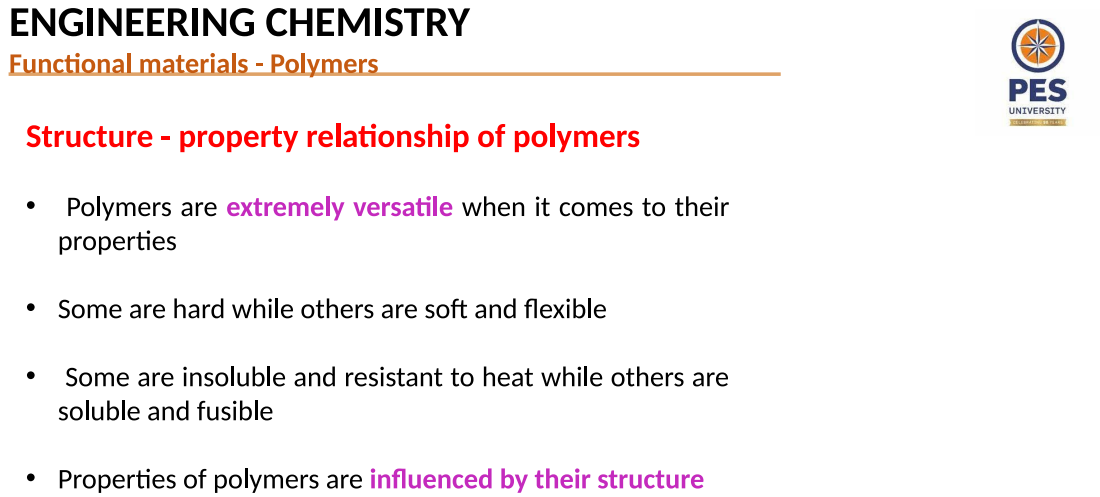


Figure G000056. Structure property relationship · source slide page 3.

EV0000360

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

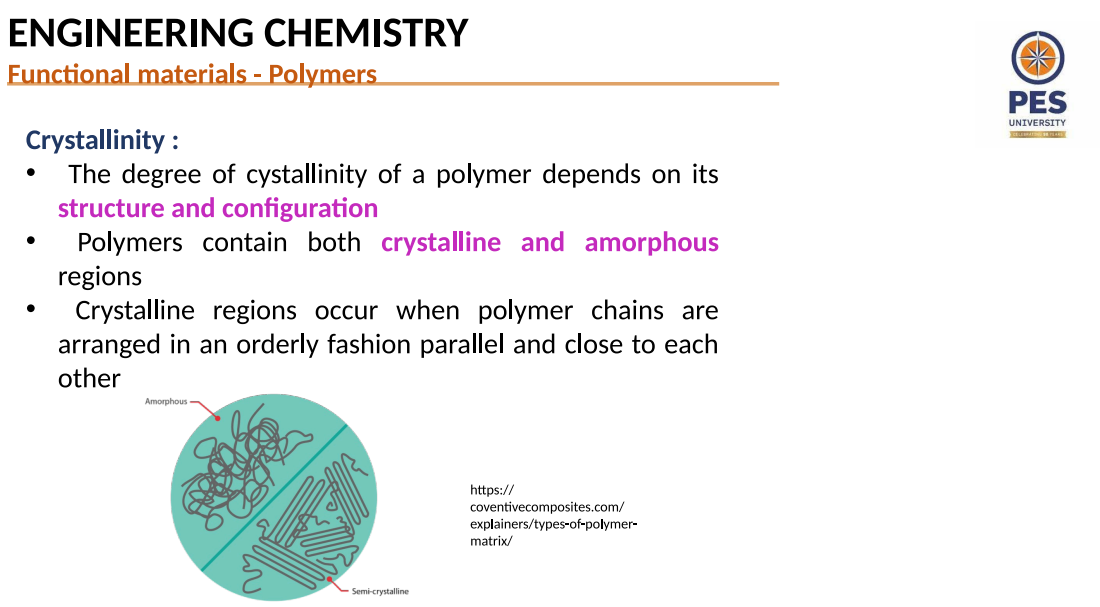


Figure G000057. Structure property relationship · source slide page 4.

EV0000363

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

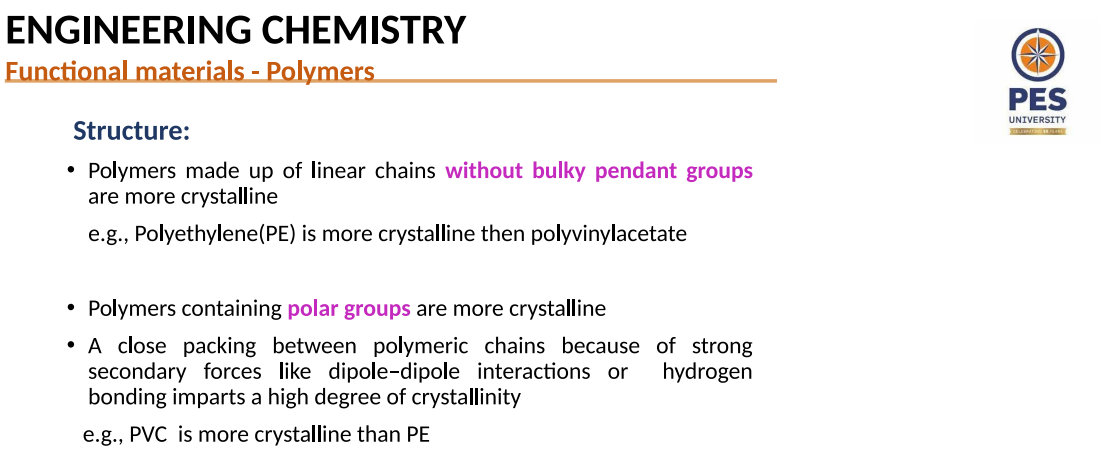


Figure G000058. Structure property relationship · source slide page 5.

EV0000366

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

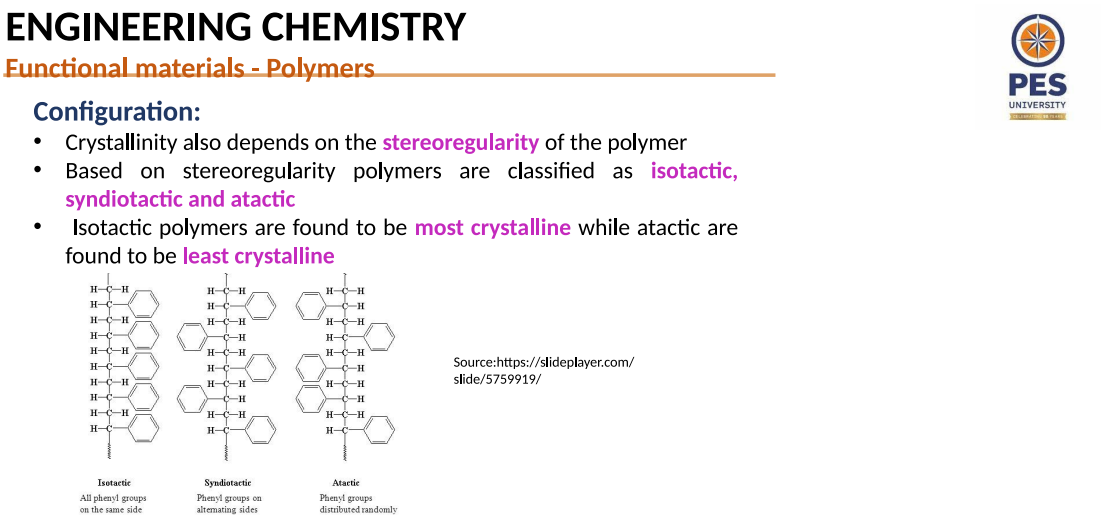


Figure G000059. Structure property relationship · source slide page 6.

EV0000369

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

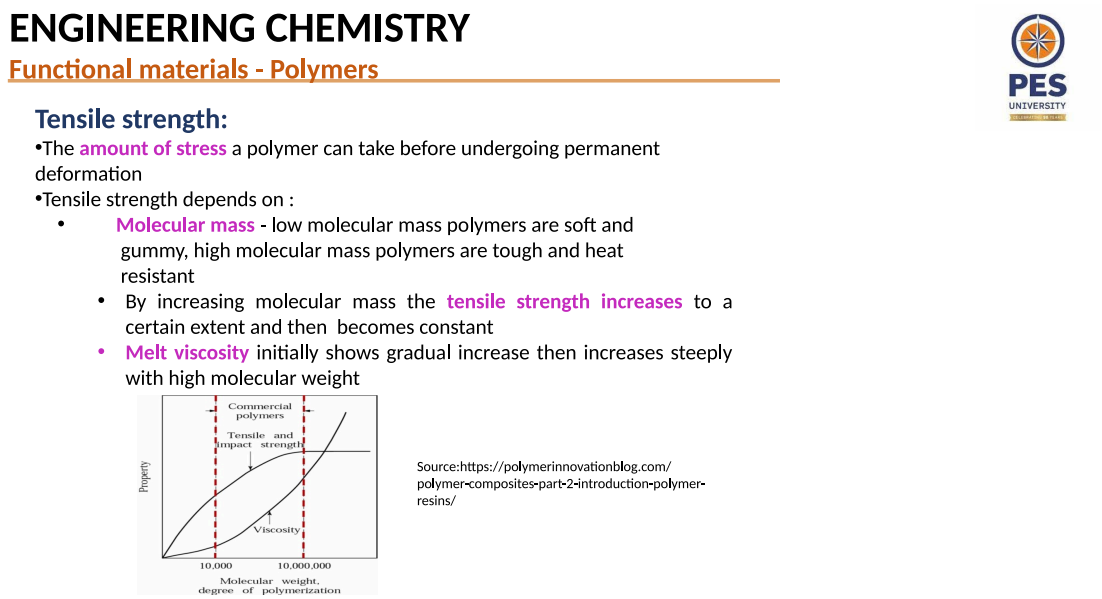


Figure G000060. Structure property relationship · source slide page 7.

EV0000372

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

- Tensile strength increases when polymer has **polar groups**
- Force of attraction between chains prevents polymer chains from **slipping against each other** e.g., PVC, Nylon
- **Cross linking** in a polymer also increases its tensile strength

Figure G000061. Structure property relationship · source slide page 8.

EV0000375

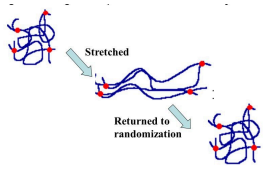
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Elasticity :

- The property of a polymer by which it **deforms on applying force and regains original shape on release of force** is called elasticity
- Elastomers have **long coiled and entangled chains** which straighten and orient themselves on stretching. When stress is removed they go back to coiled form



Source:https://slideplayer.com/slide/8755303/

- For a polymer to show elasticity the individual chains **should not break** on prolonged stretching

Figure G000062. Structure property relationship · source slide page 9.

EV0000378

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

- Presence of **polar groups** in the polymer chain results in strong secondary force of attraction between the chains making them **stiff and inflexible**
- **Non polar groups** have weak van der Waal's attraction between them and hence they are **more elastic**
- Presence of **bulky, aromatic and cyclic groups** in the polymer **decreases** the elasticity of the polymer

Figure G000063. Structure property relationship · source slide page 10.

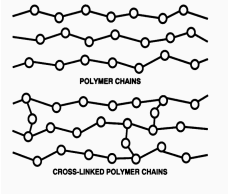
EV0000381

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

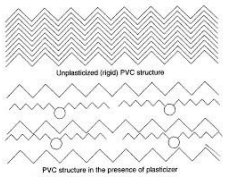
Functional materials - Polymers

- In elastic materials such as natural rubber, elasticity can be improved by **introducing cross linking** at suitable molecular positions like in vulcanisation



Source:https://revision4gcses.wordpress.com/science/chemistry-2/polymers/

- In non-elastic materials, elasticity is achieved by using **plasticisers** , for example (Ph_3PO_4), which enter between the polymer chains and make a hard polymer more flexible



Source:https://www.diva-portal.org/smash/get/diva2:1312811/FULLTEXT01.pdf

Figure G000064. Structure property relationship · source slide page 11.

EV0000384

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Chemical resistance:

- Resistance to **swell, dissolve and get degraded** in the presence of a solvent or chemical
- It depends on the **chemical nature of monomeric units** and their **molecular arrangement**

- **Presence of polar and non-polar groups**: Like dissolves like - A polymer is more soluble in a solvent of similar chemical structure
  - Polymers having **polar groups** (–OH or –COOH groups) are usually attacked or **dissolved by polar liquids** such as water or alcohols. (e.g. PVA dissolves in water)
  - Polymers with **non-polar groups** such as $-CH_3$ and $-C_6H_5$ are not easily attacked by polar solvents but they easily **swell and sometimes dissolve in non-polar solvents** such as petrol, benzene and carbon tetrachloride

Figure G000065. Structure property relationship · source slide page 12.

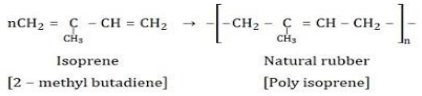
EV0000387

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

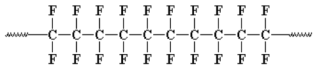
Functional materials - Polymers

- **Residual unsaturation** leads to oxidative degradation: polymers with residual double bonds get attacked and degraded in air and UV light
  e.g.,natural rubber



Source:https://www.embibe.com/study/natural-rubber-vulcanization-of-natural-rubber-concept?entity_code=KTBP50

- **Packing:** Dense packing does not allow the solvent to penetrate between the layers hence resistance increases, for example Teflon



Source:https://pslc.ws/macrog/ptfe.htm

- **Molecular mass:** For a given polymer, the swelling character decreases with the increase in the molecular mass
- **Degree of cross linking:** Greater the degree of cross linking lesser is the solubility

Figure G000066. Structure property relationship · source slide page 13.

EV0000390

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Plastic deformation :

Property of a polymer by which on application of heat it becomes soft and flexible and on cooling it becomes hard and brittle is called **plasticity or plastic deformation**

Based on their **response to heat** two kinds of polymers are:

• **Thermoplastic** - A thermoplastic polymer is one which softens on heating and becomes hard on cooling. It can be remoulded into new shapes, e.g., Polyethylene, polypropylene, polystyrene

•**Thermosetting** - A thermosetting polymer becomes hard on heating. It cannot be remoulded, its structure instead undergoes a total degradation on heating, e.g.,Bakelite, Aniline aldehyde resin, urea formaldehyde resin

Figure G000067. Structure property relationship · source slide page 14.

EV0000393

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

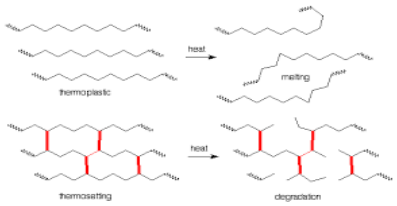
ENGINEERING CHEMISTRY

Functional materials - Polymers

The reason for the **difference** in the thermal behaviour is:

•Thermoplastics have **relatively weak forces of attraction** like van der Waals forces, dipole-dipole attraction or H-bonding between the chains, which are overcome when the material is heated

• Thermosetting plastics have the **cross-linking** between the chains is by strong covalent bonds cannot be broken easily. On strong heating will result in charring and degradation



Source:http://chemistry2.csudh.edu/rpendarvis/Polymer.html

Figure G000068. Structure property relationship · source slide page 15.

EV0000396

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000069. Structure property relationship · source slide page 16.

EV0000399

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000353 · EV0000354 · EV0000355
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 43 content: Structure-property relationship EV0000356 · EV0000357 · EV0000358
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Structure - property relationship of polymers Polymers are extremely versatile when it comes to their properties Some are hard while others are soft and flexible Some are insoluble and resistant to heat while others are soluble and fusible Properties of polymers are influenced by their structure EV0000359 · EV0000360 · EV0000361
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Crystallinity : The degree of cystallinity of a polymer depends on its structure and configuration Polymers contain both crystalline and amorphous regions Crystalline regions occur when polymer chains are arranged in an orderly fashion parallel and close to each other [Image text, OCR] Amorphous LU https://coventivecomposites.com/explainers/types-of-polymer-matrix/ EV0000362 · EV0000363 · EV0000364

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Structure: Polymers made up of linear chains without bulky pendant groups are more crystalline e.g., Polyethylene(PE) is more crystalline then polyvinylacetate Polymers containing polar groups are more crystalline A close packing between polymeric chains because of strong secondary forces like dipole–dipole interactions or hydrogen bonding imparts a high degree of crystallinity e.g., PVC is more crystalline than PE EV0000365 · EV0000366 · EV0000367
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Configuration: Crystallinity also depends on the stereoregularity of the polymer Based on stereoregularity polymers are classified as isotactic, syndiotactic and atactic Isotactic polymers are found to be most crystalline while atactic are found to be least crystalline High crystallinity leads to sharper melting point, greater rigidity, strength and density [Image text, OCR] Polymer Structure V STEREOCHEMISTRY (Tacticity) +CH-CH+n Inthe previous picture you seeallthe phenyl groups arelocated onthe same sideofthe polymer chain.But they don't have to be this way.To illustrate let's look ata chain of polystyrene from above.You can see that thependant phenyl groups canbe either on the right or left side of the chain. Polystyrene! H H H- H H H H H H H H- H- H H H H H H H H H H- H H- H H H H H H H H H H H H Isotactic Syndiotactic Atactic All phenyl groups Phenyl groups on Phenyl groups on thesame side altemating sides distributedrandomly Source:https://slideplayer.com/slide/5759919/ EV0000368 · EV0000369 · EV0000370
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Tensile strength: The amount of stress a polymer can take before undergoing permanent deformation Tensile strength depends on : Molecular mass - low molecular mass polymers are soft and gummy, high molecular mass polymers are tough and heat resistant By increasing molecular mass the tensile strength increases to a certain extent and then becomes constant Melt viscosity initially shows gradual increase then increases steeply with high molecular weight [Image text, OCR] Commercial polymers Tensile and mpact strength Viscosity 10,000 10,000,000 Molecularweight, degree of polymerization Source:https://polymerinnovationblog.com/polymer-composites-part-2-introduction-polymer-resins/ EV0000371 · EV0000372 · EV0000373
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Tensile strength increases when polymer has polar groups Force of attraction between chains prevents polymer chains from slipping against each other e.g., PVC, Nylon Cross linking in a polymer also increases its tensile strength EV0000374 · EV0000375 · EV0000376
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Elasticity : The property of a polymer by which it deforms on applying force and regains original shape on release of force is called elasticity Elastomers have long coiled and entangled chains which straighten and orient themselves on stretching. When stress is removed they go back to coiled form For a polymer to show elasticity the individual chains should not break on prolonged stretching [Image text, OCR] ELASTOMERS Elastomer is defined as a long chain polymer which under stress undergoes elongation by several times and regains its original shape when the stress is fully released Stretched Returnedto randomization Source:https://slideplayer.com/slide/8755303/ EV0000377 · EV0000378 · EV0000379

10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Presence of polar groups in the polymer chain results in strong secondary force of attraction between the chains making them stiff and inflexible Non polar groups have weak van der Waal's attraction between them and hence they are more elastic Presence of bulky, aromatic and cyclic groups in the polymer decreases the elasticity of the polymer EV0000380 · EV0000381 · EV0000382
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers In elastic materials such as natural rubber, elasticity can be improved by introducing cross linking at suitable molecular positions like in vulcanisation In non-elastic materials, elasticity is achieved by using plasticisers , for example (Ph3PO4), which enter between the polymer chains and make a hard polymer more flexible Source:https://revision4gcses.wordpress.com/science/chemistry-2/polymers/ Source:https://www.diva-portal.org/smash/get/diva2:1312811/FULLTEXT01.pdf EV0000383 · EV0000384 · EV0000385
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Chemical resistance: Resistance to swell, dissolve and get degraded in the presence of a solvent or chemical It depends on the chemical nature of monomeric units and their molecular arrangement Presence of polar and non-polar groups: Like dissolves like - A polymer is more soluble in a solvent of similar chemical structure Polymers having polar groups (–OH or –COOH groups) are usually attacked or dissolved by polar liquids such as water or alcohols. (e.g. PVA dissolves in water) Polymers with non-polar groups such as –CH3 and –C6H5 are not easily attacked by polar solvents but they easily swell and sometimes dissolve in non-polar solvents such as petrol, benzene and carbon tetrachloride EV0000386 · EV0000387 · EV0000388
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Residual unsaturation leads to oxidative degradation: polymers with residual double bonds get attacked and degraded in air and UV light e.g.,natural rubber Source:https://www.embibe.com/study/natural-rubber-vulcanization-of-natural-rubber-concept?entity_code=KTBP50 Packing: Dense packing does not allow the solvent to penetrate between the layers hence resistance increases, for example Teflon Molecular mass: For a given polymer, the swelling character decreases with the increase in the molecular mass Degree of cross linking: Greater the degree of cross linking lesser is the solubility Source:https://pslc.ws/macrog/ptfe.htm EV0000389 · EV0000390 · EV0000391
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Plastic deformation : Property of a polymer by which on application of heat it becomes soft and flexible and on cooling it becomes hard and brittle is called plasticity or plastic deformation Based on their response to heat two kinds of polymers are: Thermoplastic - A thermoplastic polymer is one which softens on heating and becomes hard on cooling. It can be remoulded into new shapes, e.g., Polyethylene, polypropylene, polystyrene Thermosetting - A thermosetting polymer becomes hard on heating. It cannot be remoulded, its structure instead undergoes a total degradation on heating, e.g.,Bakelite, Aniline aldehyde resin, urea formaldehyde resin EV0000392 · EV0000393 · EV0000394

15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers The reason for the difference in the thermal behaviour is: Thermoplastics have relatively weak forces of attraction like van der Waals forces, dipole-dipole attraction or H-bonding between the chains, which are overcome when the material is heated Thermosetting plastics have the cross-linking between the chains is by strong covalent bonds cannot be broken easily. On strong heating will result in charring and degradation [Image text, OCR] hermoplastic temoseting degracaton Source:http://chemistry2.csudh.edu/rpendarvis/Polymer.html EV0000395 · EV0000396 · EV0000397
16. Source slide page 16 (extracted text): THANK YOU Department of Science and Humanities EV0000398 · EV0000399 · EV0000400

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.708] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module seven on functional materials, that is, polymers. [00:00:10.852] We will talk about the structure property relationship. The properties we consider will consider will be crystallinity, tensile strength, elasticity, chemical resistance and plastic deformation. [00:00:26.052] What do we understand by the term the structure property relationship? The properties of polymers are influenced by their structure. Polymers when it comes to properties are extremely versatile. [00:00:41.508] Some can be hard, whereas others can be very soft and flexible. Some could be insoluble and resistant to heat, while others could be soluble and fusible. So what we have to understand is how is the structure of the polymer dictating its properties. [00:01:00.836] Let us take up the first property, crystallinity. Now the degree of crystallinity of a polymer depends on its structure as well as configuration. [00:01:12.900] Now when we talk of crystallinity in a polymer, we know that it's not like a lattice which is getting formed, like how we have in uh uh different uh compounds, in organic compounds, a perfect lattice, crystal lattice. But here, normally the polymer will be in the form of this amorphous uh matrix. [00:01:33.476] But in some polymers, we find that within this matrix also there could be some kind of alignment like this where the polymer chains come very close to each other. So you will have these small areas inside the polymer, and this is referred to as a [00:01:50.928] Crystallite. So overall. [00:01:55.652] crystallite. Now overall if there are so many crystallites inside the polymer then we can say that the polymer is crystalline. Otherwise if those are missing then that is going to be amorphous in nature. So basically what we have to look for is [00:02:12.512] Are the chains coming close to each other, is it possible for the chains to come close to each other and that would discur uh uh decide the crystallinity in a polymer. [00:02:24.484] What could be the reason for a polymer to be crystalline? It depends on the structure to a great extent. [00:02:32.816] Now polymers which are made up of linear chains without any bulky pendant groups are found to be more crystalline. For example, polyethylene. Now when we have these polyethylene chains, what do we have? CH2, CH2, CH2. Uh combine each of these is CH2, CH2, CH2. Like this you have [00:02:50.544] all these chains and since there is no obstruction for the chains to come close to each other we can say that this will be more crystalline as compared to a polyvenyl acetate. Polyvinyl acetate would be like this CH [00:03:07.972] O C O C H three. So again, C H two. C H O C O. [00:03:16.932] CH3. Now this is just one chain. So you can imagine there will be another chain like this, and therefore these groups are referred to as bulky pendant groups. When you have such bulky pendant groups, obviously there will be an obstruction for the chains to come close to each other. Therefore [00:03:34.704] such a polymer would definitely be less

crystalline or more amorphous as compared to when there are no bulky pendant groups. [00:03:44.740] Second reason is the presence of polar groups. Whenever there are polar groups, then it is found the polymer becomes more crystalline. Why? Because whenever you have uh these polar groups in the [00:03:58.820] Polymer chain, we know that this could give rise to dipole dipole interactions or hydrogen bonding. Now when the polymer chains have such strong attraction as compared to when there would be only Vanderball process, then definitely the polymer chains are going to come close to each other. Now when you have [00:04:18.884] Polyvinyl fluoride, we know that the chain is like this, extending. And similarly you will have another chain here. So we find that because of the polar nature [00:04:34.180] We know that there is polarity here, H gets a small positive, and this bond is going to slightly shift here. Similarly, here. So therefore, there is attraction between the uh between the chains. There is a dipole dipole interaction which is happening between the chains, and therefore the [00:04:51.952] chains can come closer whereas in polyethylene it is just as we saw just now CH2 CH2 CH2 and there is no such attraction except for the Van der Waals forces of attraction. Therefore we find that in the case of polyethylene it is

EV0000078 · EV0000079 · EV0000080 · EV0000081

Transcript block 2 (source interval 000300-000599 seconds): [00:05:07.680] much uh the crystallinity will be less as compared to polynomial fluoride. Now this also depends on configuration. The crystallinity also depends on configuration. Now [00:05:19.968] Ste uh is uh that is stereoregularity, crystallinity. When we say configuration, we are talking of stereoregularity, that is how are the substituents arranged in space. Now based on stereo regularity we can have three types of polymers isotactic, symdiotactic, and a tactic. [00:05:37.648] Now, if you look at this styrene structure over here, in the case of isotactic, all the styrene substituents are on the same side of the backbone. Here in syndioctic they are occupying alternate positions, whereas in A-tactic, they are just occupying any random position. [00:05:57.616] Now based on this we ca we find that when it's isotactic [00:06:02.916] The polymer is found to be most crystalline, whereas in A- tactic it is found to be least crystalline. What is the reason? Crystallin T is because you are the chains can come close to each other. We can see in isotactic they can come much much closer to each other as compared to syntyotactic and a tactic. [00:06:20.688] Now high crystallinity will lead to sharper melting point, greater rigidity, strength and density. [00:06:27.524] Coming to the next property, tensile strength. What would be tensile strength? The amount of stress a polymer can take before undergoing permanent deformation. Now, tensile strength is going to depend on the molecular mass. [00:06:44.644] Now, if you have low molecular mass polymers, they will be soft and gummy, whereas if you have high molecular mass, then they will be tough and heat resistant. So [00:06:54.848] By increasing the molecular weight we can understand that the tensile strength is going to increase. Like you can see in this graph, the tensile strength, when you increase the degree of polymerization or molecular weight, the tensile strength is increasing to a limit, and then after that it becomes almost constant. [00:07:12.496] Increasing further would not make too much difference in the tensile stem. But one more property which we have to look at is the melt viscosity. As we increase the degree of polymerization of the molecular weight, the melt viscosity also goes on increasing. So commercially, when we have to make the polymer. [00:07:31.620] Remember that polymer. [00:07:34.128] Something which we just test in the lab, they are for commercial purpose. So if the mat viscosity is very high, then the uh maneuvering of the process is going to be that much more difficult. Therefore, we need to make a compromise and therefore somewhere in this region where the tensile strength is also high. [00:07:51.824] And the valve viscosity is not very

high. This is the range which is used for fabrication of the polymers, I mean manufacture of the polymers, preparation of the polymers. [00:08:06.436] Now tensile strengt increases when polymer has polar groups. Now when there are polar groups, what happens? Now if you have polar groups like we saw in the case of [00:08:19.300] PDC just now we know that there is a dipole dipole attraction between the molecules, and when there is dipole dipole attraction, obviously the chains will not slip each other slip against each other. Whereas like in the case of polyethylene, what did we notice in polyethylene? I just draw like this. [00:08:37.072] This is the carbon chain which we have CH2, CH2, CH2. Because there's just Van der Waals forces. [00:08:43.168] Tensile strength when you're applying a force on it, the chains will slip against each other. Whereas if you have some kind of attraction, like in the case of nylon, between the chains we find that there is hydrogen bonding, there is a attraction that is [00:09:00.304] Hydrogen bonding between the chains. Now these hydrogen the hydrogen bonding is quite strong. So this hydrogen bonding is not going to allow the chains to slip against each other. So if we have polar groups, we find that the tensile strength is high. And of course, if we have cross linking, then [00:09:17.936] Definitely the inner polymer will also increase with the molecular weight and therefore there will be increase in the tensile strength. So therefore when we have cross linking like this, we find [00:09:33.828] Cross linking means there is covalent bond formation between the chains network three dimension network and therefore definitely it is going to this is going to result in increase in tensile strength. [00:09:53.828] Cross linking also increases the tensile strength. Elasticity.

EV0000082 · EV0000083 · EV0000084 · EV0000085

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.712] This is the next property we need to consider. The property of a polymer by which it deforms on applying force and regains its original shape when we when the force is released, and that is known as elasticity. Now, why do certain polymers behave like this? There is a [00:10:19.264] Uh it it goes to the structure again. They have long coiled and entangled chains, and when you [00:10:26.724] Apply force, you are straightening them and they get oriented like this. So if they are long coiled chains like this entangled, but when you apply force they get stretched like this, therefore they become hard, and then when you release again they go back to the [00:10:44.528] coil form. So such is the property of elastomers. So for a polymer to show elasticity the individual chains should not break on prolonged stretching. Now presence of polar groups of course is going to result in polymers which will be [00:11:04.368] very stiff and inflexible because of the strong force of attraction which will exist between the chains. Now if nonpolar groups are there then it is just weak wanderwald forces and therefore they will be much more [00:11:18.212] We can say elastic in nature and presence of bulky aromatic and cyclic groups of is also going to result in decreasing the elasticity of a polymer. If we have an elastic material, [00:11:33.348] can we increase its elasticity? That can be done with the help [00:11:38.784] of cross linking as is done in the case of vulcanization. What is vulcanization? Suppose this is uh a polymer chain of natural rubber, what is done in vulcanization? It is treated with sulfur so that the sulfur linkages will be uh [00:11:55.136] Obtained in between the polymer chains, and since these are covalent bonds, obviously it is going to give much more strength, and when we apply [00:12:04.048] uh force the chains are not going to uh slip against each other. So if we have elastic met like natural rubber, there is only Van der Waals forces which exist between chains. But if we introduce a small amount of cross linking [00:12:19.488] uh using sulfur we find that is called balconized rubber and we find that these chains are not going to slip against each other and definitely the tensile strength can be uh very increased to a good extent. Of course we

should not introduce too much of vulcanization because instead of becoming elastic it is going to [00:12:37.136] To become stiff because you have increased too much of force of attraction covalent, too many covalent bonds between the chains that is going to result in making it more stiff. So a small amount of uh vulcan, a small amount of cross-linking is introduced between the chains in the case where [00:12:54.704] We have elastic materials. But if the material is non-elastic, meaning it's very stiff, can we make it more elastic? Yes, that is done with the help of plasticizers. Now if you see PVC here, there the chains are very close to each other. [00:13:10.340] What are plasticises? They are small organic molecules which are [00:13:16.576] made to enter between the chains. These small molecules like for example triphenyl phosphate. Now this can enter between the chains and when it enters between the chains it's understood that the the distance between the polymer chains will increase and therefore it will make [00:13:34.224] make it less stiff and more flexible. [00:13:38.980] The next property is chemical resistance. So how would we define that? The resistant to resistance to swell, dissolve, and get degraded in the presence of a solvent or a chemical, and it will depend on the chemical nature of the monomeric units which are being used. [00:13:56.752] And their molecular arrangement. Now what factors do we consider when we talk of chemical resistance? If there are polar or non-polar groups. Now we know this very common fact that light dissolves light. [00:14:15.632] So a polymer is more soluble in a solvent of similar chemical structure. Now there are polymers which are polar groups like OH and COH groups if they have, then they are usually attacked by polar liquids such as water or alcohols. [00:14:33.392] For example, PVA polyvinyl alcohol will dissolve in water. Why? Because it can easily form hydrogen bonds with water. It's all like dissolves like. Similarly, if we have nonpolar groups such as C H3, C6H5, etc. then these are not going to be attacked by polar solvents. Instead, they will easily [00:14:50.960] swell and sometimes dissolve in non-polar solvents such as petrol, benzene, carbon tetrachloride, etc. So like dissolves like is applicable for polymers also. The second one could be

EV0000086 · EV0000087 · EV0000088 · EV0000089

Transcript block 4 (source interval 000900-001199 seconds): [00:15:07.044] is the residual unsaturation. Whenever there is residual unsaturation, it leads to oxidative degradation. So what does it mean? If there is a double bond in the polymer. See addition polymerization when we are doing monomers will always have double bonds. Like for example isoprene. [00:15:26.896] That is natural rubber, but when uh that is the uh monomer unit in natural rubber. Now when natural rubber is formed by polymerization of isoprene, we get polyisoprene. But what do we notice here in the polymer itself there is still a double bond. When you have such a situation [00:15:44.592] Because of the nature of this double bond. In the polymer there is still a double bond, and we know that pi electrons can easily be uh take part in reaction. Therefore, such polymers will undergo oxidative degradation and [00:16:01.344] Because of this residual uh bond which is there, residual double bond which is there or unsaturation which is there, it gets easily degraded in air and UV light. [00:16:12.800] So here we see there are double bonds, the monomer, that is fine, but in the polymer also there is still a double bond and this would lead to uh degradation of the polymer very quickly. And of course [00:16:25.232] Packing. How is dense is the packing? Because when there is dense packing in the molecule, then it will not allow the solvent to penetrate between the layers, as in the case of say Teflon. We know that Teflon, this is the [00:16:42.244] Single chain. Now when we have Teflon chains like this. [00:16:46.148] What happens is they are so close to one another. Why? Because of the polarity. Each of these bonds is polar in nature. CF, CF, CF, all these bonds are polar in nature, and therefore the chains are going to be pulled very close to each

other. So therefore, it will not allow any solvent. [00:17:03.920] To get in between the chains. Therefore, we find that Teflon if water cannot enter, so it acts as non-stick. So it will neither allow water nor oil, any kind of liquid cannot enter between the layers. So packing, the dense packing is resulting in the [00:17:23.408] Polymer being more chemical resistant. And of course, molecular mass for a given polymer, as the molecular mass is increasing, we find that the swelling character decreases, and of course, if there is decross linking, if there is a high uh amount of cross-linking in the polymer, the solubility definitely is going to be less. [00:17:43.748] Now coming to the last property [00:17:47.364] That is plastic deformation. So what is plasticity or plastic deformation? Some polymers have this property that when we apply heat they become soft and flexible, but on cooling they again return to their original shape. This is known as plasticity. [00:18:05.136] Now, based on the response to heat, you can have two types of plastics, that is the polymers, one is known as the thermoplastic, another is known as the thermosetting plastic. A thermoplastic polymer would be one which softens on heating and becomes hard on [00:18:22.224] cooling and it can be remoulded into new shapes and that is as the case is of polyethylene, polypropylene, etc. But what is a thermosetting polymer it becomes hard on [00:18:36.928] Uh it initially it will be uh fusible, it becomes uh fusible, but the moment you cool it first first time itself it is going to set into a hard mass and after that it cannot be uh changed anymore, it cannot be re molded anymore. Such polymers are known as [00:18:54.640] uh thermosetting uh polymers and it uh they if at all you heat it then it is going to undergo [00:19:04.688] Total degradation that means it will not uh melt, rather it is going to degrade into some other compound. So the examples would be bacolite, aniline andehydrasin, urea formaldehyde resin. These are all thermo examples of thermosetting plastics. What is the reason? [00:19:25.060] Why are they behaving like this? It has to do with the structure. Now when you have uh thermoplastic, there are relatively weak forces of attraction. There could be underwhelm forces, it could be dipole dipole interaction or hydrogen band bonding between the chains. So when you are heating [00:19:41.680] then you can easily overcome the uh force of attraction and therefore you see here when you have a thermoplastin, then when you heat these chains become loose, you have overcome the force of attraction and therefore they start to uh melt and the moment you reduce the heat or you c cool it, they will come back to the original shape.

EV0000090 · EV0000091 · EV0000092 · EV0000093

Transcript block 5 (source interval 001200-001499 seconds): [00:20:00.676] And in the case of thermosetting, we know that in a thermosetting it's a three-dimensional cross-linked structure which is there. And there are these covalent bonds which are existing between the chains. And now when you heat it, we cannot overcome this force. [00:20:18.448] Of covalent bond because it's very high. Even if you do, then it is going to break into different types of molecules. So you cannot have a thermosetting plastic once it is set into a hard mass. Once these cross link structures are formed, we will not be [00:20:36.016] We are able to get them into. [00:20:39.236] uh get them again uh remould them again and again. So that is the difference and the structural difference is most important. Here weak forces of attraction exist whereas here there are covalent bonds and we cannot uh when we heap them we cannot overcome these strong forces of attraction. [00:20:57.008] That is a covalent making the covalent bond. So this is this is about the structure property relation. What we understand is that the properties of the polymers are dictated by the structure of the polymer. [00:21:14.116] So in the next class we are going to talk about some commercial polymers and we will take up butile rubber. Thank you.

EV0000094 · EV0000095 · EV0000096 · EV0000097

Full transcript: 3203 normalized words, 88 timestamped segments, 1283.946667 seconds duration. Low-confidence segments flagged: 0.

EV0000078 · EV0000079 · EV0000080 · EV0000081

7 Synthesis and properties of epoxy resin

This lesson is indexed from the source video titled Synthesis and properties of epoxy resin. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000098 · EV0000099

Graphs and diagrams

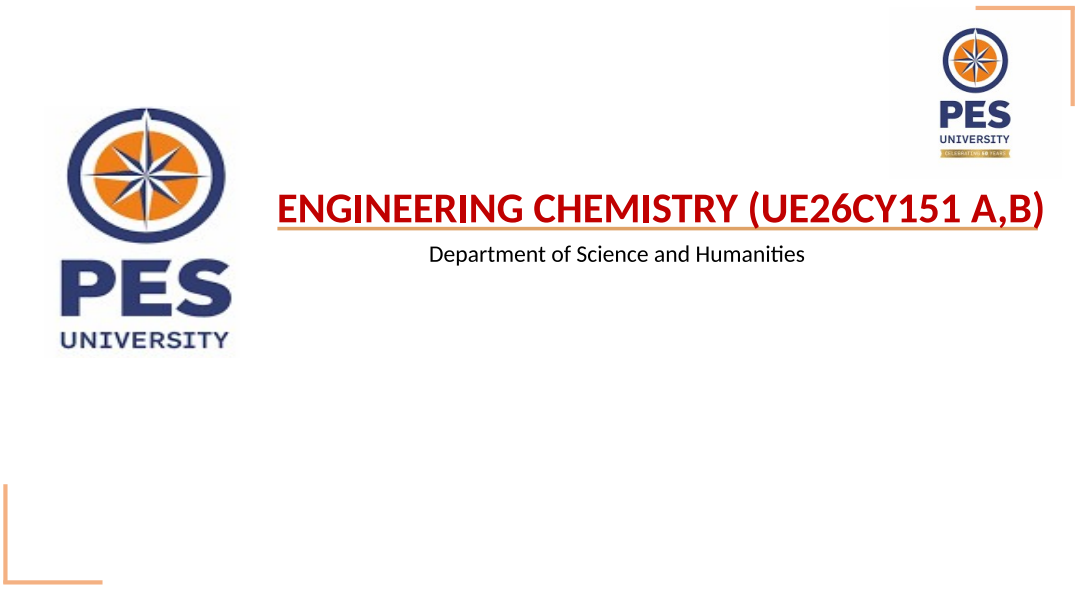


Figure G000076. Synthesis and properties of epoxy resin · source slide page 1.

EV0000420

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

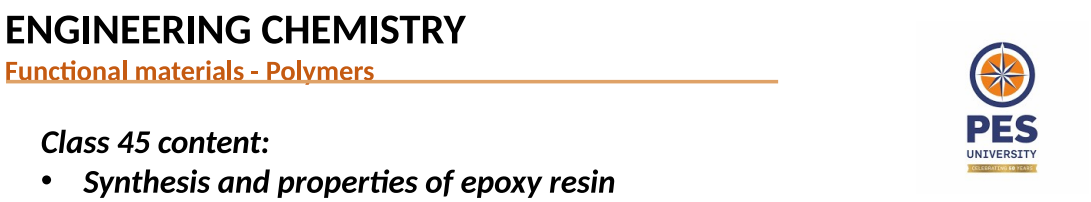


Figure G000077. Synthesis and properties of epoxy resin · source slide page 2.

EV0000423

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

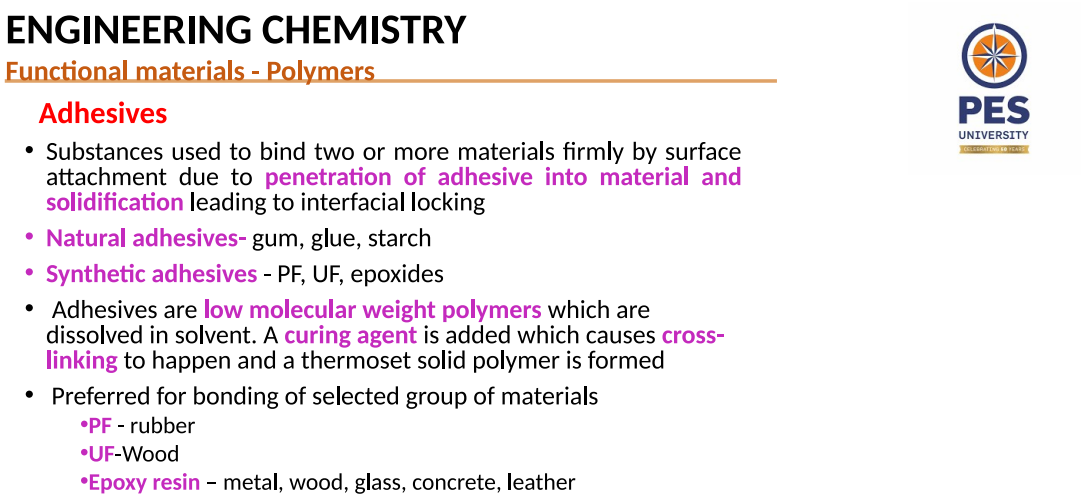


Figure G000078. Synthesis and properties of epoxy resin · source slide page 3.

EV0000426

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000079. Synthesis and properties of epoxy resin · source slide page 4.

EV0000429

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

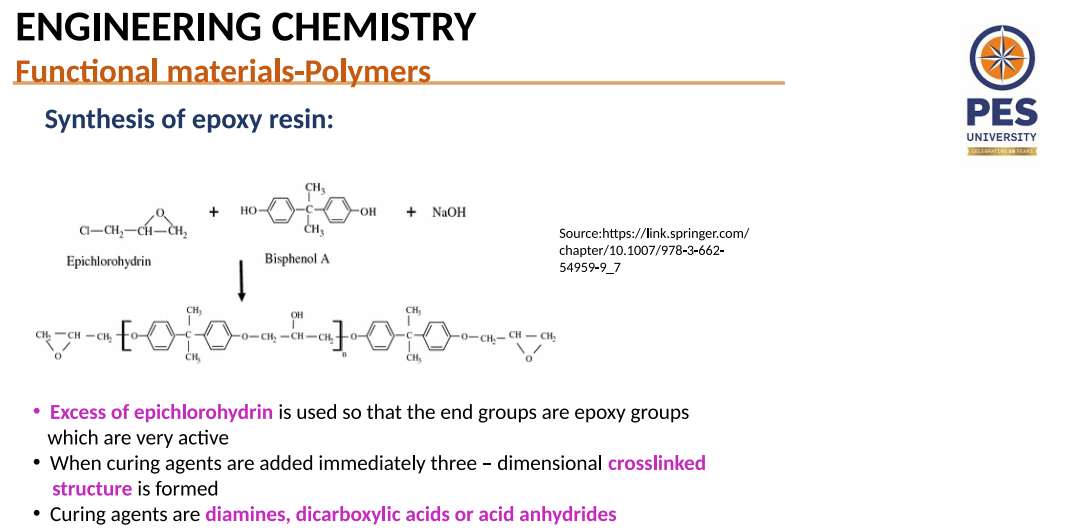


Figure G000080. Synthesis and properties of epoxy resin · source slide page 5.

EV0000432

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

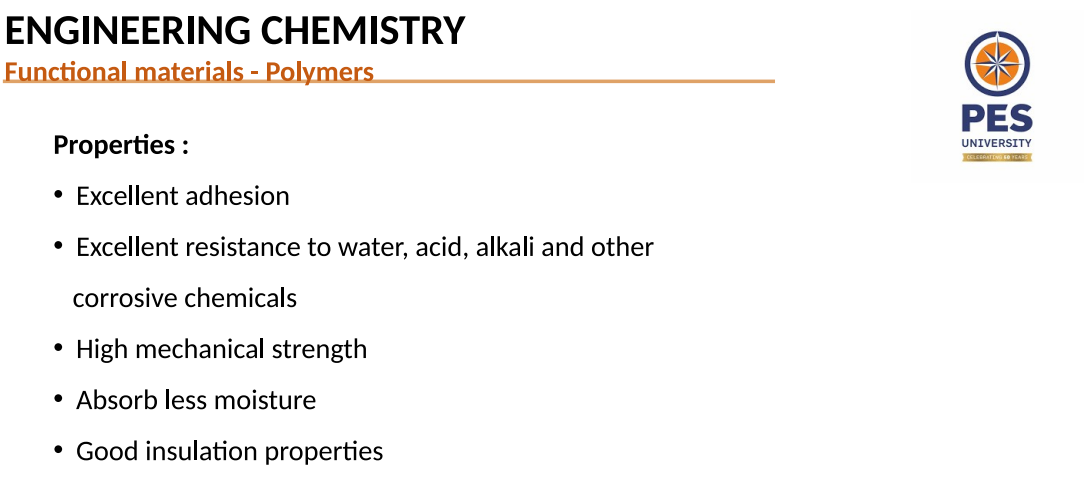


Figure G000081. Synthesis and properties of epoxy resin · source slide page 6.

EV0000435

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

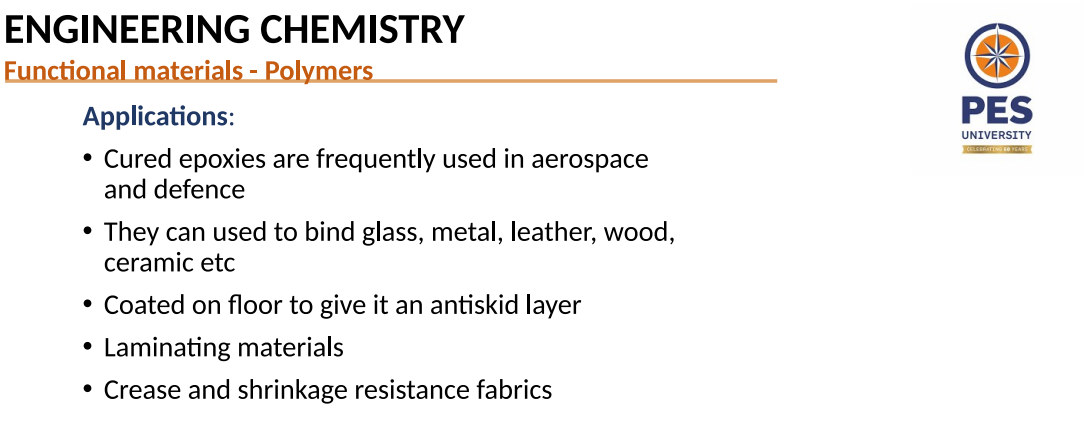


Figure G000082. Synthesis and properties of epoxy resin · source slide page 7.

EV0000438

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000083. Synthesis and properties of epoxy resin · source slide page 8.

EV0000441

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000419 · EV0000420 · EV0000421
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 45 content: Synthesis and properties of epoxy resin EV0000422 · EV0000423 · EV0000424
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Adhesives Substances used to bind two or more materials firmly by surface attachment due to penetration of adhesive into material and solidification leading to interfacial locking Natural adhesives- gum, glue, starch Synthetic adhesives - PF, UF, epoxides Adhesives are low molecular weight polymers which are dissolved in solvent. A curing agent is added which causes cross-linking to happen and a thermoset solid polymer is formed Preferred for bonding of selected group of materials PF - rubber UF-Wood Epoxy resin – metal, wood, glass, concrete, leather EV0000425 · EV0000426 · EV0000427
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Source: https://www.oatext.com/bisphenol-a-in-odontology-recent-data.php#gsc.tab=0 Epoxy resins: They contain the epoxy group, which is a 3 membered ring of an O atom bonded to carbon atoms Commercially epoxy resins are known as Araldite or Epon Synthesis : Epoxy resins can be obtained by the condensation reaction between bisphenol A and epichlorohydrin EV0000428 · EV0000429 · EV0000430

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials-Polymers Synthesis of epoxy resin: [Image text, OCR] HO OH NaOH CI-CH-CH- -CH Epichlorohydrin Bisphenol A CH; OH CH CH -CH O-CH CH CH, CH, O Source:https://link.springer.com/chapter/10.1007/978-3-662-54959-9_7 Excess of epichlorohydrin is used so that the end groups are epoxy groups which are very active When curing agents are added immediately three – dimensional crosslinked structure is formed Curing agents are diamines, dicarboxylic acids or acid anhydrides EV0000431 · EV0000432 · EV0000433
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Properties : Excellent adhesion Excellent resistance to water, acid, alkali and other corrosive chemicals High mechanical strength Absorb less moisture Good insulation properties EV0000434 · EV0000435 · EV0000436
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Applications: Cured epoxies are frequently used in aerospace and defence They can used to bind glass, metal, leather, wood, ceramic etc Coated on floor to give it an antiskid layer Laminating materials Crease and shrinkage resistance fabrics EV0000437 · EV0000438 · EV0000439
8. Source slide page 8 (extracted text): THANK YOU Department of Science and Humanities EV0000440 · EV0000441 · EV0000442

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.612] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module 7 on functional materials on polymers, and today we are going to study about adhesives and specifically about epoxy resin. [00:00:15.460] Let us first define adhesives, yes, substances which are used to bind two or more materials firmly by surface attachment due to the penetration of the adhesive into the material and solidification which leads to interfacial locking. [00:00:32.624] So, when we have two surfaces and we apply the liquid adhesive onto the surface and bring them together, what exactly happens is the adhesive is going to penetrate into the top layers of the two surfaces, and then it is going to [00:00:49.040] uh bind together by a solidification process. Only when it solidifies then the two surfaces get locked to each other. [00:00:59.108] So there could be natural adhesives, we are all aware, of gum, glue, starch, etc., or there are many synthetic adhesives also available these days. The examples would be phenol formaldehydrasen, urea formal dehydration, and epoxytes. [00:01:14.756] How do these actually work? Now adhesives are basically low molecular weight polymers which are dissolved in a solvent. [00:01:25.508] After that, the most important step is to add a curing agent. What does this curing agent do? It is going to trigger cross-linking into the adhesive. [00:01:39.536] And because of cross-linking three-dimensional network which is formed, solidification happens, and what you get is a thermoset solid polymer, and that is how the two surfaces got bound together. So there are two parts always in an [00:01:54.864] Addition one is the low molecular weight polymer, which is soluble, and the other one, which we usually refer to as hardener, is actually known as a curing agent, and that is responsible for the uh solidification of the [00:02:11.428] Polymer and that happens due to three-dimensional cross-linking network getting form. Now [00:02:19.024] There are certain materials for which you have to use a particular type of adhesive. For example, for rubber we use phenol formaldehydrin. For wood, it's usually urea formalhydrasin, but epoxy resin is a very very good resin because. [00:02:36.688] because it can be used to attach metal, wood, glass, concrete, leather, any kind of surfaces, epoxy resin can act as a

very good adhesive. [00:02:50.436] So let us study in more detail about what exactly is there in the structure of epoxy resin that is making it so effective as an adhesive. So basically, as we know from the name, there is an epoxy group. What is an epoxy group? It is a three-membered ring. [00:03:07.376] of an O atom bonded to carbon atoms. So when we have three member ring, we know that this is going to be very very reactive. [00:03:18.372] Three member rings are usually very reactive. Now this epoxy resin is commercially available to us in the form of araldite or epon. Let us see the synthesis. [00:03:29.728] Now when you want to make an epoxy resin, it is a condensation reaction which will take place between bisphenol A and epichlorohydrate. So before we talk about the condensation between the two monomers. [00:03:43.696] Let us see how to synthesize these monomers in the first place. Now, bisphenol A is the first monomer which is being used. Now, bisphenol, the name itself suggests there are two phenol groups and A stands for acetone. [00:03:57.520] So two phenol molecules and one acetone molecule. When these react, what we get is the bisphenol A. [00:04:07.108] The other monomer is epi chloroth. Now in epichlorohydrate we start with propene and we are going to chlorinate it. We add Cl2, what we get is the allyl chloride. [00:04:22.052] That is now treated with HOCl. So this is now going to give you 13 dichloro 2 propanol or 12 dichloro 1 propanol. Now in this case, once now you do uh treat it with the base, what will happen is [00:04:39.520] Dehydration is going to happen. It's sorry, it's Cl will be removed, and you get a three-membered ring over there. So one of them will be now it will become this three-membered ring, and you have the [00:04:52.960] uh Cl over here. So this is known as Epichlorohydrine and you get NaCl because you are treating with it NaOH NaOH plus this HCl so you get the NaCl. So this is how epi chlorohydrine is made.

EV0000098 · EV0000099 · EV0000100 · EV0000101

Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.696] So we are the starting point would be probene, or sometimes we say with the starting point is ally chloride. Now that we have learned the synthesis of the monomers bisphenol A and epi chlorohydrate, let us see how we synthesize the epoxy resin. [00:05:27.268] So we have taken this epichlorohydrate and this is the this phenol A. And what do we notice here that there is an OH group over here and a Cl. [00:05:38.464] So we know that they can be condensed easily. These two molecules can undergo condensation, and this would happen in the presence of NaOH. So when NaOH is used, [00:05:50.272] HCl is getting eliminated. So there will be an HCl molecule which is getting removed here. And what would happen as a result of this condensation? We can see here that there is an attachment between the bisphenol A and [00:06:06.560] And the epi chlorhydrate. So this molecule has basically condensed with this molecule with the elimination of HCl, but something else has also happened. This ring has opened up because if you look at this monoma. [00:06:21.380] There are two hydroxyl groups on both sides, therefore it would classify as a monomalase. In the first class I mentioned, at least two functional groups should be there. But in this epiclorohydrate, we find there is just one CL which can take part in the condensation, whereas here we don't find anything, but yes. [00:06:39.168] That is the reason why we have the ring here. Now this ring being very reactive, it is going to open up when this [00:06:48.768] uh uh condensation will happen and therefore this ring is going to open up like this and we will find that we get the ch oh you can see here the ch oh is formed and now this ch2 is free to condense with the next [00:07:06.224] bisphenol A ring. So this is how the condensation goes on. So between the two one side uh both of these will take part in the condensation. So the repeat unit is going to be the bisphenol A and this and this is the mechanism by which it happens. [00:07:23.920] So here now this we have Cl condensation on

one side and the other side it is CH2 where the ring is opened up and there's a free bond here that is going to react with the next bisphenol A. So this is the mechanism, and therefore we get this as the [00:07:40.036] main uh repeat unit in the epoxy uh polymer. Now an important fact which we have to know that whenever a this [00:07:51.104] Epoxy resin is formed, an excess of epichlorohydrine is used. What would be the reason for that? The reason is that the N groups we want them to be epoxy groups. Now, how would we [00:08:04.880] Uh make sure that the uh that happens by using an excess of epi chlorohyain. When we use excess of epichlorohyden, because there is no more bisphenol, further reaction with bisphenole is not possible. If there was more bisphenol as we understood the mechanism just now, this ring would open. [00:08:22.512] We'll get a CHOH and the CH2 combined would have combined with another bisphenole, but there is no bisphenol, so the ring remains intact, and in the end, we have ensured that both the side chains, the end of terminal chains, terminal groups are epoxy groups. So why would we do [00:08:40.080] That because we know that epoxy is a very reactive group, and when we add the curing agent, immediately there will be a [00:08:51.204] Three dimensional cross link structure which is formed, and we know that when between these two chains we have cross-link structure getting formed, the polymer is going to solidify. So that is the reason why we use excess of epiclorohydrate, and of course, curing agents, what are we going to [00:09:08.976] Use we use diamines, di carboxylic acid, or acid anhydrides. So using these curing agents. [00:09:16.688] Because there are these reactive epoxy groups, immediately cross link structure will be formed, which will result in solidification, and hence the two surfaces are going to get bound. So this is how the [00:09:31.088] uh synthesis of epoxy resin is done and this is how it works in binding those two surfaces together. What are the properties of these epoxy resins? [00:09:44.420] They have, of course, excellent addition, excellent resistance to water, acid, alkali, and other corrosive chemicals. So they are extremely resistant. And they also have very high mechanical strength. They absorb less moisture.

EV0000102 · EV0000103 · EV0000104 · EV0000105

Transcript block 3 (source interval 000600-000899 seconds): [00:10:00.880] and they have very good insulation properties. So they have wonderful properties, epoxy resins have very good properties and they're all listed here. Now what would be the applications for these epoxy resins? [00:10:16.464] Now cured epoxy resins are frequently used in aerospace and defense. They are also used to bind glass, metal, leather, wood, ceramic, etc. Of course, that is the uh reason why they come under the class of adhesives. Then when they are coated on the floor, they give [00:10:34.096] and anti-skid layer. They are used for laminating materials and for crease and shrinkage resistance in fabric. So that is why epoxy resin acts as a very good adhesive. [00:10:51.524] So in the next class we are going to talk of another important class of polymers or rather polymer composites. So we will take up Kepler in the next class. Thank you.

EV0000106 · EV0000107 · EV0000108 · EV0000109

Full transcript: 1615 normalized words, 46 timestamped segments, 669.162667 seconds duration. Low-confidence segments flagged: 0.

EV0000098 · EV0000099 · EV0000100 · EV0000101

8 Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1)

This lesson is indexed from the source video titled Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000110 · EV0000111

Graphs and diagrams

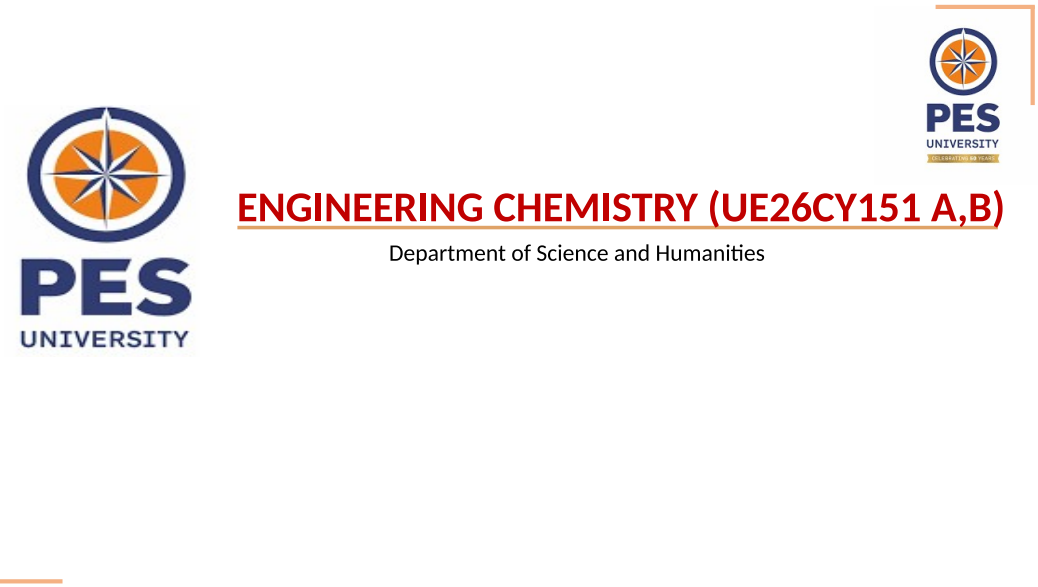


Figure G000084. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 1.

EV0000444

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

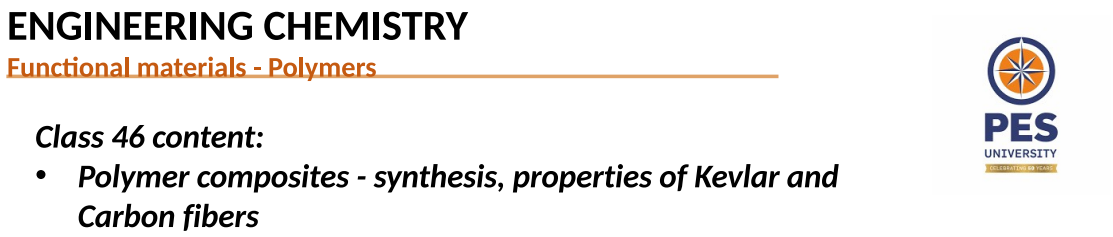


Figure G000085. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 2.

EV0000447

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

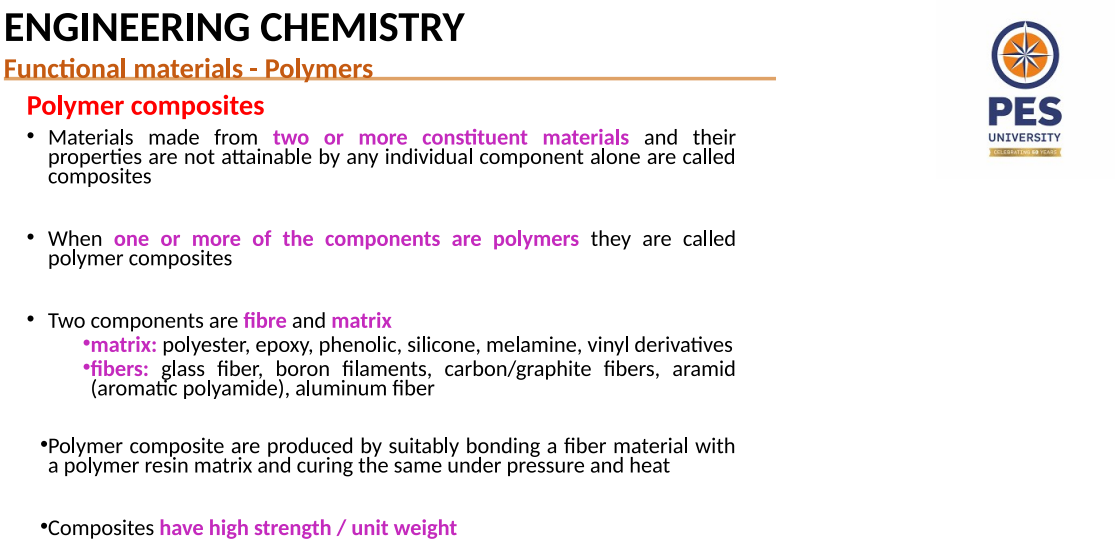


Figure G000086. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 3.

EV0000450

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

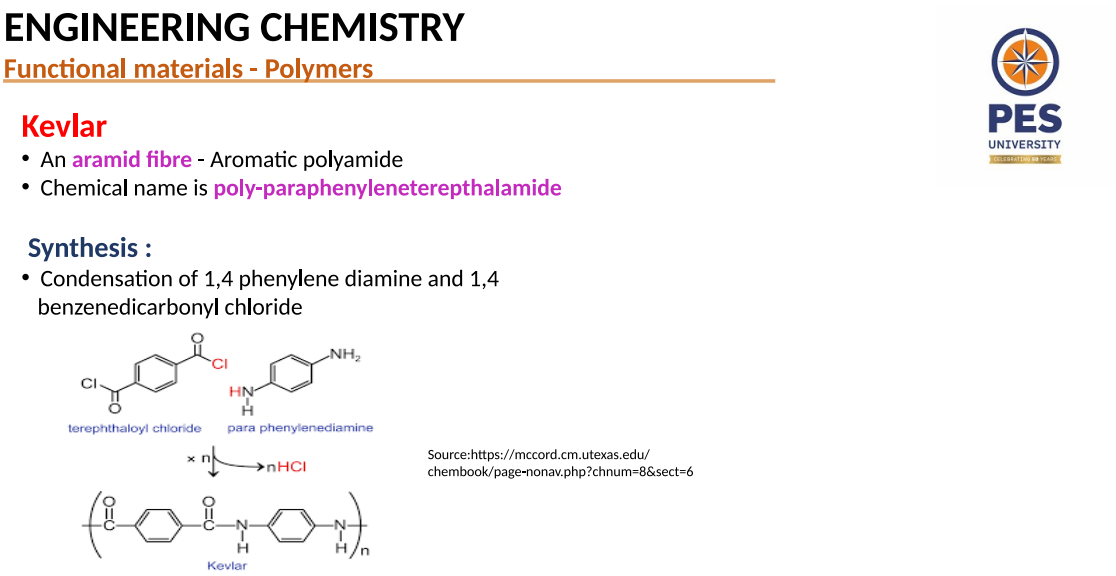


Figure G000087. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 4.

EV0000453

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

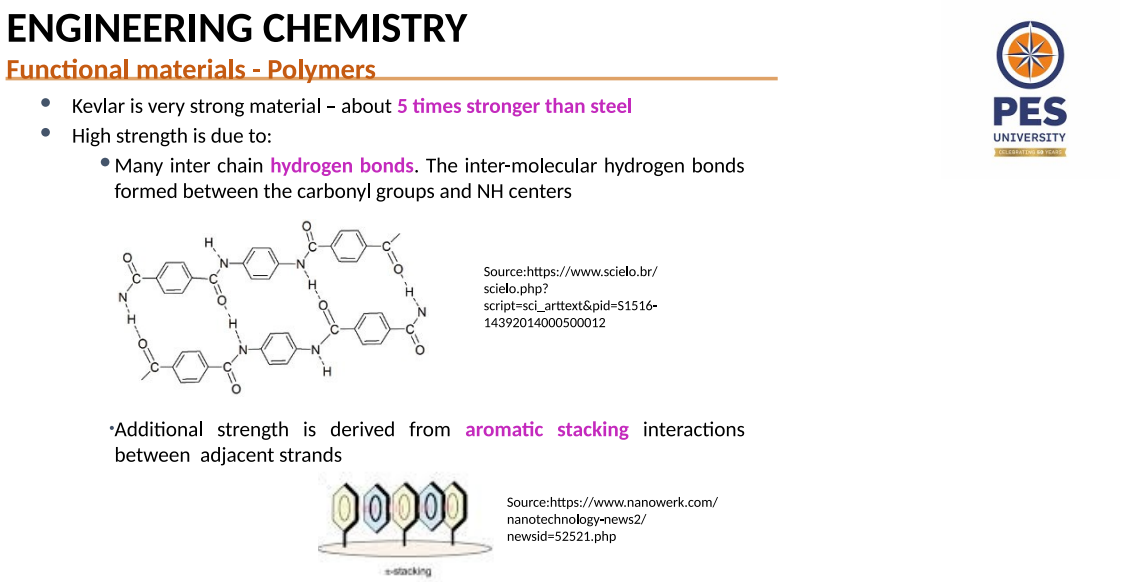


Figure G000088. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 5.

EV0000456

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

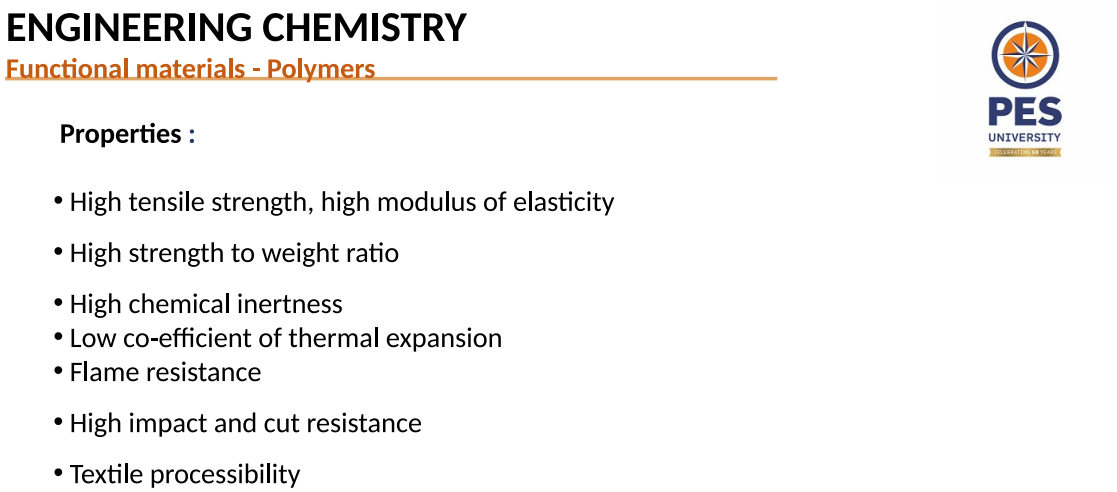


Figure G000089. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 6.

EV0000459

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

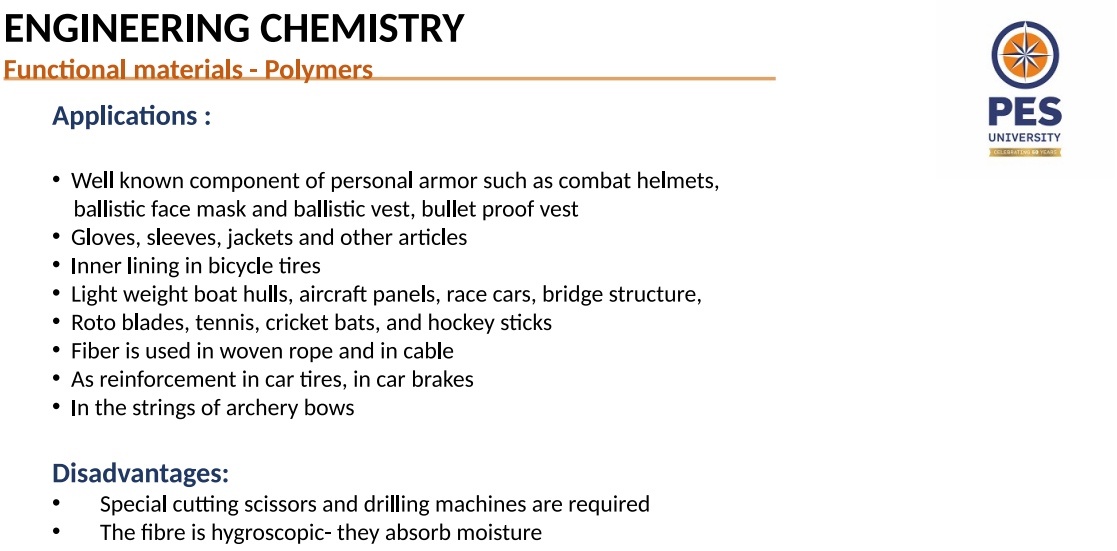


Figure G000090. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 7.

EV0000462

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

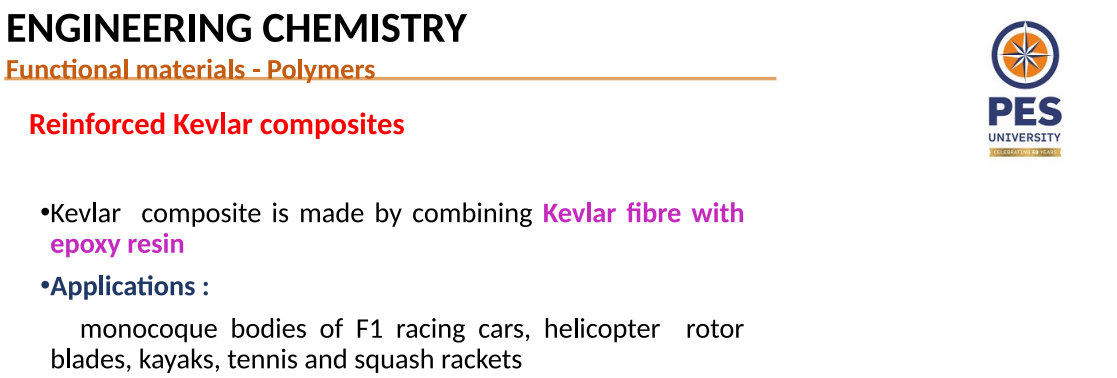


Figure G000091. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 8.

EV0000465

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

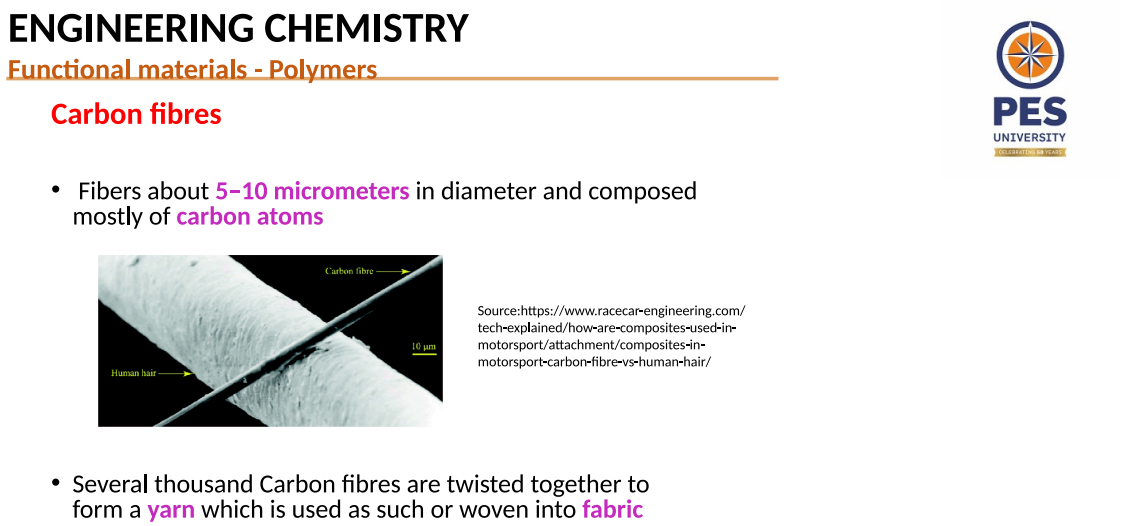


Figure G000092. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 9.

EV0000468

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

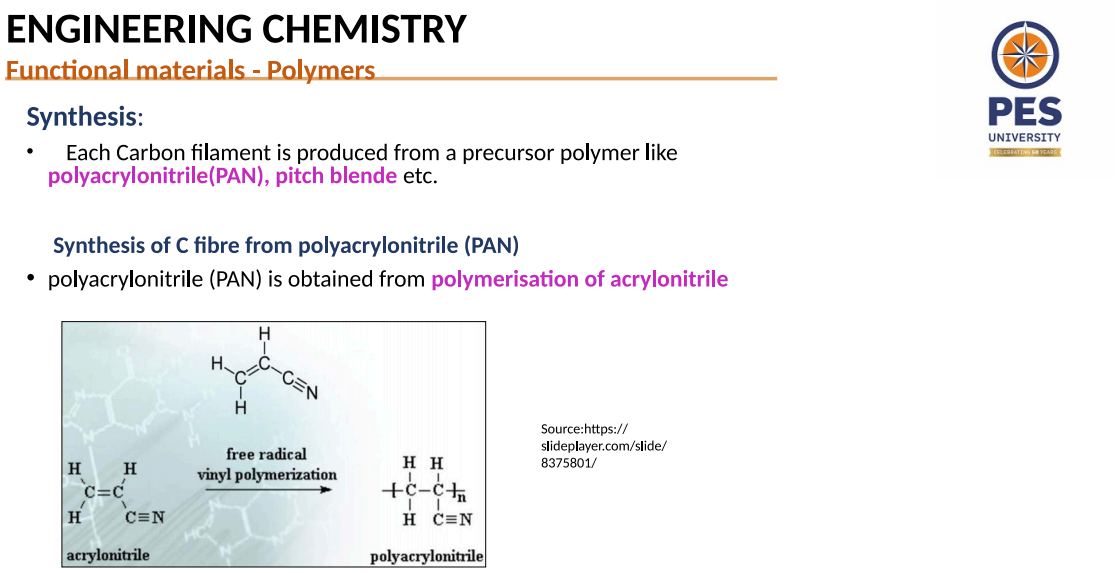


Figure G000093. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 10.

EV0000471

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

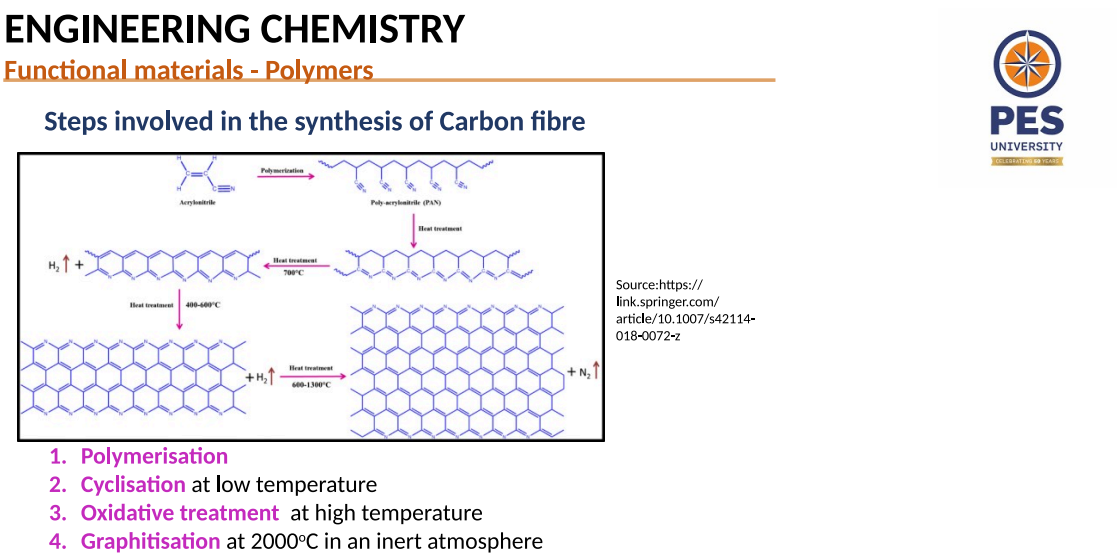


Figure G000094. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 11.

EV0000474

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

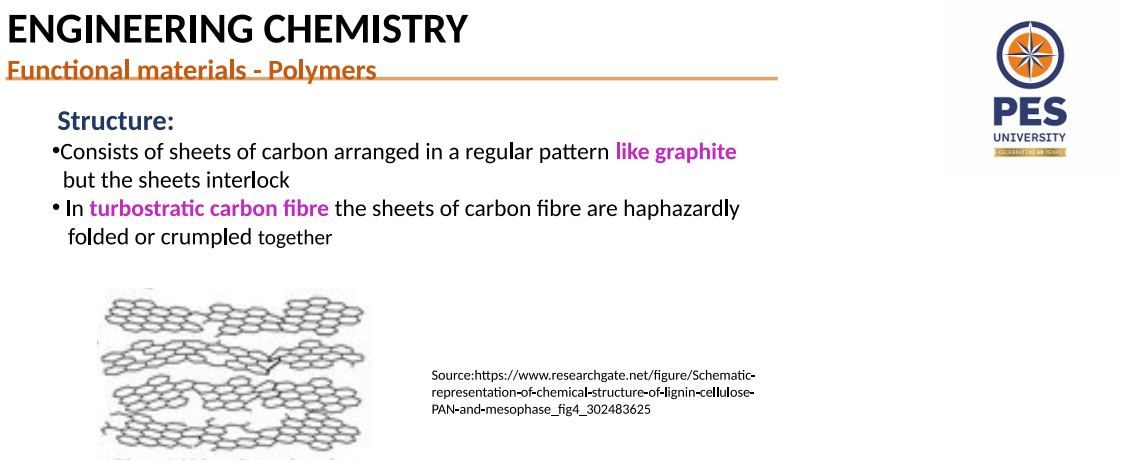


Figure G000095. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 12.

EV0000477

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

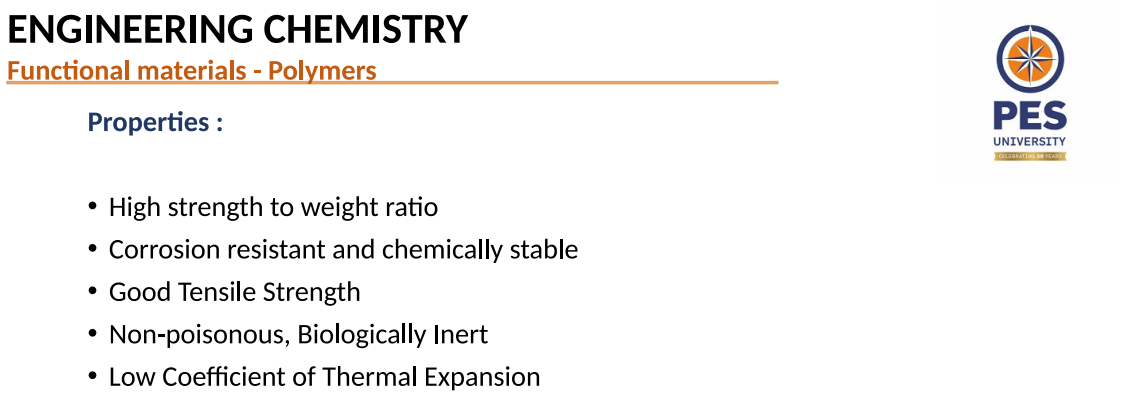


Figure G000096. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 13.

EV0000480

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

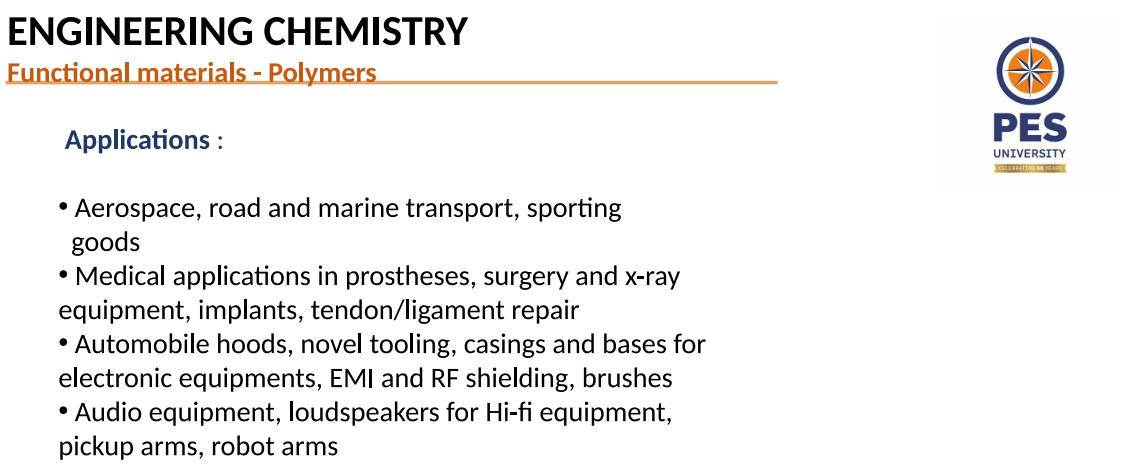


Figure G000097. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 14.

EV0000483

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

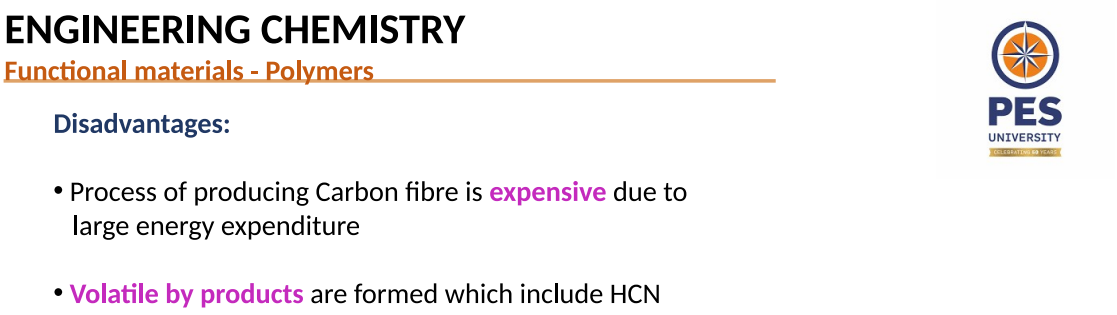


Figure G000098. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 15.

EV0000486

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

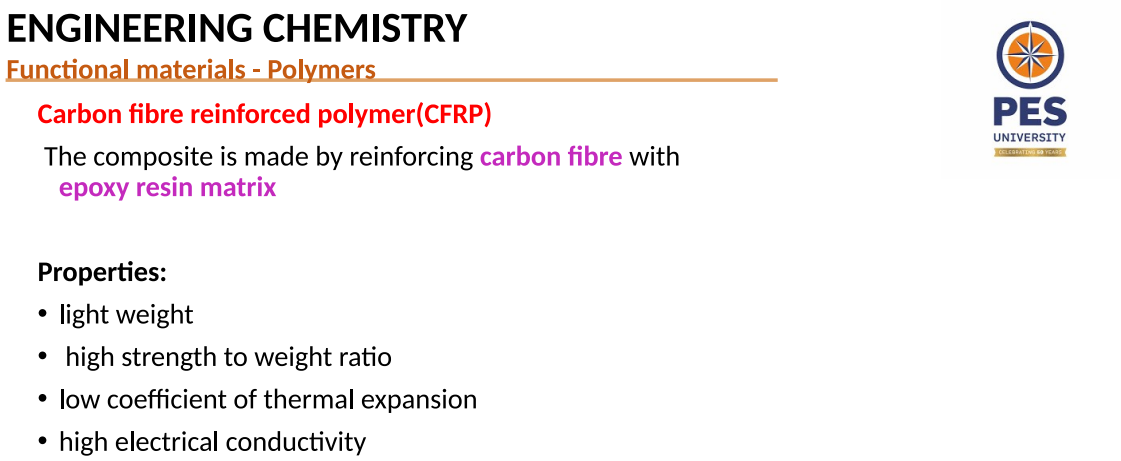


Figure G000099. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 16.

EV0000489

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

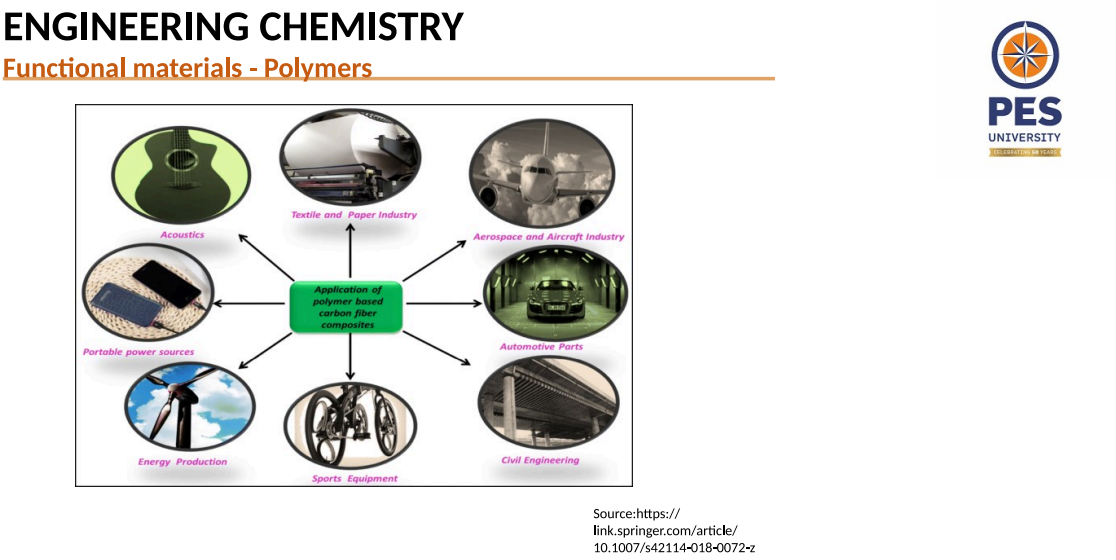


Figure G000100. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 17.

EV0000492

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000101. Polymer composites synthesis properties of Kevlar and Carbon fibers · source slide page 18.

EV0000495

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000443 · EV0000444 · EV0000445
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 46 content: Polymer composites - synthesis, properties of Kevlar and Carbon fibers EV0000446 · EV0000447 · EV0000448
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Polymer composites Materials made from two or more constituent materials and their properties are not attainable by any individual component alone are called composites When one or more of the components are polymers they are called polymer composites Two components are fibre and matrix matrix: polyester, epoxy, phenolic, silicone, melamine, vinyl derivatives fibers: glass fiber, boron filaments, carbon/graphite fibers, aramid (aromatic polyamide), aluminum fiber Polymer composite are produced by suitably bonding a fiber material with a polymer resin matrix and curing the same under pressure and heat Composites have high strength / unit weight EV0000449 · EV0000450 · EV0000451
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Kevlar An aramid fibre - Aromatic polyamide Chemical name is poly-paraphenyleneterepthalamide Synthesis : Condensation of 1,4 phenylene diamine and 1,4 benzenedicarbonyl chloride Source:https://mccord.cm.utexas.edu/chembook/page-nonav.php?chnum=8&sect=6 EV0000452 · EV0000453 · EV0000454

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Kevlar is very strong material – about 5 times stronger than steel High strength is due to: Many inter chain hydrogen bonds. The inter-molecular hydrogen bonds formed between the carbonyl groups and NH centers Additional strength is derived from aromatic stacking interactions between adjacent strands [Image text, OCR] Figure3.Structureofpoly(p-phenyleneterephthalamide) Source:https://www.scielo.br/scielo.php?script=sci_arttext&pid=S1516-14392014000500012 Source:https://www.nanowerk.com/nanotechnology-news2/newsid=52521.php EV0000455 · EV0000456 · EV0000457
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Properties : High tensile strength, high modulus of elasticity High strength to weight ratio High chemical inertness Low co-efficient of thermal expansion Flame resistance High impact and cut resistance Textile processibility EV0000458 · EV0000459 · EV0000460
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Applications : Well known component of personal armor such as combat helmets, ballistic face mask and ballistic vest, bullet proof vest Gloves, sleeves, jackets and other articles Inner lining in bicycle tires Light weight boat hulls, aircraft panels, race cars, bridge structure, Roto blades, tennis, cricket bats, and hockey sticks Fiber is used in woven rope and in cable As reinforcement in car tires, in car brakes In the strings of archery bows Disadvantages: Special cutting scissors and drilling machines are required The fibre is hygroscopic- they absorb moisture EV0000461 · EV0000462 · EV0000463
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Reinforced Kevlar composites Kevlar composite is made by combining Kevlar fibre with epoxy resin Applications : monocoque bodies of F1 racing cars, helicopter rotor blades, kayaks, tennis and squash rackets EV0000464 · EV0000465 · EV0000466
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Carbon fibres Fibers about 5–10 micrometers in diameter and composed mostly of carbon atoms Several thousand Carbon fibres are twisted together to form a yarn which is used as such or woven into fabric [Image text, OCR] Carbon fibre 10m Human hair Source:https://www.racecar-engineering.com/tech-explained/how-are-composites-used-in-motorsport/attachment/composites-in-motorsport-carbon-fibre-vs-human-hair/ EV0000467 · EV0000468 · EV0000469
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Synthesis: Each Carbon filament is produced from a precursor polymer like polyacrylonitrile(PAN), pitch blende etc. Synthesis of C fibre from polyacrylonitrile (PAN) polyacrylonitrile (PAN) is obtained from polymerisation of acrylonitrile [Image text, OCR] H C≡N free radical H HH H vinylpolymerization +C-Ctn C=C H C=N H C=N acrylonitrile polyacrylonitrile Source:https://slideplayer.com/slide/8375801/ EV0000470 · EV0000471 · EV0000472
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Steps involved in the synthesis of Carbon fibre [Image text, OCR] Polymerization Acrylonitrile Poly-acrylonitrile (PAN) Heat treatmest H2 Heat treatment 700°C Heat treatment 400-600°C +H Heat treatment + N 600-1300°C Source:https://link.springer.com/article/10.1007/s42114-018-0072-z Polymerisation Cyclisation at low temperature Oxidative treatment at high temperature Graphitisation at 2000oC in an inert atmosphere EV0000473 · EV0000474 · EV0000475

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Structure: Consists of sheets of carbon arranged in a regular pattern like graphite but the sheets interlock In turbostratic carbon fibre the sheets of carbon fibre are haphazardly folded or crumpled together Source:https://www.researchgate.net/figure/Schematic-representation-of-chemical-structure-of-lignin-cellulose-PAN-and-mesophase_fig4_302483625 EV0000476 · EV0000477 · EV0000478
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Properties : High strength to weight ratio Corrosion resistant and chemically stable Good Tensile Strength Non-poisonous, Biologically Inert Low Coefficient of Thermal Expansion EV0000479 · EV0000480 · EV0000481
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Applications : Aerospace, road and marine transport, sporting goods Medical applications in prostheses, surgery and x-ray equipment, implants, tendon/ligament repair Automobile hoods, novel tooling, casings and bases for electronic equipments, EMI and RF shielding, brushes Audio equipment, loudspeakers for Hi-fi equipment, pickup arms, robot arms EV0000482 · EV0000483 · EV0000484
15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Disadvantages: Process of producing Carbon fibre is expensive due to large energy expenditure Volatile by products are formed which include HCN EV0000485 · EV0000486 · EV0000487
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Carbon fibre reinforced polymer(CFRP) The composite is made by reinforcing carbon fibre with epoxy resin matrix Properties: light weight high strength to weight ratio low coefficient of thermal expansion high electrical conductivity EV0000488 · EV0000489 · EV0000490
17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers [Image text, OCR] TextileandPaperIndustry Acoustics AerospaceandAircraftIndustry Application of polymerbased carbon fiber composites Portablepowersources AutomotiveParts EnergyProduction Civil Engineering SportsEquipment Source:https://link.springer.com/article/10.1007/s42114-018-0072-z EV0000491 · EV0000492 · EV0000493
18. Source slide page 18 (extracted text): THANK YOU Department of Science and Humanities EV0000494 · EV0000495 · EV0000496

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.772] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module seven on functional materials on polymers, and in today's class we are going to study about polymer composites. [00:00:17.156] We will look at Kevlar synthesis properties and applications and then we will study about reinforced Kevlar composites. Firstly, let us understand what is a composite. [00:00:32.100] In a composite we will have two or more constituent materials [00:00:37.604] and their properties are not attainable by each of these components acting alone. It's only when they are mixed together and form a single material that is known as a composite. [00:00:54.148] Now what would be a polymer composite? When one or more of these components are polymers, we call them as polymer composites. So what are the two components? There is always a fiber and a matrix. The matrix would be [00:01:09.776] Polyester, epoxy, phenolic, silicone, melamine, or a vinyl derivative. And the fiber could be a glass fibre, boron filaments, carbon, graphite fibers, aromate, that is aromatic polyamide, or aluminium fibre. So [00:01:27.312] With these two components, the

material which is made is known as the polymer composite. [00:01:34.276] Now how do we produce them? They are produced by suitably bonding the fibre material with the polymeric coalescent matrix and curing the same under pressure and heat. [00:01:48.592] Now why are these composites so popular? Because they have very high strength per unit weight. So that makes them very useful materials for a variety of applications. [00:02:07.012] Let us study about Kevlar, which is a very important fibre. It is no it is an aromate fibre. What do we mean by aromate fibre? It is aromatic. [00:02:19.588] polyamide. Now the chemical name for this would be polyparaphenylin pterythalamide. Now how does it get that name? [00:02:30.832] It is because of the components which are being used, because of the monomers which are being used to make the polymer. Now it is a condensation polymer. [00:02:43.684] Condensation of two monarchs that is one for phenyline diamine and one for benzene di carbone fluoride. Now one for benzene di carbide chloride is basically coming from thalic acid. So we know that thalic acid, this is thalic acid, teri thalide. [00:03:01.456] thalic acid because these are in the para position the COOH COH groups are in the parapositions so we call it as steal acid and instead of the thalic acid we are using the acid chloride so this becomes steady thaloal chloride. [00:03:17.040] That is because the reactivity will be much more enhanced when we use the teritaloid fluoride. Now what is the other monoma being used? It is 14 phenyline dye. [00:03:29.648] Now that also would be para-phenylin diamine. Again, these two are in the para position. The amine groups are in the para position. So we have an NH2 here, NH2 here, and we have COCL, COCL. So as expected between the two monomers, we have this condensation. [00:03:48.484] polymerization which will take place, the condensate condensation reaction would [00:03:53.616] Result in elimination of an HCl molecule. So we can see very clearly when HCl is eliminated from here, we are going to get a CONH linkage, and that is known as the amide linkage. So like this, these two monomers are because they have [00:04:11.344] Two functional groups each. So if this condensation happens, this side is still open for the next uh diamine to attach, and similarly this NH2 is open for the next. [00:04:24.064] COCL group to attach. So therefore the polymerization continues this way, and what we get is amide linkages between these monomers and we get Keblar. So this polymer is known as Keblar. How did we get it? It is a condensation polymer. [00:04:43.556] Now let us see why Kevlar is such a strong material. Kevlar is known to be five times stronger than steel on an equal weight basis. Equal weight.

EV0000110 · EV0000111 · EV0000112 · EV0000113

Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.692] basis. All of us know that Kebler is used to make bulletproof jackets. So what is the reason, what is there in the structure of Kevlar that it is making it so strong. [00:05:17.892] Well there are two reasons for that. The first reason is that there is extensive intermolecular hydrogen bonding which is taking place between the Kevlar chains. Now you can see here there's what this is one chain and this the other chain and this is the third chain. [00:05:35.664] chain and so on so many chains are there in Kebler now between these chains there is formation of hydrogen bond now we know that between the uh polymer units itself the when we join the units monomer units the kind of [00:05:53.232] leakage which we are getting is an amide linkage. Now amide leakage would be NHCO. Now between two units here we see there is an NH and CO. Then again here we have CO NH. Now this is again [00:06:10.724] NH C O. So this is getting repeated, and here again in the same way in the next chain also we see there is C O N H N H C O C O N H N H C O. This is one thing we need to uh pay attention to that once we have [00:06:28.064] NH group here and CO here, the next one cannot

be NH and CO, it has to be CO and NH because it is formed from the diamine, so NH and NH and then again we have CO that is the amide linkage here. Now this will be this molecule is will have will be [00:06:45.744] from the thalic acid so COCO so this we should pay attention to [00:06:50.832] And then so we see that there are amide linkages and but between the cablar chains now we have strong hydrogen bonding which is taking place. How is it happening between the carbonyl groups and the NH centers. So you can see here. [00:07:09.328] that the hydrogen and the oxygen the hydrogen of the amine and the oxygen of the CO are coming very close, they are very in close proximity and that would result in formation of hydrogen bond. So there are so many hydrogen bonds [00:07:26.960] possible because repeatedly uh every unit there is a CONH group and these N of one the hydrogen of one of the uh chains and the [00:07:39.232] O of the other chain they are coming in close proximity, and that will result in formation of hydrogen bond. So you can see that there is extensive hydrogen bonding between the chains, and this is what is responsible for Kebler being so strong. Now we also [00:07:56.976] know that in nylon 66 also we have similar amide linkages but nylon is not as strong as Keblard what could be the reason for that the reason is that in nylon between the uh the amide linkages there is an allephatic chain [00:08:16.772] C6. Now that C6 has been replaced by an aromatic ring here. Now because of this aromatic ring, this is giving a lot of firmness to Kepler. If it's an elephatic chain, we know it will be more flexible. Whereas now it is in the form of an aromatic ring. [00:08:34.544] And that is giving firmness to Kevlar. So, this is the reason why Keblar is so strong. [00:08:42.208] And another reason for this is aromatic stacking. Now we can see here that the benzene rings are lying one above the other. So you can see here if this kind of arrangement happens, then there is a weak interaction between these [00:08:57.696] benzene rings and this gives an additional strength to uh Kevla. So when we have this aromatic stacking, these being planar rings, there will be the this will give a lot of firmness to the structure. [00:09:11.844] To the polymer, and therefore we find that keevlar is very strong. So, why is Kevlar five times stronger than steel on an equal weight basis? The answer is there are two reasons. One is the extensive hydrogen bonding. [00:09:28.656] between chains of kevlar and additional strength is coming from aromatic stacking. [00:09:36.388] What are the properties of Kevlar? We saw the structure and based on that we know that properties are going to depend on the structure. So it is going to have a very high tensile strength, a high modulus of elasticity. [00:09:52.260] It has high strength to weight ratio and very high chemical inertness, low coefficient of thermal expansion. It also has flame resistance and high impact and cut resistance and textile processability.

EV0000114 · EV0000115 · EV0000116 · EV0000117

Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.956] Based on all these properties now we can understand that there will be a lot of applications of Kevlar. [00:10:19.268] So it is a well known component of personal armor, all of us know about it, and uh such as combat helmets, ballistic face masks, bulletproof jackets. Then gloves, sleeves, jackets, etcetera are made of Kevlar to protect users from cuts and heat. [00:10:37.088] And then inner lining in bicycle tires, then lightweight boot hulls, aircraft panels, race cars, bridge structure. Now this fiber is woven into ropes and it is used for offshore drilling. So [00:10:54.816] Earlier they used to use steel pipes. Now we know that when this is in happening offshore there will be in the ocean and therefore there'll be a lot of salinity and corrosion is uh very rampant over there. So these Kevlar fibers now have replaced those steel. [00:11:12.464] ropes in offshore drilling. Now they are also used as reinforcement in car tires, in car brakes and in the strings of archery pose. [00:11:23.748] Now, all

these good properties have led to so many applications, but there are a few disadvantages. And [00:11:32.832] What are they? Special cutting scissors and drilling machines are required because it's a very strong material. So in order to give it a shape, we need special cutting scissors and drilling machines. And one more problem here is that the fibre is hygroscopic. [00:11:50.480] So it absorbs moisture, hence moisture resistant substituents have to be added inside to make it as less hygroscopic as possible. Now what we discussed so far is about the Keblar fiber. [00:12:09.540] We have still not formed the composite because when we talk of a composite, we know there are two components. So Kepler fiber itself has so much strength. Now if this is made into a composite, so how will we make it into a composite? Obviously, we need a [00:12:27.632] uh matrix for that and that matrix is usually an epoxy resin. So it is made by combining the blad fiber with the epoxy resin matrix. And now the material will become so strong that it is going to [00:12:45.296] now have many more applications. So what is the use of making a composite like this? This matrix matrix when you have and these fibers are embedded in it, then any force which is applied onto the [00:13:01.264] uh material will be dissipated the the impact is going to be taken by the metric so that makes the material even more strong. So what would be now applications of such a material? This is known as reinforced Kevlar composites. [00:13:17.136] Now here we have we can make monocope bodies of Formula One racing cars, helicopter rotor blades, kayaks. [00:13:25.056] tennis and tennis and squash rackets. So now the applications have increased because the material is still more stronger as compared to Kepler. So this is about [00:13:38.976] Kevlar on the next class we will talk of another polymer composite and that is carbon fiber. Thank you.

EV0000118 · EV0000119 · EV0000120 · EV0000121

Full transcript: 1885 normalized words, 56 timestamped segments, 827.349333 seconds duration. Low-confidence segments flagged: 0.

EV0000110 · EV0000111 · EV0000112 · EV0000113

9 Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2))

This lesson is indexed from the source video titled Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000122 · EV0000123

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.156] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module seven on functional materials, that is, polymers. [00:00:11.652] Today we will talk about another polymer composite that is carbon fiber. We talk about the synthesis, properties and applications and finally we will look at the composite of carbon fiber which is known as carbon fiber reinforced polymer. [00:00:30.020] What are carbon fibers? Basically, they are fibers which are made mostly of carbon

atoms, and the diameter is just about 5 to 10 micrometers. [00:00:42.992] When you compare this diameter with the human hair, as you can see in the picture here, the human hair is much much thicker as compared to the carbon fiber. [00:00:55.044] So they are very very thin fibers and several thousands of these fibers are twisted together to make a yarn and from that yarn the fabric is woven. [00:01:08.356] So how do we synthesize this carbon fiber? Now each carbon filament is produced from what we call as a precursor polymer like polyacrylonitrile or pitch, etc. [00:01:22.016] Now let us study how we are going to prepare this carbon fibre from polyacrylo nitrite. [00:01:29.508] So initially polyacrylonitrile is made by polymerization of acrylonitrile. So we studied in the uh when we were talking about addition polymerization that [00:01:42.736] in the in case of ethylene, if we take ethylene and we substitute with a cyanide group, what we get [00:01:49.856] On polymerization is known as polyacrylonitrile. So simple polymerization will happen and we will get polyacrylonitride. So now the next step, how we are going to use this polyacrylonitrile to uh get the carbon fiber. [00:02:08.292] Now one thing we have to remember that the carbon fiber has a graphite like structure. So when we are talking about synthesis, we have to keep in mind how to arrive at that graphite like structure. So here again the first step would be polymerization which I mentioned just now. [00:02:26.064] So we have this acrylonitri which have which has been polymerized and that gives you polyacrylonitrile. Now you can see here of course these acrylonitrile are placed in such a way that they will polymerize [00:02:44.708] So you have to place them like this. So all these acrylonitrile molecules are going to be placed like this because it is renal cyanide. So this is how these [00:03:01.092] phil uh acrylonitrile monomers are placed and then polymerization is initiated. So when polymerization happens what is going to happen? The polymerization will always take across take place across the double bond. So this double bond is now going to [00:03:18.528] uh open up and then we get a polymer which is getting formed it says double bond is gone now and we get cyclic uh the polymerization which happens like this and that's what you can see here in this picture. So you can see the uh [00:03:34.976] chain has got formed because of the opening of this double bond. Okay. So that is the first step which is [00:03:41.776] Polymerization. Now the next step is we need to cyclize it. Remember that we want a graphite like structure. So the next step would be cyclization that is done at a low temperature. Now here again we have these cyanide C triple bond N. So what is going to happen during this cyclone? [00:03:59.408] Cyclization process there will be a bond which will be formed between this carbon and this nitrogen. So basically, once here, if you see the polymerization after it happens, this chain has got formed, and now between this nitrogen, there's a triple bond. So one of the bonds is going to open. [00:04:16.976] Up and you get a bond getting formed between C and N, and here it will be left with double bond. That's exactly what you can see in this picture, and that will result in a cyclic kind of molecules, and therefore that is called as the cyclization process. So the whole [00:04:34.544] Polymer now has become cyclic is constituted of cyclic chains, or you can see here cyclic rings, and these are in the form of a chain. So this is the second step that is cyclization at low temperature. So now the next step would be [00:04:52.112] I want a double bond here. Now that [00:04:56.708] Because we want to go towards a graphite like structure. I want a double bond here. How do I get it? That is done by the oxidative treatment. So cyclization is the second step which we saw here. This is the cyclization, and the third step would be oxidative treatment, which is done at a higher temperature.

EV0000122 · EV0000123 · EV0000124 · EV0000125

Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.480] Now this was at low temperature. This is done at a higher temperature. Now what would happen due to oxidation? Obviously, it is treated with oxygen and water molecule will be eliminated, and here basically we are removing hydrogen. This [00:05:30.336] Hydrogen that is unsaturation now came up. So obviously hydrogen has been removed, and of course we are doing it. Uh we are basically causing oxidation of this um. [00:05:41.824] uh this small uh this chain polymer over here. So what we get is an unsaturation at these positions. So now the next step is the most important step where we are going to go towards the graphite like [00:06:00.400] structure. Now but before that if you do uh at a slightly lower temperature we find that these ribbons they are formed like ribbons now two three ribbons like so many of them one place one below the other they start joining together and of course again hydrogen is getting eliminated. So you're coming closer to [00:06:18.064] graphite like structure now what is missing here it is that there are still some nitrogen atoms present here so as you go on increasing the temperature give it a treatment at a higher temperature then we find that more of these will now if you have three like this now another three will join together [00:06:35.632] and you get start getting the carbon fiber with the elimination of this nitrogen. Of course the last step would be where you remove all other elements except carbon because we want a graphite like structure that is made of carbon. [00:06:54.000] So graffiti is the final process and that is done at a very high temperature and in an inert atmosphere and inert atmosphere because you're treating it taking the fibre to such a high temperature, so it will it'll become uh inflammable, it may catch fire. So we have to do it in a [00:07:11.664] inert atmosphere. So by this process starting from acrylonitrile you got polyacrylonitrile then you cause the cyclization of this chain polymer and then we call we made it [00:07:26.896] We got some unsaturation in these positions by oxidative treatment and finally we heated it and we got the ribbon like structures joined together to give you this type of structure and finally when it is graffitized, that's called graphitization at a very high temperature. [00:07:44.528] Then we get what we call as the carbon fiber. [00:07:48.720] Now this is the process by which we can get the carbon fiber. So what do we understand that this carbon fiber has to have a graphite like structure. Now what is a graphite like structure? If you remember graphite is made up of it, it has a layer structure and [00:08:08.580] These chains between these layers actually there is a uh uh there is just Vanderwall force of attraction. So so these chains actually slip. But we do that carbon fiber is one of the strongest uh uh material. [00:08:26.352] So how is that possible? Now the structure if you see these uh consists of sheets of carbon just like we have in graphite, but they are not arranged in a regular pattern. In graphite we have very regular pattern these [00:08:44.048] lay of carbon they are just one below the other and because they're all sp2 hybridized so we have layers like this and these chains between the chains is just one wall force and the these chains can slip against each other. So your carbon fiber is also a graphite like structure as you can see. But [00:09:01.024] It is known as a turbostratic arrangement. What is that? These sheets of carbon fiber are half as a leaf folded or crumpled together. So now what happens? They are all [00:09:13.248] folded and crumpled so as a result that slipping of chains is not possible and this is known as the turbostratic arrangement and that gives the carbon fiber uh a good strength. [00:09:27.044] Now uh properties we can understand yes, it has very high strength to weight ratio, it is very corrosion resistant and very chemically stable, it has good tensile strength, it is non-poisonous and biologically inert. So basically we are seeing [00:09:44.816] Say that it is also biocompatible. It is a fiber which is biocompatible and it has a low coefficient of

thermal expansion. So it has wonderful properties. Of course, the most used property is high strength to weight ratio.

EV0000126 · EV0000127 · EV0000128 · EV0000129

Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.572] What would be the applications? It's very clear such a strong material we have. It is used in aerospace, road and marine transport, it is used in sporting goods and medical applications in prostesis surgery. Now this is and also in implants tendon. [00:10:23.344] Ligamentary repair of those. [00:10:25.728] Why? Because it is biocompatible. So carbon fibers can be used for uh for skin grafting, etc. Then we also have automobile goods, novel tooling, casings and bases for electronic equipment, EMI and RF shielding brushes. [00:10:43.376] So audio equipment, loudspeakers or hi-fi equipment, pickup arms, robo arms, wherever uh you need the polymer to be very strong, uh we are going to use this carbon fiber. [00:10:57.616] But what we have learnt now is about the advantages. Now just let us look at the disadvantages. Of course, it is very expensive due to the large ex energy expenditure. So producing a carbon fibre itself is very expensive, and that's why we don't [00:11:15.376] find too many consumer goods which are using this carbon fiber, but it is understood that as the manufacturing becomes more and more and the use of carbon fiber becomes more and more then it is the price is definitely going to uh come down and we will have lot of consumer goods [00:11:32.944] Which are going to use carbon fiber, especially where a lot of strength is required. And one more disadvantage would be volatile byproducts are formed. Like for example, H Cn could be formed when we are making carbon fiber from polyacrylonitrile. So these are the disadvantages. [00:11:51.664] So, what we have talked about now is just the carbon fiber, which has such wonderful properties. So, if we want to make a composite out of it, then we can understand that this is going to further make the material very, very strong. [00:12:09.424] So, how will we make this composite? We know that when we make a composite, there has to be a fiber that is the carbon fiber, and the matrix is the epoxy resin. So we are going to combine the two, they will be cured at high temperature and pressure, and we are going to get what we call a [00:12:26.992] the carbon fiber reinforced polymer so once again it is the metrics like this and we have these fibers embedded in here so depending on what exactly you want to prepare so these embedded fibers [00:12:43.616] uh are into the matrix. So if we need high impact strength or any lot lot of force is applied on this material, the matrix is going to take the impact of that uh force and it will get dissipated. So therefore [00:13:00.388] That is how it is getting the strength. Now, what would be the properties? Of course, we know light weight and high strength to weight ratio, low coefficient of thermal expansion. [00:13:13.380] But one new property which has come here and very very important is the high electrical conductivity. Why? Because we know it has a graphite like structure and we know that graphite. [00:13:27.748] is conducting in nature. So because of that we find that it has high electrical conductivity. So what would be the applications of such a polymer composite [00:13:41.316] we know that there are going to be a number of application like in all fields, in various fields, like say sports equipment, then energy production, then consumer goods or power uh portable uh power sources then coastics [00:13:58.064] because the fibers can be used over there and uh so on like there are so many and then uh of course in aerospace industry then for automobile civil engineering so all these areas we find a lot of application for this [00:14:15.184] the polymer composite of carbon fibre. So in the next class now we are going to talk about a new class of polymers which are called conducting [00:14:30.928] Polymers so far polymers were always

thought of as insulators, but yes, there is a class of polymers which are which can actually conduct as good as that of the metal, and these are known as conducting polymers. Thank you.

EV0000130 · EV0000131 · EV0000132 · EV0000133

Full transcript: 2171 normalized words, 58 timestamped segments, 887.082667 seconds duration. Low-confidence segments flagged: 0.

EV0000122 · EV0000123 · EV0000124 · EV0000125

10 Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same

This lesson is indexed from the source video titled Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000134 · EV0000135

Graphs and diagrams

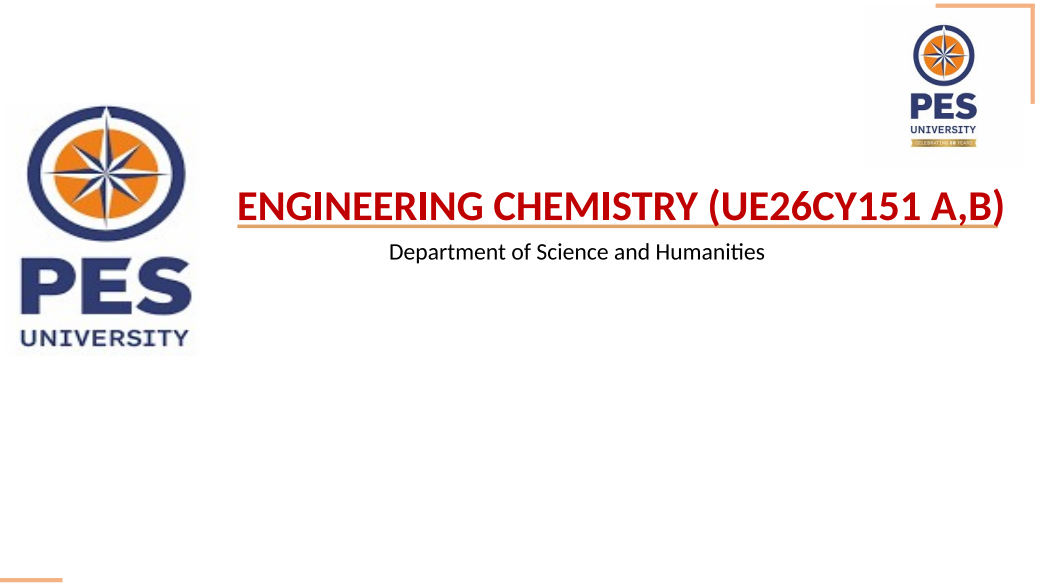


Figure G000102. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 1.

EV0000498

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

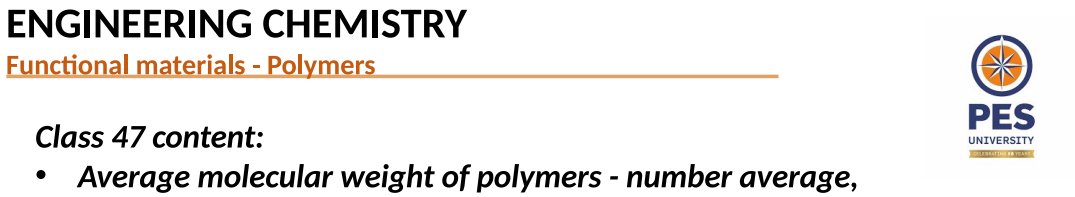


Figure G000103. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 2.

EV0000501

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

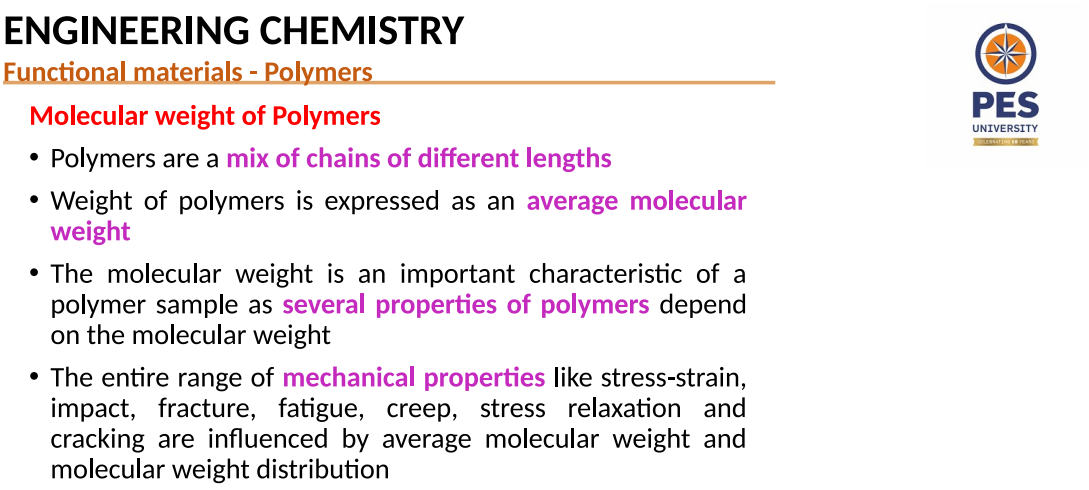


Figure G000104. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 3.

EV0000504

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

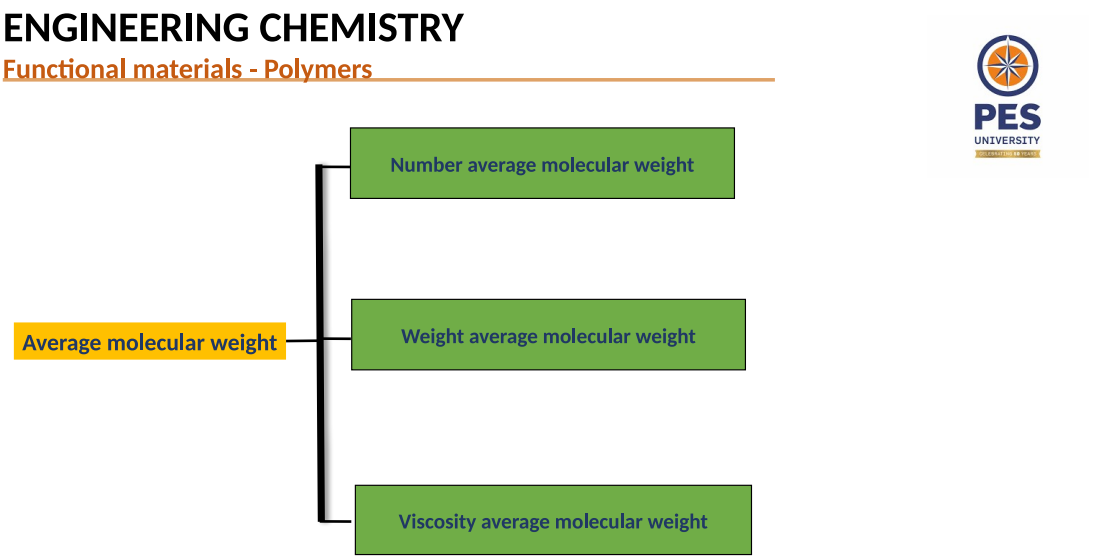


Figure G000105. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 4.

EV0000507

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

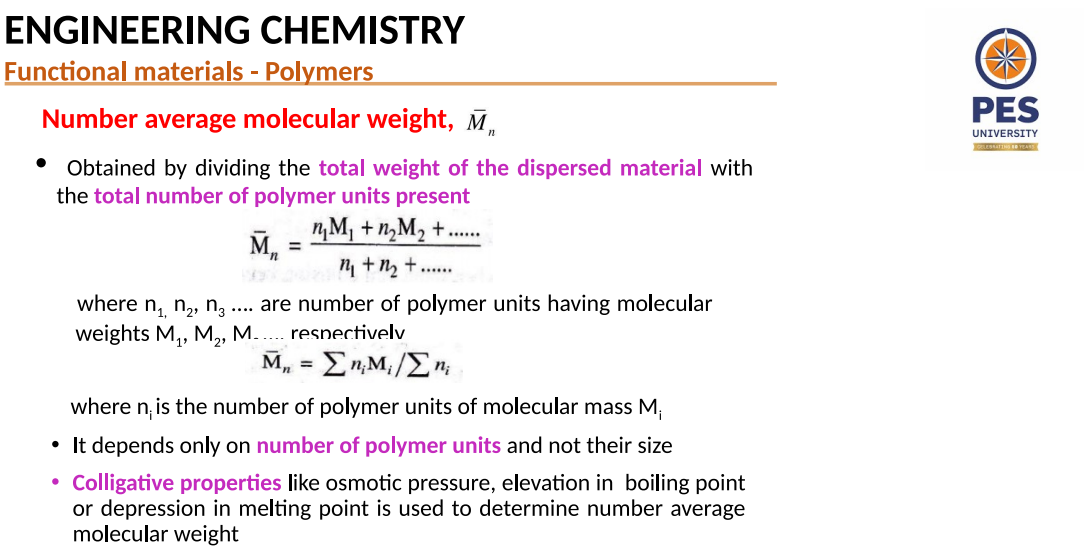


Figure G000106. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 5.

EV0000510

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

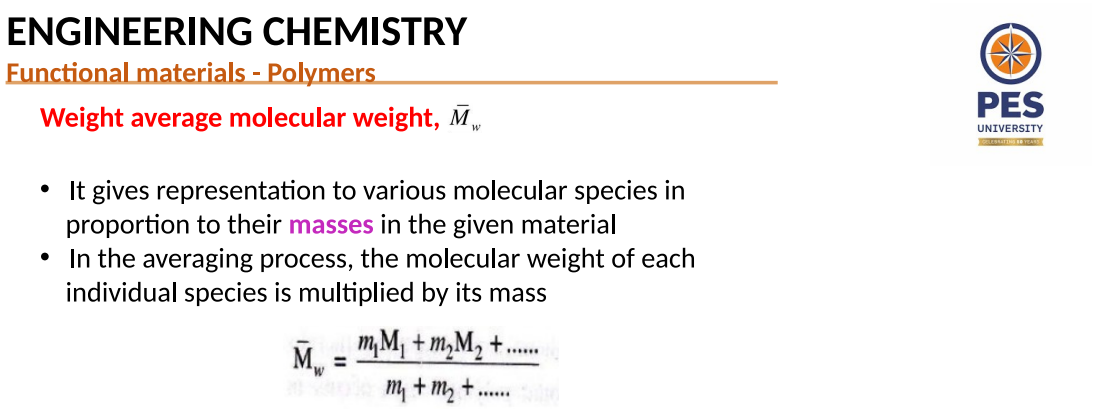


Figure G000107. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 6.

EV0000513

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

- if n_1, n_2, n_3 .... are number of polymer units having molecular weights M_1, M_2, M_3 respectively then $m_1 = n_1M_1$, $m_2 = n_2 M_2$ ....
- Weight average molecular weight can be written as

$$\bar{M}_w = \frac{n_1M_1^2 + n_2M_2^2 + ...}{n_1M_1 + n_2M_2 + ...} = \sum n_iM_i^2 / \sum n_iM_i$$

- It depends on size of the polymer units
- Properties which depend on size like sedimentation velocity or light scattering are used to determine weight average molecular weight

Figure G000108. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 7.

EV0000516

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

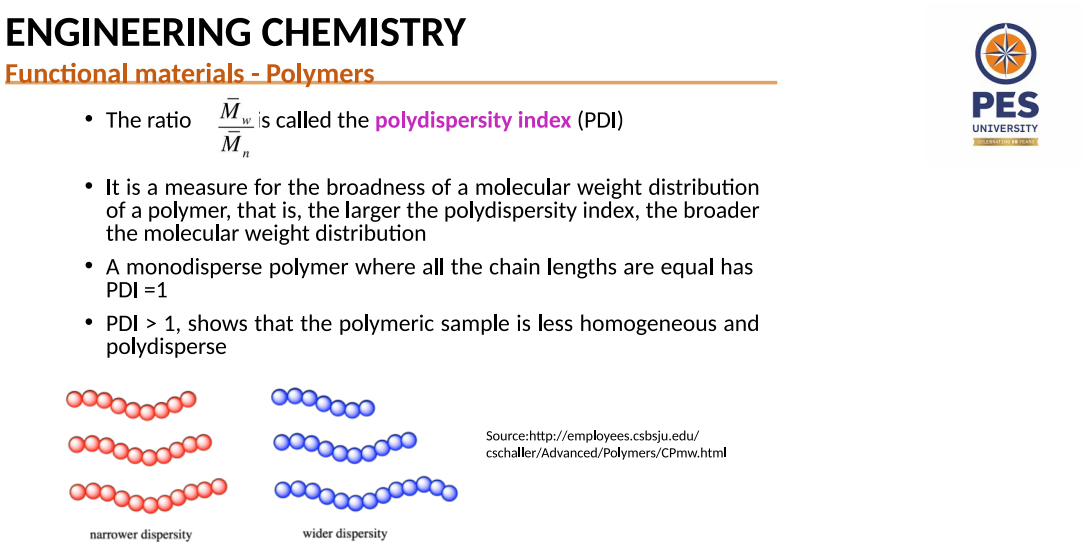


Figure G000109. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 8.

EV0000519

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

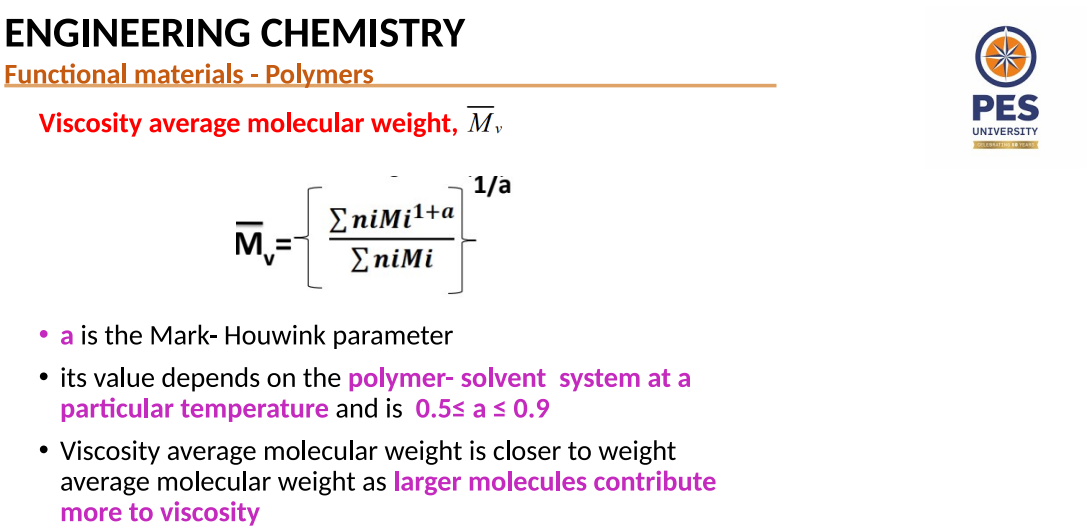


Figure G000110. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 9.

EV0000522

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

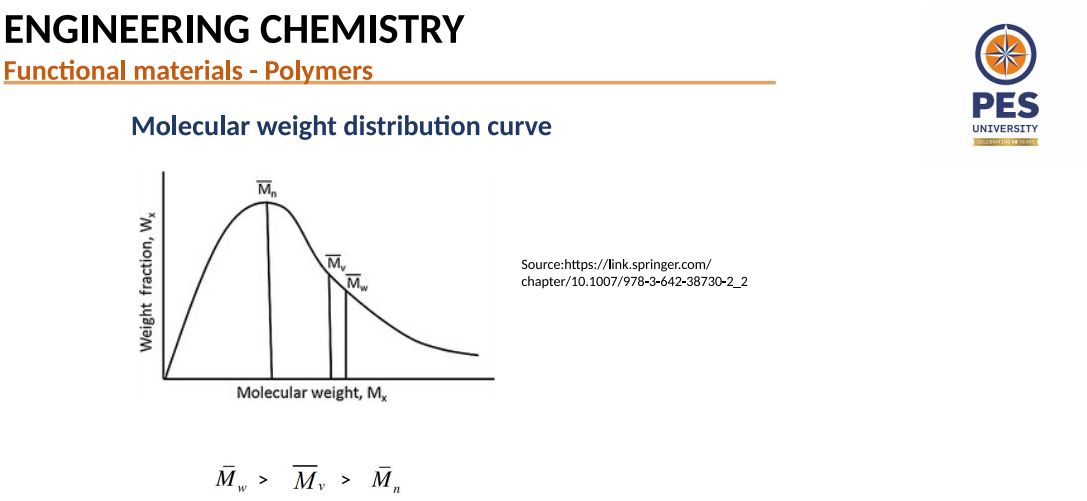


Figure G000111. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 10.

EV0000525

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

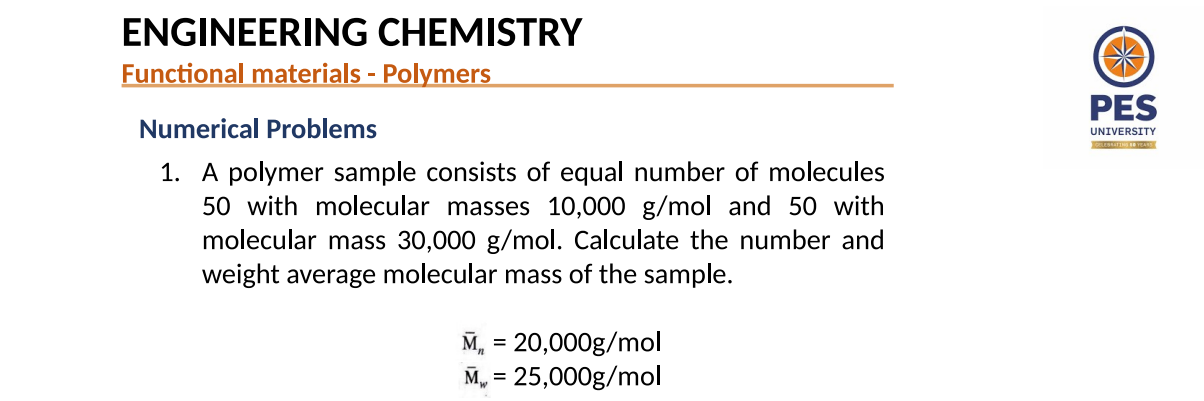


Figure G000112. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 11.

EV0000528

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

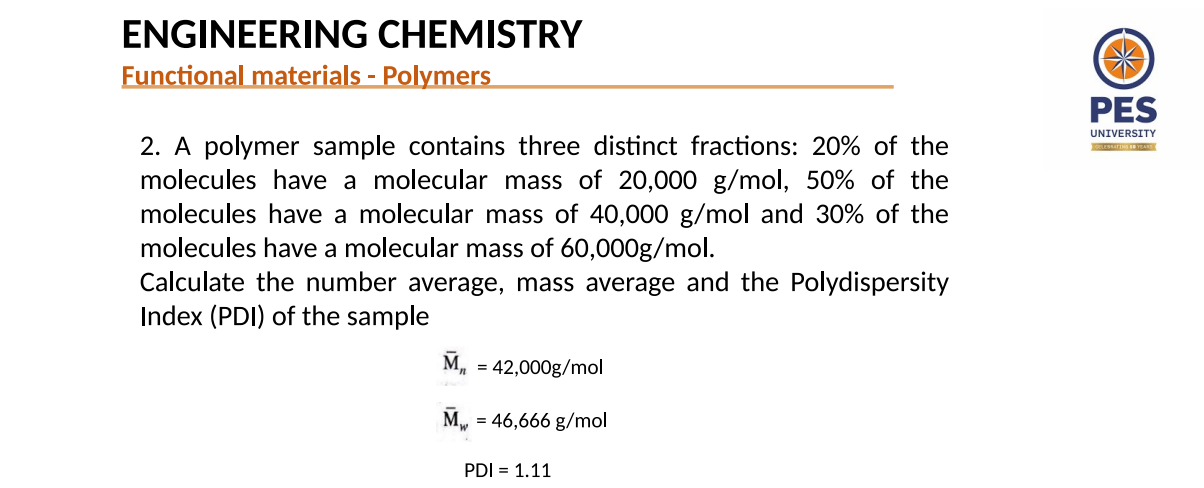


Figure G000113. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 12.

EV0000531

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

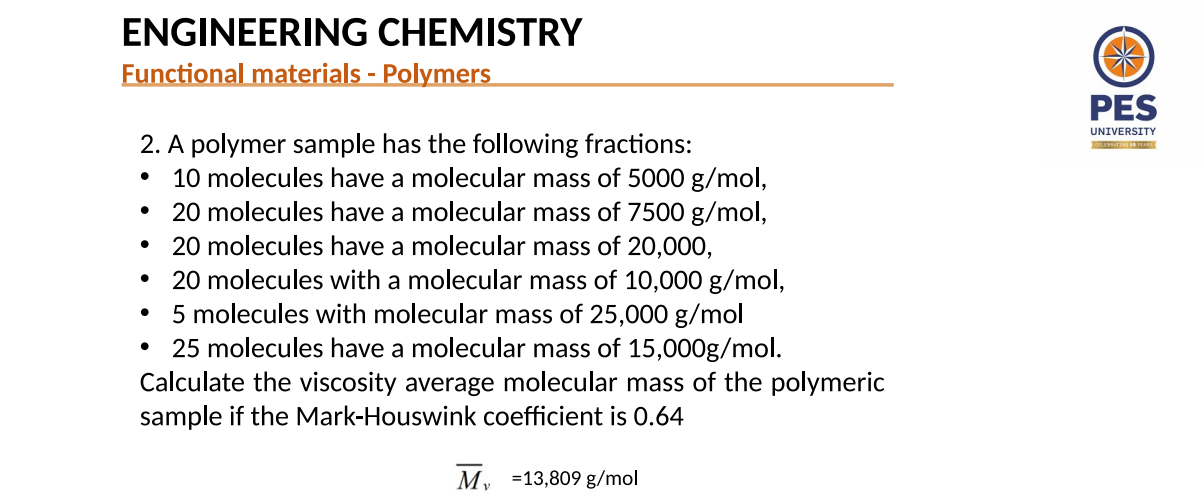


Figure G000114. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 13.

EV0000534

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000115. Average molecular weight of polymers number average weight average and viscosity average molecular weight nu · source slide page 14.

EV0000537

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000497 · EV0000498 · EV0000499
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 47 content: Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems EV0000500 · EV0000501 · EV0000502
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Molecular weight of Polymers Polymers are a mix of chains of different lengths Weight of polymers is expressed as an average molecular weight The molecular weight is an important characteristic of a polymer sample as several properties of polymers depend on the molecular weight The entire range of mechanical properties like stress-strain, impact, fracture, fatigue, creep, stress relaxation and cracking are influenced by average molecular weight and molecular weight distribution EV0000503 · EV0000504 · EV0000505
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Number average molecular weight Weight average molecular weight Average molecular weight Viscosity average molecular weight EV0000506 · EV0000507 · EV0000508

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Number average molecular weight, Obtained by dividing the total weight of the dispersed material with the total number of polymer units present where n1, n2, n3 .... are number of polymer units having molecular weights M1, M2, M3,.... respectively where ni is the number of polymer units of molecular mass Mi It depends only on number of polymer units and not their size Colligative properties like osmotic pressure, elevation in boiling point or depression in melting point is used to determine number average molecular weight EV0000509 · EV0000510 · EV0000511
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Weight average molecular weight, It gives representation to various molecular species in proportion to their masses in the given material In the averaging process, the molecular weight of each individual species is multiplied by its mass where m1, m2, m3 .... are masses of polymer units having molecular weights M1, M2, M3 respectively EV0000512 · EV0000513 · EV0000514
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers if n1, n2, n3 .... are number of polymer units having molecular weights M1, M2, M3 respectively then m1 = n1M1, m2 = n2 M2 .... Weight average molecular weight can be written as It depends on size of the polymer units Properties which depend on size like sedimentation velocity or light scattering are used to determine weight average molecular weight EV0000515 · EV0000516 · EV0000517
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers The ratio is called the polydispersity index (PDI) It is a measure for the broadness of a molecular weight distribution of a polymer, that is, the larger the polydispersity index, the broader the molecular weight distribution A monodisperse polymer where all the chain lengths are equal has PDI =1 PDI > 1, shows that the polymeric sample is less homogeneous and polydisperse [Image text, OCR] 8 888 8 ： Source:http://employees.csbsju.edu/cschaller/Advanced/Polymers/CPmw.html EV0000518 · EV0000519 · EV0000520
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Viscosity average molecular weight, a is the Mark- Houwink parameter its value depends on the polymer- solvent system at a particular temperature and is 0.5≤ a ≤ 0.9 Viscosity average molecular weight is closer to weight average molecular weight as larger molecules contribute more to viscosity EV0000521 · EV0000522 · EV0000523
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Molecular weight distribution curve [Image text, OCR] Molecularweight,M Source:https://link.springer.com/chapter/10.1007/978-3-642-38730-2_2 > > EV0000524 · EV0000525 · EV0000526
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Numerical Problems A polymer sample consists of equal number of molecules 50 with molecular masses 10,000 g/mol and 50 with molecular mass 30,000 g/mol. Calculate the number and weight average molecular mass of the sample. = 20,000g/mol = 25,000g/mol EV0000527 · EV0000528 · EV0000529

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers 2. A polymer sample contains three distinct fractions: 20% of the molecules have a molecular mass of 20,000 g/mol, 50% of the molecules have a molecular mass of 40,000 g/mol and 30% of the molecules have a molecular mass of 60,000g/mol. Calculate the number average, mass average and the Polydispersity Index (PDI) of the sample = 42,000g/mol = 46,666 g/mol PDI = 1.11 EV0000530 · EV0000531 · EV0000532
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers 2. A polymer sample has the following fractions: 10 molecules have a molecular mass of 5000 g/mol, 20 molecules have a molecular mass of 7500 g/mol, 20 molecules have a molecular mass of 20,000, 20 molecules with a molecular mass of 10,000 g/mol, 5 molecules with molecular mass of 25,000 g/mol 25 molecules have a molecular mass of 15,000g/mol. Calculate the viscosity average molecular mass of the polymeric sample if the Mark-Houswink coefficient is 0.64 =13,809 g/mol EV0000533 · EV0000534 · EV0000535
14. Source slide page 14 (extracted text): THANK YOU Department of Science and Humanities EV0000536 · EV0000537 · EV0000538

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.548] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module seven on functional materials that is polymers and this will be the last class of this module. In this class we are going to talk about molecular weight of polymers. [00:00:19.172] That is number average molecular weight, weight average molecular weight, and viscosity average molecular weight. When we talk of molecular weight of any compound, for example, water [00:00:33.764] We know that its molecular weight is eighteen, and irrespective of the source from where we are going to take the water, the molecular weight is going to still be eighteen, and each of the molecule [00:00:48.516] is going to have a molecular weight of 18, but the case of polymers is different because in a sample the molecular weight of the polymers, the chain length of the polymers is different. Each unit may have a different [00:01:06.512] Length. So polymers basically are a mix of chains of different lengths. Therefore, whenever we talk about molecular weight of a polymer, we talk about an average molecular weight. [00:01:20.580] So this molecular weight now we know is a very important characteristic of a polymer sample because several properties of polymers are found to depend on the molecular weight. [00:01:32.512] The entire range of mechanical properties which are very useful to determine the application of a polymer like stress strain, impact, fracture, fatigue, creep, stress relaxation, and cracking, they are all influenced by the average molecular weight and also the molecular weight distribution. [00:01:52.772] So the average molecular weight of a polymer can be found in different ways. There could be number average molecular weight, we can also have a weight average molecular weight and also viscosity average molecular weight. Let us look at each one of them. [00:02:09.028] Number average molecular weight is denoted by Mn bar. The bar over M stands for the average. Now this is obtained by dividing the total weight of the dispersed material that is the polymer. [00:02:24.096] with the total number of polymer units present. Now how do we determine the total weight of the dispersed material? Suppose N1, N2, N3 are the number of polymer units having molecular weights of N1, M2, N3. [00:02:38.880] Then the total weight would be obviously n1 multiplied by m1. [00:02:44.416] plus n two multiplied by m2 and n three multiplied by m three so on so forth and then the total number of polymer units because

there are so many uh polymer units n1 n two and three the total number will obviously add up [00:02:59.024] and we will have n1 plus n2 plus n three and the formula therefore will be mn bar is equal to n1 m1 plus n2 m2 and continue this way and divided by n1 plus m2 n2 plus and so on. So we can write this in this form [00:03:17.344] Summation of NIMI divided by summation of NI would give me the number average molecular weight. Of course the summation is over I and what is I? It is the particular species we are talking about. So I will be 1 for the first species, so N1, M1. [00:03:35.056] and divided by N1. And similarly, N2, M2. So we have to sum up all these over all the species, the N1, N NI, MI. [00:03:47.428] And then divide by the total number of units that is n1 plus n two plus n three and so on so forth, what we saw here. This is just writing it in a short form. So now this [00:03:59.300] number average molecular weight we can see clearly is only dependent on the number of polymer units and not on their size. And [00:04:09.488] We are aware that if we have to determine any property of a polymer or any uh substance for that matter which is based only on the number of species, then we have to consider colligative properties because we know that colligative properties [00:04:27.120] Depend only on the number of the species and not on the size or the nature of the species. And such colligative properties you are already aware, like osmotic pressure, elevation and boiling point, depression and freezing point, or uh even relative lowering of [00:04:43.920] vapor pressure. So all these could be used to determine the number average molecular weight because polligative properties depend only on number and when we are talking of number average molecular weight we are only concerned about the number of polymer units. [00:04:59.780] Let us talk about weight average molecular weight. So here it is given by NW MW bar and that bar again stands for the average. Now here what we are going to do is we are going to

EV0000134 · EV0000135 · EV0000136 · EV0000137

Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.852] use the mass we are going to consider the mass of each of these species and not just the number [00:05:22.528] Why many times the number average molecular weight does not give you a clear picture of the average uh molecular weight. We want to we understand that the higher masses would contribute more to the molecular weight than the lower mass. So we talk of another [00:05:40.208] average that is known as the weight average molecular weight. So it is here we give representation to various molecular species in proportion of their masses in the given material. [00:05:53.028] So in the averaging process now the molecular weight of each of the individual species is multiplied by its mass. There we consider number. In number average molecular weight here we are talking of the mass. So therefore mass average molecular weight on weight average molecular weight will be [00:06:10.800] M1 multiplied by capital M1 plus M2 multiplied by capital M2 and so on divided by the masses M1 plus M2 plus so on. [00:06:20.036] Now what is this M1, M2, M3? Again, they are masses of the polymer units having molecular weights M1, M2, M3 respectively. So this is how we are going to uh determine the weight average molecular weight. [00:06:35.268] Now if we want to bring in the concept of number of polymer units, can we do that? Now we if number of polymer units like N1, N2, N3 [00:06:46.304] are the number of polymer units which have molecular weights m1 m2 m3 respectively, then we know that mass m1 can be given by N1 M1. Similarly M two will be given by N2 M2. So can we substitute in the previous equation [00:07:01.952] uh deplace m1 by n one m1 so what happens to the formula of weight average molecular weight now we have [00:07:11.104] M1 is replaced by N1 M1. So therefore it becomes N1 M1 squared. Similarly, this becomes N2 M2 squared and so on. And in the denominator, we had M1 plus M2 plus M3. So now

M1 M2 we can see here it is becoming N1 M1. [00:07:30.832] plus n two n two and so on. So you are if you write in the short form it will be NIMI squared by summation of NI MI summation of NIMI squared by summation of NIMI. I stands for the that particular species. Now here we see very clearly [00:07:49.136] that it depends on the size of the polymer unit. So what type of experimental determination uh would you set up in order to determine the weight average molecular weight? Obviously some experiment which depends on the size like how we chose polligative properties for [00:08:06.768] Number average molecular weight because colligative properties depend only on number. Here we are going to look for an experiment which is going to uh be dependent on the size, and these would be sedimentation velocity or light scattering. Sedimentation velocity will depend on the [00:08:24.932] the weight of the units and light scattering would depend on the size of the each of these units and therefore these two experiments could be used to determine weight average molecular weight. [00:08:40.324] Now the ratio of the number average molecular weight and the weight average molecular weight or rather MW by Mn. [00:08:49.776] is known as the polydispersity index. So it is the ratio of weight average molecular weight by the number average molecular weight. This is known as a poly dispersity index. What does this tell us? [00:09:05.444] It is a measure of the broadness of a molecular weight distribution of a polymer, that is, the larger the polydispersity index, the broader is the molecular weight distribution. Now if [00:09:20.644] you had a mono dispersed polymer that means only one type of polymer unit. From the formula we can understand that weight average will become equal to the number average. So [00:09:34.276] How much is the weight average more than the number average? That is what we are talking about, the ratio. Weight average by the number average, how much is it different from one? That will give us the polydispersity index, meaning that if it was equal [00:09:53.760] If it all the polymer chains were of equivalent or very close to uh equal length, then we would have had PDI that is a polydispersity index is equal to one.

EV0000138 · EV0000139 · EV0000140 · EV0000141

Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.736] But if they are different, then the ratio m w bar by mn bar, that is a poly dispersed index would be greater than one. So that indicates how dispersed is the sample. So [00:10:22.212] Now looking at this picture here we see that these polymer chains there's very narrow dispersity, so therefore we can uh uh that is how we when we calculate the poly dispersity index, this will come very close to one, whereas here we see the polymer chains are of different uh lengths. So such [00:10:39.984] which is a uh sample where we have wide dispersity and in this case it is definitely going to be more than one. So the polydispersity index gives us an idea of how dispersed is the [00:10:53.008] uh sample and as a rule weight average is always found to be more than the number average. [00:11:01.764] Now there's a third type of average which we are going to study that is the viscosity average MV bar. So here it's very obvious that we are going to study use viscosity measurements to determine the molecular weight. Now there's a very famous uh [00:11:19.536] equation that is the viscosity is given by K M, this is the viscosity average, and A. This is known as the mark hoving equation. [00:11:34.212] So here K and A are constant, they're known as Mark Hobbing parameters. So this A here, if you see, is used in this equation. So if you want to uh express the viscosity average molecular weight in terms of NIMI. [00:11:51.812] the molecular weight of each of the species and the number of uh polymer units of that particular species, then the uh equation is given this. [00:12:02.096] Viscosity average molecular weight is given by summation of NIMI one plus A divided by summation of NIMI and the whole thing is raised to the power one by A. Now what is A here? It is the same

which is used in the Markov which comes in the Markoving. [00:12:19.792] uh equation and this is how we determine using uh viscosity uh experimentally we can [00:12:28.320] Determine the viscosity average molecular weight. So this is the expression, and A is the Mark Hoving parameter as I mentioned just now. And this value of A is going to be dependent on the polymer solvent system and also the temperature. And usually the value lies. [00:12:45.968] between 0.5 and 0.9. Now if the value was 1, we can see very clearly that the viscosity average would be equal to the weight average molecular weight. [00:13:03.812] Now viscosity average molecular weight is usually found to be very close to the weight average molecular weight. The reason? Because we know that larger molecules will contribute more to viscosity. [00:13:20.068] So how is the molecular weight distribution curve? Now we studied about three types of average the number average molecular weight, the weight average molecular weight, and the viscosity average molecular weight. And we found that weight weight average is [00:13:34.912] always higher than the uh number average molecular weight. So if we want to show the distribution of the uh polymer uh units then say this is the weight fraction which is being given here up uh against the molecular weight then [00:13:52.624] This peak here, when we get a bell shaped curve like this, the peak will represent the number average molecular weight, and since this is the molecular weight scale, we find weight average is always more than the number average molecular weight. So we find it here. So this is the weight average and viscosity average happens to [00:14:10.192] to lie between the number average and the weight average molecular weight and this viscosity average you can see here is somewhere here in between and but it is also closer to the weight average molecular weight. [00:14:26.368] B so that is the reason uh that reason I explained just now that because viscosity [00:14:33.316] The larger molecules contribute more to the viscosity of the system and therefore viscosity average molecular weight will be closer to the weight average molecular weight. So we can make this statement. [00:14:51.088] number average molecular weight. So with that we come to the end of module seven and in the next class learn about organic light emitting diodes. Thank you

EV0000142 · EV0000143 · EV0000144 · EV0000145

Full transcript: 2192 normalized words, 61 timestamped segments, 905.834667 seconds duration. Low-confidence segments flagged: 0.

EV0000134 · EV0000135 · EV0000136 · EV0000137

11 Conducting polymers - examples, types of doping

This lesson is indexed from the source video titled Conducting polymers - examples, types of doping. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000146 · EV0000147

Graphs and diagrams

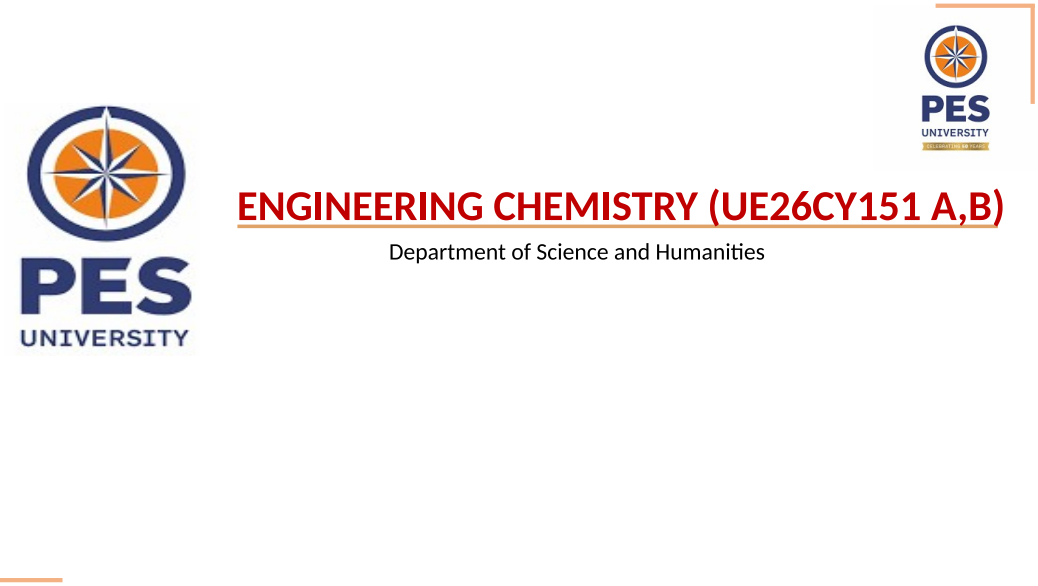


Figure G000116. Conducting polymers examples types of doping · source slide page 1.

EV0000540

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

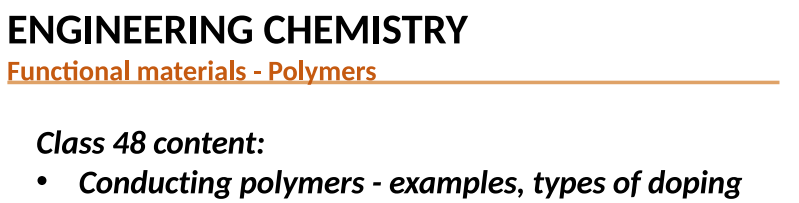


Figure G000117. Conducting polymers examples types of doping · source slide page 2.

EV0000543

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

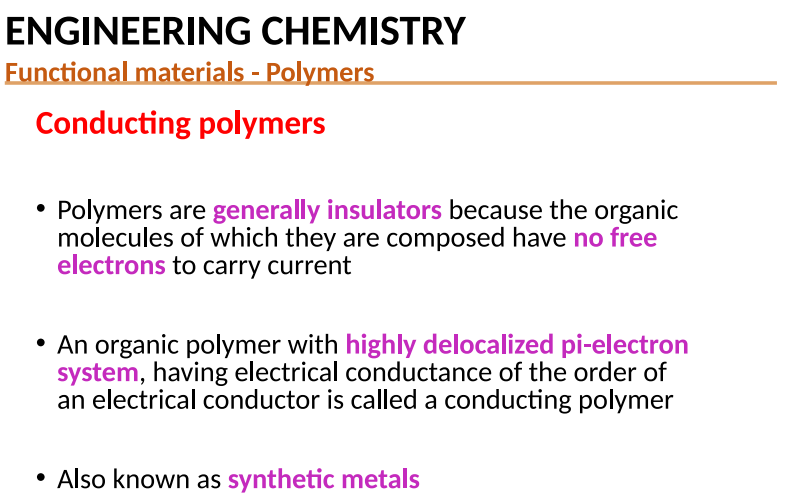


Figure G000118. Conducting polymers examples types of doping · source slide page 3.

EV0000546

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

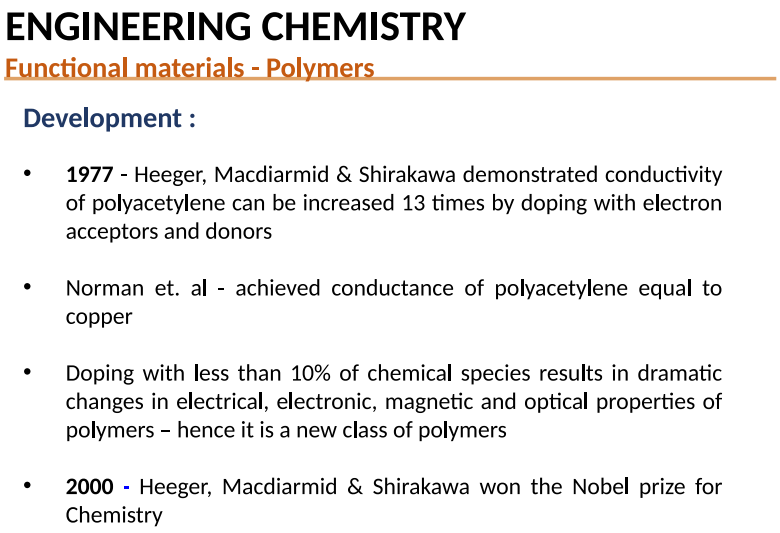


Figure G000119. Conducting polymers examples types of doping · source slide page 4.

EV0000549

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

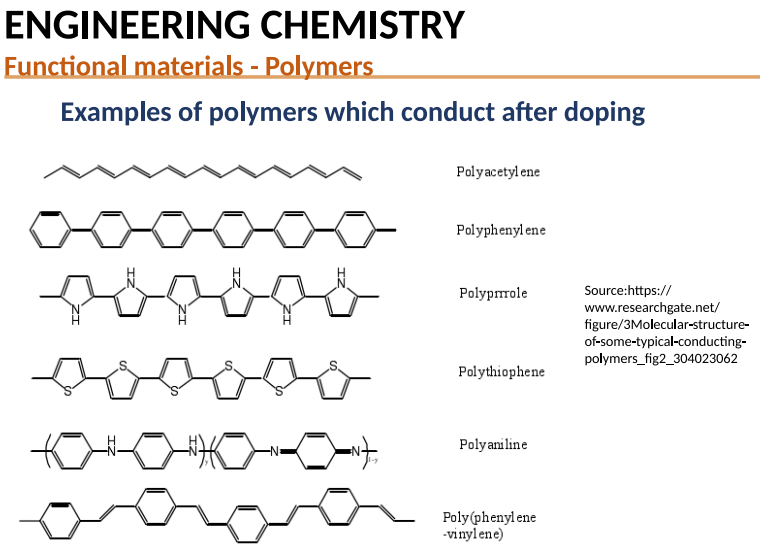


Figure G000120. Conducting polymers examples types of doping · source slide page 5.

EV0000552

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

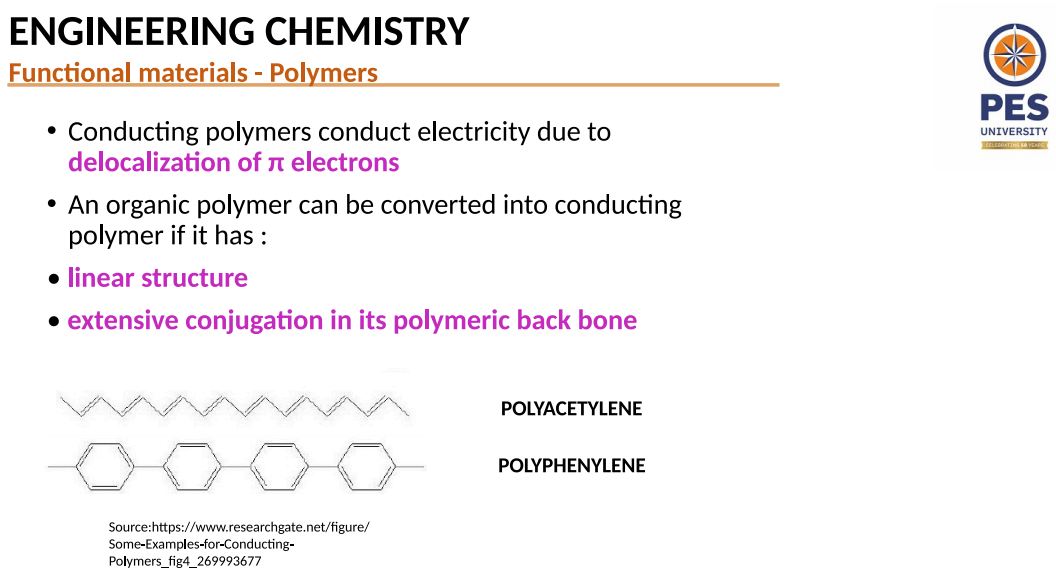


Figure G000121. Conducting polymers examples types of doping · source slide page 6.

EV0000555

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

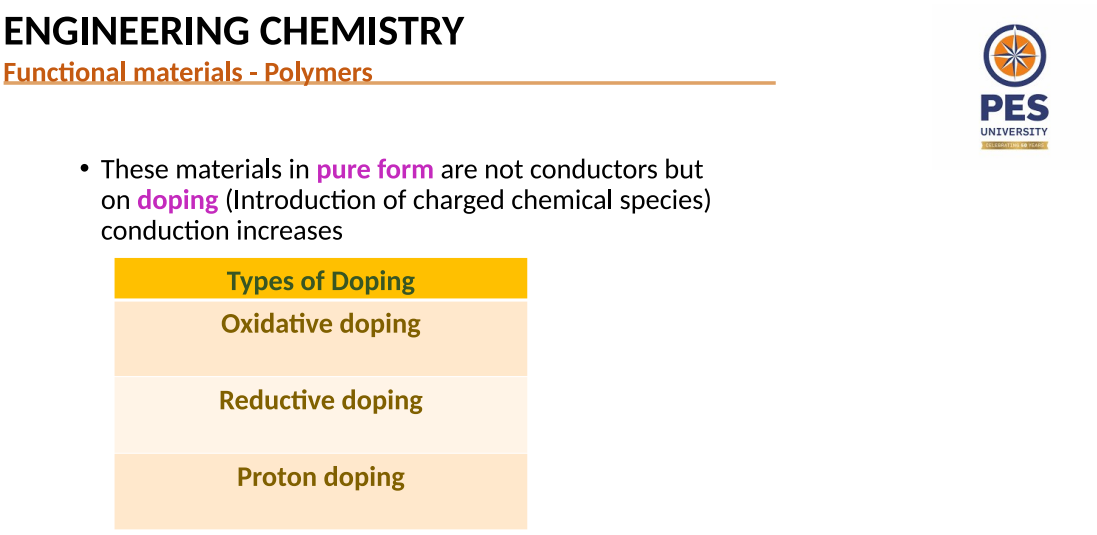


Figure G000122. Conducting polymers examples types of doping · source slide page 7.

EV0000558

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

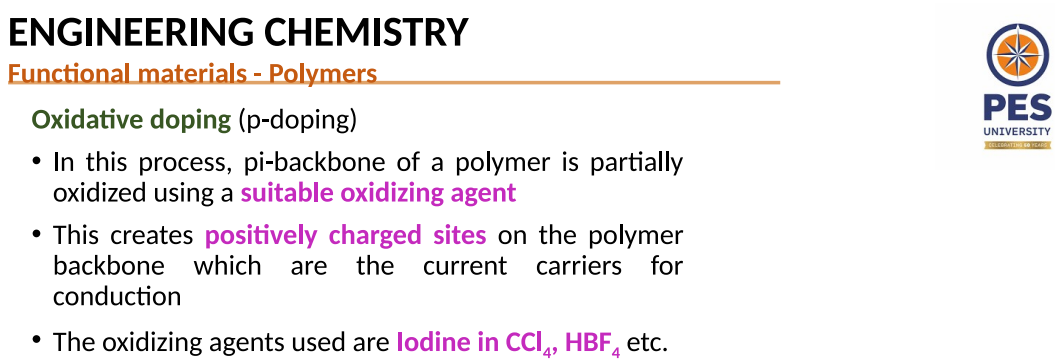


Figure G000123. Conducting polymers examples types of doping · source slide page 8.

EV0000561

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

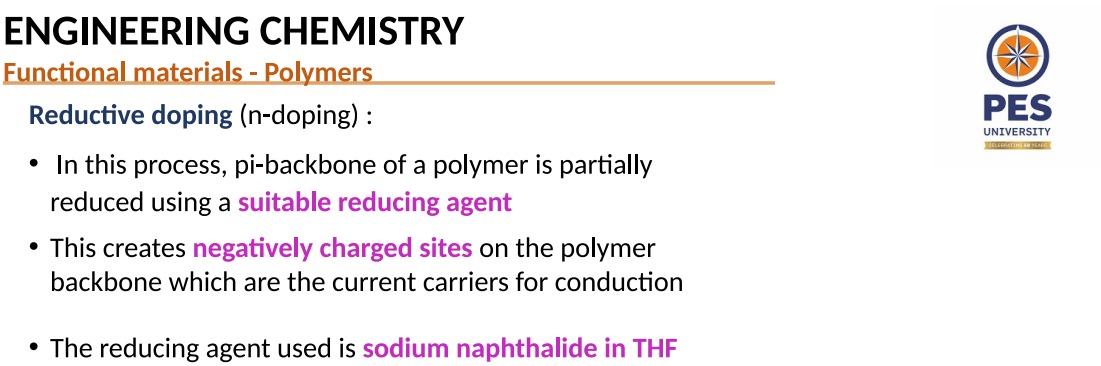


Figure G000124. Conducting polymers examples types of doping · source slide page 9.

EV0000564

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

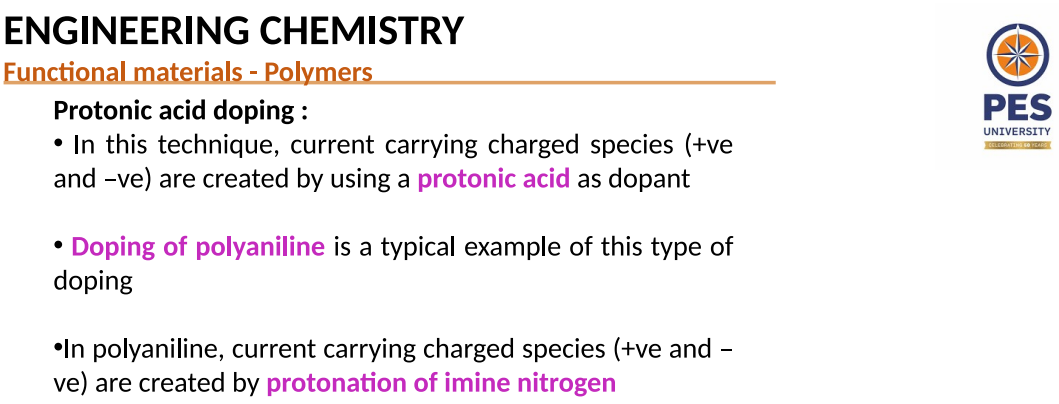


Figure G000125. Conducting polymers examples types of doping · source slide page 10.

EV0000567

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

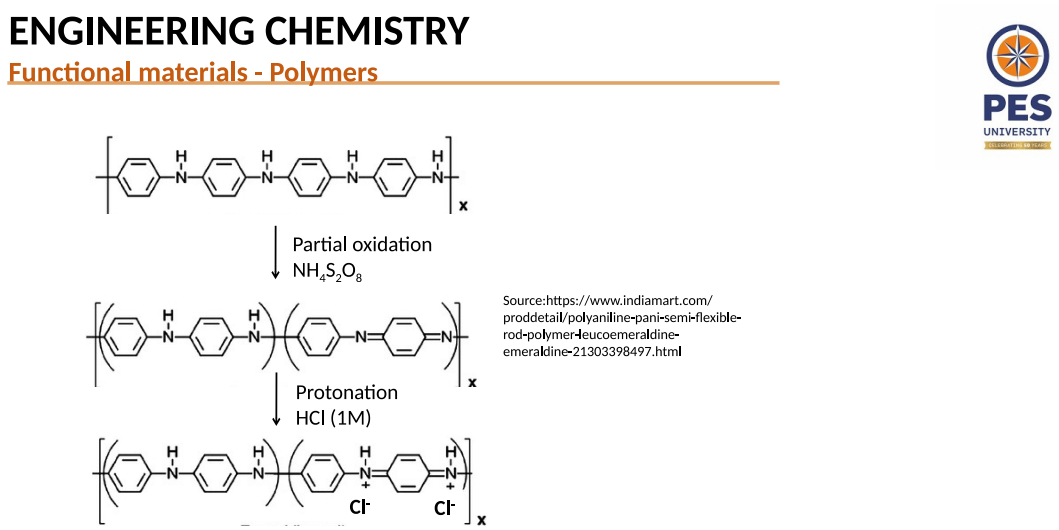


Figure G000126. Conducting polymers examples types of doping · source slide page 11.

EV0000570

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

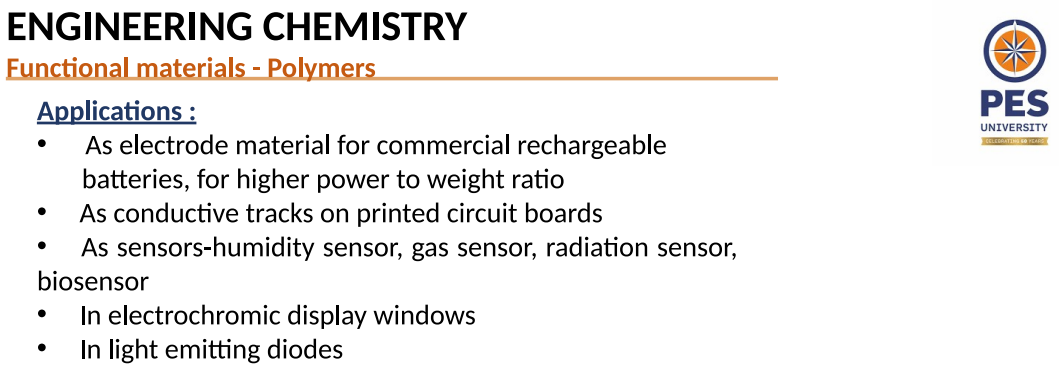


Figure G000127. Conducting polymers examples types of doping · source slide page 12.

EV0000573

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000128. Conducting polymers examples types of doping · source slide page 13.

EV0000576

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000539 · EV0000540 · EV0000541
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 48 content: Conducting polymers - examples, types of doping EV0000542 · EV0000543 · EV0000544
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Conducting polymers Polymers are generally insulators because the organic molecules of which they are composed have no free electrons to carry current An organic polymer with highly delocalized pi-electron system, having electrical conductance of the order of an electrical conductor is called a conducting polymer Also known as synthetic metals EV0000545 · EV0000546 · EV0000547
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Development : 1977 - Heeger, Macdiarmid & Shirakawa demonstrated conductivity of polyacetylene can be increased 13 times by doping with electron acceptors and donors Norman et. al - achieved conductance of polyacetylene equal to copper Doping with less than 10% of chemical species results in dramatic changes in electrical, electronic, magnetic and optical properties of polymers – hence it is a new class of polymers 2000 - Heeger, Macdiarmid & Shirakawa won the Nobel prize for Chemistry EV0000548 · EV0000549 · EV0000550
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Examples of polymers which conduct after doping [Image text, OCR] Polyacetylene Polyphenylene Polyprrrole Polythiophene Polyaniline Poly(phenyl ene -vinylene) Source:https://www.researchgate.net/figure/3Molecular-structure-of-some-typical-conducting-polymers_fig2_304023062 EV0000551 · EV0000552 · EV0000553
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Conducting polymers conduct electricity due to delocalization of π electrons An organic polymer can be converted into conducting polymer if it has : • linear structure • extensive conjugation in its polymeric back bone POLYACETYLENE POLYPHENYLENE Source:https://www.researchgate.net/figure/Some-Examples-for-Conducting-Polymers_fig4_269993677 EV0000554 · EV0000555 · EV0000556
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers These materials in pure form are not conductors but on doping (Introduction of charged chemical species) conduction increases Types of Doping Oxidative doping Reductive doping Proton doping EV0000557 · EV0000558 · EV0000559
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Oxidative doping (p-doping) In this process, pi-backbone of a polymer is partially oxidized using a suitable oxidizing agent This creates positively charged sites on the polymer backbone which are the current carriers for conduction The oxidizing agents used are Iodine in CCl4, HBF4 etc. EV0000560 · EV0000561 · EV0000562
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Reductive doping (n-doping) : In this process, pi-backbone of a polymer is partially reduced using a suitable reducing agent This creates negatively charged sites on the polymer backbone which are the current carriers for conduction The reducing agent used is sodium naphthalide in THF EV0000563 · EV0000564 · EV0000565

10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Protonic acid doping : In this technique, current carrying charged species (+ve and –ve) are created by using a protonic acid as dopant Doping of polyaniline is a typical example of this type of doping In polyaniline, current carrying charged species (+ve and –ve) are created by protonation of imine nitrogen EV0000566 · EV0000567 · EV0000568
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Partial oxidation NH4S2O8 Source:https://www.indiamart.com/proddetail/polyaniline-pani-semi-flexible-rod-polymer-leucoemeraldine-emeraldine-21303398497.html Protonation HCl (1M) Cl- Cl- EV0000569 · EV0000570 · EV0000571
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Applications : As electrode material for commercial rechargeable batteries, for higher power to weight ratio As conductive tracks on printed circuit boards As sensors-humidity sensor, gas sensor, radiation sensor, biosensor In electrochromic display windows In light emitting diodes EV0000572 · EV0000573 · EV0000574
13. Source slide page 13 (extracted text): THANK YOU Department of Science and Humanities EV0000575 · EV0000576 · EV0000577

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.740] Hello everyone, welcome back to this course on engineering chemistry. [00:00:06.052] We will continue with module seven on functional materials that is polymers and today's class we are going to talk about conducting polymers, a brief introduction, the types of dopamine, the applications of conducting polymers. [00:00:22.916] Polymers are generally known for their insulating property because the organic molecules of which they are composed have no free electrons to carry current. [00:00:35.332] So when conducting polymers were discovered, it was a very big revolution in the polymer industry. [00:00:44.740] So, how would you define a conducting polymer? An organic polymer with highly delocalized pi electron system and having an electrical conductance of the order of an electrical conductor. Such a polymer was referred to as a conducting polymer. For example, [00:01:04.516] We know that copper has very copper metal has very high conductivity of the order of 10 raised to the power of eight sements per meter, whereas Teflon has [00:01:20.772] conductance of 10 to the power of minus 8 siemens per meter. And if you have doped polyaniline, it is found to have 10 to the power of 5 sements per meter. So if you look at doped polyaniline [00:01:43.428] then the conductance is found to be ten to the power of five sevens per meter. Now this is very close, or rather closer to that of [00:01:53.312] copper which is 10 to the power of 8 whereas teflon as I mentioned just now is 10 to the power of minus 8. So Teflon is used as an insulator. Now when you see duped polyaniline we understand that the conductivity is very close to that of a metal. So these [00:02:10.960] These are also known as synthetic metals. So let us talk about the development of conducting polymers because just recently in the year 2000. [00:02:24.224] Heger, McDermott, and Shirakawa won the Nobel Prize for Chemistry for the du discovery of conducting polymers. So in nineteen seventy seven, Heger, McDermott, and Shirakawa demonstrated that the conductivity of polyacetylene can be increased [00:02:41.872] thirteen times by doping with electron acceptors or and donors. Now normal etal achieved a conductance of polyacetylene to that of copper. [00:02:53.412] Now it was found that doping with less than ten percent of chemical species in these polymers resulted in dramatic changes in electrical, electronic, magnetic, and optical properties of

the polymers, and hence these were clapped as a new [00:03:11.184] class of polymers known as conducting polymers. So do all polymers conduct? No. The answer is definitely no. And let us have a look at some examples of such polymers. [00:03:28.160] Now these polymers listed here are known to conduct after doping. For example, polyacetylene, then we have polyphenylin, then this is polypyro. [00:03:42.052] poly thiophene, polyaniline, and here we can see polyphenylin vinyline. So there is something special about all these polymers which become conducting after they are doped. So let us see what is that special feature. [00:04:00.740] Now these conducting polymers conduct electricity due to delocalization of the pi electrons. So what would be requirement the requirement for an organic polymer to be converted into a conducting polymer? [00:04:16.176] what we can see they have a linear structure. All those polymers which we listed just now they have a linear structure and here's I've taken just two examples. So that of polyacetylene and polyphenyline we can see that they are linear and apart from that one more important property [00:04:33.840] Which should be there in their structure, and that is extensive conjugation in its polymer backbone. What would that mean? That they should have alternate. [00:04:44.512] single and double bond. So then it is possible to have delocalization of the pi electron. So all these conducting polymers they are [00:04:54.640] uh have they have these special features like a linear structure and also extensive conjugation in their polymer backbone.

EV0000146 · EV0000147 · EV0000148 · EV0000149

Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.372] Now these materials in pure form they are not found to be conductors but only on doping. When what is doping what are we doing here? We are introducing a charged chemical species, we find that the conduction increases uh manifold. [00:05:24.164] Now doping can be done in three ways. There could be oxidative doping, there could be reductive doping, or there could also be proton doping. So let us look at each one of them little closely. Now when we talk of oxidative doping, it is also known as P type doping. [00:05:42.724] Now what is done here is [00:05:45.440] The pi backbone of the polymer is partially oxidized with a suitable oxidizing agent. Now we know that what does an oxidizing agent do? It is going to take away electron from the polymer backbone. And now we understand why we chose a [00:06:03.184] Pi uh system where where because we know that pi electrons uh they can easily be removed from the they are more mobile or uh [00:06:12.736] as we say as compared to the uh sigma bar sigma electrons, sigma bond the electrons which are forming the sigma bond. See these plants are much more free to move. So therefore they can be easily removed from the system. That's why we chose the pi [00:06:30.448] electron system here. So now when we are going to use an oxidizing agent, it is going to remove one electron from the pi electron pi bond. [00:06:41.924] So what happens? This creates a positively charged site on the polymer backbone and now these become the current carriers for conduction. [00:06:52.080] So what are the oxidizing agents which are used normally iodine and CCL4, HBO HBA4, etc. Now these agents what do they do? They are going to remove an electron from the pi system, hence we get a positive charge over here. [00:07:09.744] In order to compensate for the positive charge, we know that you have learned during different mechanisms that these electrons will move to the next bond, these will move to the next bond, hence the electrons are moving in one direction and the hole is moving a positive charge or the whole is moving in another direction just like it happens in. [00:07:27.312] So such a doping is called as [00:07:31.248] P type doping and what is moving along the chain when a potential is applied is the electron or this positive charge, they are moving in opposite direction, and that is how the polymer becomes conducting when we do oxidative doping. So here we can see

[00:07:48.912] The mechanism how it is happening. Now in reductive doping, we are going to do n-type doping, meaning that we are going to give electrons to the [00:07:59.600] uh pi electron system. So in this process the pi backbone of the polymer is partially reduced by using a suitable reducing agent. [00:08:09.264] Now when we use a reducing agent, it is going to give electrons to the system. Now when you give electron to the system, obviously it creates negatively charged sites on the polymer backbone and now they become current carriers for the [00:08:24.272] Conduction. So when a potential is applied, so again because these electrons can be delocalized, so these electrons now move in the opposite direction, and therefore this negative charge moves in one direction, and again the uh the hole is going to move in the direction. [00:08:41.936] the opposite direction. So because of this pi electron system, the alternate extensive conjugation which we have in this polymer backbone, this type of movement is possible. [00:08:54.128] And that is how now we have negatively charged sites which are moving and hence we call it as N type of doping and this can be done with reducing agents like for example sodium naphtalide in tetrahydrofural. So this is the mechanism again. So this is called N type of doping. [00:09:13.668] The third type of doping is known as protonic acid doping. Now, as the name suggests, we are going to use protonic acid as a dopant. So basically, H plus is used as a dopant here. And what is here the current carrying [00:09:31.664] charge species that is the positive and the negative species are again created and that is with the help of a protonic acid. Now example best example would be doping of polyaniline and in polyaniline [00:09:48.208] The current carrying charge species that the that is the positive and the negative species are created by protonation of imine nitrogen. Let us just have a look at what the structures look like. Now this is polyanine, which you can see here.

EV0000150 · EV0000151 · EV0000152 · EV0000153

Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.784] Partial oxidation is done with the help of ammonium peroxy disulfate. Now only partial oxidation is done wherein you see now these nitrogens here have become immune nitrogens and you get a structure like this. So you will have alternate this will be alternated along the [00:10:22.528] entire uh polyanyline change uh chain and now what we are going to do the next step is a protonation that is done with H Cl that is one molar H Cl. So this is where we are introducing the [00:10:34.656] H plus i that's why it's called protonic acid doping. Now this enine nitrogen is the one which will get protonated and we get that. [00:10:44.388] Now this becomes NH plus. This gets the plus charge and now it is balanced with the help of Cl minus. [00:10:56.004] Cl minus so this is now becoming a plus n h plus n h plus and now this is balanced with the help of Cl minus and here also Cl minus because we added H Cl. So this is how it got doped, and now because of the delocalized pi electron which are available here, we can [00:11:15.856] uh have these charges moving along the uh along the chain. So again it is not the proton which moves it is the charge which is moving along the chain and this is known as the emeraldine salt. Emeraldine [00:11:33.860] salt and this is green in color and this structure is what is conducting polyanine so you can write here conducting polyanum [00:11:52.868] So as such, in this case we don't find it to be conducting the first uh structure. When it undergoes partial oxidation, what we get is known as the amoraldine base. [00:12:09.572] This is also not conducting. It's only after protonation when the MRL wing salt is formed, then [00:12:16.656] This is this is found to be green in color, and this is the structure of conducting polyanuline. And how did we make it conducting? So polyanuline was made conducting by this uh doping process with a protonic acid, and this [00:12:31.184] uh is known as protonic acid doping. So we have three types of doping oxidative doping, uh reductive doping and protonic acid doping. So this is about the

mechanism of doping. Now we come to applications. We know now that polymer [00:12:50.980] are conducting in nature some of these polymers and they can be made conducting by doping. So the best application for them would be as electrode material for commercial rechargeable batteries. [00:13:06.244] Now we know that the electrolyte can be conducting in nature, ionic conductance is found there. But when we talk of electrodes, we are talking of electronic conduction. Now what we saw in conducting polymers [00:13:20.736] is electronic conduction because the electron could move through the polymer chain. So therefore they will be used as electrode material and because they are lightweight we expect a very high power to weight ratio. [00:13:35.876] Then they can be used as conductive tracts on PCBs normally we see copper being printed there. So that can be replaced by these conducting polymers. Then they can be used as sensors. [00:13:47.392] Like humidity sensor, gas sensor, radiation sensor, biosensors, etc. Then electrochromic display window, specially polyaniline, is known to have different colours in different atmospheres. So we can have a completely [00:14:03.168] uh reduced form of polyanlene and you can have a completely oxided form of polyanuline and then we know the conducting polyanine is green in color. Now the color of each of these [00:14:14.432] is different and therefore it can be used in electrochromic display windows to tell us what kind of atmosphere is there inside the [00:14:23.840] uh system which is which is being studied. Then in information storage devices as film membranes for gas separations in light emitting diodes and they are important materials in microelectronics applications, then electro catalysis [00:14:40.848] fuel cell electrodes and reinforced composites etc. So that is about the conducting polymers. And now in the next class we are talking about the molecular weight of polymers. Thank you.

EV0000154 · EV0000155 · EV0000156 · EV0000157

Full transcript: 2059 normalized words, 62 timestamped segments, 898.112 seconds duration. Low-confidence segments flagged: 0.

EV0000146 · EV0000147 · EV0000148 · EV0000149

12 Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism

This lesson is indexed from the source video titled Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000158 · EV0000159

Graphs and diagrams

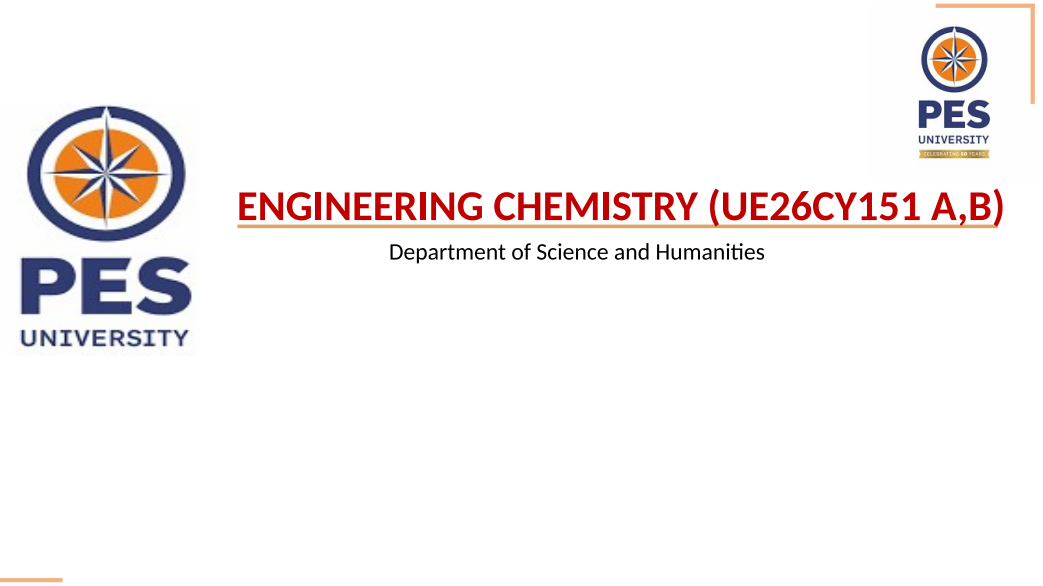


Figure G000129. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 1.

EV0000579

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

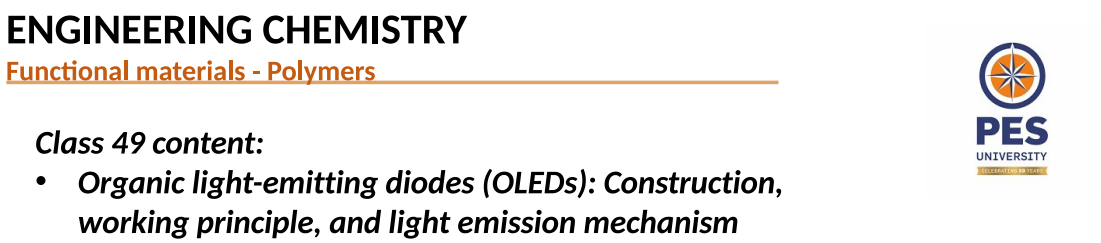


Figure G000130. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 2.

EV0000582

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

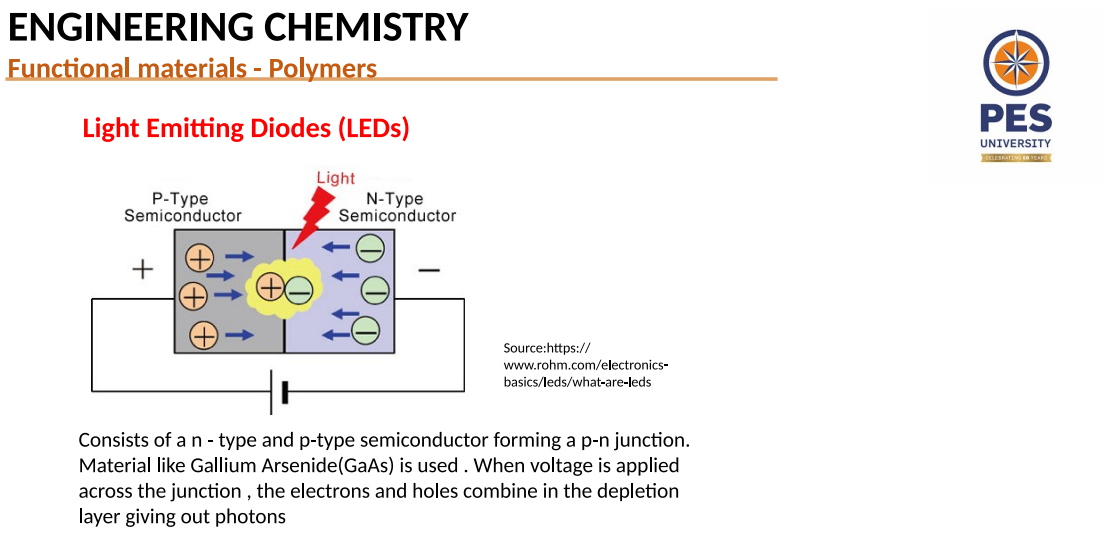


Figure G000131. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 3.

EV0000585

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

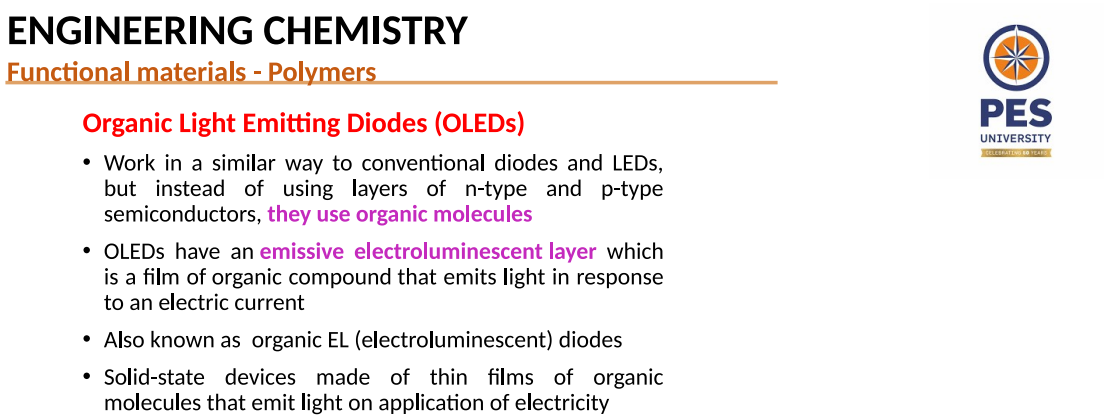


Figure G000132. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 4.

EV0000588

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

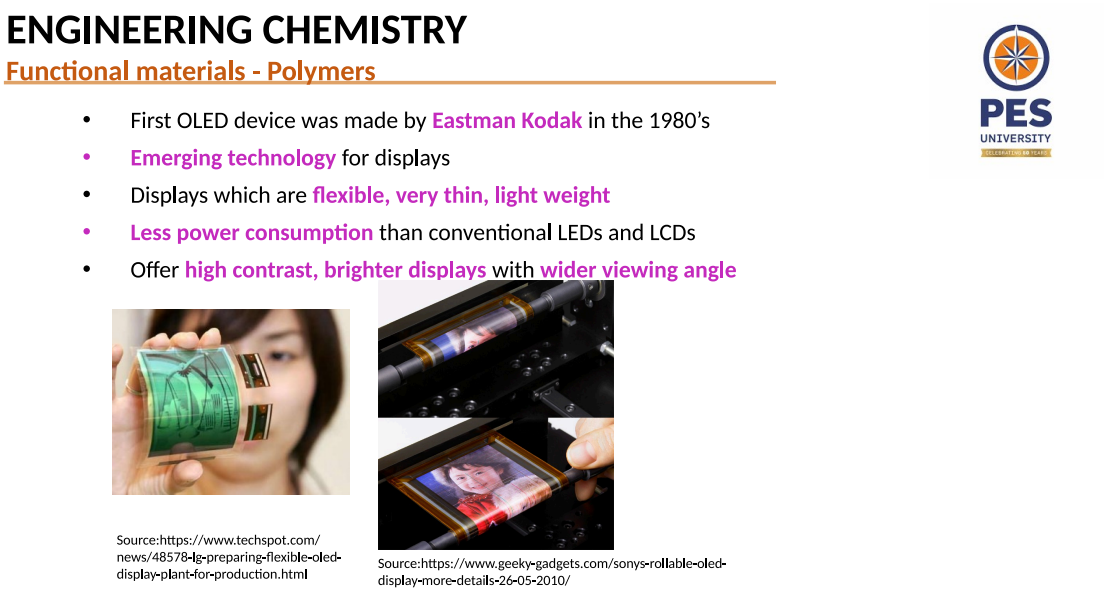


Figure G000133. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 5.

EV0000591

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

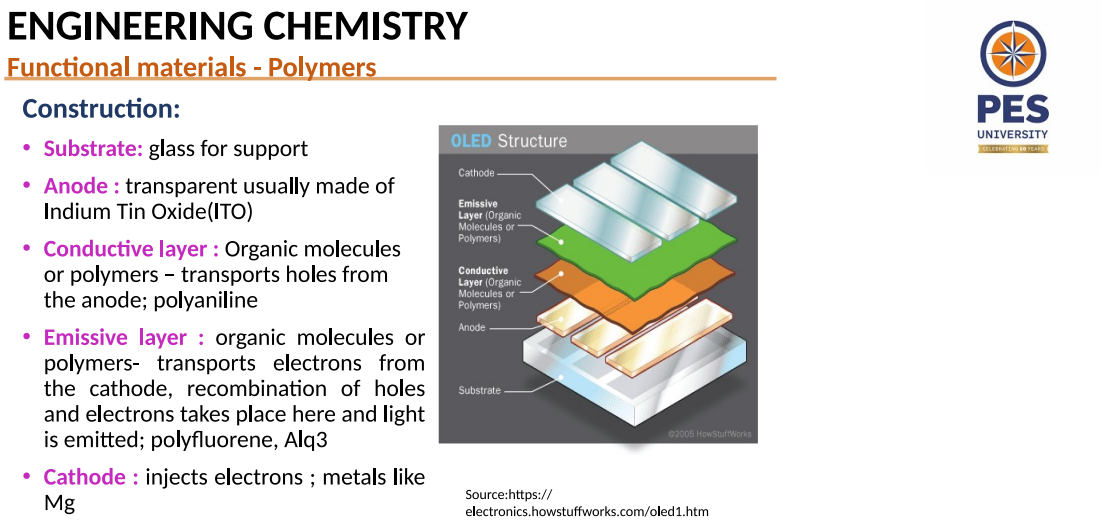


Figure G000134. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 6.

EV0000594

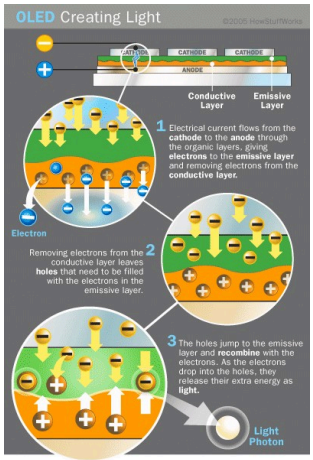
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Working:

- A voltage is applied across the anode and cathode of the OLED
- Electrons flows from cathode to anode through the organic layers
- Electrons are transported to the emissive layer from cathode
- Holes are transported to the conductive layer from anode
- Electrons combine with the holes thereby emitting light
- Recombination happens in the emissive layer
- Colour of light depends on the organic molecules used and intensity depends on current density



Source:https://www.researchgate.net/figure/Basic-OLED-working_fig2_308332988

Figure G000135. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 7.

EV0000597

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

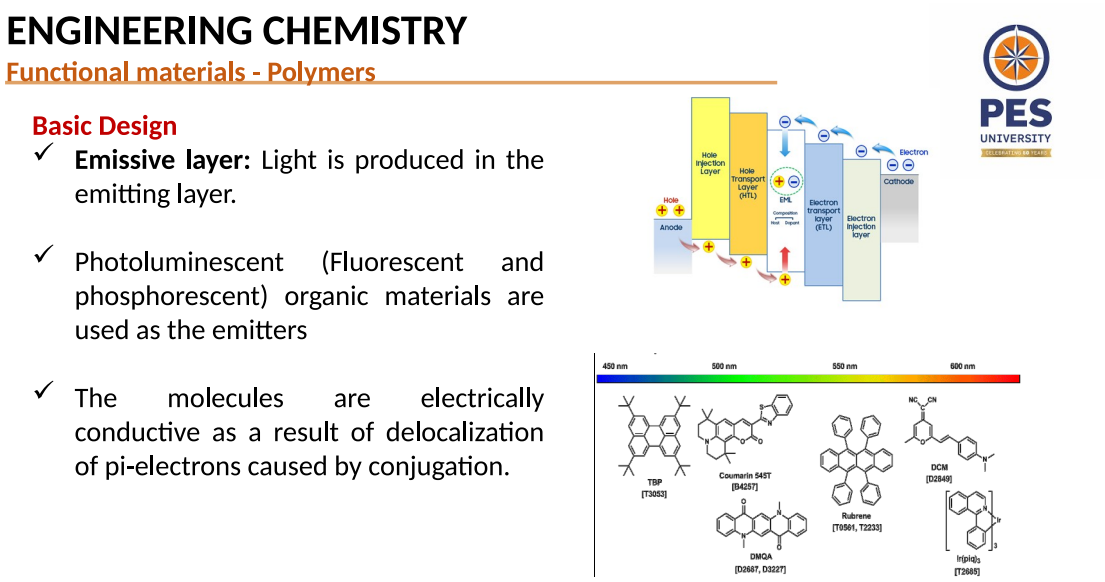


Figure G000136. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 8.

EV0000600

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

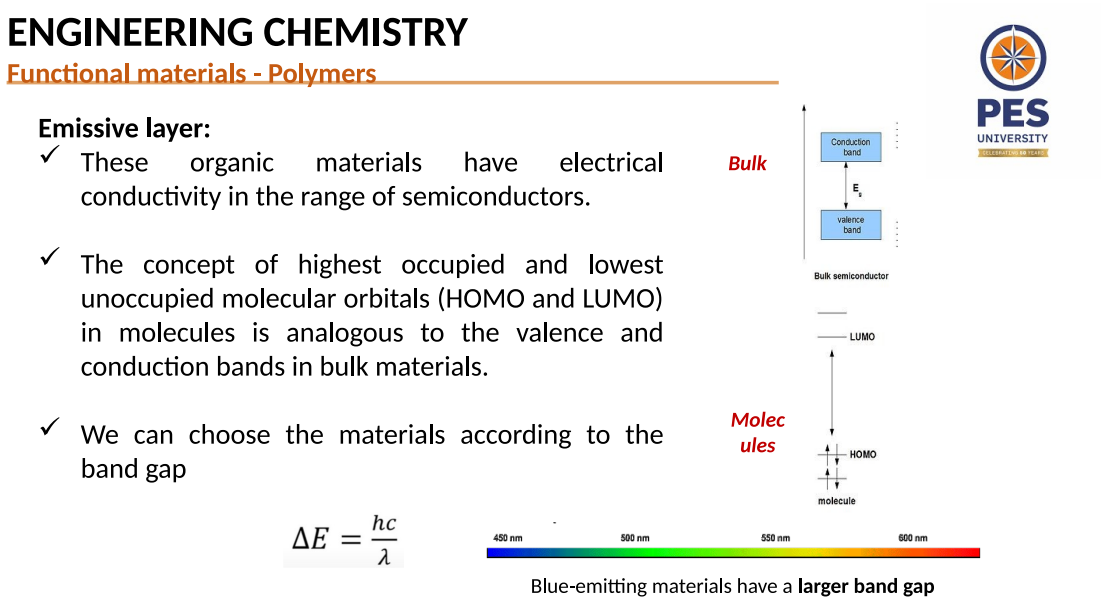


Figure G000137. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 9.

EV0000603

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Why Multi-Layered Structure?

- Inorganic semiconductors have a higher conductivity than the organic materials, so a layered structure is not needed.
- Organic materials have a lower conductivity than the inorganic counterparts
- HOMOs and LUMOs of the materials in the layered structure act as small steps and guide the charge carriers to the emissive layer.
- The layered structure helps maintain a smoother carrier flow.

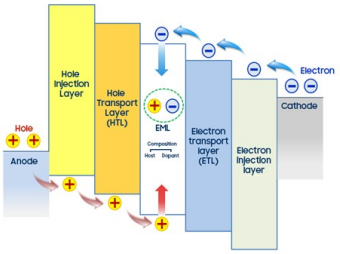


Figure G000138. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 10.

EV0000606

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

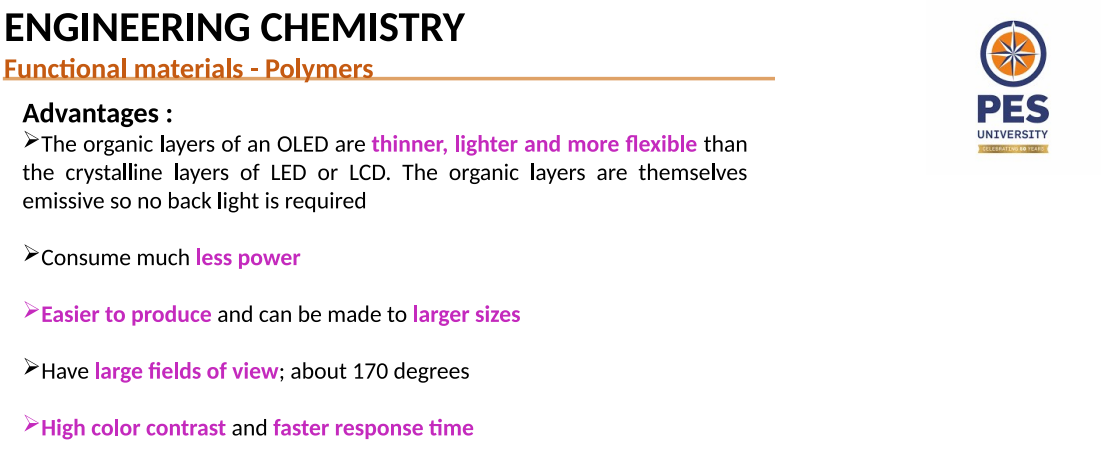


Figure G000139. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 11.

EV0000609

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000140. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 12.

EV0000612

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

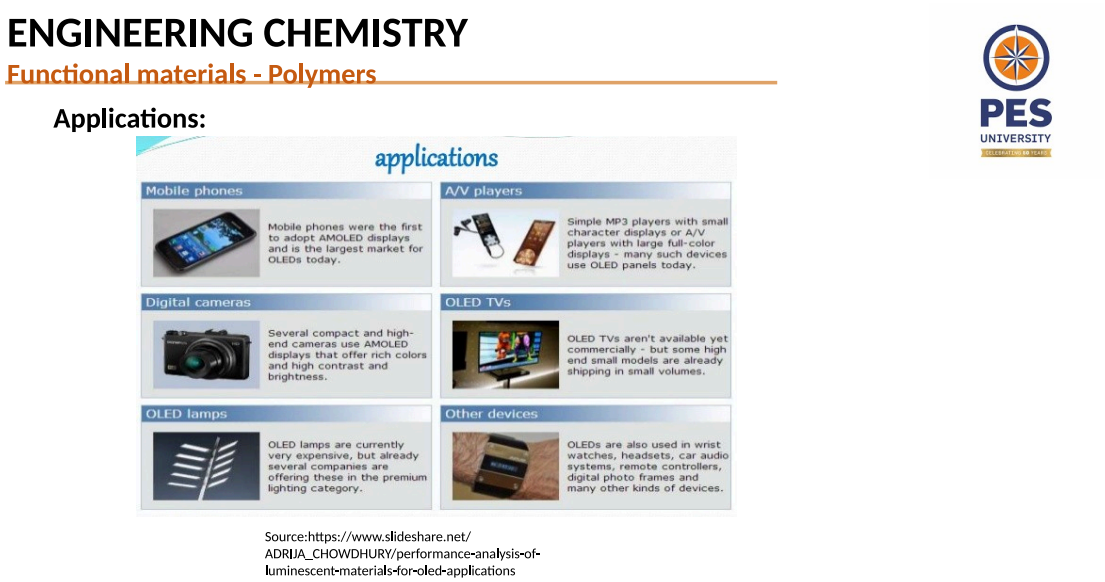


Figure G000141. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 13.

EV0000615

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000142. Organic light emitting diodes OLEDs Construction working principle and light emission mechanism · source slide page 14.

EV0000618

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000578 · EV0000579 · EV0000580
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 49 content: Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism EV0000581 · EV0000582 · EV0000583
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Light Emitting Diodes (LEDs) [Image text, OCR] Light P-Type N-Type Semiconductor Semiconductor Source:https://www.rohm.com/electronics-basics/leds/what-are-leds Consists of a n - type and p-type semiconductor forming a p-n junction. Material like Gallium Arsenide(GaAs) is used . When voltage is applied across the junction , the electrons and holes combine in the depletion layer giving out photons EV0000584 · EV0000585 · EV0000586
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Organic Light Emitting Diodes (OLEDs) Work in a similar way to conventional diodes and LEDs, but instead of using layers of n-type and p-type semiconductors, they use organic molecules OLEDs have an emissive electroluminescent layer which is a film of organic compound that emits light in response to an electric current Also known as organic EL (electroluminescent) diodes Solid-state devices made of thin films of organic molecules that emit light on application of electricity EV0000587 · EV0000588 · EV0000589

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers First OLED device was made by Eastman Kodak in the 1980's Emerging technology for displays Displays which are flexible, very thin, light weight Less power consumption than conventional LEDs and LCDs Offer high contrast, brighter displays with wider viewing angle Source:https://www.techspot.com/news/48578-lg-preparing-flexible-oled-display-plant-for-production.html Source:https://www.geeky-gadgets.com/sonys-rollable-oled-display-more-details-26-05-2010/ EV0000590 · EV0000591 · EV0000592
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Construction: Substrate: glass for support Anode : transparent usually made of Indium Tin Oxide(ITO) Conductive layer : Organic molecules or polymers – transports holes from the anode; polyaniline Emissive layer : organic molecules or polymers- transports electrons from the cathode, recombination of holes and electrons takes place here and light is emitted; polyfluorene, Alq3 Cathode : injects electrons ; metals like Mg [Image text, OCR] OLED Structure Cathode Emissive Layer(Organic Molecules or Polymers) Conductive Layer(Organic Molecules or Polymers) Anode Substrate 02005 HowStuffWorks Source:https://electronics.howstuffworks.com/oled1.htm EV0000593 · EV0000594 · EV0000595
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Working: A voltage is applied across the anode and cathode of the OLED Electrons flows from cathode to anode through the organic layers Electrons are transported to the emissive layer from cathode Holes are transported to the conductive layer from anode Electrons combine with the holes thereby emitting light Recombination happens in the emissive layer Colour of light depends on the organic molecules used and intensity depends on current density [Image text, OCR] OLED Creating Light 2005HowSturtWorks ATL CATHODE CATHODE ANODE Conductive Emissive Layer Layer Electrical current flows from the cathode to the anode through the organic layers,giving electrons to the emissive layer andremoving electrons from the conductive layer. Electron Removing electrons from the conductive layer leaves holes that need to be filled with the electrons in the emissive layer. The holes jump to the emissive layerand recombine with the electrons.As the electrons drop into the holes,they release their extra energy as light. Light Photon Source:https://www.researchgate.net/figure/Basic-OLED-working_fig2_308332988 EV0000596 · EV0000597 · EV0000598
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers [Image text, OCR] Hole Electron Injectlon Layer Hole Transport + Cathode Layer Hole (HTL) EML ++ Electron Compostion transport Dopant layer Electron Anode Host (ETL) Injectlon layer Basic Design Emissive layer: Light is produced in the emitting layer. Photoluminescent (Fluorescent and phosphorescent) organic materials are used as the emitters The molecules are electrically conductive as a result of delocalization of pi-electrons caused by conjugation. [Image text, OCR] OLED Dopants 450nm 500nm 550nm 600nm DCM Coumarin545T TBP [D2849] [B4257] [T3053] Rubrene [T0561,T2233] DMQA Ir(piq)3 [D2687,D3227] [T2685] EV0000599 · EV0000600 · EV0000601

9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Bulk Molecules Emissive layer: These organic materials have electrical conductivity in the range of semiconductors. The concept of highest occupied and lowest unoccupied molecular orbitals (HOMO and LUMO) in molecules is analogous to the valence and conduction bands in bulk materials. We can choose the materials according to the band gap Blue-emitting materials have a larger band gap EV0000602 · EV0000603 · EV0000604
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Why Multi-Layered Structure? Inorganic semiconductors have a higher conductivity than the organic materials, so a layered structure is not needed. Organic materials have a lower conductivity than the inorganic counterparts HOMOs and LUMOs of the materials in the layered structure act as small steps and guide the charge carriers to the emissive layer. The layered structure helps maintain a smoother carrier flow. [Image text, OCR] Hole Electron Injectlon Layer Hole Transport + Cathode Layer Hole (HTL) EML ++ Electron Compostion transport Dopant layer Electron Anode Host (ETL) Injectlon layer EV0000605 · EV0000606 · EV0000607
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Advantages : The organic layers of an OLED are thinner, lighter and more flexible than the crystalline layers of LED or LCD. The organic layers are themselves emissive so no back light is required Consume much less power Easier to produce and can be made to larger sizes Have large fields of view; about 170 degrees High color contrast and faster response time EV0000608 · EV0000609 · EV0000610
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Disadvantages: Have shorter life span Manufacturing process is expensive Water can easily damage them EV0000611 · EV0000612 · EV0000613
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Applications: [Image text, OCR] applications Mobile phones AVplayers Mobile phoneswere the first Simple MP3players with small toadoptAMOLED displays characterdisplaysorA/V andis thelargestmarketfor players with large full-color OLEDs today. displays-manysuchdevices use OLEDpanels today. Digital cameras OLEDTVS Several compactandhigh- endcamerasuse AMOLED OLEDTVsaren'tavailableyet displays thatofferrich colors commercially-butsomehigh andhigh contrastand end small models are already brightness. shipping in small volumes. OLED lamps Other devices OLEDlampsare currently OLEDsarealsoused inwrist very expensive， but already watches,headsets,caraudio severalcompanies are systems,remote controllers, offering thesein the premium digitalphoto framesand lighting category. many otherkindsofdevices. 26May2014 23 Source:https://www.slideshare.net/ADRIJA_CHOWDHURY/performance-analysis-of-luminescent-materials-for-oled-applications EV0000614 · EV0000615 · EV0000616
14. Source slide page 14 (extracted text): THANK YOU Department of Science and Humanities EV0000617 · EV0000618 · EV0000619

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.612] Hello everyone, welcome back to this course on engineering chemistry. [00:00:06.628] We will study module 7a today about organic light emitting diodes. This is the only class in this topic, and we are going to study about the construction, working, the types of OLEDs, the advantages, disadvantages, and applications.

[00:00:26.692] First, let us understand the working of light emitting diodes, because the same concept can be extended to organic light emitting diodes. [00:00:37.380] So an LED as it is popularly known has an N type and a P type semiconductor which is forming a PN junction. The most common material is gallium arsenide, which is used to make these LEDs. [00:00:56.324] Further doping is done and we have different types of colored LEDs. Now when voltage is applied across the junction, the electrons and the holes they move towards each other as you can see in the picture here. [00:01:13.312] And they recombine in the depletion layer. When they recombine, they give out a photon and that is what we see as light. So that is the how an LED works. Now let us understand what is an O LED. [00:01:31.364] So as the name suggests, we are going to use some organic molecules. So they work in a similar way to the conventional diodes and the LEDs, but instead of using layers N type and P type of semiconductors, [00:01:47.312] They use organic molecules and hence the name organic light emitting diodes. The phenomena which is happening in OLEDs is called electroluminescence. Therefore, OLEDs [00:02:03.172] have an emissive electroluminescent layer, which is a film of an organic compound that emits light in response to an electric current. Hence they are also known as organic electroluminescent diodes. [00:02:20.292] So let us have a clear definition of what is an OLED. So we can define them as solid state devices which are made up of thin films of organic molecules that emit light on application of electricity. [00:02:40.708] The first OLED device was made by Eastman Kodak in the nineteen eighties. So we know Kodak Labs, it was first made there and it's an emerging technology for displays. All of us have seen [00:02:57.764] OLED TVs, so much advertisement for OLED mobile displays. So why are they so important? Because these displays can be flexible, they can be very thin and of light weight. [00:03:13.924] Moreover, the power consumption is less as compared to the conventional LEDs and the L C Ds. Apart from that, they offer a high contrast, brighter displays with wider viewing angle. [00:03:29.796] So being flexible is a very big advantage which OLEDs give us. So you can see in the picture here how nicely the display is foldable. And here you can actually roll it. So these kind of advantages make O L E Ds [00:03:47.536] a topic of research and it is an emerging technology for displays. Let us understand the construction. [00:03:58.756] So we have a layered structure as you can see in the picture here. First of all, there is a substrate which is made of glass and that is basically for support. On the substrate, we can see the anode. This is usually transparent and made of [00:04:16.804] indium tin oxide it is ITO and then we have two organic layers here as you can see this orange layer over here is known as the conductive layer and it is going to transport holes from the [00:04:34.468] anode and this is usually made of polyaniline and also there are other kind of material organic molecules which are used to make this conductive layer. [00:04:48.580] And then we also have the next layer which is again an organic uh mole made of organic molecules and this is known as the emissive layer. [00:04:59.012] And here we have these this is very very important because this is where the light emission happens. So it helps in transporting electrons from the cathode and a recombination of electrons and poles takes place here.

EV0000158 · EV0000159 · EV0000160 · EV0000161

Transcript block 2 (source interval 000300-000599 seconds): [00:05:15.776] and light is emitted. One example of a polymer which is used. So we should understand that it could be organic molecules or polymers which can make up these conductive layer, this conductive layer and the emissive layer. So in emissive layer we use polyflorine [00:05:34.276] And the cathode of course is used to inject electrons and it is usually made with metals like magnesium which have a low work function. Now

let us understand the working of an OLED. [00:05:51.268] A voltage is applied across the anode and cathode of the OLED. So let us look at this picture very carefully here. We can see that the cathode will be connected to the negative terminal, hence there are [00:06:06.960] Electrons coming from the cathode. So these electrons now move into the emissive layer. The green layer you can see here, these are moving to the emissive layers from emissive layer from the cathode. And at the same time, electrons. [00:06:23.728] uh which have moved out of this [00:06:27.556] orange layer over here which is the conductive layer. So therefore what happens? Holes have now moved into the conductive layer. So holes are getting transported from the anode into the conductive layer and electrons are getting transported from the cathode into the emissive layer. [00:06:44.740] Now these electrons and holes they will combine and thereby emit and emit a photon. Now this recombination happens closer to the emissive layer. You can see here closer to the emissive layer and therefore [00:07:02.208] light is being emitted from the emissive layer. Now the color of the light is going to depend on the organic molecule which is being used. [00:07:15.716] And the intensity of the light, that is the brightness of the light, is going to depend on the current density. [00:07:24.228] So we can understand that there are electrons which are moving from the cathode into the emissive layer, and there are holes which are moving to the conductive layer from the anode. These two are going the holes in the electrons are going to [00:07:40.688] combine and this combination happens closer to the emissive layer and that is how whenever there's a combination of a electron and a hole then there will be emission of light. So basically what is happening the combination of electron and hole what we understand. Now the electrons are at [00:07:58.320] And the holes are at a lower energy level. So we can say the electrons are in the lowest unoccupied molecule molecular orbitals and the holes are now in the [00:08:11.312] uh Homo that is highest occupied molecular orbital. So when these electrons come back from a higher energy level to the lower energy level, we know that a photon is given out or energy is given out and that is the radiation. We have learned this in right from [00:08:28.976] spectros molecular spectroscopy when we were studying. Whenever there is an electron transition from a higher level to the lower level, the energy is uh given out in the form of a radiation, and here it is in the visible uh region. So we are seeing the [00:08:43.728] uh light. So this color of this light obviously will depend on the energy gap and of course that is going to depend on the organic molecules which we are using and the intensity of course will depend on the current density. [00:08:59.556] Now let us study different types of O LEDs which are available. [00:09:05.796] We have passive matrix OLED, which is known as the emolet, and we have the active matrix matrix OLED, that is the emolet. Then we have transparent OLEDs, we have foldable O LED, we have top emitting OLED and white O LED. [00:09:26.148] Let us look at each one of them carefully what are these different types of O LEDs which we are referring to. The first one is a passive matrix O LED, and here we have strips of these cathode. [00:09:43.460] and anode and the organic layers. The anode strips are arranged perpendicular to the cathode strips and the intersection of the cathode and anode they form the pixels from where the light is emitted.

EV0000162 · EV0000163 · EV0000164 · EV0000165

Transcript block 3 (source interval 000600-000899 seconds): [00:10:00.452] There has to be an external circuitry here which applies current to selected strips of anode and cathode, thereby determining which pixel should be turned on and which pixel should be turned off. [00:10:16.100] So because of this external circuitry here, the power consumption is more compared to the other OLEDs, and therefore these are mostly used for small screens. [00:10:30.180] Coming to active matrix OLEDs which are more popular and many of the mobile displays now they use this amulet.

Now they what do they have? They have a full layer of cathode, organic molecules, and anode. [00:10:46.852] Here the anode layer has a thin film transistor TFT array at the back plate. There is a back plate of TFT and that forms a matrix. [00:10:59.492] Now this TFT array itself is the circuitry and that determines which pixels get turned on to form the image. [00:11:10.564] Now amulets therefore consume less power than the P amulets because the TFT array requires less power as compared to the external circuitry and therefore they are very efficient for large displays. [00:11:28.356] like we have electronic uh signages or we have these big TVs, large screen TVs. Transparent OLEDs. They consist of a transparent substrate, cathode and anode. [00:11:46.692] So when a transparent OLED display is turned on, it allows light to pass in both the directions as you can see here. And a transparent OLED display can be either active matrix type or a passive matrix type. [00:12:01.600] And it is used in transparent projector screens and glasses. The next one is a top editing OLED. So basically you can see here it's only in one direction. So they have a substrate that is either opaque or reflective. [00:12:20.132] They are best suited for active metrics design and they are used for making small card displays. [00:12:28.900] Then foldable O LEDs. They have the substrate made of very flexible metal foils or plastics, as you can see here, and they are very lightweight, ultra thin and [00:12:42.032] and because they are foldable they can reduce the breakage of displays. So we understand they can be used in cell phones and PDAs. Personal digital assistance. [00:12:56.676] Now coming to the last one that is white OLED. Now these emit white light that is brighter, more uniform, and more energy efficient than that emitted by fluorescent lights. [00:13:10.340] They give you true color quality like of incandescent lighting, and therefore they can be used for large and efficient lighting systems. Let us see what are the advantages of OLEDs. [00:13:26.372] The organic lay of an OLED are thinner, lighter, and more flexible than the crystalline layers of LED or L CD. We know that in the market we have L C D or [00:13:40.484] TVs and then came the LED TVs. And now there's a lot of talk about O LED screens. [00:13:49.604] They are brighter than LEDs because the organic layer of an O LED are much thinner than the corresponding inorganic crystal layers which are used in LED. [00:14:02.820] And so the conductive and the emissing emissive lay of an OLED can be multi-layered. So you get a very bright picture. And of course they consume much less power than the L C D's. [00:14:18.820] and then easier to produce and can be made in large sizes and they have large fields of view. [00:14:27.172] In an L C D we know that if we go at an angle we will not be able to see the picture. That is not the case with O L E D. Up to 170 degrees the viewing angle has now become very large and there's a very high color contrast and faster response time. [00:14:48.324] So what are the disadvantages? They have shorter lifespan and the manufacturing process is expensive. And one of the most important problem we face with OLEDs is water can easily damage them.

EV0000166 · EV0000167 · EV0000168 · EV0000169

Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.004] What are the applications? They are used in mobile phone displays. And then we have digital cameras where they can be used. And lighting. As we said, white bullets, they can be used for lighting. [00:15:24.836] Then we have audio video players and then we have OLED TVs and then also different [00:15:36.900] Other type of devices like wristwatches, so displays in wristwatches, etc. And if they are foldable, you can put it right over your wrist. So these are the applications of OLEPs. [00:15:51.492] So that brings us to the end of module seven A. And in the next class we will be talking about green chemistry, which is the last class of unit five. Thank you.

EV0000170 · EV0000171 · EV0000172 · EV0000173

Full transcript: 2084 normalized words, 64 timestamped segments, 967.872 seconds duration. Low-confidence segments flagged: 0.

EV0000158 · EV0000159 · EV0000160 · EV0000161

13 Green chemistry: Introduction, twelve principles of green chemistry with examples

This lesson is indexed from the source video titled Green chemistry: Introduction, twelve principles of green chemistry with examples. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000174 · EV0000175

Graphs and diagrams

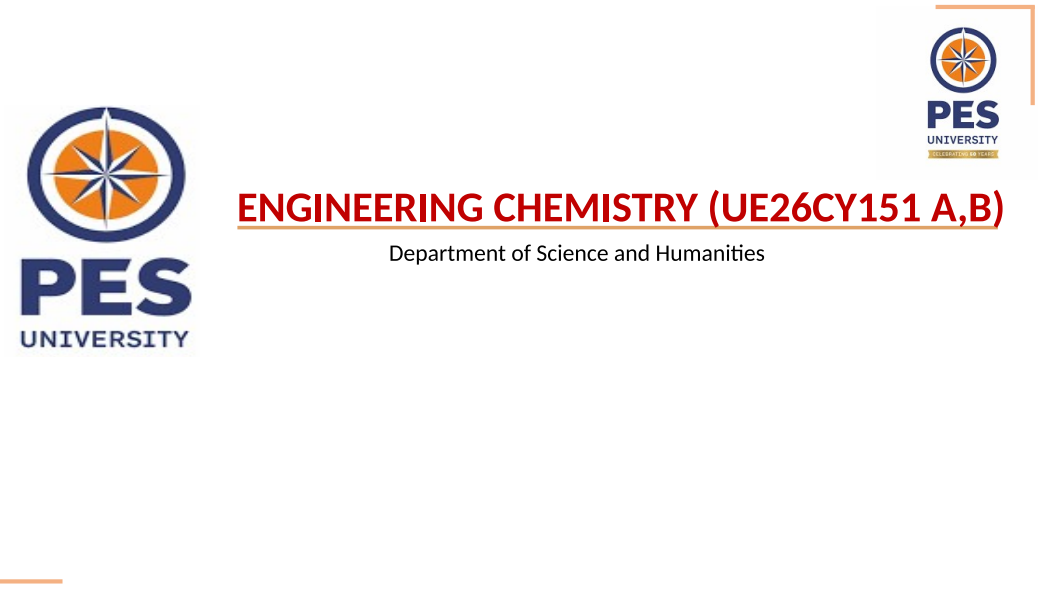


Figure G000143. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 1.

EV0000621

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

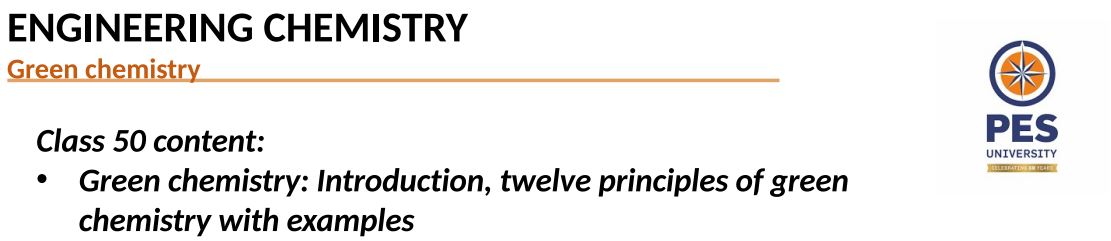


Figure G000144. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 2.

EV0000624

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

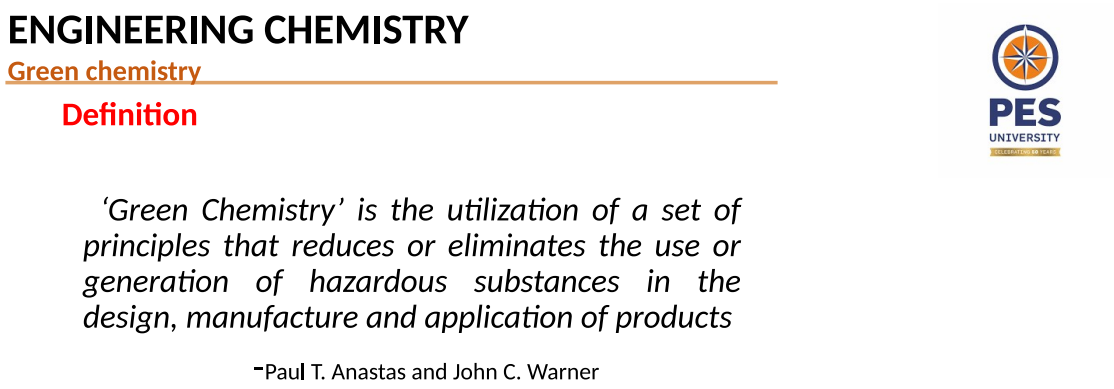


Figure G000145. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 3.

EV0000627

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

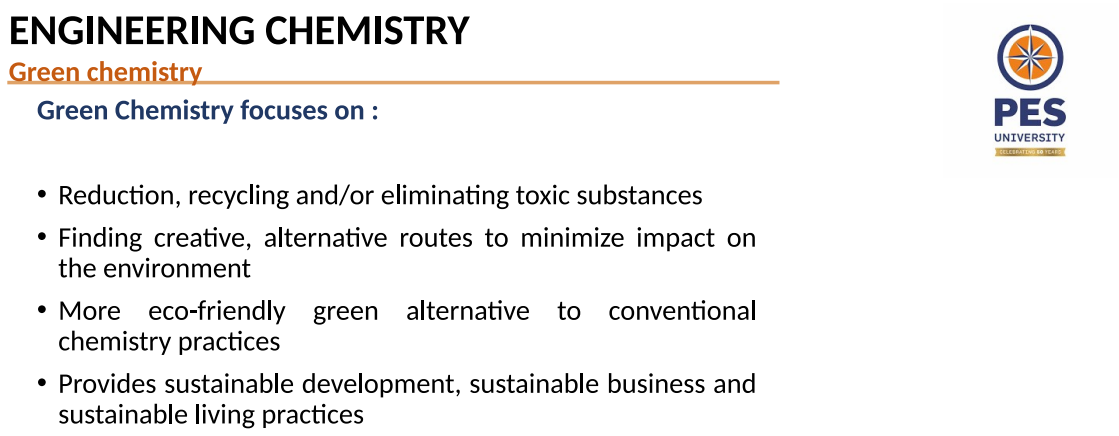


Figure G000146. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 4.

EV0000630

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

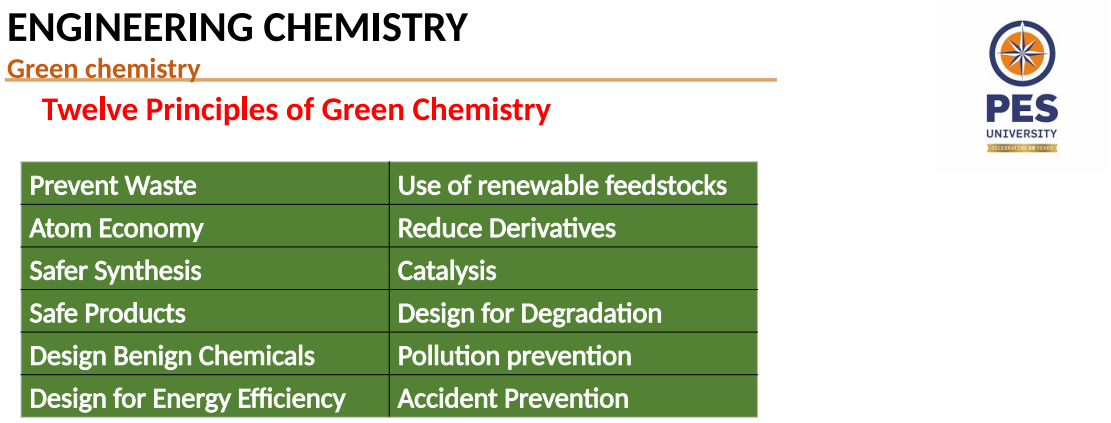


Figure G000147. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 5.

EV0000633

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

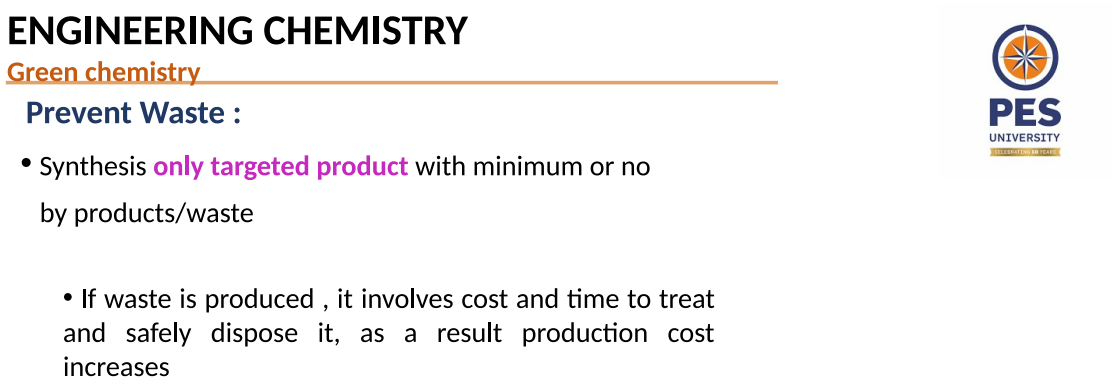


Figure G000148. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 6.

EV0000636

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

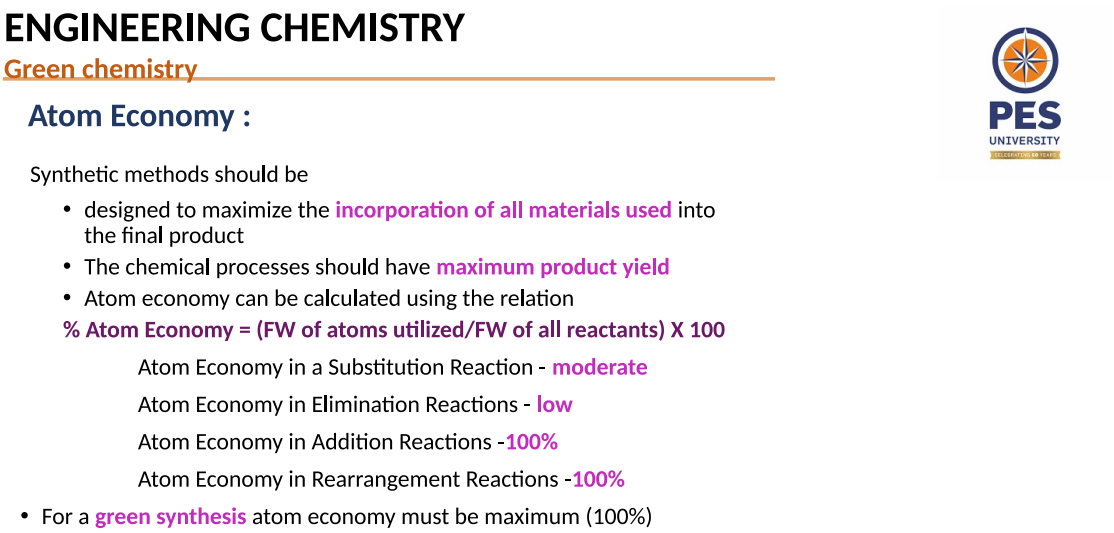


Figure G000149. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 7.

EV0000639

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

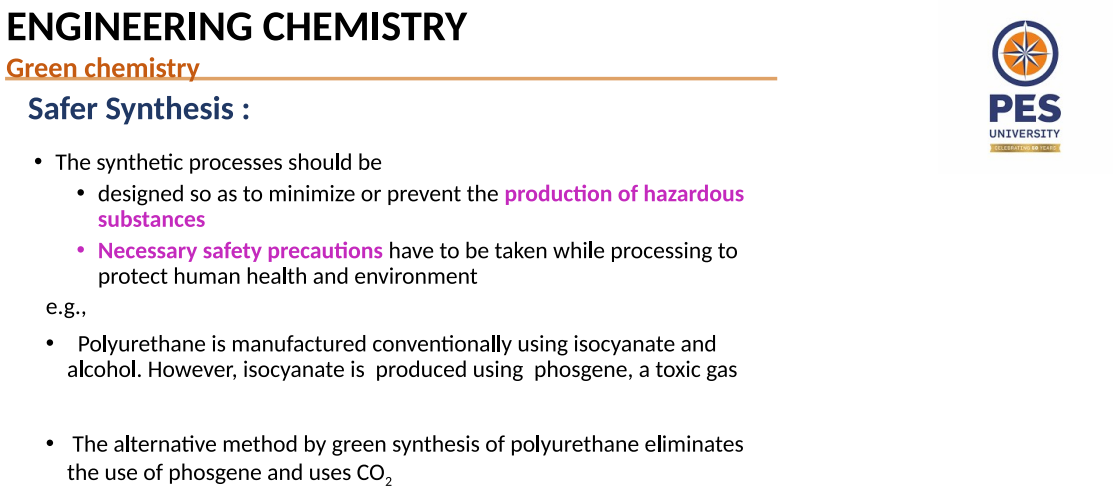


Figure G000150. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 8.

EV0000642

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

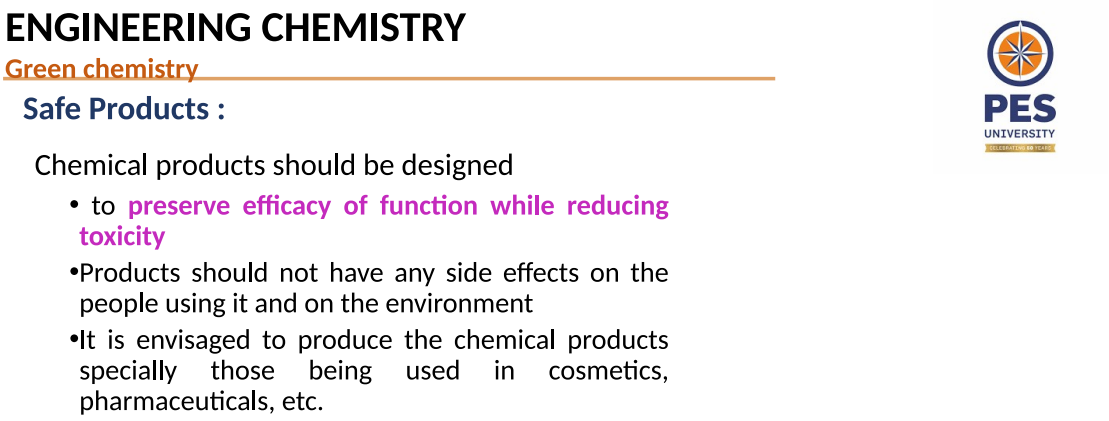


Figure G000151. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 9.

EV0000645

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

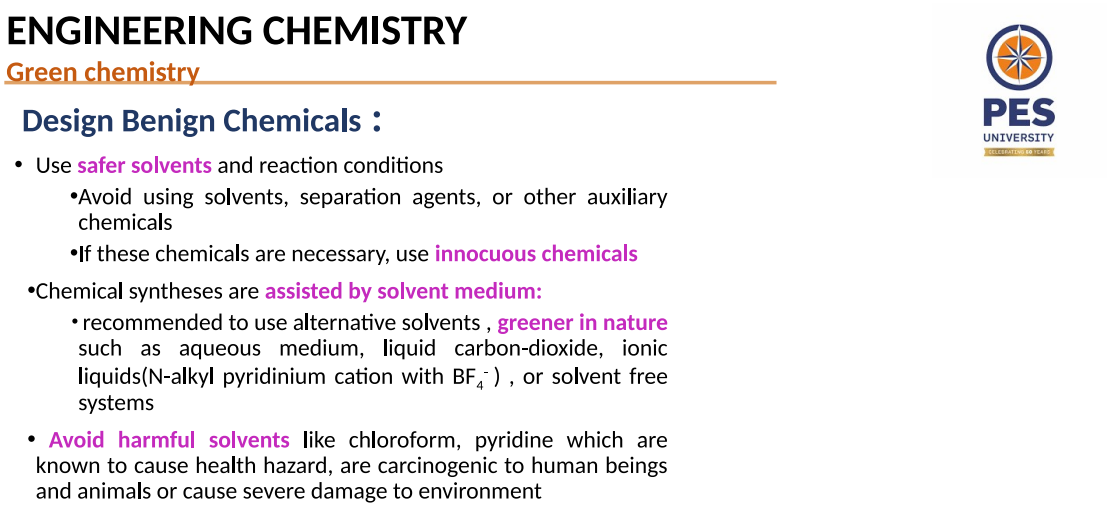


Figure G000152. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 10.

EV0000648

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

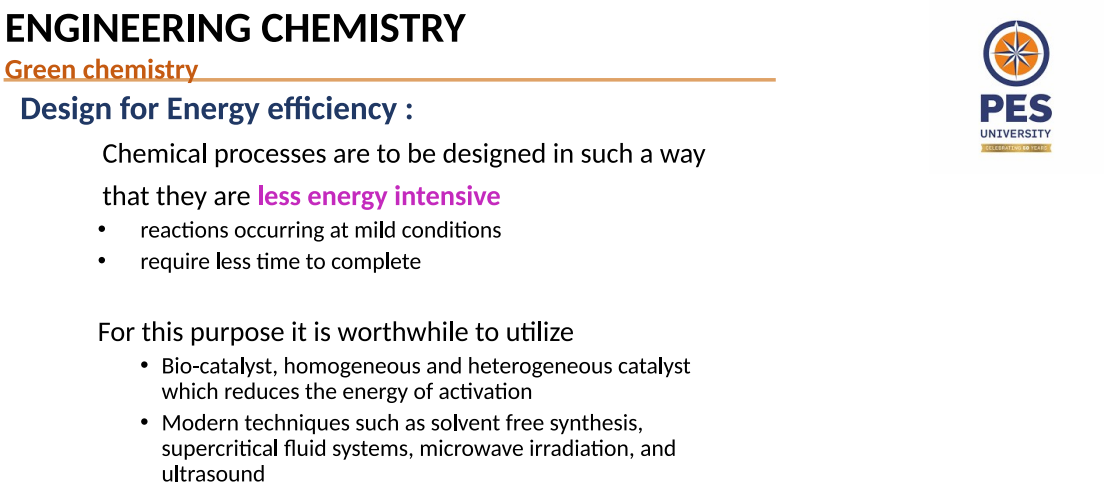


Figure G000153. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 11.

EV0000651

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

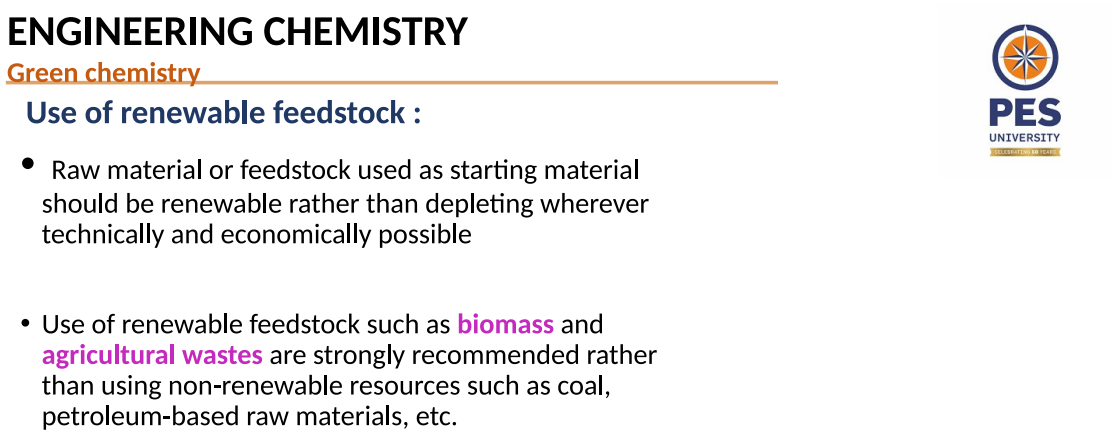


Figure G000154. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 12.

EV0000654

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

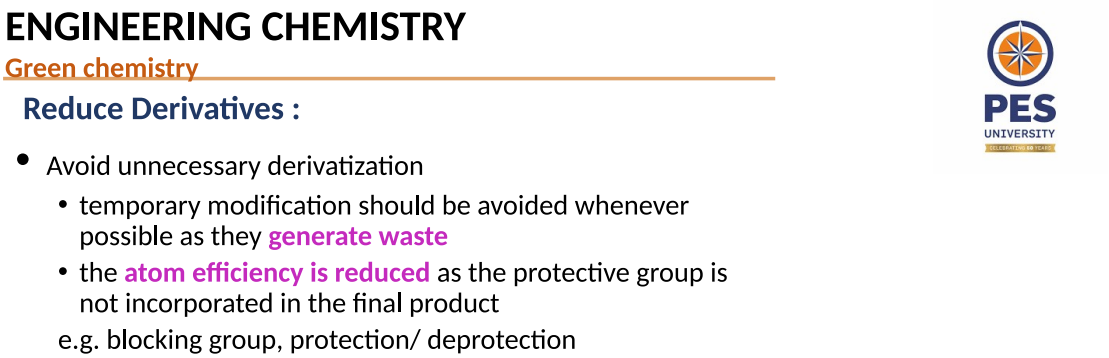


Figure G000155. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 13.

EV0000657

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

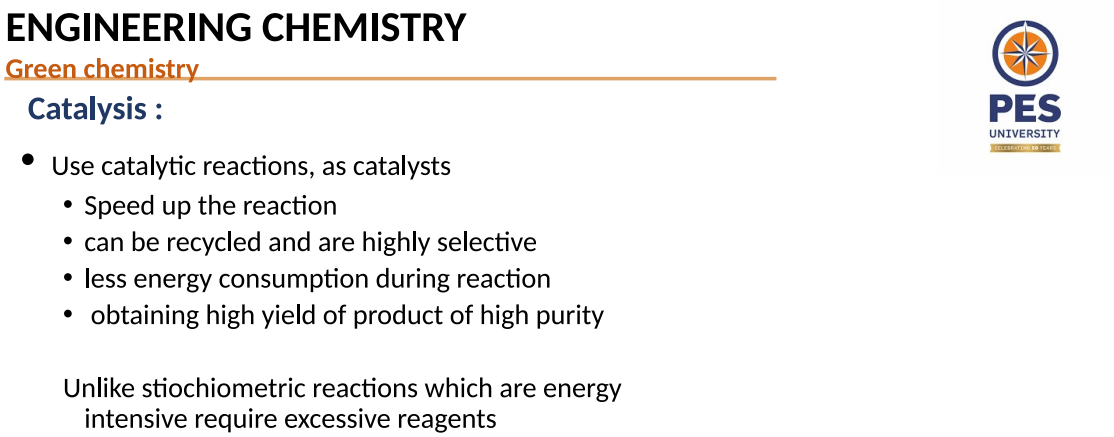


Figure G000156. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 14.

EV0000660

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

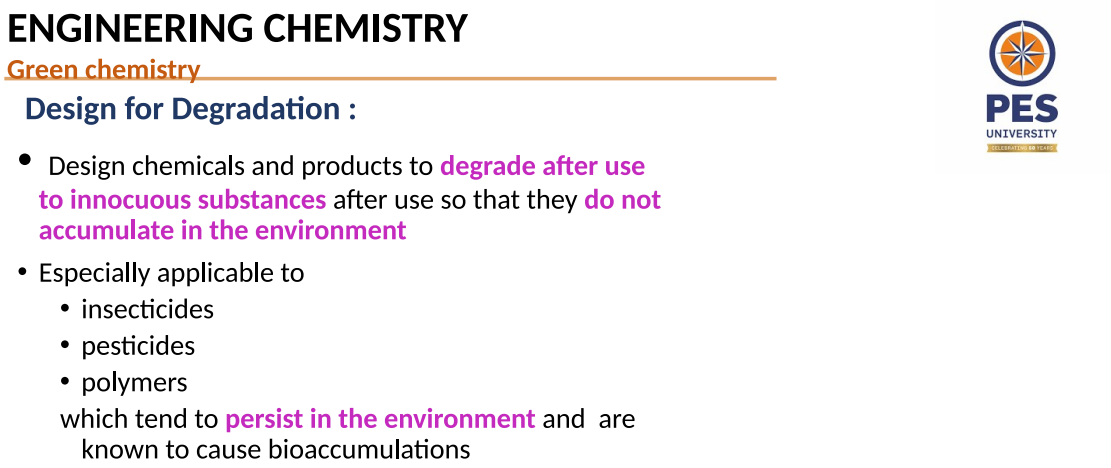


Figure G000157. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 15.

EV0000663

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

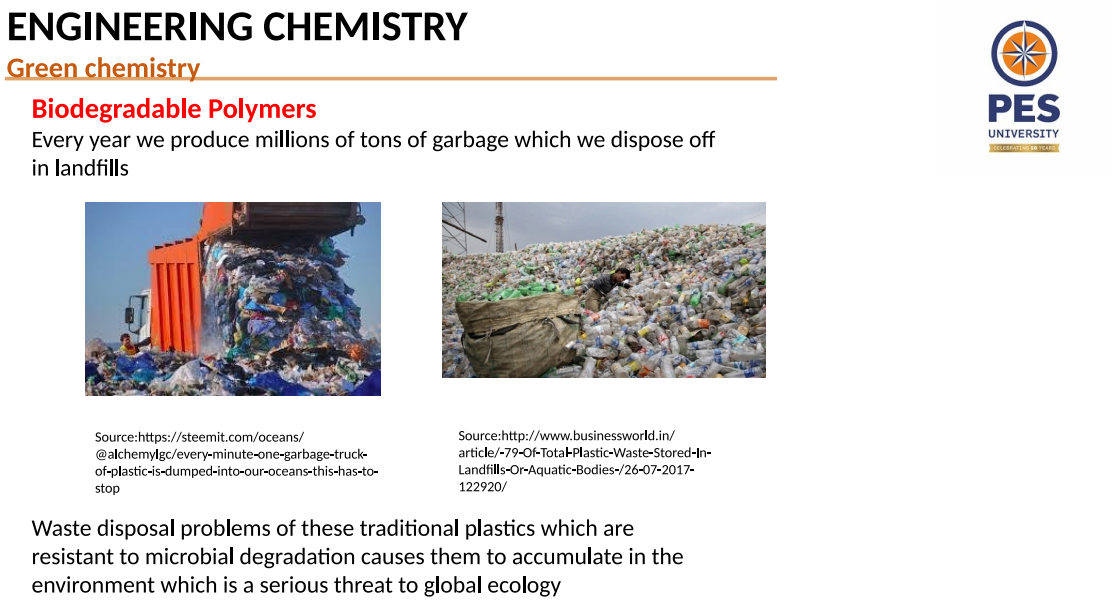


Figure G000158. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 16.

EV0000666

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

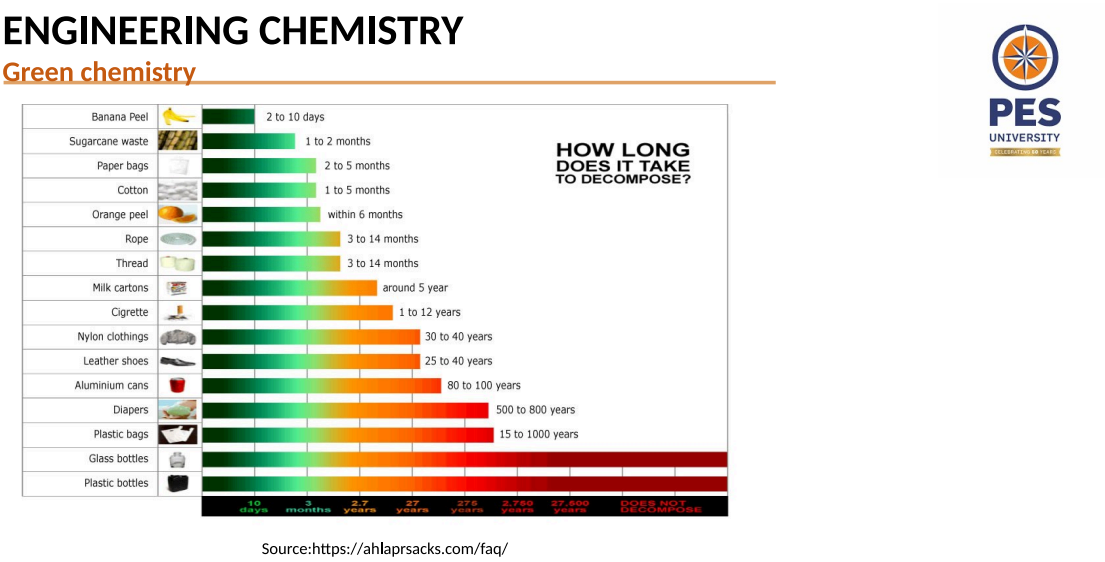


Figure G000159. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 17.

EV0000669

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

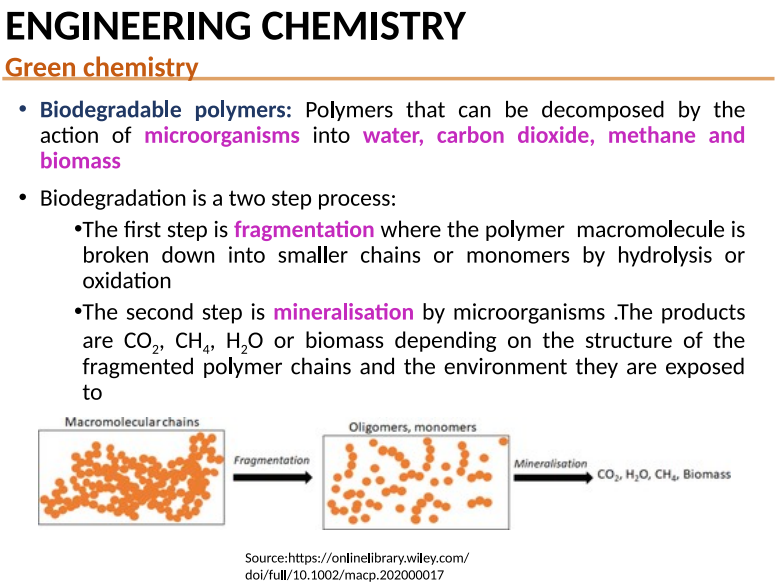


Figure G000160. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 18.

EV0000672

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

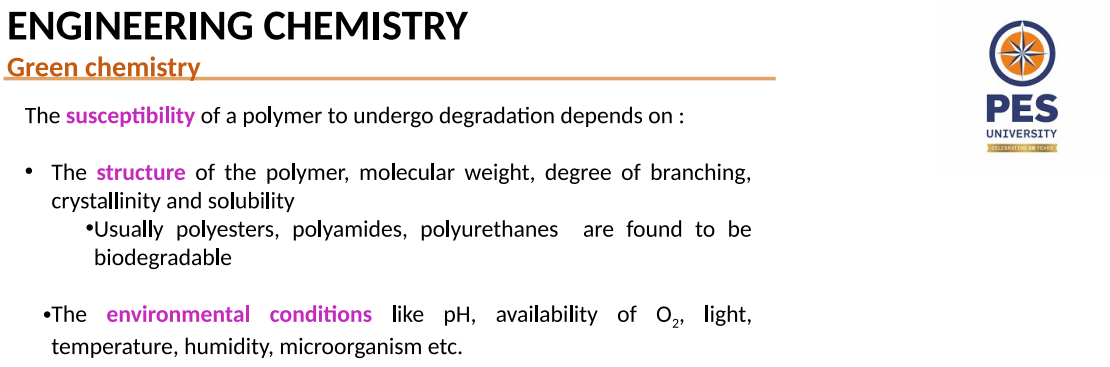


Figure G000161. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 19.

EV0000675

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

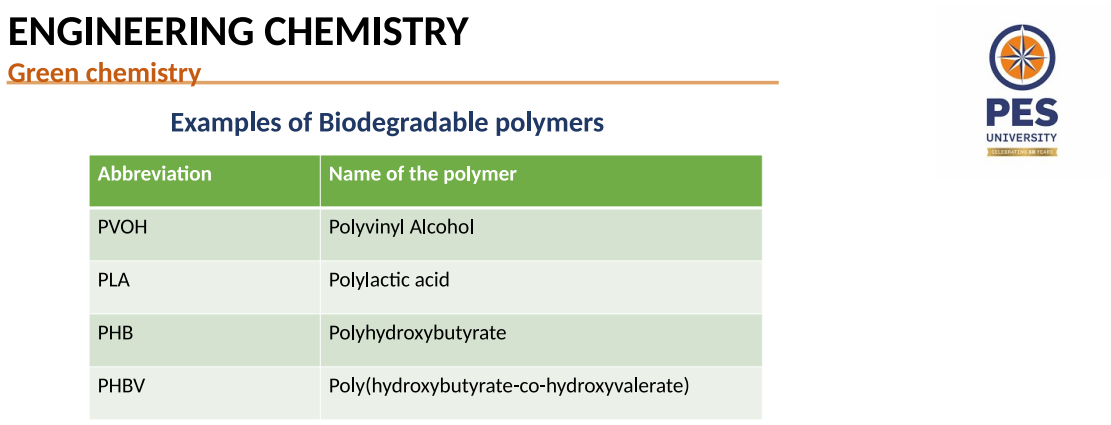


Figure G000162. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 20.

EV0000678

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

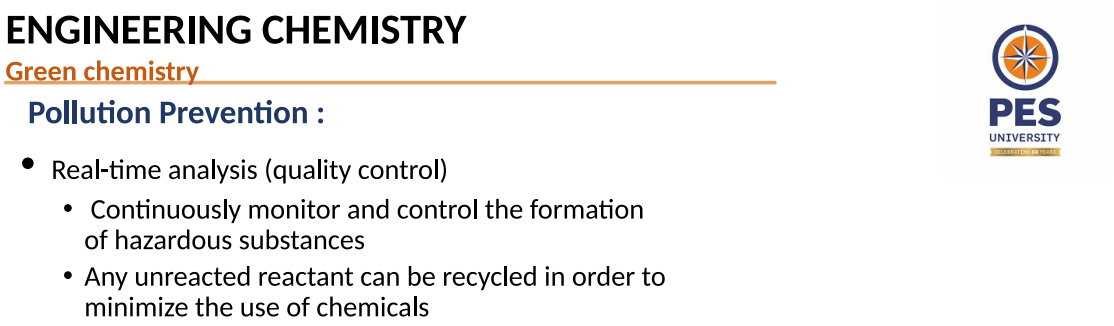


Figure G000163. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 21.

EV0000681

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

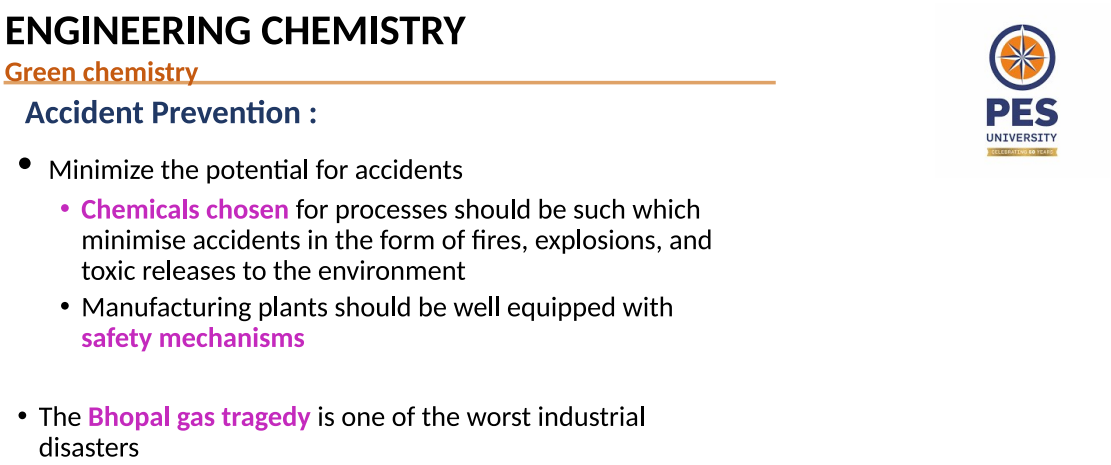


Figure G000164. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 22.

EV0000684

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

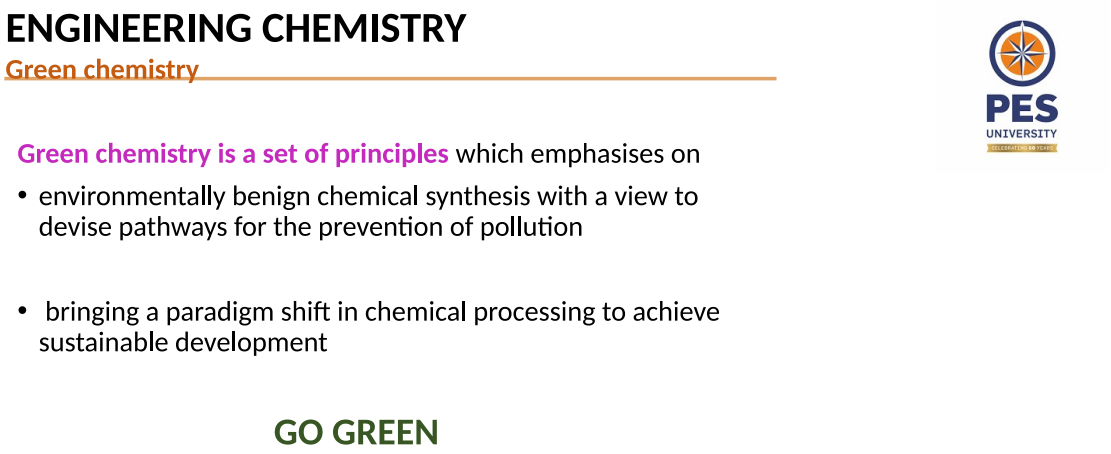


Figure G000165. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 23.

EV0000687

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000166. Green chemistry Introduction twelve principles of green chemistry with examples · source slide page 24.

EV0000690

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000620 · EV0000621 · EV0000622
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Green chemistry Class 50 content: Green chemistry: Introduction, twelve principles of green chemistry with examples EV0000623 · EV0000624 · EV0000625
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Green chemistry Definition ‘Green Chemistry’ is the utilization of a set of principles that reduces or eliminates the use or generation of hazardous substances in the design, manufacture and application of products -Paul T. Anastas and John C. Warner EV0000626 · EV0000627 · EV0000628
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Green chemistry Green Chemistry focuses on : Reduction, recycling and/or eliminating toxic substances Finding creative, alternative routes to minimize impact on the environment More eco-friendly green alternative to conventional chemistry practices Provides sustainable development, sustainable business and sustainable living practices EV0000629 · EV0000630 · EV0000631
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Green chemistry Twelve Principles of Green Chemistry Prevent Waste | Use of renewable feedstocks Atom Economy | Reduce Derivatives Safer Synthesis | Catalysis Safe Products | Design for Degradation Design Benign Chemicals | Pollution prevention Design for Energy Efficiency | Accident Prevention EV0000632 · EV0000633 · EV0000634

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Green chemistry Prevent Waste : Synthesis only targeted product with minimum or no by products/waste If waste is produced , it involves cost and time to treat and safely dispose it, as a result production cost increases EV0000635 · EV0000636 · EV0000637
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Green chemistry Atom Economy : Synthetic methods should be designed to maximize the incorporation of all materials used into the final product The chemical processes should have maximum product yield Atom economy can be calculated using the relation % Atom Economy = (FW of atoms utilized/FW of all reactants) X 100 Atom Economy in a Substitution Reaction - moderate Atom Economy in Elimination Reactions - low Atom Economy in Addition Reactions -100% Atom Economy in Rearrangement Reactions -100% For a green synthesis atom economy must be maximum (100%) EV0000638 · EV0000639 · EV0000640
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Green chemistry Safer Synthesis : The synthetic processes should be designed so as to minimize or prevent the production of hazardous substances Necessary safety precautions have to be taken while processing to protect human health and environment e.g., Polyurethane is manufactured conventionally using isocyanate and alcohol. However, isocyanate is produced using phosgene, a toxic gas The alternative method by green synthesis of polyurethane eliminates the use of phosgene and uses CO2 EV0000641 · EV0000642 · EV0000643
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Green chemistry Safe Products : Chemical products should be designed to preserve efficacy of function while reducing toxicity Products should not have any side effects on the people using it and on the environment It is envisaged to produce the chemical products specially those being used in cosmetics, pharmaceuticals, etc. EV0000644 · EV0000645 · EV0000646
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Green chemistry Design Benign Chemicals : Use safer solvents and reaction conditions Avoid using solvents, separation agents, or other auxiliary chemicals If these chemicals are necessary, use innocuous chemicals Chemical syntheses are assisted by solvent medium: recommended to use alternative solvents , greener in nature such as aqueous medium, liquid carbon-dioxide, ionic liquids(N-alkyl pyridinium cation with BF4-) , or solvent free systems Avoid harmful solvents like chloroform, pyridine which are known to cause health hazard, are carcinogenic to human beings and animals or cause severe damage to environment EV0000647 · EV0000648 · EV0000649
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Green chemistry Design for Energy efficiency : Chemical processes are to be designed in such a way that they are less energy intensive reactions occurring at mild conditions require less time to complete For this purpose it is worthwhile to utilize Bio-catalyst, homogeneous and heterogeneous catalyst which reduces the energy of activation Modern techniques such as solvent free synthesis, supercritical fluid systems, microwave irradiation, and ultrasound EV0000650 · EV0000651 · EV0000652
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Green chemistry Use of renewable feedstock : Raw material or feedstock used as starting material should be renewable rather than depleting wherever technically and economically possible Use of renewable feedstock such as biomass and agricultural wastes are strongly recommended rather than using non-renewable resources such as coal, petroleum-based raw materials, etc. EV0000653 · EV0000654 · EV0000655

13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Green chemistry Reduce Derivatives : Avoid unnecessary derivatization temporary modification should be avoided whenever possible as they generate waste the atom efficiency is reduced as the protective group is not incorporated in the final product e.g. blocking group, protection/ deprotection EV0000656 · EV0000657 · EV0000658

14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Green chemistry Catalysis : Use catalytic reactions, as catalysts Speed up the reaction can be recycled and are highly selective less energy consumption during reaction obtaining high yield of product of high purity Unlike stiochiometric reactions which are energy intensive require excessive reagents EV0000659 · EV0000660 · EV0000661

15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Green chemistry Design for Degradation : Design chemicals and products to degrade after use to innocuous substances after use so that they do not accumulate in the environment Especially applicable to insecticides pesticides polymers which tend to persist in the environment and are known to cause bioaccumulations EV0000662 · EV0000663 · EV0000664

16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Green chemistry Biodegradable Polymers Every year we produce millions of tons of garbage which we dispose off in landfills Waste disposal problems of these traditional plastics which are resistant to microbial degradation causes them to accumulate in the environment which is a serious threat to global ecology Source:http://www.businessworld.in/article/-79-Of-Total-Plastic-Waste-Stored-In-Landfills-Or-Aquatic-Bodies-/26-07-2017-122920/ Source:https://steemit.com/oceans/@alchemylgc/every-minute-one-garbage-truck-of-plastic-is-dumped-into-our-oceans-this-has-to-stop EV0000665 · EV0000666 · EV0000667

17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Green chemistry [Image text, OCR] BananaPeel 2to10days Sugarcanewaste 1to2months HOWLONG Paperbags 2to5months DOESITTAKE TODECOMPOSE? Cotton 1to5months Orangepeel within6months Rope 3to14months Thread 3to14months Milkcartons around5year Cigrette 1 to12years Nylon clothings 30 to40years Leathershoes 25to40years Aluminium cans 80 to100years Diapers 500to800years Plasticbags 15to1000years Glassbottles Plasticbottles 10 3 2.7 27 275 2e750 27.500 BEESNPTSE days months Source:https://ahlaprsacks.com/faq/ EV0000668 · EV0000669 · EV0000670

18. Source slide page 18 (extracted text): ENGINEERING CHEMISTRY Green chemistry Biodegradable polymers: Polymers that can be decomposed by the action of microorganisms into water, carbon dioxide, methane and biomass Biodegradation is a two step process: The first step is fragmentation where the polymer macromolecule is broken down into smaller chains or monomers by hydrolysis or oxidation The second step is mineralisation by microorganisms .The products are CO2, CH4, H2O or biomass depending on the structure of the fragmented polymer chains and the environment they are exposed to Source:https://onlinelibrary.wiley.com/doi/full/10.1002/macp.202000017 EV0000671 · EV0000672 · EV0000673

19. Source slide page 19 (extracted text): ENGINEERING CHEMISTRY Green chemistry The susceptibility of a polymer to undergo degradation depends on : The structure of the polymer, molecular weight, degree of branching, crystallinity and solubility Usually polyesters, polyamides, polyurethanes are found to be biodegradable The environmental conditions like pH, availability of O2, light, temperature, humidity, microorganism etc. EV0000674 · EV0000675 · EV0000676
20. Source slide page 20 (extracted text): ENGINEERING CHEMISTRY Green chemistry Examples of Biodegradable polymers Abbreviation | Name of the polymer PVOH | Polyvinyl Alcohol PLA | Polylactic acid PHB | Polyhydroxybutyrate PHBV | Poly(hydroxybutyrate-co-hydroxyvalerate) EV0000677 · EV0000678 · EV0000679
21. Source slide page 21 (extracted text): ENGINEERING CHEMISTRY Green chemistry Pollution Prevention : Real-time analysis (quality control) Continuously monitor and control the formation of hazardous substances Any unreacted reactant can be recycled in order to minimize the use of chemicals EV0000680 · EV0000681 · EV0000682
22. Source slide page 22 (extracted text): ENGINEERING CHEMISTRY Green chemistry Accident Prevention : Minimize the potential for accidents Chemicals chosen for processes should be such which minimise accidents in the form of fires, explosions, and toxic releases to the environment Manufacturing plants should be well equipped with safety mechanisms The Bhopal gas tragedy is one of the worst industrial disasters EV0000683 · EV0000684 · EV0000685
23. Source slide page 23 (extracted text): ENGINEERING CHEMISTRY Green chemistry Green chemistry is a set of principles which emphasises on environmentally benign chemical synthesis with a view to devise pathways for the prevention of pollution bringing a paradigm shift in chemical processing to achieve sustainable development GO GREEN EV0000686 · EV0000687 · EV0000688
24. Source slide page 24 (extracted text): THANK YOU Department of Science and Humanities EV0000689 · EV0000690 · EV0000691

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.548] Hello everyone, welcome back to this course on engineering chemistry. [00:00:06.244] Today we will look at module eight and that is about green chemistry and this is the only class in this module. So we are going to discuss about green chemistry. We will study the definition and the twelve principles of green chemistry. [00:00:24.452] Green chemistry is defined as the utilization of a set of principles that reduces or eliminates the use of [00:00:34.288] use or generation of hazardous substances in the design, manufacture, and application of products. So how is it different from environmental chemistry? We all are aware of environmental chemistry where we talk about removing eliminating [00:00:52.048] pollution removing pollution from the environment. But here green chemistry talks about [00:00:59.076] Removing pollution, yes, but not generating it in the first place itself. So it's a concept where we talk of reducing or eliminating the use [00:01:13.060] of hazardous substances and even generation of hazardous substances where is starting from the design to the manufacture and application of products. So this was definition was coined by Paul T. Anistos and John C. Warner. So what are [00:01:32.548] The what is the focus of green chemistry? So green chemistry talks about reduction, recycling, and or eliminating toxic substances. Then [00:01:43.632] we have to find creative alternative routes to minimize impact on the environment. So it is basically a concept and when we have the principle, we keep it in mind and we are going to use that principle

and try how best we can uh minimize [00:02:01.264] impact of all these pollutants on the environment and it is a more eco-friendly green alternative to conventional chemistry uh practices there are certain practices which have been going on for so long but now we are talking of green alternative [00:02:18.832] to those practices. And it also, of course, provides sustainable development, sustainable businesses, and sustainable living practices. [00:02:28.740] So as I mentioned, there are twelve principles of green chemistry. Prevent waste, atom economy, safe synthesis, safe products, design benign chemicals, design for energy efficiency, use of renewable feed stocks, reduce derivatives, catalysis, [00:02:46.384] Designed for degradation, pollution prevention, and accident prevention. Now let us look at each one of them and understand what the principle stands for. First [00:02:59.552] Prevent waste. Of course the first thing we have to talk about uh when we talk about pollution is that not to generate waste in the first place because once the waste is produced that it involves an extra cost and time to treat and safely dispose it. [00:03:17.232] And therefore the production cost also increases. So what we have to keep in mind here is that we synthesize only targeted product and minimum or no byproducts or waste material. [00:03:34.468] The second principle is atom economy. Now, what does it mean that synthetic methods should be used [00:03:44.452] In fact, in such a way the design should be of any reaction should be such that all the materials, in fact all the atoms which are present in the reactants should be incorporated into the products. That is atom economy. Now earlier [00:04:02.992] The emphasis was more on yield. Yes, here also we want a good yield, but at the same time it is not just the yield we are interested in. We want that all the atoms so that there is no uh extra atoms which are not being used or getting wasted. [00:04:20.656] So all the atoms in the reactants have to be included into the products, and this can be calculated for each reaction using this relationship. So percentage atom economy is equal to formula weight of the atoms which have been utilized and the [00:04:38.224] formula weight of all the reactants which have been used. So formula weight of atoms utilized divided by the formula weight of all reactants multiplied by 100 would give us the atom economy. Now let us take up some reactions. Like for example there's a substitution reaction. So what we are doing [00:04:55.792] Some of the atoms are getting substituted by another atom. So this would result in a moderate atom economy.

EV0000174 · EV0000175 · EV0000176 · EV0000177

Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.000] Whereas if there was an elimination reaction, definitely some small molecules are getting eliminated and or even uh larger molecules sometimes and then what happens to your atom economy? It has become low. If you have an addition reaction, we know that [00:05:21.796] Suppose there's a double bond and the reactants are getting incorporated into it, yes then we get hundred percent atom economy, and even if there is a rearrangement reaction again uh it is just rearrangement of the atoms which is taking place again we have a hundred percent atom economy. So if you have [00:05:39.568] Talking of green synthesis, then of course we understand atom economy must be maximum, and that is 100%. [00:05:48.740] Safer synthesis is the third principle. Now synthetic processes should be such that they are designed such a way to minimize or prevent the production of hazardous substances. [00:06:03.696] As far as possible, we should use only uh substances which are innocuous and necessary safety precautions have to be taken while processing to protect human health and the environment. For example, polyurethane [00:06:22.576] is a very popular polymer and it was manufactured conventionally conventionally using isocyanate and alcohol. But to prepare the isocyanate we had to use [00:06:35.248] phosgene and we know that phosgene is a very toxic gas,

COCl2. It's a toxic gas. Now this principle of safe synthesis is forbidding us to use toxic [00:06:50.608] or hazardous substances. So the alternative now has been found and the green synthesis of polyurethane eliminates the use of first gene and instead CO2 is being used. So this is what we mean by safer synthesis. [00:07:07.428] Now safe products. Now chemical products should be designed in such a way that the efficacy is preserved. [00:07:16.324] It functions as efficiency and as efficiently, but the toxicity is reduced. Now many of these products are generally found to have side effects. Now if we can think of ways [00:07:30.704] by which they will be efficient, but at the same time the side effects are reduced or brought to the minimum, then we see that we have been successful in creating safe products for the people using it, especially say cosmetics. [00:07:46.912] or in pharmaceuticals this plays a very important role because many of these drugs which are being used for as medicine we know that there is lot of side effects. So if they can be effective to [00:08:00.416] uh cure a particular disease but at the same time not have any side effects. So this is the principle which is uh ask us to find out alternatives which we would call it as a green alternative. [00:08:14.532] Then design benign chemicals. Now what happens is many of these chemical reactions we know are s uh carried out in a solvent medium. [00:08:25.728] Uh so always there is it is they are assisted by the solvents. So what it says is that use safer solvents and reaction conditions. So wherever possible do not use solvents, separation separation agents or other auxiliary uh chemicals. And if it's [00:08:43.376] All we have to use, then they should be harmless, innocuous chemicals which are harmless. So most of the time, as I said just now, we use a lot of solvents, and some of these solvents are hazardous. So in fact, some of them have been banned. So what is the recommended solvent here? [00:09:02.628] We can use alternative solvents, find alternative solvents greener in nature. So we can have aqueous medium, of course, we know that is the safest, or we can have liquid carbon dioxide, and what is becoming popular now is these ionic liquids. So many of the [00:09:20.400] researchers are using these ionic liquids now organic solvents are usually avoided because they are volatile in nature and they can affect the health as well as the environment but when you have ionic liquids now these have the properties to dissolve [00:09:37.968] At the same time they will not be volatile. One good example would be N alkyl peridinium cation with BF4 minus. So what would it look like? We have pyridine which is like this. [00:09:52.068] with three bonds and there is a plus because we have attach an [00:09:58.656] alkyl group here. Now this is the positive ion and it is neutralized by BF4 minus. So this is known as a N alkyle. So at the end now there is an R which is attached here. So this would be an N alkyle

EV0000178 · EV0000179 · EV0000180 · EV0000181

Transcript block 3 (source interval 000600-000899 seconds): [00:10:16.400] This alkyl group could be ethyl, methyl, uh and so on. And this [00:10:22.992] Positive is now because of the N having an extra bond over here, this becomes positive and this is neutralized with the B at four minus. So this is kno if this is a very uh popular ionic liquid and or we should use solvent free system. [00:10:38.816] At any cost we have to avoid harmful solvents like chloroform and peritene, etc., which are known to cause health hazards, and some of them could be carcinogenic to human beings and animals and also cause severe damage to the environment. [00:10:57.348] The next principle says design for energy efficiency. Now we find that many processes require a lot of energy. Now if we can somehow find alternatives where which are less energy intensive. [00:11:14.512] Say the reaction could be occurring at milder conditions or it requires less time to complete. So we have to find these alternatives where we can save energy and [00:11:25.424] For this purpose, it is worthwhile to utili something like biocatalyst or homogeneous or heterogeneous catalyst, which will

reduce the energy of activation for the reactions and hence we will consume less energy. And modern techniques such as solvent free super [00:11:43.056] And more importantly these days microwave irradium and ultrasound. For example, biodiesel. Biodiesel when it is made in the lab, the normal way, it takes nothing less than [00:12:00.452] say one hour one and a half hours. But the same thing when we do in the microwave when we use microwave radiation to uh to have that reaction or to uh perform the uh reaction when then that reaction takes place at a very fast rate and [00:12:18.224] And within 15 minutes we can have the same reaction happening which was taking one and a half hours. So this way we have reduced the time as well as the energy consumed, and similarly ultrasound is also being used to make it more. [00:12:35.072] uh less energy intensive and saving more energy. [00:12:40.548] Use of renewable feedstock. I think all of us are aware of this. Raw material or feedstock used as starting material for any reaction should be a renewable so it's should come from a renewable source rather than one which is depleting. Wherever it is technically as well as [00:12:58.320] economically possible and for renewable feedstock the most common thing being looked at is biomass and agricultural waste which are strongly recommended rather than using the non-reduable sources like coal and petroleum based raw materials which we know are uh depleting very fast. [00:13:19.236] Now talking of reduced derivatives. [00:13:22.788] This is very important because many times when reactions are carried out, we make derivatives. So what this principle says is avoid unnecessary derivatization. So temporary modification should be avoided whenever possible because what happens because of that? [00:13:40.560] Going to generate waste, and at the same time, we know whenever we have these extra byproducts getting form the atom efficiency is reduced as the protective group is not getting. [00:13:52.144] Incorporated in the final product. For example, we could be using some group to block or as a protection and then again deprotection. Let us take an example. Suppose we have aniline. [00:14:06.084] uh like this and I want to nitrate it. The moment I do nitration I will get ortho as well as the para substituted but what I want is just the ortho or the para. So what do I do for that? I usually take aniline [00:14:24.612] and I'm going to treat it with CH3CO C L, so thereby making it into NH. [00:14:34.916] COCH3. So because of this, what happened now? We are blocking, we are protecting the NH2 group or making it less reactive. And now when we do nitration, what we observe is that either we get the para [00:14:55.588] NHCO CH3. So that is NO2, or we get the ortho product.

EV0000182 · EV0000183 · EV0000184 · EV0000185

Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.180] So NHCO CH3 and we get NO2 over here. So either the ortho product or the [00:15:15.728] Para product that is what we wanted. But then we wanted an ananine group, I mean NH2 group over here. So what we are going to do further now we will do hydrolysis and try to get rid of the COCH3 from there so that we get NH2 and NO2. [00:15:33.184] uh in the paraposition and then NH2 nitroaniline that is para nitroaniline and ortho nitro aline aniline and not the disubstituted aniline. But what happened because of this you uh carried out so many reactions first of all and then you blocked this [00:15:50.336] uh position and finally you have to unblock it like protection and deprotection now because of that a lot of wastage happens so we have to look for alternatives wherein directly aniline can be converted into orthodonitromine or paranitry [00:16:07.984] Anuline as we desire. This is an example where we talk about we have to reduce derivatives so that we can bring down the wastage. [00:16:20.356] of chemicals and uh also uh the time consumed in the reaction. Now another way of doing uh carrying out reactions is by catalysis and this is definitely a greener method. So we have to use catalytic reactions because catalytic [00:16:38.128] This we know are going to speed up the

reaction, they can be recycled and are highly selective in nature. So we can get the exact product what we are looking for, and then of course there is less energy consumption during the reaction. [00:16:55.696] we can get very high yield and also not only high yield but we'll get very pure products. So therefore wherever it is possible we should look for reactions which can be carried out with the help of a catalyst. So catalytic reactions would definitely [00:17:13.264] B of greener options because when we use tychometric reactions, they are highly energy intensive and they also require excessive reagents. [00:17:25.508] The next principles talk about design for degradation. Now we have to design chemicals and products which degrade after use to harmless inoc that is inoculant substances and so that they do not accumulate in the [00:17:45.188] environment. So here specially this is applicable to insecticides, pesticides, polymers. We all understand that insecticides and pesticides they will remain [00:17:58.304] uh in and they are required for the plants but nevertheless they remain in the soil and gets absorbed into the earth and from the water there when uh it comes into the water bod and finally it is going to come back to us and affect us in a uh bad way. [00:18:15.952] They're going to cause harm. So these are hazards. So therefore, uh insecticides, pesticides, polymers, etc. We have to look for uh degradable uh alternatives for this so that they do not persist in the environment and are [00:18:33.520] known to cause these generally will persist in the environment and they are known to cause bioaccumulations. Therefore, we have to find alternatives which are degradable so that they do not persist in the environment. Then pollution prevention. [00:18:53.700] Here we are talking of [00:18:56.240] Real time analysis. Of course, all these points which we have mentioned so far about are about pollution prevention, but here we are talking of real-time analysis that is quality control. So what we have to do is continuously monitor and control the form. [00:19:13.872] The definition of hazardous substances. [00:19:16.576] So in C2 as the reaction is happening, we are controlling we are monitoring it and also trying to control the formation of hazardous substances and any unreacted uh reactant, it has to be recycled in order to [00:19:34.256] order to minimize the use of chemicals. So whatever is getting uh formed and or or is unreacted it should go back so that again it can be used and then that way we can reduce the use of chemicals. [00:19:51.140] Accident prevention. This is the last principle, that is the twelfth principle. Now we have to reduce the potential of accidents. The chemicals chosen for the processes should be such that

EV0000186 · EV0000187 · EV0000188 · EV0000189

Transcript block 5 (source interval 001200-001499 seconds): [00:20:06.948] We can minimize accidents in the form of fires, explosions, and toxic releases to the environment. [00:20:15.044] Also, manufacturing plants should be well equipped with safety mechanisms. All of us are aware of the mobile gas tragedy. It is one of the worst industrial disasters and [00:20:29.552] after so many years people are still suffering because of the leakage of a toxic gas into the environment. So such uh accidents should be prevented. So right in the beginning when the [00:20:46.640] designing is happening there itself we should be very careful to choose the right kind of processes and also there should be safety mechanisms in place. So these are the twelve principles of green chemistry. [00:21:02.148] So what exactly is green chemistry what we have understood? It is a set of principles which emphasizes on environmentally benign chemical synthesis with a view to devise pathways for the prevention of pollution. [00:21:18.468] and then bringing a paradigm shift in chemical processing to achieve sustainable development. [00:21:26.224] Now for all of us it is very very important. It is a a concept which we should always keep in mind whichever field we are working in, that we have to save the environment, we have to keep it clean and safe. [00:21:43.712] for us.

So go green. That brings us to the end of module eight and we have completed unit five. It also brings us to the end of this course on engineering chemistry. I hope you enjoyed the course. Thank you.

EV0000190 · EV0000191 · EV0000192 · EV0000193

Full transcript: 2969 normalized words, 90 timestamped segments, 1323.733333 seconds duration. Low-confidence segments flagged: 0.

EV0000174 · EV0000175 · EV0000176 · EV0000177

14 Source slide deck: Commercial polymers synthesis and properties of Plexiglass

This entry preserves the complete slide deck titled Commercial polymers synthesis and properties of Plexiglass as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_04, to avoid an unsupported match. EV0000401 · EV0000402

Graphs and diagrams

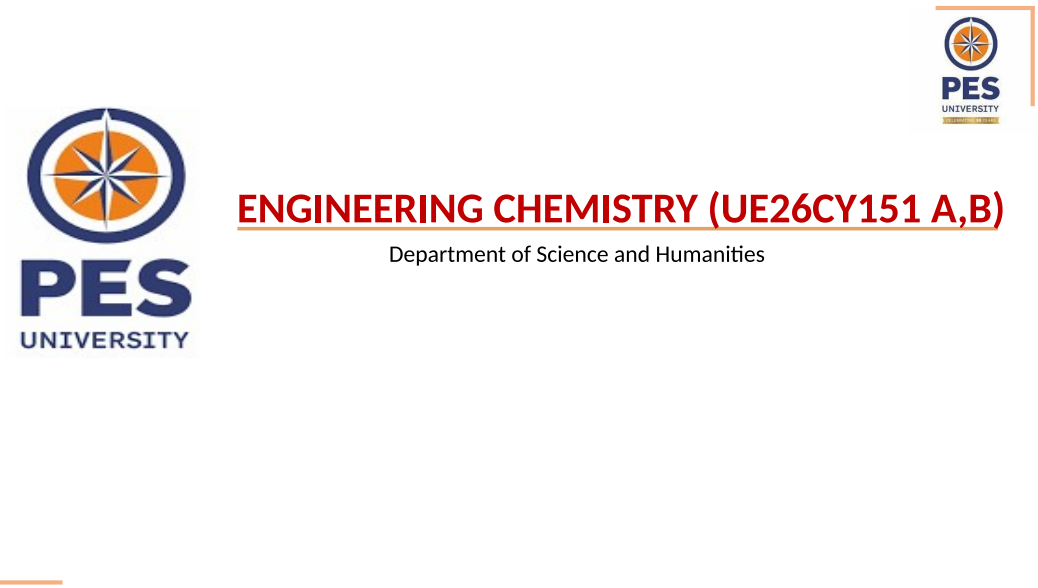


Figure G000070. Commercial polymers synthesis and properties of Plexiglass · source slide page 1.

EV0000402

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

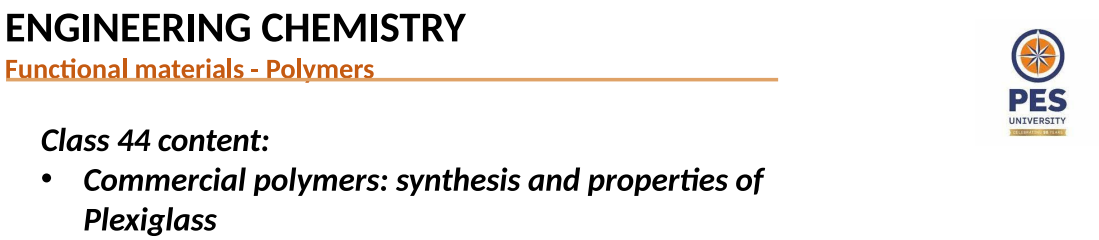


Figure G000071. Commercial polymers synthesis and properties of Plexiglass · source slide page 2.

EV0000405

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Functional materials - Polymers

Polymethyl Methacrylate (PMMA) or Plexi Glass

- It is an addition polymer
- Plexi glass is obtained by radical polymerization in bulk, emulsion or suspension from the monomer methyl methacrylate (MMA)
- Hydrogen peroxide is used as initiator
- The monomer is obtained from acetone.

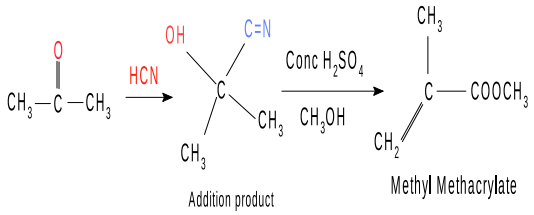


Figure G000072. Commercial polymers synthesis and properties of Plexiglass · source slide page 3.

EV0000408

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

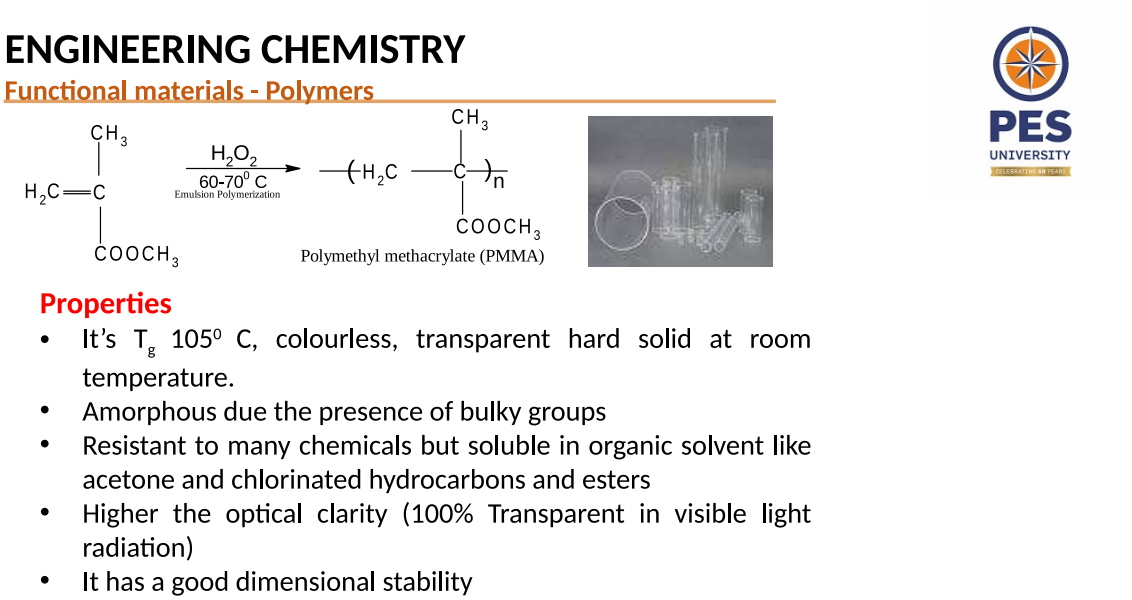


Figure G000073. Commercial polymers synthesis and properties of Plexiglass · source slide page 4.

EV0000411

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000074. Commercial polymers synthesis and properties of Plexiglass · source slide page 5.

EV0000414

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000075. Commercial polymers synthesis and properties of Plexiglass · source slide page 6.

EV0000417

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000401 · EV0000402 · EV0000403
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Class 44 content: Commercial polymers: synthesis and properties of Plexiglass EV0000404 · EV0000405 · EV0000406
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Polymethyl Methacrylate (PMMA) or Plexi Glass It is an addition polymer Plexi glass is obtained by radical polymerization in bulk, emulsion or suspension from the monomer methyl methacrylate (MMA) Hydrogen peroxide is used as initiator The monomer is obtained from acetone. EV0000407 · EV0000408 · EV0000409
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Properties It's Tg 1050 C, colourless, transparent hard solid at room temperature. Amorphous due the presence of bulky groups Resistant to many chemicals but soluble in organic solvent like acetone and chlorinated hydrocarbons and esters Higher the optical clarity (100% Transparent in visible light radiation) It has a good dimensional stability EV0000410 · EV0000411 · EV0000412
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Functional materials - Polymers Applications Due to its optical clarity it is used to make aircraft windows, signal boards etc For Making artificial teeth and contact lens Used in paints and adhesives Making transparent bottles and wash basins. Good substitute for glass but has poor scratch resistance. EV0000413 · EV0000414 · EV0000415

6. Source slide page 6 (extracted text): THANK YOU Department of Science and Humanities EV0000416 · EV0000417 · EV0000418

SHORT-NOTE RECAP

Source deck preserved: Commercial polymers synthesis and properties of Plexiglass. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review. EV0000401

15 Source slide deck: Experiential learning Based on nanomaterials

This entry preserves the complete slide deck titled Experiential learning Based on nanomaterials as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_04, to avoid an unsupported match. EV0000692 · EV0000693

Graphs and diagrams

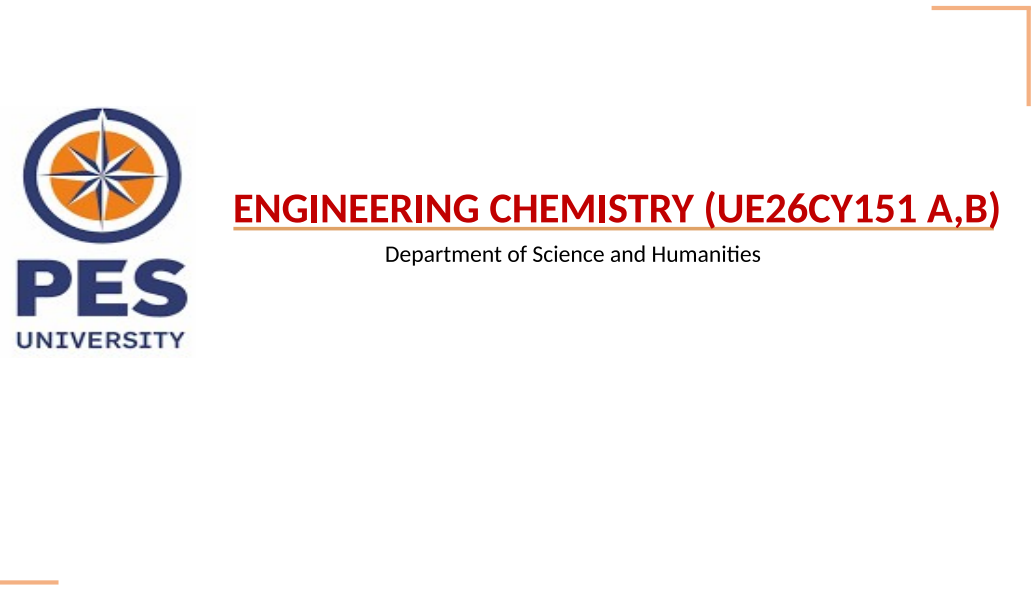


Figure G000167. Experiential learning Based on nanomaterials · source slide page 1. EV0000693

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

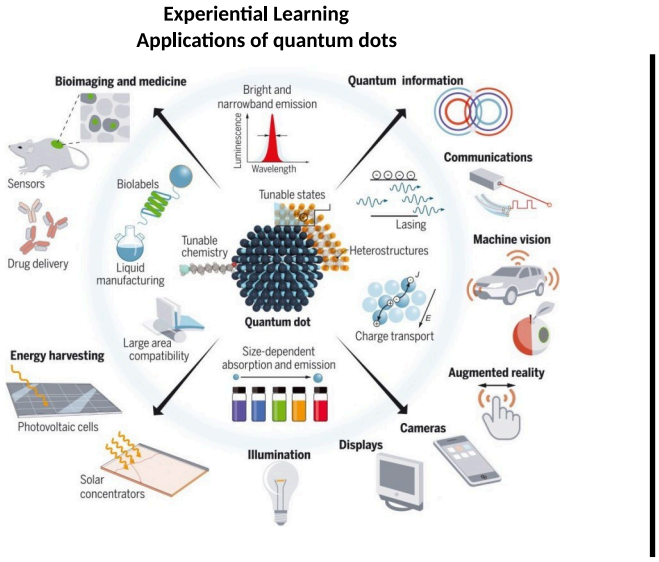


Figure G000168. Experiential learning Based on nanomaterials · source slide page 2.

EV0000696

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000169. Experiential learning Based on nanomaterials · source slide page 3.

EV0000699

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000692 · EV0000693 · EV0000694
2. Source slide page 2 (extracted text): Experiential Learning Applications of quantum dots [Image text, OCR] Bioimagingandmedicine Quantuminformation Brightand narrowbandemission Communications Wavelength 000o Sensors Biolabels Tunable states W Lasing Tunable Machinevision chemistry Heterostructures Drugdelivery Liquid manufacturing Quantumdot Chargetransport Large area Energyharvesting compatibility Size-dependent absorptionandemission Augmentedreality Photovoltaiccells Cameras Displays llumination Solar concentrators EV0000695 · EV0000696 · EV0000697
3. Source slide page 3 (extracted text): THANK YOU Department of Science and Humanities EV0000698 · EV0000699 · EV0000700

SHORT-NOTE RECAP

Source deck preserved: Experiential learning Based on nanomaterials. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000692

16 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Engineering Chemistry, Unit_04 (Functional materials and green chemistry) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000701 · EV0000702

Important points and why

1. Archive page 46 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. OCR/sparse page: visual confirmation is required. Overview of Hydrogen Storage Methods Form Advantages Disadvantages O Gaseous Form - Less infrastructure required - Needs extra energy for compression - More cost-effective - Occupies a large space - Can be compressed to high - Requires leak-proof tanks pressure Liquid Form - Exists at a very low temperature - Reinforced and insulated storage tank required (20K / -253°C) - Energy-intensive cooling and compression (~30% loss) - Stored in cryogenic tanks - Low volume Solid Form - Hydrogen can be released on - Currently under intensive Research & Development demand (e.g., by heat) (R&D) - Ease of handling - Enhanced safety s s s Key challenges of the materials and their solutions 1. Weak Binding Forces: Hydrogen storage materials have high specific surface areas but weak binding forces (e.g., Van der Waals forces). 2. Metal Decorations: Metal decorations are effective in increasing the bonding energy of hydrogen on sorption-based materials. 3. Hydrogen Spillover: e Hydrogen storage in carbon materials is enhanced by decorating these materials with transition metals. ¢ Definition: Hydrogen spillover refers to the migration of hydrogen atoms from the metal surface to anon-metal support or absorbent. EV0000701
2. Archive page 1 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.9167; source text/OCR may be partial. (4.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - JULY - 2023 UE22CY151B - Engineering Chemistry Total Marks : 100.0 1.a. State Frank-Condon principle. Draw the three possibilities of vibrational - electronic transitions energy level diagram and comment on the spectrum obtained by each of the transition. 1.b. i. "Nanomaterials are classified based on dimensions" mentions the types with suitable examples (4M) ii. What do you mean by Blue shift? In nanomaterials, engergy gap and blue shift are interelated. Illustrate with example (2M) 1.c. Comment on Scattering of light, Raman scattering, stokes lines and anti-stokes lines. (4M) EV0000702
3. Archive page 7 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (4.0 Marks) (4.0 Marks) (5.0 Marks) 5.b. i. Write the reaction of synthesis of epoxy resine starting from Epichlorohydrin and Bisphenol A - 2M ii. Mention the curing agents added to epoxy resine, what is role played by it? - 2M 5.c. A polymer has the following molar mass distribution 1. 50 molecules of molecular weight 5000 g.mol 2. 75 molecules of molecular weight 6000 g.mole Calulate number average and weight average moleculare weight of the polymer. 4M 5.d. i. "PVC is more crystalline than PE" Justify the statement (2M) ii. Mention any three principles of Green Chemistry with example (3M) -1 -1 EV0000703

4. Archive page 5 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (5.0 Marks) 4.d. Why is viscometer placed in the water bath during flow rate measurements in viscosity avergage molecular mass determination experiment? 5.a. (i) Why is poly vinyl chloride more crystalline than polyethylene? (2 marks) (ii) How does the addition of triphenyl phosphate (Ph PO) to a non elastic polymer affect its flexibilty? Why? (2 marks) (iii) Is PMMA an addition or condensation polymer? Mention one of its application. (2 marks) 5.b. (i) Name the monomeric units from which epoxy resin is synthesised and give the reaction corresponding to the synthesis of epoxy resin from these monomers (reaction depicting the synthesis of monomeric units is not required). 3 Marks (ii) Which of the two monomeric units is used in excess during the synthesis of epoxy resin? Why? (2 Marks) (iii) Give one important application of epoxy resin. (1 Mark) 5.c. (i) Calculate the viscosity average molecular mass of the polymer sample which has 70 molecules of molecular mass 800, 100 molecles of molecular mass 2500 and 130 molecules of molecular mass 3500. Given the Mark Houwink coefficient in the viscosity measurements is 0.65. Mention of formula and final answer is required) (3 marks) (ii) As per the principles of Green Chemistry, why should derivatization be reduced during a chemical synthesis? (2 marks) 3 4 EV0000704
5. Archive page 5 from 1__1rBrYYqw3s.pdf; unit match candidate score 1.4167; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (6.0 Marks) (4.0 Marks) 4.a. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber ? iii) Write any two principles of Green chemistry. 4.b. i) Give the synthesis of kevlar from 1, 4 phenylene diamine. ii) "Kevlar is 5 times stronger than steel", justify the statement. iii) Give any one application and one disadvantage of kevlar. 4.c. Calculate the number average, weight average and viscosity average molecular weight of a polymer having 10 molecules of molecular weight 3000, 15 molecules of molecular weight 3500 and 25 molecules of molecular weight 4000. (Given : a = 0.65). 4.d. The optical properties of nano materials change with size of the nanopartcles. Explain using i) Quantum confinement ii) Surface plasmon resonance EV0000705
6. Archive page 5 from 2__11pj9jsL0V.pdf; unit match candidate score 1.3333; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (6.0 Marks) (4.0 Marks) 4.a. Using acetone and hydrogen cyanide as the starting materials, outline the reactions involved in the synthesis of PMMA. Give any two applications of PMMA. 4.b. i) What are Biodegradable polymers? Explain with a suitable example. ii) Write the steps involved in the biodegradation of polymers. Give any two advantages. 4.c. What are Organic Light Emitting Diodes? Explain the construction, working principle, and applications of OLEDs. 4.d. Account for the following i) The strength of nanomaterials is greater than bulk material. ii) Design for Degradation and Catalysis. (Green Chemistry Principles) EV0000706
7. Archive page 6 from 3__1MwY1CTl_j.pdf; unit match candidate score 1.1667; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 6/7 (4.0 Marks) (6.0 Marks) (8.0 Marks) 4.a. Classify the nanomaterials based on their dimensions with suitable examples 4.b. (a) Write the differences between addition and condensation polymerization processes (2M) (b) Discuss the chemical synthesis of polymethyl methacrylate with proper reactants and reagents (4M) 4.c. (a) Discuss the construction and working principles of organic light emitting diodes (4M) (b) Describe the various doping processes of conducting polymers (4M) EV0000707

8. Archive page 6 from 4__1eKMFaJ2lx.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 6/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber ? iii) Write any two principles of Green chemistry. 4.b. i) Give the synthesis of kevlar from 1, 4 phenylene diamine. ii) "Kevlar is 5 times stronger than steel", justify the statement. iii) Give any one application and one disadvantage of kevlar. 4.c. Calculate the number average, weight average and viscosity average molecular weight of a polymer having 10 molecules of molecular weight 3000, 15 molecules of molecular weight 3500 and 25 molecules of molecular weight 4000. (Given : a = 0.65). EV0000708
9. Archive page 7 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 7/7 (4.0 Marks) (3.0 Marks) 4.d. The optical properties of nano materials change with size of the nanopartcles. Explain using i) Quantum confinement ii) Surface plasmon resonance 4.e. i) Define viscosity co-efficient. ii) Write expression of Lambert- Beer's law and explain the terms. EV0000709
10. Archive page 6 from 5__1pIeTzocnU.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 6/7 Discuss the structure-property relationship with regard to the following properties of polymers (i) Stereoregularity (ii) Tensile strength (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. 4.b. Give the synthesis of epoxy resin by the polymerization of monomers. How does curing impart a high degree of hardness to the polymer? Give any two applications of epoxy resin. 4.c. Complete the following table for OLEDs : Anode Cathode Emissive layer Conductive layer Advantage Application EV0000710
11. Archive page 7 from 5__1pIeTzocnU.pdf; unit match candidate score 0.5; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 7/7 (i) When the particle size is of nano order there is a decrease in the conductance of the species when compared to bulk materials. Give reasons. (ii)State any one principle of Green Chemistry and give relevant examples. (4.0 Marks) (3.0 Marks) 4.d. 4.e. 25 ml of sewage water sample is mixed with 25 ml of K Cr O solution and the unreacted K Cr O consumed 9 ml of 0.25N FAS to give the colour change with ferroin indicator. Under similar conditions, in blank titration 15ml of FAS is used up. Calculate the COD (Chemical Oxygen Demand) of the sample in mg/L (ppm). EV0000711
12. Archive page 6 from 6__1H6QH_PVNf.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 6/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber ? iii) Write any two principles of Green chemistry. 4.b. i) Give the synthesis of kevlar from 1, 4 phenylene diamine. ii) "Kevlar is 5 times stronger than steel", justify the statement. iii) Give any one application and one disadvantage of kevlar. 4.c. Calculate the number average, weight average and viscosity average molecular weight of a polymer having 10 molecules of molecular weight 3000, 15 molecules of molecular weight 3500 and 25 molecules of molecular weight 4000. (Given : a = 0.65). EV0000712
13. Archive page 7 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 7/7 (4.0 Marks) (3.0 Marks) 4.d. The optical properties of nano materials change with size of the nanopartcles. Explain using i) Quantum confinement ii) Surface plasmon resonance 4.e. i) Define viscosity co-efficient. ii) Write expression of Lambert- Beer's law and explain the terms. EV0000713

14. Archive page 5 from 7__16pEjcSTVO.pdf; unit match candidate score 1.0833; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.d. Why should viscometer be placed in water bath during flow rate measurements in viscosity average molecular mass determination experiment. 5.a. (i) Define Glass transition temperature of a polymer? What does it indicate about the polymer. (ii) Give any three points of difference between addition and condensation polymer. 5.b. (i) Give the reaction of synthesis of Kevlar from the monomeric units. (ii) What is the reason for the high strength of kevlar (iii) Give one application of kevlar. 5.c. (i) Explain any one principle of Green Chemistry with suitable example. (ii) If a polymeric sample has 100 molecules of molecular mass 1000g/mol, 200 molecules of molecular mass 2000 g/mol and 500 molecules of molecular mass 5000g/mol. Calculate the number average and mass average molecular mass of the sample. EV0000714
15. Archive page 6 from 8__1015pA2nzY.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 6/7 (5.0 Marks) (6.0 Marks) (6.0 Marks) 4.d. Describe the process of tinning. Mention one advantage of tinning. 5.a. i)Distinguish between addition and condensation polymerization. ii)write the relevent reaction of Nylon 6,6. 5.b. Calculate the number average, weight average and viscosity average of a polymer having 10 molecules of molecular weight is 3000, 15 molecules of molecular weight is 3500 and 25 molecules of molecular weight is 4000. (Given a=0.65) EV0000715
16. Archive page 7 from 8__1015pA2nzY.pdf; unit match candidate score 0.5833; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 7/7 (6.0 Marks) (2.0 Marks) 5.c. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber? iii) Write any two principles of Green chemistry. 5.d. "Kevlar is 5 times stronger than steel" justify the statement. EV0000716
17. Archive page 8 from 9__14fBQb1-hv.pdf; unit match candidate score 0.75; source text/OCR may be partial. 11/8/24, 1:02 PM ESA - DEC 2023 - UE22CY151B/UE22CY151A (set- 1) about:blank 8/9 (4.0 Marks) (4.0 Marks) (5.0 Marks) 5.b. i. Kevlar is very strong material – about 5 times stronger than steel- Justify the statement ii. Mention any two disadvantages of Kevlar 5.c. A polymer has the following molar mass distribution 1. 50 molecules of molecular weight 5000 g/mole 2. 75 molecules of molecular weight 6000 g/mole Calculate number average and weight average molecular weigt of the polyymer (4M) 5.d. i. Polyethylene(PE) is more crystalline then polyvinylacetate Justify the statement (2M) ii. Mention any three green chemistry Principles with example (2M) EV0000717

18. Archive page 5 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 5/6 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. Account for the following- (i) Polyvinylchloride is more crystalline than polyethene. (2marks) (ii) During the synthesis of butyl rubber, 5-10% isoprene is used. (2marks) (iii) Bakelite and polypropylene differ completely in the way they respond to heat. (2marks) 4.b. Give a polymer as an example for the below mentioned applications - (i) A polymer used to impart shrinkage/crease resistance to fabrics (Ii) A polymer used in making seals for vacuum applications (iii) A polymer used in manufacture of aquariums and aircraft canopies. (iv) A polymer used in the conductive layer of an OLED (Organic Light Emitting Diode) (v) A polymer used in applications in prostheses, implants, tendon/ligament repair. (v) A polymer that is a component of personal armor such as combat helmets. 4.c. Account for the following (i) Bending of bulk Cu happens readily but nano sized Cu particles are super hard. (2marks) (ii) Gold spheres of > 50 nm absorb at λ = 575 nm whereas gold spheres of 2-5 nm absorb at λ = 517 nm. (2marks) (iii) Bulk gold is catalytically inactive but gold nanoparticles are catalytically active. (2marks) EV0000718

19. Archive page 6 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 6/6 (7.0 Marks) 4.d. (i) Why should the use of blocking group and unnecessary derivatization be avoided during a chemical reaction. (2 marks) (ii) Elaborate on 'Safer Synthesis' principle of 'Green Chemistry'. (2 marks) (iii) 4 g of Ferrous ammonium sulphate (FAS) is accurately weighed out, half a test tube conc. H SO is added to it and it is made upto mark in a 250 ml standard flask and taken in the burette. 25 ml of waste paper is mixed with acidified dichromate and this consumes 12.5 ml of the standard FAS for complete reaction in the estimation of COD of waste water sample. If the blank titre value when no waste water sample is taken is 45ml of the same standard FAS, Calculate the COD of one litre of water sample . Given the Gram equivalent weight of FAS is 392.13. (3 marks) 2 4 EV0000719

20. Archive page 6 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 1.25; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 6/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. Account for, i) High strength of kevlar fibre. (2M) ii) High electrical conductance of conducting polyaniline. (4M) Explain the above with suitable structures. 4.b. i)Give the synthesis of carbon fibre from polyacrylonitrile (with reactions). (4M) ii) Calculate the weight average molecular weight of a polymer having 10 molecules of molecular weight is 3000, 15 molecules of molecular weight is 3500 and 25 molecules of molecular weight is 4000. (2M) 4.c. i) Describe the construction and working of OLEDs. (4M) ii) Write any two principles of green chemistry. (2M) EV0000720

SHORT-NOTE RECAP

20 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000701

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9960; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9969; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9970; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Structure-property relationship transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9971; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9975; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9955; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9971; only the JSON segmentation supplies the displayed timestamps.
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea (converted PDF; original hash ae5d11c83d7ee0572e21b2a94bbb3623e085fa917470c7d72dff1a365eb13c76)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s (converted PDF; original hash b05ecebc066c8c1af6db2b16f0d65865d70cbadf9ab0b7c9ab61c058936838cb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Polymers Introduction types of polymerizations definition of glass transition temperature	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Polymers Introduction types of polymerizations definition of glass transition temperature	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Polymers Introduction types of polymerizations definition of glass transition temperature (converted PDF; original hash 1ad6fc41336a211ac5c12645b0d6b7c0ef8d13c9f0d4bdd2502112a3c03b4f12)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Structure property relationship	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Structure property relationship	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Structure property relationship (converted PDF; original hash 93190733cb7dd6e8fc21d21da279c07476ce73574251c19ef48cef845c3a81cd)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Commercial polymers synthesis and properties of Plexiglass	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Commercial polymers synthesis and properties of Plexiglass	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Commercial polymers synthesis and properties of Plexiglass (converted PDF; original hash 57bca0020e382c0b12e473ceb72e9250bfa79c292485bd1c1968ef1349466906)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Synthesis and properties of epoxy resin	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Synthesis and properties of epoxy resin	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Synthesis and properties of epoxy resin (converted PDF; original hash 19fbeebb19cb44c83573666f8878a6a091c1664eec36ee1544b0d9df00364b9d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Polymer composites synthesis properties of Kevlar and Carbon fibers	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Polymer composites synthesis properties of Kevlar and Carbon fibers	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Polymer composites synthesis properties of Kevlar and Carbon fibers (converted PDF; original hash 2560c8f6143cbf4231c5c6dfa9794b01a13c27aa4e20369f322f680e0b94a93b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Average molecular weight of polymers number average weight average and viscosity average molecular weight nu	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Average molecular weight of polymers number average weight average and viscosity average molecular weight nu	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Average molecular weight of polymers number average weight average and viscosity average molecular weight nu (converted PDF; original hash ae4d397b497d8ca2e63777121805fdc0902e941301d0376f18ab85c8238855d1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Conducting polymers examples types of doping	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Conducting polymers examples types of doping	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Conducting polymers examples types of doping (converted PDF; original hash ca326f637ca66e6900f3cf0d5fd9e089f7a5e25ff2ec2719b20b860aa87aa94f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Organic light emitting diodes OLEDs Construction working principle and light emission mechanism	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Organic light emitting diodes OLEDs Construction working principle and light emission mechanism	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Organic light emitting diodes OLEDs Construction working principle and light emission mechanism (converted PDF; original hash a9b807e70297b390226c96b5b111ec2155b7d8d040bb5fb99cf1fc594aa8b61d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Green chemistry Introduction twelve principles of green chemistry with examples	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Green chemistry Introduction twelve principles of green chemistry with examples	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Green chemistry Introduction twelve principles of green chemistry with examples (converted PDF; original hash ec2cb5da0a96641aa41cc7de99dfe13473d90ebceb1004effb11919c74e3caf1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Based on nanomaterials	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Based on nanomaterials	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Based on nanomaterials (converted PDF; original hash a633a07de184f9133999395e1afcbe5124c4399b4824bea6778640e31355d34b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 10__1vFS6E1bKV.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/10__1vFS6E1bKV.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2__11pj9jsL0V.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/2__11pj9jsL0V.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 3__1MwY1CTl_j.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/3__1MwY1CTl_j.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 4__1eKMFaJ2lx.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/4__1eKMFaJ2lx.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 5__1pIeTzocnU.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/5__1pIeTzocnU.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 6__1H6QH_PVNf.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/6__1H6QH_PVNf.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 8__1015pA2nzY.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/8__1015pA2nzY.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 9__14fBQb1-hv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/9__14fBQb1-hv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-EC[2025]__1ZtWpoG9Dv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-EC[2025]__1ZtWpoG9Dv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV00000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV00000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9960; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 0.9960; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 0.9960; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 0.9960; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9969; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000022	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000023	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000024	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9969; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000025	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000026	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000027	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000028	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9969; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000029	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000030	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000031	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000032	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 0.9969; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000033	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000034	SRC00006/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000035	SRC00007/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000036	SRC00008/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 0.9969; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000037	SRC00009/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000038	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000039	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000040	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9970; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000044	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 0.9970; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000045	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000046	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000047	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000048	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 0.9970; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000049	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000050	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000051	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 0.9970; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000057	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000058	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000065	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000066	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000067	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000068	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0078. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000069	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000070	SRC00014/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0079 to s0099. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000071	SRC00015/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0079 to s0099. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000072	SRC00016/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0079 to s0099. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00017/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0079 to s0099. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00014/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00015/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00016/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0101. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00017/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000079	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000080	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000081	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000086	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000087	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000088	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000089	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000090	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000091	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000092	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000093	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00018/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00019/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00020/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0088. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00021/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 00	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 01	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 02	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 03	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 04	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 05	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 06	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000107	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000112	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000113	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000114	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000115	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000116	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000117	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000118	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000119	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000120	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000121	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9975; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000126	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000127	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 28	SRC00032/t00030 0-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare at similarity 0.9975; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 29	SRC00033/t00030 0-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 30	SRC00030/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 31	SRC00031/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 32	SRC00032/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 0.9975; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 33	SRC00033/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 34	SRC00034/t00000 0-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000135	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9955; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000140	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9955; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000141	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000142	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9955; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000146	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000147	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000148	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 49	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 50	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 51	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 52	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 53	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 54	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 55	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000156	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000163	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000168	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000169	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000170	SRC00042/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000171	SRC00043/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000172	SRC00044/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0064. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00045/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000174	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000175	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000176	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000177	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000181	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000182	SRC00046/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000183	SRC00047/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000184	SRC00048/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000185	SRC00049/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000186	SRC00046/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000187	SRC00047/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000188	SRC00048/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000189	SRC00049/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000190	SRC00046/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000191	SRC00047/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000192	SRC00048/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000193	SRC00049/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000194	SRC00050/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000195	SRC00052/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000196	SRC00051/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00001 97	SRC00050/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 98	SRC00052/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00001 99	SRC00051/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 00	SRC00050/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 01	SRC00052/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000202	SRC00051/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000203	SRC00050/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000204	SRC00052/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000205	SRC00051/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000206	SRC00050/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000207	SRC00052/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000208	SRC00051/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000209	SRC00050/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000210	SRC00052/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000211	SRC00051/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000212	SRC00050/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000213	SRC00052/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000214	SRC00051/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000215	SRC00050/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000216	SRC00052/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000217	SRC00051/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000218	SRC00050/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000219	SRC00052/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000220	SRC00051/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000221	SRC00050/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000222	SRC00052/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000223	SRC00051/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000224	SRC00050/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000225	SRC00052/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000226	SRC00051/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000227	SRC00050/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000228	SRC00052/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000229	SRC00051/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000230	SRC00050/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000231	SRC00052/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000232	SRC00051/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000233	SRC00050/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000234	SRC00052/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000235	SRC00051/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000236	SRC00053/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000237	SRC00055/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000238	SRC00054/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000239	SRC00053/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000240	SRC00055/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000241	SRC00054/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000242	SRC00053/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000243	SRC00055/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000244	SRC00054/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000245	SRC00053/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000246	SRC00055/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000247	SRC00054/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000248	SRC00053/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000249	SRC00055/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000250	SRC00054/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000251	SRC00053/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000252	SRC00055/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000253	SRC00054/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000254	SRC00053/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000255	SRC00055/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000256	SRC00054/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000257	SRC00053/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000258	SRC00055/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000259	SRC00054/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000260	SRC00053/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000261	SRC00055/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000262	SRC00054/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000263	SRC00053/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000264	SRC00055/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000265	SRC00054/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000266	SRC00053/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000267	SRC00055/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000268	SRC00054/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000269	SRC00053/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000270	SRC00055/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000271	SRC00054/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000272	SRC00053/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000273	SRC00055/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000274	SRC00054/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000275	SRC00053/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000276	SRC00055/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000277	SRC00054/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000278	SRC00053/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000279	SRC00055/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000280	SRC00054/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000281	SRC00053/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000282	SRC00055/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000283	SRC00054/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000284	SRC00053/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000285	SRC00055/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000286	SRC00054/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000287	SRC00053/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000288	SRC00055/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000289	SRC00054/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000290	SRC00053/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000291	SRC00055/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000292	SRC00054/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000293	SRC00053/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000294	SRC00055/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000295	SRC00054/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000296	SRC00053/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 97	SRC00055/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 98	SRC00054/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 99	SRC00056/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 00	SRC00058/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 01	SRC00057/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 02	SRC00056/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000303	SRC00058/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000304	SRC00057/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000305	SRC00056/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000306	SRC00058/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000307	SRC00057/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000308	SRC00056/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000309	SRC00058/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000310	SRC00057/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000311	SRC00056/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000312	SRC00058/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000313	SRC00057/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000314	SRC00056/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000315	SRC00058/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000316	SRC00057/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000317	SRC00056/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000318	SRC00058/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000319	SRC00057/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000320	SRC00056/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000321	SRC00058/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000322	SRC00057/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000323	SRC00056/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000324	SRC00058/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000325	SRC00057/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000326	SRC00056/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000327	SRC00058/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000328	SRC00057/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000329	SRC00056/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000330	SRC00058/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000331	SRC00057/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000332	SRC00056/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000333	SRC00058/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000334	SRC00057/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000335	SRC00056/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000336	SRC00058/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000337	SRC00057/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000338	SRC00056/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000339	SRC00058/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000340	SRC00057/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000341	SRC00056/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000342	SRC00058/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000343	SRC00057/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000344	SRC00056/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 45	SRC00058/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 46	SRC00057/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 47	SRC00056/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 48	SRC00058/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 49	SRC00057/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 50	SRC00056/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000351	SRC00058/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000352	SRC00057/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000353	SRC00059/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000354	SRC00061/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000355	SRC00060/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000356	SRC00059/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000357	SRC00061/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000358	SRC00060/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000359	SRC00059/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000360	SRC00061/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000361	SRC00060/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000362	SRC00059/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000363	SRC00061/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000364	SRC00060/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000365	SRC00059/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000366	SRC00061/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000367	SRC00060/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000368	SRC00059/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000369	SRC00061/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000370	SRC00060/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000371	SRC00059/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000372	SRC00061/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000373	SRC00060/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000374	SRC00059/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003375	SRC00061/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003376	SRC00060/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00003377	SRC00059/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003378	SRC00061/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003379	SRC00060/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00003380	SRC00059/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000381	SRC00061/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000382	SRC00060/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000383	SRC00059/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000384	SRC00061/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000385	SRC00060/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000386	SRC00059/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000387	SRC00061/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000388	SRC00060/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000389	SRC00059/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000390	SRC00061/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000391	SRC00060/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000392	SRC00059/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000393	SRC00061/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000394	SRC00060/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000395	SRC00059/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000396	SRC00061/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000397	SRC00060/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000398	SRC00059/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000399	SRC00061/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000400	SRC00060/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000401	SRC00062/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000402	SRC00064/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000403	SRC00063/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000404	SRC00062/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000405	SRC00064/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000406	SRC00063/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000407	SRC00062/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000408	SRC00064/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000409	SRC00063/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000410	SRC00062/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000411	SRC00064/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000412	SRC00063/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000413	SRC00062/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000414	SRC00064/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000415	SRC00063/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000416	SRC00062/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000417	SRC00064/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000418	SRC00063/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000419	SRC00065/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000420	SRC00067/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000421	SRC00066/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000422	SRC00065/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000423	SRC00067/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000424	SRC00066/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000425	SRC00065/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000426	SRC00067/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000427	SRC00066/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000428	SRC00065/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000429	SRC00067/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000430	SRC00066/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000431	SRC00065/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000432	SRC00067/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000433	SRC00066/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000434	SRC00065/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000435	SRC00067/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000436	SRC00066/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000437	SRC00065/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000438	SRC00067/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000439	SRC00066/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000440	SRC00065/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000441	SRC00067/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000442	SRC00066/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000443	SRC00068/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000444	SRC00070/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000445	SRC00069/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000446	SRC00068/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000447	SRC00070/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000448	SRC00069/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000449	SRC00068/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000450	SRC00070/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000451	SRC00069/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000452	SRC00068/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000453	SRC00070/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000454	SRC00069/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000455	SRC00068/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000456	SRC00070/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000457	SRC00069/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000458	SRC00068/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000459	SRC00070/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000460	SRC00069/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000461	SRC00068/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000462	SRC00070/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000463	SRC00069/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000464	SRC00068/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004065	SRC00070/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004066	SRC00069/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00004067	SRC00068/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004068	SRC00070/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004069	SRC00069/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00004070	SRC00068/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000471	SRC00070/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000472	SRC00069/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000473	SRC00068/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000474	SRC00070/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000475	SRC00069/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000476	SRC00068/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000477	SRC00070/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000478	SRC00069/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000479	SRC00068/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000480	SRC00070/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000481	SRC00069/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000482	SRC00068/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000483	SRC00070/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000484	SRC00069/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000485	SRC00068/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000486	SRC00070/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000487	SRC00069/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000488	SRC00068/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000489	SRC00070/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000490	SRC00069/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000491	SRC00068/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000492	SRC00070/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000493	SRC00069/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000494	SRC00068/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004 95	SRC00070/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 96	SRC00069/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 97	SRC00071/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 98	SRC00073/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 99	SRC00072/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 00	SRC00071/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000501	SRC00073/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000502	SRC00072/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000503	SRC00071/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000504	SRC00073/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000505	SRC00072/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000506	SRC00071/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000507	SRC00073/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000508	SRC00072/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000509	SRC00071/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000510	SRC00073/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000511	SRC00072/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000512	SRC00071/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000513	SRC00073/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000514	SRC00072/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000515	SRC00071/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000516	SRC00073/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000517	SRC00072/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000518	SRC00071/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000519	SRC00073/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000520	SRC00072/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000521	SRC00071/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000522	SRC00073/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000523	SRC00072/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000524	SRC00071/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000525	SRC00073/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000526	SRC00072/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000527	SRC00071/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000528	SRC00073/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000529	SRC00072/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000530	SRC00071/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000531	SRC00073/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000532	SRC00072/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000533	SRC00071/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000534	SRC00073/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000535	SRC00072/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000536	SRC00071/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000537	SRC00073/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000538	SRC00072/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000539	SRC00074/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000540	SRC00076/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000541	SRC00075/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000542	SRC00074/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000543	SRC00076/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000544	SRC00075/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000545	SRC00074/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000546	SRC00076/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000547	SRC00075/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000548	SRC00074/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000549	SRC00076/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000550	SRC00075/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000551	SRC00074/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000552	SRC00076/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000553	SRC00075/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000554	SRC00074/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000555	SRC00076/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000556	SRC00075/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000557	SRC00074/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000558	SRC00076/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000559	SRC00075/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000560	SRC00074/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000561	SRC00076/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000562	SRC00075/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000563	SRC00074/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000564	SRC00076/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000565	SRC00075/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000566	SRC00074/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000567	SRC00076/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000568	SRC00075/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000569	SRC00074/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000570	SRC00076/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000571	SRC00075/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000572	SRC00074/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000573	SRC00076/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000574	SRC00075/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000575	SRC00074/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000576	SRC00076/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000577	SRC00075/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000578	SRC00077/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 79	SRC00079/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 80	SRC00078/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 81	SRC00077/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 82	SRC00079/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 83	SRC00078/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 84	SRC00077/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000585	SRC00079/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000586	SRC00078/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000587	SRC00077/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000588	SRC00079/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000589	SRC00078/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000590	SRC00077/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000591	SRC00079/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000592	SRC00078/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000593	SRC00077/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000594	SRC00079/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000595	SRC00078/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000596	SRC00077/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000597	SRC00079/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000598	SRC00078/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000599	SRC00077/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000600	SRC00079/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000601	SRC00078/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000602	SRC00077/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000603	SRC00079/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000604	SRC00078/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000605	SRC00077/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000606	SRC00079/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000607	SRC00078/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000608	SRC00077/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000609	SRC00079/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000610	SRC00078/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000611	SRC00077/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000612	SRC00079/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000613	SRC00078/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000614	SRC00077/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000615	SRC00079/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000616	SRC00078/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000617	SRC00077/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000618	SRC00079/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000619	SRC00078/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000620	SRC00080/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000621	SRC00082/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000622	SRC00081/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000623	SRC00080/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000624	SRC00082/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000625	SRC00081/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000626	SRC00080/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000627	SRC00082/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000628	SRC00081/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000629	SRC00080/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000630	SRC00082/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000631	SRC00081/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000632	SRC00080/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000633	SRC00082/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000634	SRC00081/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000635	SRC00080/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000636	SRC00082/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000637	SRC00081/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000638	SRC00080/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000639	SRC00082/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000640	SRC00081/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000641	SRC00080/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000642	SRC00082/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000643	SRC00081/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000644	SRC00080/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000645	SRC00082/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000646	SRC00081/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000647	SRC00080/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000648	SRC00082/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000649	SRC00081/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000650	SRC00080/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000651	SRC00082/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000652	SRC00081/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000653	SRC00080/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000654	SRC00082/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000655	SRC00081/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000656	SRC00080/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000657	SRC00082/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000658	SRC00081/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000659	SRC00080/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000660	SRC00082/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000661	SRC00081/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000662	SRC00080/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000663	SRC00082/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000664	SRC00081/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000665	SRC00080/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000666	SRC00082/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000667	SRC00081/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000668	SRC00080/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000669	SRC00082/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000670	SRC00081/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000671	SRC00080/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000672	SRC00082/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000673	SRC00081/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000674	SRC00080/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000675	SRC00082/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000676	SRC00081/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000677	SRC00080/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000678	SRC00082/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000679	SRC00081/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000680	SRC00080/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000681	SRC00082/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000682	SRC00081/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000683	SRC00080/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000684	SRC00082/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000685	SRC00081/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000686	SRC00080/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000687	SRC00082/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000688	SRC00081/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.
EV0000689	SRC00080/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000690	SRC00082/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000691	SRC00081/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000692	SRC00083/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000693	SRC00085/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000694	SRC00084/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000695	SRC00083/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000696	SRC00085/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000697	SRC00084/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000698	SRC00083/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000699	SRC00085/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000700	SRC00084/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000701	SRC00086/q46	covered	Original archive PDF page 46; candidate unit association is text-based and unconfirmed.
EV0000702	SRC00087/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV0000703	SRC00087/q7	covered	Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
EV0000704	SRC00088/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000705	SRC00089/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000706	SRC00090/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000707	SRC00091/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000708	SRC00092/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000709	SRC00092/q7	covered	Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
EV0000710	SRC00093/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000711	SRC00093/q7	covered	Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
EV0000712	SRC00094/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV0000713	SRC00094/q7	covered	Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
EV0000714	SRC00095/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000715	SRC00096/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000716	SRC00096/q7	covered	Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
EV0000717	SRC00097/q8	covered	Original archive PDF page 8; candidate unit association is text-based and unconfirmed.
EV0000718	SRC00098/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000719	SRC00098/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000720	SRC00099/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.

C Revision and retrieval practice